CHAPTER – 1 INDICES, LOGATRITHM, RATIO-PROPORTION

INDICES

Q.01. $x^2 \times x^3$ = (a) x^{-5} (b) x^5 (c) x^6 (d) None

Q.02. $\frac{x^6}{x^2}$ = (a) x^5 (b) x^4 (c) x^6 (d) None

Q.03. $(x^3)^2$ = (a) x^3 (b) x^{-6} (c) x^6 (d) None

Q.04. $\sqrt{x}$ = (a) $x^{\frac{1}{2}}$ (b) $x^{\frac{-1}{2}}$ (c) X (d) None

Q.05. $\sqrt[2]{x}$ = (a) $x^{\frac{\sqrt{X}}{2}}$ (b) $x^{\frac{1}{2}}$ (c) $x^{\frac{-1}{2}}$ (d) None

Q.06. $\sqrt[3]{x}$ = (a) $3x^{\frac{1}{3}}$ (b) $x^{\frac{-1}{3}}$ (c) $x^{\frac{1}{3}}$ (d) None

Q.07. $\sqrt[5]{x}$ = (a) $x^{\frac{1}{5}}$ (b) $5x^{\frac{1}{5}}$ (c) $x^{\frac{-1}{5}}$ (d) None

Q.08. $\sqrt{x+y}$ = (a) $\sqrt{x-y}$ (b) $\sqrt{x} + \sqrt{y}$ (c) xy (d) None

Q.09. $\sqrt{x-y}$ = (a) $\sqrt{x} - \sqrt{y}$ (b) $\sqrt{x} + \sqrt{y}$ (c) xy (d) None

Q.10. $\sqrt{x \times y}$ = (a) $\sqrt{x+y}$ (b) $\sqrt{x} \times \sqrt{y}$ (c) xy (d) None

Q.11. $\sqrt{\frac{x}{y}}$ = (a) $\frac{2\sqrt{x}}{\sqrt{y}}$ (b) $\frac{-\sqrt{x}}{\sqrt{y}}$ (c) $\frac{\sqrt{x}}{\sqrt{y}}$ (d) None

Q.12. $\sqrt{x} \times \sqrt{x}$ = (a) $\sqrt{x}$ (b) x (c) –x (d) None

Q.13. $(x + y)^n$ = (a) x + y (b) $x^n + y^n$ (c) xy (d) None

Q.14. $(x \times y)^n$ = (a) $x^n \times y^n$ (b) $x \times y$ (c) xy (d) None

Q.15. $\left(\frac{x}{y}\right)^n$ = (a) $\frac{-x^n}{y^n}$ (b) $\frac{x^n}{y^n}$ (c) $\frac{x}{y}$ (d) None

Q.16. $\frac{x+y}{n}$ = (a) $\frac{x}{n} + \frac{y}{n}$ (b) $\frac{n}{x+y}$ (c) xy (d) None

Q.17. $\frac{x-y}{n}$ = (a) $\frac{n}{x-y}$ (b) $\frac{x}{n} - \frac{y}{n}$ (c) xy (d) None

Q.18. $\frac{x \times y}{n}$ = (a) $\frac{x}{n} \times \frac{y}{n}$ (b) $\frac{x}{n} - \frac{y}{n}$ (c) xy (d) None

Q.19. $\frac{x \div y}{n}$ = (a) $\frac{x}{n} - \frac{y}{n}$ (b) $\frac{x}{n} \div \frac{y}{n}$ (c) xy (d) None

Q.20. x° = (a) 1 (b) x (c) 2x° (d) None

Q.21. x^{-n} = (a) $\frac{-1}{x^n}$ (b) $\frac{n}{x^n}$ (c) $\frac{1}{x^n}$ (d) None

Q.22. $\left(\frac{x}{y}\right)^{-n}$ = (a) $\left(\frac{y}{x}\right)^{n}$ (b) $\left(-\frac{y}{x}\right)^{n}$ (c) $\frac{y}{x}$ (d) None

Q.23. x^n = (a) $\frac{1}{x^n}$ (b) $\frac{1}{x^{-n}}$ (c) $\frac{-1}{x^{-n}}$ (d) None

Q.24. $(8)^{\frac{1}{3}}$ = (a) 8 (b) 2 (c) 0 (d) None

Q.25. $(32)^{\frac{1}{5}}$ = (a) 0 (b) 2 (c) 32 (d) None

Q.26. $2 \times (256)^{-\frac{1}{8}}$ = (a) 1 (b) 0 (c) 256 (d) None

Q.27. $\left(\frac{8}{27}\right)^{\frac{1}{3}}$ = (a) $\frac{-2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) None

Q.28. $\left(\frac{81 \times x^4}{y^{-8}}\right)^{\frac{1}{4}}$ = (a) xy (b) $3xy^2$ (c) xy^2 (d) None

Q.29. $\left[\frac{2p^2 q^3}{3xy}\right]^{\circ}$ = (a) 1 (b) $3xy$ (c) –1 (d) None

Q.30. $x^{a-b} \times x^{b-c} \times x^{c-a}$ =
(a) xy (b) 1 (c) 0 (d) None

Q.31. $2^{\frac{1}{2}} \times 4^{\frac{3}{4}}$ = (a) 4 (b) 2 (c) 1 (d) None

Q.32. $\left(\frac{8}{27}\right)^{-\frac{1}{3}} \times \left(\frac{32}{243}\right)^{-\frac{1}{5}}$ =
(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{9}{4}$ (d) None

Q.33. $16 \times x^{-3} \times y^2 \times 8^{-1} \times x^3 \times y^{-2}$ =
(a) 1 (b) 2 (c) – 2 (d) None

Q.34. $x^{a-b} \times x^{b-c} \times x^{c-a} \times x^{-a-b}$ =
(a) $\frac{1}{x^{(a+b)}}$ (b) $\frac{1}{x^{(a-b)}}$ (c) $\frac{1}{(a+b)}$ (d) None

Q.35. $(125)^{\frac{2}{3}} \times \sqrt{25} \times \sqrt[3]{5^3} \times 5^{\frac{1}{2}}$ =
(a) $625\sqrt{4}$ (b) $625\sqrt{5}$ (c) $625\sqrt{6}$ (d) None

Q.36. $\left[\{2^{\frac{1}{2}} \times 4^{\frac{3}{4}} \times 8^{\frac{5}{6}} \times 16^{\frac{7}{8}} \times 32^{\frac{9}{10}}\}^4\right]^{\frac{3}{25}}$ =
(a) 64 (b) 51 (c) 69 (d) None

Q.37. $[1 - \{1 - (1-x^2)^{-1}\}^{-1}]^{\frac{-1}{2}}$
(a) x (b) 1 (c) 0 (d) None

Q.38. $\left(\frac{x^a}{x^b}\right)^{(a+b)} \times \left(\frac{x^b}{x^c}\right)^{(b+c)} \times \left(\frac{x^c}{x^a}\right)^{(c+a)} =$

(a) – 1 (b) 1 (c) 0 (d) None

Q.39. $\left(\frac{x^a}{x^b}\right)^{a^2+ab+b^2} \times \left(\frac{x^b}{x^c}\right)^{b^2+bc+c^2} \times \left(\frac{x^c}{x^a}\right)^{(c^2+ac+a^2)}$

(a) $\frac{x^a}{x^b}$ (b) – 1 (c) 1 (d) None

Q.40. If $a^x = b$, $b^y = c$, and $c^z = a$, then find the value of xyz

(a) 1 (b) 0 (c) xyz (d) None

Q.41. If $x = P^{\frac{1}{3}} - P^{-\frac{1}{3}}$ then, which of the following option is correct.

(a) $x^2 + 3x = p + \frac{1}{p}$ (b) $x^3 + 3x = p - \frac{1}{p}$

(c) $x^3 + 3x = 0$ (d) None of these.

Q.42. If $x = 3^{\frac{1}{3}} + 3^{-\frac{1}{3}}$, then find the value of $3x^3 - 9x$

(a) 3 (b) 9 (c) 10 (d) None

Q.43. $X = 5^{\frac{1}{3}} + 5^{-\frac{1}{3}}$, then find the value of $5x^3 - 15x$

(a) 28 (b) 26 (c) 30 (d) None

Q.44. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{7}$, then find the value $\frac{a+b+c}{c}$

(a) 2 (b) 3 (c) 7 (d) None

Q.45. If $\frac{x}{2} = \frac{y}{3} = \frac{z}{7}$, then find the value of $\frac{2x+4z-5y}{2y}$

(a) $\frac{-17}{6}$ (b) $\frac{11}{6}$ (c) $\frac{17}{6}$ (d) None

Q.46. $2^x = 3^y = 6^{-z}$ then find the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

(a) 1 (b) 0 (c) $\frac{1}{2}$ (d) None

Q.47. $(5.678)^x = (0.5678)^y = 10^z$ then find the value of $\frac{1}{x} - \frac{1}{y} - \frac{1}{z}$

(a) 0 (b) 10^2 (c) 10 (d) None

Q.48. $(4.8)^x = (0.48)^y = 1000$, then find the value of $\frac{1}{x} - \frac{1}{y}$

(a) 1/3 (b) 1/2 (c) 0 (d) None

Q.49. $2^a = 3^b = 12^c$, then find the value of $\frac{1}{c} - \frac{1}{b} - \frac{2}{a}$

(a) 1 (b) 0 (c) – 1 (d) None

Q.50. $3^a = 5^b = (75)^c$, then find the value of $ab - c(2a+b)$

(a) (2a + b) (b) 1 (c) 0 (d) None

Q.51. $x^{1/p} = y^{1/q} = z^{1/r}$ and xyz = 1 , then find the value of p + q + r

(a) 0 (b) 1 (c) – 1 (d) None

Q.52. $\frac{x}{b+c-a} = \frac{y}{c+a-b} = \frac{z}{a+b-c}$ find the value of $(b-c)x + (c-a)y + (a-b)z$

(a) (a + b) (b) 0 (c) (a – b) (d) None

Q.53. $\left(x^{\frac{1}{a-b}}\right)^{\frac{1}{a-c}} \times \left(x^{\frac{1}{b-c}}\right)^{\frac{1}{b-a}} \times \left(x^{\frac{1}{c-a}}\right)^{\frac{1}{c-b}}$ =

(a) 1 (b) $\frac{1}{c-b}$ (c) $\frac{1}{c \mp b}$ (d) None

Q. 54. $\sqrt[a+b]{\frac{x^{a^2}}{x^{b^2}}} \times \sqrt[b+c]{\frac{x^{b^2}}{x^{c^2}}} \times \sqrt[c+a]{\frac{x^{c^2}}{x^{a^2}}}$ =

(a) (a + b) (b) 1 (c) (a – b) (d) None

Q.55. $\left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \times \left(\frac{x^c}{x^a}\right)^{\frac{1}{ca}} \times \left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}}$ =

(a) $\frac{1}{ab}$ (b) 1 (c) – 1 (d) None

Q.56. $\left(x^{\frac{b+c}{c-a}}\right)^{\frac{1}{a-b}} \times \left(x^{\frac{c+a}{a-b}}\right)^{\frac{1}{b-c}} \times \left(x^{\frac{a+b}{b-c}}\right)^{\frac{1}{c-a}}$ =

(a) $\frac{1}{b+c}$ (b) $\frac{1}{b-c}$ (c) 1 (d) None

Q.57. $\left(\frac{x^a}{x^{-b}}\right)^{(a^2-ab+b^2)} \times \left(\frac{x^b}{x^{-c}}\right)^{(b^2-bc+c^2)} \times \left(\frac{x^c}{x^{-a}}\right)^{(c^2-ca+a^2)}$

(a) $x^{2(a^3+b^3+c^3)}$ (b) $x^{2(a^3-b^3+c^3)}$

(c) $x^{2(a^3-b^3-c^3)}$ (d) None

Q.58. Find the square root of $(3 + \sqrt{5})$

(a) $\sqrt{\frac{5}{2}} + \sqrt{\frac{1}{2}}$ (b) $\sqrt{\frac{5}{2}} - \sqrt{\frac{1}{2}}$ (c) $\sqrt{\frac{5}{2}} + \sqrt{\frac{3}{2}}$ (d) $\sqrt{\frac{3}{2}} + \sqrt{\frac{1}{2}}$

Q.59. Find the square root of $(11 - \sqrt{120})$

(a) $\sqrt{6} + \sqrt{5}$ (b) $\sqrt{6} - \sqrt{5}$ (c) $2\sqrt{3} - 3\sqrt{2}$ (d) $2\sqrt{3} + 3\sqrt{2}$

Q.60. Find the square root of $x + \sqrt{x^2 - y^2}$

(a) $\frac{1}{\sqrt{2}}\left[\sqrt{x+y} + \sqrt{x-y}\right]$ (b) $\frac{1}{\sqrt{2}}\left[\sqrt{x+y} - \sqrt{x-y}\right]$

(c) $\left[\sqrt{x+y} + \sqrt{x-y}\right]$ (d) $\left[\sqrt{x+y} - \sqrt{x-y}\right]$

Q.61. $\sqrt{x^3\sqrt{x^3\sqrt{x^3}}}$ is :-

(a) x^7 (b) x^8 (c) $x^{21/8}$ (d) $\frac{1}{x^{21/8}}$

Q.62. (;fn) If $3^x = 4^y = 12^z$, then z is equal to (rks z dk eku gksxk)

(a) $\frac{x+y}{xy}$ (b) $\frac{xy}{x+y}$ (c) $\frac{y}{x-y}$ (d) $\frac{x}{x-y}$

Q.63. If $\left(\frac{2}{3}\right)^{x} \times \left(\frac{3}{2}\right)^{2x} = \left(\frac{27}{8}\right)$ then x is equal to :-
(a) 1 (b) 2 (c) 3 (d) 4

Q.64. If $a^x \div a^2 = 1$ then the value of x is:-
(a) 4 (b) 2 (c) 6 (d) 0

Q.65. If x = 81, then the value of $(x^{1/4}-1).(x^{1/4}+1)$ will be
(a) 4 (b) 8 (c) x^0 (d) None

Q.66. The value of $\frac{2^n+2^{n-1}}{2^{n+1}-2^n}$ is :-
(a) 2 (b) 8 (c) 3/2 (d) 2/3

Q.67. If $2^x = 3^y = 6^{-z}$, then $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is :-
(a) 0 (b) 1 (c) 36 (d) –1

Q.68. If $2^x - 2^{x-1} = 4$, then the value of x^x is :-
(a) 2 (b) 1 (c) 64 (d) 27

Q.69. Find x, if $x^{-1} = 3^{-1} - 4^{-1}$
(a) 1/12 (b) 1/4 (c) 12 (d) 13

Q.70. If $4^{2x} - 2^{2x} = 12$ then x is :-
(a) 4 (b) 2 (c) 1 (d) 0

MISCELLANEOUS QUESTIONS OF *INDICES*

Q.01. Value of $\sqrt[4]{16^{-3}}$
(a) 8 (b) 1/8 (c) 4 (d) 1/4

Q.02. The value of $\left(\frac{243}{32}\right)^{-4/5}$
(a) 4/9 (b) 16/81 (c) 2/3 (d) 3/2

Q.03. If $2\times 3^{x-1} = 162$, then x is :-
(a) 2 (b) 4 (c) 5 (d) 0

Q.04. If $6^{x+3} = (36)^{3x-4}$ then x is :-
(a) 5 (b) 11/5 (c) 5/11 (d) 4/3

Q.05. Value of $\sqrt{a\sqrt{b\sqrt{c\sqrt{d}}}}$ is :-
(a) $a^{1/2}\, b^{1/2}\, c^{1/2}\, d^{1/2}$ (b) $a^{1/2}\, b^{1/4}\, c^{1/8}\, d^{1/16}$
(c) $(abcd)^{1/2}$ (d) $(abcd)^{1/8}$

Q.06. $\sqrt{p^3.q^{-2/3}.r^{-7/6}} \div \sqrt{p^4.q^{4/3}.r^{5/6}}$ can be simplified to

(a) $\frac{1}{pq}\sqrt{\frac{1}{r}}$ (b) $\frac{1}{qr}\sqrt{\frac{1}{p}}$ (c) $\frac{1}{pr}\sqrt{\frac{1}{q}}$ (d) $\frac{1}{r}\sqrt{\frac{1}{pq}}$

Q.07. $\left[(a^m)^{m-\frac{1}{m}}\right]^{\frac{1}{m+1}}$ =

(a) a^m (b) a^{m+1} (c) a^{m-1} (d) a^{m+2}

Q.08. If $9 \times 81^x = (1/27)^{x-2}$ then x is equal to

(a) 1 (b) 2 (c) 4/7 (d) 4

Q.09. If $\frac{9^n\,(3^2)(3^{-\frac{n}{2}})^{-2} - 27^n}{3^{3m}(2^3)} = \frac{1}{27}$, then

(a) m – n = 2 (b) m – n = 1 (c) m – n = –2 (d) m – n = –1

Q.10. The value of $5^{1/4} \times (125)^{0.25}$ is:-

(a) $\sqrt{5}$ (b) 5 (c) $\sqrt[3]{5}$ (d) 25

Q.11. $\frac{2^{n+4} - 2.2^n}{2.2^{n+3}} + 2^{-3}$ is equal to :-

(a) 2^{n+1} (b) $-2^{n+1} + \left(\frac{1}{8}\right)$ (c) $\left(\frac{9}{8}\right) - 2^n$ (d) 1

Q.12. If $5\sqrt{5} \times 5^3 \div 5^{-3/2} = 5^{a+2}$, then the value of a is :-

(a) 4 (b) 5 (c) 6 (d) 8

Q.13. $\sqrt{a^5}.a^{3/4}.\sqrt[4]{a^{-5}}.1/a^{-3}$ =

(a) a^5 (b) a^2 (c) $\frac{1}{a^5}$ (d) a

Q.14. The value of $\sqrt[8]{b^2.\sqrt[4]{b^{-3}}}$

(a) b^{32} (b) $b^{\frac{1}{32}}$ (c) $b^{\frac{5}{32}}$ (d) b

Q.15. $\left(\frac{1}{1+x^{n-m}} + \frac{1}{1+x^{m-n}}\right)$ =

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) x^{m+n}

Q.16. If $2^{x+4} - 2^{x+2} = 3$, then *x* is equal to :-

(a) 0 (b) 2 (c) -1 (d) -2

Q.17. If $3^x = 4^y = 12^{-z}$, then $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ is equal to :-

(a) 0 (b) 1 (c) $\frac{3}{2}$ (d) $\frac{-1}{2}$

Q.18. If m and n are whole numbers such that $m^n = 121$,
Then the value of $(m-1)^{n+1}$ is :-

(a) 1 (b) 10 (c) 121 (d) 1000

Q.19. If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then values of a and b are:-

(a) 2, 3 (b) $-2, 3$ (c) 2, -3 (d) 3, 2

Q.20. If $32 \times 8^{n+3} = 2^m$, then m is equal to :-
(a) n + 12 (b) 2n + 14 (c) 3n + 12 (d) 3n + 14

Q.21. If $X = 4^{1/3} + 4^{-1/3}$, then $4X^3 - 12X$ is given by
(a) 12 (b) 13 (c) 15 (d) 17

Q.22. If $(1.001)^x = (0.1001)^y = 10^z$ then :-
(a) $\frac{1}{x} - \frac{1}{y} + \frac{1}{z} = 1$ (b) $\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = 0$
(c) $\frac{1}{x} - \frac{1}{y} + \frac{1}{z} = -1$ (d) None of these

Q.23. If $a^m = b^n = (ab)^{mn}$ then (m + n) is :-
(a) 1 (b) 2 (c) 3 (d) 4

Q.24. If $x^{x\sqrt{x}} = \left(x\sqrt{x}\right)^x$ then x is equal to
(a) 1/2 (b) $\frac{3}{2}$ (c) $\frac{9}{4}$ (d) None of these

Q.25. $64^{-\frac{1}{2}} - (-32)^{-\frac{4}{5}} = x$, then the value of x is :-
(a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{1}{16}$ (d) $\frac{3}{16}$

Q.26. The value of $\frac{(243)^{0.13} \times (243)^{0.07}}{(7)^{0.25} \times (49)^{0.075} \times (343)^{0.2}}$ is :-
(a) $\frac{3}{7}$ (b) $\frac{7}{3}$ (c) $\frac{5}{7}$ (d) $\frac{2}{7}$

Q.27. Given that $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is :-
(a) 1.45 (b) 1.88 (c) 2.9 (d) 3.7

Q.28. The quantity $\frac{(0.6)^0 - (0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1}\left(\frac{3}{2}\right)^3 + \left(-\frac{1}{3}\right)^{-1}}$ is equal to :-
(a) $-\frac{3}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

Q.29. If $2^x = 4^y = 8^z$ and $\left(\frac{1}{2x} + \frac{1}{4y} + \frac{1}{6z}\right) = \frac{24}{7}$ then the value of z is :-
(a) $\frac{7}{16}$ (b) $\frac{7}{32}$ (c) $\frac{7}{48}$ (d) $\frac{7}{64}$

Q.30. If $m = x^a$, $n = x^b$, $p = x^c$, then $m^{b-c}.n^{c-a}.p^{a-b}$ equals to :-
(a) 0 (b) 1 (c) –1 (d) 2

Q.31. On simplification $\left[\frac{x^{\frac{a}{a-b}}}{x^{\frac{a}{a+b}}} \times \frac{x^{\frac{b}{b-a}}}{x^{\frac{b}{b+a}}}\right]^{a+b}$ reduces to : -
(a) 1 (b) –1 (c) 0 (d) None

LOGARITHM

Q.01. log6 + log5
(a) log30 (b) log20 (c) log11 (d) None

Q.02. log $\frac{32}{4}$
(a) 2.log2 (b) 3.log2 (c) 4.log2 (d) None

Q.03. log $(1\times 2\times 3)$
(a) log1+log2+log3 (b) log6 (c) log(1+2+3) (d) All

Q.04. 2.logx = 4.log3 then find the value of x
(a) 8 (b) 9 (c) 0 (d) None

Q.05. If log2 = 0.3010 and log3 = 0.4771 then find the value of log6
(a) 0.7782 (b) 0.7780 (c) 0.7781 (d) None

Q.06. log x + log y = log (x + y) Then y = ?
(a) y = x – 1 (b) y = $\frac{x}{x-1}$ (c) y = $\frac{x-1}{x}$ (d) y = $\frac{x}{x+1}$

Q.07. 2.log 5 + log 8 $-\frac{1}{2}$ log 4 =
(a) 2 (b) 4 (c) 6 (d) None

Q.08. log$[1-\{1-(1-x^2)^{-1}\}^{-1}]^{\frac{-1}{2}}$
(a) 2log (b) log y (c) log x (d) None

Q.09. log $\sqrt[4]{729\ \sqrt[3]{9^{-1}\times 27^{-4/3}}}$
(a) log 2 (b) log 3 (c) log 4 (d) None

Q.10. $\log_2 8$
(a) log 3 (b) log 2 (c) 3 (d) None

Q.11. $\log_2 64$
(a) 2 (b) 6 (c) 9 (d) None

Q.12. $\log_{\sqrt{2}} 64$
(a) 12 (b) 18 (c) 21 (d) None

Q.13. $\log_{2\sqrt{3}} 1728$
(a) 28 (b) 3 (c) 6 (d) None

Q.14. $\log_9 \left(\frac{1}{81}\right)$
(a) 2 (b) – 2 (c) 1 (d) None

Q.15. $\log_9 0.\overline{3}$
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{-1}{2}$ (d) None

Q.16. $\log_2 0.0625$

(a) – 4 (b) 4 (c) 2 (d) None

Q.17. $\log_2 \log_2 \log_2 16$
(a) 0 (b) 1 (c) – 1 (d) None

Q.18. $\log_2 [\log_2 \{\log_3 (\log_3 27^3)\}]$
(a) 1 (b) 0 (c) – 1 (d) None

Q.19. $(\log_b a \times \log_c b \times \log_a c)^3$ =
(a) 1 (b) $\log_c b$ (c) – 1 (d) None

Q.20. $\log_2 x + \log_4 x + \log_{16} x = \frac{21}{4}$ then find the value of x
(a) 4 (b) 6 (c) 8 (d) None

Q.21. If $\log_{10} 2$ = x, $\log_{10} 3$ = y, then find the value of $\log_{10} 60$ =
(a) x + y + 1 (b) x + y – 1 (c) 1 – x + y (d) None

Q.22. $\log_{10} 2$ = x, $\log_{10} 3$ = y then find value of $\log_{10} 1.2$
(a) 2x – y + 1 (b) 2x + y –1 (c) 2x + y + 1 (d) None

Q.23. log x = m + n, log y = m – n, then find the value of $\log\left(\frac{10x}{y^2}\right)$
(a) 1 + m + 3n (b) 1 – n + 3m (c) 1 – m + 3n (d) None

Q.24. If log2 = 0.3010, then find the value of $\log_8 25$
(a) 1.04817 (b) 1.54817 (c) 1.54148 (d) None

Q.25. $\log\left(\frac{a+b}{3}\right) = \frac{1}{2}[\log a + \log b]$ then, a/b + b/a = ?
(a) 7 (b) – 7 (c) 0 (d) None

Q.26. If $a^3 + b^3 = 0$, then find the value of $\log(a+b) - \frac{1}{2}[\log a + \log b + \log 3] =$
(a) log a (b) log b (c) 0 (d) None

Q.27. $\frac{\log a}{y-z} = \frac{\log b}{z-x} = \frac{\log c}{x-y}$, then abc = ?
(a) 1 (b) 21 (c) 0 (d) None

Q.28. $\frac{\log a}{y-z} = \frac{\log b}{z-x} = \frac{\log c}{x-y}$, then $a^{y+z} \times b^{z+x} \times c^{x+y}$ =
(a) 4 (b) 0 (c) 1 (d) None

Q.29. $x^{2a-3} \times y^{2a} = x^{6-a} \times y^{5a}$ then find the value of $a.\log\left(\frac{x}{y}\right)$ = ?
(a) log x (b) 3 log x (c) 2 log x (d) None

Q.30. If $x = \frac{e^n - e^{-n}}{e^n + e^{-n}}$ then find the value of n
(a) $\frac{1}{2}\log_e\left(\frac{1-x}{1+x}\right)$ (b) $\log_e\left(\frac{1+x}{1-x}\right)$ (c) $\frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right)$ (d) None

Q.31. If $\log a = \frac{1}{2}\log b = \frac{1}{5}\log c$, then find the value of $a^4.b^3.c^{-2}$

(a) 3 (b) 8 (c) 1 (d) None

Q.32. $a^{\log b - \log c} \times b^{\log c - \log a} \times c^{\log a - \log b}$ =
(a) 1 (b) 10 (c) 2 (d) None

Q.33. $(bc)^{\log^{b/c}} \times (ac)^{\log^{c/a}} \times (ab)^{\log^{a/b}}$
(a) – 1 (b) 1 (c) 0 (d) None

Q.34. $\frac{1}{\log_{ab}(abc)} + \frac{1}{\log_{bc}(abc)} + \frac{1}{\log_{ca}(abc)}$
(a) – 1 (b) 1 (c) 2 (d) None

Q.35. If $x = \log_a bc$, $y = \log_b ca$, $z = \log_c ab$, then $xyz - x - y - z$ = ?
(a) 2 (b) 1 (c) 0 (d) None

Q.36. If log2 = 0.3010 then the number of digits in 2^{64} is :–
(a) 10 (b) 20 (c) 19 (d) None

MISCELLANEOUS QUESTIONS OF *LOGARITHM*

Q.01. Formula $\log_b a \times \log_a b = 1$ is called
(a) Product formula (b) Quotient formula
(c) Reciprocal formula (d) power formula

Q.02. Logarithm of 1 to ---------- is equal to zero.
(a) base 0 (b) base 1 (c) base e (d) any base

Q.03. Logarithm of any number to the ---------- is 1.
(a) base 0 (b) base 1 (c) same base (d) any base

Q.04. If $\log_4(x^2 + x) - \log_4(x + 1) = 2$, then
(a) x = 2 (b) x = 4 (c) x = 5 (d) x = 16

Q.05. If $a.\log_2 x = b.\log_4 x$, then
(a) a = 2b (b) $a^2 = b$ (c) b = 2a (d) $b^2 = a$

Q.06. Find the value of $\log_3(27)^3$
(a) 27 (b) 9 (c) 3 (d) 81

Q.07. What is the value of $\frac{\log_{10}125}{\log_{10}25}$?
(a) 5 (b) 3/2 (c) 2 (d) 6

Q.08. If $\log_8 m + \log_8 2 = \frac{2}{3}$, then what is the value of m
(a) 1 (b) 3/2 (c) 2 (d) 0

Q.09. log 144 is equal to
(a) 2 log 4 + 2 log 2 (b) 4 log 2 + 2 log 3 (c) 3 log 2 + 4 log 3 (d) 3 log 2 – 4 log 3

Q.10. If $\log \frac{x+y}{5} = \frac{1}{2}(\log x + \log y)$, then $\frac{x}{y} + \frac{y}{x}$ =

(a) 20 (b) 23 (c) 22 (d) 21

Q.11. If $\frac{\log x}{2} = \frac{\log y}{3} = \frac{\log z}{5}$ then yz in term of x is :-
(a) x (b) x^2 (c) x^3 (d) x^4

Q.12. $\log_5 5 . \log_4 9 . \log_3 2$ =
(a) 2 (b) 1 (c) 5 (d) 3/2

Q.13. If $p^x = q$, then
(a) $\log_p x = q$ (b) $\log_x q = p$ (c) $\log_p q = x$ (d) $\log_q p = x$

Q.14. If $\log_{10} x = 7$, then x is given by
(a) 10^{10} (b) 10^7 (c) 7^{10} (d) None of these

Q.15. If $\log_a 3 = \frac{1}{3}$, then a is given by
(a) 27 (b) 81 (c) 72 (d) None of these

Q.16. If $\log_{10000} x = -1/4$, then x is given by
(a) 1/100 (b) 1/10 (c) 1/20 (d) None of these

Q.17. The value of $\log_{10}(0.00001)$ is given by
(a) –5 (b) – 6 (c) – 7 (d) None of these

Q.18. $\log_{27}(1/81)$ = ?
(a) – 4/3 (b) 3/4 (c) 1/3 (d) 4/3

Q.19. If $\log_5(x^2 + x) - \log_5 x = 2$, then the value of x is
(a) 24 (b) 25 (c) 23 (d) 120

Q.20. If $\frac{\log 125}{\log 5}$ = x then x is equal to
(a) 2 (b) 3 (c) 4 (d) 1/2

Q.21. $\log_9 27 - \log_{27} 9$ is equal to
(a) 6/5 (b) 5/6 (c) 3 (d) 3^2

Q.22. $\log_2(1/64)$ = ?
(a) 6 (b) 7 (c) – 6 (d) None of these

Q.23. $\log_5 5 . \log_4 9 . \log_3 2$ simplifies to
(a) 2 (b) 1 (c) 5 (d) None of these

Q.24. The value of $\log_2 \log_3 \log_3 27^3$ is :-
(a) 1 (b) 0 (c) 2 (d) 3

Q.25. The value of $2.\log_{10} 5 + \log_{10} 8 - \frac{1}{2}\log_{10} 4$ is equal to
(a) 10 (b) 8 (c) 2 (d) 0

Q.26. If $\log_2[\log_3(\log_2 x)] = 1$, then x =
(a) 128 (b) 512 (c) 256 (d) 64

Q.27. If $\log_{10}2 = x$ and $\log_{10}4 = y$, then $\log_{10}80$ =
(a) x – y + 1 (b) x + y + 1 (c) x – y – 1 (d) 2x – y – 1

Q.28. If $\log a = \frac{1}{2}\log b = \frac{1}{5}\log c$, then value of $a^4.b^3.c^{-2}$ is :-
(a) 0 (b) 1 (c) – 1 (d) 2

Q.29. $\log_2 0.125$ =
(a) – 4 (b) – 5 (c) – 1 (d) – 3

Q.30. If 2.log x + log y = log (x + y), then the correct relation between x and y is :-
(a) $y = \frac{x}{(x-1)}$ (b) $y = \frac{x^2}{(x^2-1)}$ (c) $y = \frac{x}{(x^2-1)}$ (d) $y = \frac{x^2}{(x^2+1)}$

Q.31. $\frac{1}{\log_{\frac{a}{b}} x} + \frac{1}{\log_{\frac{b}{c}} x} + \frac{1}{\log_{\frac{c}{a}} x}$ =
(a) 0 (b) 1 (c) – 1 (d) 2

Q.32. If $\log(a^9) + \log(a) = 10$, then a =
(a) 0 (b) 10 (c) – 1 (d) – 10

Q.33. If $\frac{1}{2}\log a = \frac{1}{3}\log b = \frac{1}{5}\log c$ then $a^4 - bc$ =
(a) 0 (b) 1 (c) – 1 (d) 3

Q.34. If $\frac{1}{4}\log_2 a = \frac{1}{6}\log_2 b = -\frac{1}{24}\log_2 c$, then a^3b^2c =
(a) 0 (b) 1 (c) – 1 (d) 2

Q.35. If $\frac{1}{log_a m} + \frac{1}{log_b m} + \frac{1}{log_c m} = \frac{1}{log_z m}$ then z =
(a) abc (b) a + b + c (c) a(b + c) (d) (a + b)c

Q.36. For any 3 consecutive integers a, b, c the equation log(1+ac) – 2.logb = 0 is:-
(a) True (b) False
(c) Cannot be determined (d) None of these.

Q.37. If $\log_{1/2}[\log_x (\log_4 32)] = 2$, then x =
(a) 5/2 (b) 25/4 (c) 625/16 (d) 2/5

Q.38. If $x = \log_{2a} a$, $y = \log_{3a} 2a$ and $z = \log_{4a} 3a$, then the value of yz(2–x) is:-
(a) 1 (b) – 1 (c) 2 (d) – 1

Q.39. If $a^2 + b^2 = c^2$, then $\frac{1}{\log_{c+a} b} + \frac{1}{\log_{c-a} b} =$
(a) 1 (b) 2 (c) – 1 (d) – 2

Q.40. If $x^2 + y^2 = 11xy$, then 2.log(x – y) =
(a) log 3 + log x + log y (b) 3 log 3 + log x + log y
(c) 2 log 3 + log x + log y (d) None of these

Q.41. If $\log_{10} 3 = 0.477$, the number of digits in 3^{40} is :-
(a) 18 (b) 19 (c) 20 (d) 21

Q.42. If $\log_a (ab) = x$, then $\log_b (ab)$ is :-
(a) $\frac{1}{x}$ (b) $\frac{x}{x+1}$ (c) $\frac{x}{x-1}$ (d) $\frac{x}{1-x}$

Q.43. If $\log \frac{m}{n} + \log \frac{n}{m} = \log (m + n)$, then
(a) m + n = 1 (b) $\frac{m}{n} = 1$ (c) m – n = 1 (d) $m^2 – n^2 = 1$

Q.44. If $\log(x + y) = \log\left(\frac{3x-3y}{2}\right)$, then $\log x – \log y =$
(a) log 2 (b) log 3 (c) log 5 (d) log 6

Q.45. If $\log x : 3 = \log y : 4 = \log z : 5$, then $zx =$
(a) 2y (b) y^2 (c) 8y (d) 4y

Q.46. If $\frac{1}{log_a t} + \frac{1}{log_b t} + \frac{1}{log_c t} = \frac{1}{log_z t}$ then the value of z is :-
(a) abc (b) a + b + c (c) a (b + c) (d) (a + b) c

Q.47. If $\log 2 = a$, $\log 3 = b$, $\log 7 = c$ and $6^x = 7^{x+4}$, then x =
(a) $\frac{4ab}{b+a-c}$ (b) $\frac{4c}{a+b-c}$ (c) $\frac{4b}{c-a-b}$ (d) $\frac{4a}{a+b-c}$

RATIO

Q.01. Inverse of the ratio 11:15
(a) 11:15 (b) 15:11 (c) 1:5 (d) None

Q.02. Duplicate ratio of 3:4
(a) 9:16 (b) 3:4 (c) 16:9 (d) None

Q.03. Sub-Duplicate ratio of 25:36
(a) 2:3 (b) 25:36 (c) 5:6 (d) None

Q.04. Triplicate ratio of 2:3
(a) 4:9 (b) 8:27 (c) 6:12 (d) None

Q.05. Sub-Triplicate ratio of 8:27
(a) 2:3 (b) 8:27 (c) 4:9 (d) None

Q.06. Find the ratio compounded of 2:3, 9:4, 5:6 and 8:10
(a) 0 (b) – 1 (c) 1 (d) None

Q.07. Find the ratio compounded of 4:9 and duplication ratio of 3:4
(a) $\frac{4}{9}$ (b) $\frac{1}{4}$ (c) $\frac{-1}{4}$ (d) None

Q.08. Ratio compounded of 4:9, duplicate ratio of 3:4, 2:3 and triplicate ratio of 9:7 is :-
(a) $\frac{234}{686}$ (b) $\frac{242}{686}$ (c) $\frac{243}{686}$ (d) None

Q.09. Find the ratio compounded of duplicate ratio of 4:5, triplicate ratio of 1:3, Sub-duplicate ratio of 81:256 and sub-triplicate ratio of 125:512 is :-
(a) 1:118 (b) 1:120 (c) 1:115 (d) None

Q.10. If p:q is the sub-duplicate ratio of $(p-x^2):(q-x^2)$, then x^2 =
(a) $\frac{pq}{p+q}$ (b) $\frac{pq}{p-q}$ (c) $\frac{p+q}{pq}$ (d) None

Q.11. If the duplicate ratio of (2S− P) : (3T− P) is 2S : 3T , then
(a) $P^2 = 3ST$ (b) $P^2 = 6ST$ (c) $P^2 = ST$ (d) None

Q.12. If a : b = 3 : 4 then find (2a + 3b) : (3a + 4b)
(a) $\frac{-18}{25}$ (b) $\frac{12}{25}$ (c) $\frac{18}{25}$ (d) None

Q.13. If x : y = 3 : 4 then find the value of $(x^2 y + x y^2) : (x^3 + y^3)$
(a) $\frac{12}{13}$ (b) $\frac{11}{13}$ (c) $\frac{14}{13}$ (d) None

Q.14. p : q = 2 : 3 and x : y = 4 : 5 then 5px + 3qy : 10px + 4qy = ?
(a) 14:28 (b) 17:28 (c) 15:28 (d) None

Q.15. The angles of a Triangle are in the ratio 2:7:11 then find the angles of the triangle.
(a) $18°, 63°, 99°$ (b) $60°, 60°, 60°$ (c) $30°, 60°, 90°$ (d) None

Q.16. The ratio of the two numbers is 2 : 3, if 4 is subtracted from each then the new ratio is 3 : 5 . Find the numbers.
(a) 12, 24 (b) 16, 24 (c) 4, 18 (d) None

Q.17. Divide Rs. 324 into two parts in the ratio 11 : 7 then the two parts are -
(a) 195 & 126 (b) 189 & 126 (c) 198 & 126 (d) None

Q.18. The ratio of the two numbers is 7:10 and their difference is 105 then find the numbers.
(a) 215, 320 (b) 245, 350 (c) 275, 380 (d) None

Q.19. P, Q and R are the three cities. The ratio of the temperature of P and Q is 11:12 and the ratio of the temperature of P and R is 9:8 , then find the ratio of the temperature of Q and R.
(a) $\frac{12}{22}$ (b) $\frac{-27}{22}$ (c) $\frac{27}{22}$ (d) None

Q.20. A person A earn Rs. 80 in 7 hrs. and an another person B earn Rs. 90 in 12 hrs. Then find the ratio of their income.
(a) 12:21 (b) 32:21 (c) 27:21 (d) None

Q.21. Find the number which when subtracted from each terms of the ratio 19:31, converted it into 1:4
(a) 15 (b) 18 (c) 21 (d) None

Q.22. The ratio of the income of the two persons is 4:5 and the ratio of their expenses is 7:9 , if they both saves `50 each per day then find their daily income.
(a) 600 & 800 (b) 200 & 600 (c) 400 & 500 (d) None

Q.23. The ratio of the speed of two trains is 7:8, if the second train runs 400 km in 5 Hours then find the speed of first train.
(a) 51km/h (b) 72km/h (c) 70km/h (d) None

PROPORTION

Q.01. Find the fourth proportional to 4, 6 and 8
(a) 12 (b) 18 (c) 21 (d) None

Q.02. Find the fourth proportion to 2a, a^2 and c.
(a) $\frac{-ac}{2}$ (b) $\frac{ac}{2}$ (c) $\frac{2ac}{2}$ (d) None

Q.03. If the four numbers $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}$ and $\frac{1}{x}$ are in proportion then find the value of x
(a) $\frac{1}{2}$ (b) $\frac{-15}{2}$ (c) $\frac{15}{2}$ (d) None

Q.04. The numbers 14, 16, 35, 42 are not in proportion, then find the fourth number for which they will be in proportion.
(a) 42 (b) 40 (c) 37 (d) None

Q.05. The numbers 12, 16, x, 20 are in proportion, then find x
(a) 9 (b) 12 (c) 15 (d) None

Q.06. If the numbers 4, x, 9, $13\frac{1}{2}$ are in proportion, then find x.
(a) 6 (b) 9 (c) 12 (d) None

Q.07. Find the third proportional to 12, 18
(a) 21 (b) 25 (c) 27 (d) None

Q.08. Find the mean proportional between 25 and 81
(a) 81 (b) 45 (c) 25 (d) None

Q.09. Find the mean proportional between $12x^2$ and $27y^2$
(a) 18xy (b) 27xy (c) 12xy (d) None

Q.10. Find the mean proportional between 1.4 gm and 5.6 gm
(a) 1.4 gm (b) 2.6 gm (c) 2.8 gm (d) None

Q.11. If $A = \frac{B}{2} = \frac{C}{5}$ then A:B:C =
(a) 5 : 2 : 1 (b) 1 : 2 : 5 (c) 5 : 1 : 2 (d) None

Q.12. If A : B = 3 : 2 and B : C = 3 : 5, then find A:B:C
(a) 9 : 6 : 10 (b) 6 : 9 : 10 (c) 10 : 9 : 6 (d) None

Q.13. $\frac{5x-3y}{5y-3x} = \frac{3}{4}$ Then find x : y

(a) $\frac{-27}{29}$ (b) $\frac{20}{29}$ (c) $\frac{27}{29}$ (d) None

Q.14. Divide 750 in the three parts in the ratio 4 : 5 : 6

(a) 200 : 250 : 300 (b) 300 : 250 : 200 (c) 250 : 200 : 300 (d) None

Q.15. The sum of the ages of the three persons is 150 years. 10 years ago the ratio of their ages were 7 : 8 : 9. Find their present ages.

(a) 50, 55, 40 (b) 45, 50, 55 (c) 55, 45, 50 (d) None

Q.16. If a : b = 4 : 1 , then find the value of $\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}$

(a) $\frac{1}{2}$ (b) $\frac{-5}{2}$ (c) $\frac{5}{2}$ (d) None

Q.17. The ratio of the two numbers is 3:4, if 6 be added to each number then their new ratio will be 4:5. Find the numbers.

(a) 18 and 24 (b) 18 and 20 (c) 24 and 16 (d) None

Q.18. If $P + \sqrt{3}.Q + \sqrt{5}.R + \sqrt{15}.S = \frac{1}{1+\sqrt{3}+\sqrt{5}}$, then find the value of P.

(a) $\frac{1}{11}$ (b) $\frac{7}{11}$ (c) $\frac{-7}{11}$ (d) None

Q.19. If $a = \sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}$ then find the value of $a^3 + 3a - 2$

(a) – 1 (b) 1 (c) 0 (d) None

Q.20. If $\sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c} = 0$, then find the value of $\left(\frac{a+b+c}{3}\right)^3$

(a) 1 (b) abc (c) 0 (d) None

Q.21. There are total 23 coins of ` 1, ` 2 and ` 5 in a bag. If their total value is ` 43, the ratio of the coin of ` 1 and ` 2 is 3:2, then find the number of coin of ` 1.

(a) 12 (b) 8 (c) 3 (d) None

MISCELLANEOUS QUESTIONS OF *RATIO & PROPORTION*

Q.01. What should be added to the terms of 3:5 so as to get 2 : 3.

(a) 1 (b) 2 (c) 3 (d) 4

Q.02. In ratio, quantities to be compared must be in the

(a) Same units (b) different units

(c) Same qualities (d) different qualities.

Q.03. Which Ratio is inverse ratio of a : b

(a) a : b (b) a : a (c) b : a (d) None

Q.04. If x is the mean proportional between a and b , then

(a) ab = x (b) ax = bx (c) x = ab (d) $x = \sqrt{ab}$

Q.05. In the ratio a:b, a and b are called the terms of the ratio; 'a' is the ---------- and 'b' is the ----------
(a) Consequent (b) Antecedent (c) (a) or (b) (d) None

Q.06. One ratio is the ---------- of the another, if their product is 1.
(a) Proportion (b) inverse (c) Same (d) None

Q.07. If a:b = c:d then in the expression $\frac{a+b}{b} = \frac{k}{d}$ find the value of k
(a) c (b) c + d (c) c – d (d) a + b

Q.08. The ratio of X:Y is 7:9 and Y:Z is 6:7. The ratio of X:Z is :-
(a) 2:3 (b) 3:2 (c) 1:2 (d) 2:1

Q.09. A, B, C, D are four numbers such that A:B = 2:3, B:C = 4:5, C:D = 5:8, then A:D is:-
(a) 2:3 (b) 3:2 (c) 1:3 (d) 3:1

Q.10. The third proportional to 12 and 30 is :-
(a) 40 (b) 45 (c) 50 (d) 75

Q.11. The fourth proportional to 3, 4 and 15 is :-
(a) 20 (b) 18 (c) 17 (d) 15

Q.12. If x : $2\frac{1}{3}$ :: 21 : 50, then the value of x is :-
(a) $\frac{27}{50}$ (b) $\frac{49}{50}$ (c) $\frac{1}{50}$ (d) $1\frac{1}{49}$

Q.13. The sum of the squares of three numbers is 116 and their ratio is 2:3:4. The numbers are
(a) 2, 3, 4 (b) 4, 6, 8 (c) 4, 9, 16 (d) 8, 12, 16

Q.14. The ratio of the number of boys and girls in a school is 2:5. If there are 350 students in the school, the numbers of girls in the school are
(a) 200 (b) 275 (c) 225 (d) 250

Q.15. Rs.407 are to be divided among A, B and C so that their shares are in the ratio of 1/4:1/5:1/6 then respective shares of A, B, C are
(a) Rs.165, Rs.132, Rs.110 (b) Rs.165, Rs.110, Rs.132
(c) Rs.132, Rs.110, Rs.165 (d) Rs.110, Rs.132, Rs.165

Q.16. If A : B = 2 : 3, B : C = 4 : 5, C : D = 3 : 7 then A : B : C : D is :-
(a) 24 : 36 : 45 : 105 (b) 12 : 36 : 45 : 105
(c) 24 : 36 : 65 : 105 (d) 24 : 48 : 45 : 105

Q.17. The fraction that bears the same ratio to 4/9 that 3/11 does to 5/33 is :-
(a) $\frac{4}{5}$ (b) $\frac{5}{6}$ (c) $\frac{6}{5}$ (d) $\frac{7}{5}$

Q.18. In 40 litters mixture of glycerin and water, the ratio of glycerin and water is 3:1. The quantity of water added in the mixture in order to make this ratio 2:1 is :-
(a) 15 litters (b) 10 litters (c) 8 litters (d) 5 litters

Q.19. 5 Mangoes and 4 Oranges costs as much as 3 Mangoes and 7 Oranges. The ratio of the cost of one Mango to that of one Orange is :-
(a) 4 : 3 (b) 1 : 3 (c) 3 : 2 (d) 5 : 2

Q.20. In what ratio should tea worth Rs.10 per kg be mixed with tea worth Rs.14 per kg. so that the average price of the mixture may be Rs.11 per kg. ?
(a) 2 : 1 (b) 3 : 1 (c) 3 : 2 (d) 4 : 1

Q.21. Which ratio among 3:5 and 7:11 is greater?
(a) 3:5 $\geq$ 7:11 (b) 3:5 > 7:11 (c) 3:5 < 7:11 (d) None

Q.22. The total number of students in EXCEL is 360. The ratio of number of boys to the number of girls in EXCEL is 3:5. If 15 new girls are admitted to the EXCEL, find how many new boys should be admitted so that the ratio of the number of boys to the number of girls becomes 4:5.
(a) 75 (b) 57 (c) 55 (d) 45

Q.23. If a : b = 2 : 3 and b : c = 4 : 9 then a : b : c =
(a) 2:4:9 (b) 8:12:27 (c) 2:6:9 (d) 4:6:18

Q.24. If x = 1/3, y = 1/4, and z = 1/5, then x : y : z =
(a) 5 : 4 : 3 (b) 15 : 12 : 9 (c) 20 : 15 : 12 (d) 20 : 12 : 9

Q.25. a, b and c are weights of three persons A, B and C. the ratio of weights of A and B is 7 : 9 and that of A and C is 9 : 8 the ratio between the weights of B and C is :-
(a) 82 : 72 (b) 81 : 56 (c) 69 : 79 (d) 78 : 87

Q.26. If (a – k) : (b – k) is the duplicate ratio of a : b then k is :-
(a) b / (a + b) (b) a / (a + b) (c) ab / (a – b) (d) ab / (a + b)

Q.27. If (a – p) : (2b – p) is the sub-duplicate ratio of a : 2b then
(a) $p^2 = 2ab$ (b) p = 2ab (c) 2p = 3ab (d) None of these

Q.28. The ratio of the prices of two flats was 4:5. Three years later, when the price of 1st increased by 10% and that of 2nd by ` 6,000 the ratio of their prices becomes 11:15. Find the original prices of the two flats.
(a) 24000, 30000 (b) 25000, 30000 (b) 40000, 50000 (d) None of these

Q.29. The ratio of the present age of a father to that of his son is 7:4. Ten years hence the ratio would be 11:7 find their present ages.
(a) 56, 32 (b) 49, 28 (c) 42, 24 (d) 35, 20

Q.30. The salaries of A, B, C are in the ratio 2:3:4. If the increments of 10%, 20% and 30% are allowed in their respective salaries, then what will be the new ratio of their salaries?
(a) 11:18:26 (b) 12:18:26 (c) 12:18:24 (d) 11:18:24

Q.31. If Rs.460 be divided into three parts, proportional to 1/2 : 2/3 : 3/4, then the first part is :-
(a) 110 (b) 130 (c) 120 (d) 140

Q.32. A sum of Rs.9300 is divided among A, B, C in such a way that shares of A and B are in the ratio 1 : 3 and those of B and C are in the ratio 4 : 5 , the amount received by C is :-

(a) 1200 (b) 3600 (c) 4500 (d) 5000

Q.33. Some one-rupee, 50-paisa and 25-paisa coins make up Rs.20 and their numbers are in the Proportion of 4 : 1 : 2. Find the number of each type of coins.
(a) 4, 1, 2 (b) 8, 2, 4 (c) 16, 4, 8 (d) 20, 5, 10

Q.34. If A : B = 8 : 15, B : C = 5 : 8 and C : D = 4 : 5, then A : D is equal to :-
(a) 2 : 7 (b) 4 : 15 (c) 8 : 15 (d) 15 : 4

Q.35. The ratio of the prices of two types of cars was 16 : 23. Two years later when the price of the first has increased by 10% and that of the second by ` 477, the ratio of the prices becomes 11 : 20. Find the original prices of the two types of cars.
(a) (848, 1219) (b) (748, 1319) (c) (948, 1119) (d) None of these

Q.36. The ratio of $4^{3.5} : 2^5$ is same as :-
(a) 2 : 1 (b) 4 : 1 (c) 7 : 5 (d) 7 : 10

Q.37. If 2A = 3B and 4B = 5C, then A : C is :-
(a) 4 : 3 (b) 15 : 8 (c) 8 : 15 (d) 3 : 4

Q.38. Ratio between the two numbers is 3 : 4 , if their L.C.M. is 180. The numbers are:-
(a) 36,48 (b) 45,60 (c) 15,20 (d) None of these.

Q.39. If A : B : C = 2 : 3 : 4, then $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$ is equal to :-
(a) 4 : 9 : 16 (b) 8 : 9 : 12 (c) 8 : 9 : 15 (d) 8 : 9 : 24

Q.40. 94 is divided into two parts in such a way that the fifth part of the first and the eighth part of the second are in the ratio 3 : 4, the first part is :-
(a) 27 (b) 30 (c) 36 (d) 48

Q.41. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there are ` 30 in all, how many 5 p coins are there?
(a) 50 (b) 100 (c) 150 (d) 200

Q.42. The salaries of A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be the new ratio of their salaries?
(a) 3 : 5 : 10 (b) 10 : 11 : 20
(c) 23 : 33 : 60 (d) Cannot be determined.

Q.43. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is :-
(a) 2 : 5 (b) 3 : 5 (c) 4 : 5 (d) 6 : 7

Q.44. Rs.1980 are divided among A, B, C so that half of A's part, one-third of B's part and one-sixth of C's part are equal. Then B's part is:-
(a) Rs. 660 (b) Rs. 360 (c) Rs. 1080 (d) Rs. 540.

Q.45. The ratio of third proportional to 12 & 30 and the mean proportional between 9 and 25 is :-
(a) 2 : 1 (b) 5 : 1 (c) 7 : 15 (d) 9 : 14

Q.46. It is given that $y \propto \frac{1}{x^3 - x}$ and for x = 2 the value of y is $\frac{1}{6}$. If x = 1, then the value of y will be : -
(a) 1 (b) 0 (c) – 1 (d) None of these

Q.47. x varies inversely as square of y. Given that y = 2 for x = 1. The value of x for y = 6 will be equal to : -
(a) 2 (b) 9 (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

Q.48. A piece of string is 40 centimeters long. It is cut into three pieces. The longest piece is 3 times as long as the middle-sized and the shortest pieces is 23 centimeters shorter than the longest piece. The length of the shortest piece (in cm) is:
(a) 27 cm (b) 5 cm (c) 4 cm (d) 9 cm

Q49. A jar contains black and white marbles. If there are ten marbles in the jar, then which of the following could not be the ratio of black to white marbles?
(a) 9 : 1 (b) 7 : 3 (c) 1 : 10 (d) 1 : 4

Q.50. If one star equals four circles and three circles equal four diamonds, then the ratio of star & diamonds is :-
(a) 3/16 (b) 1/3 (c) 3/4 (d) 16/3

Q.51. If (a + b) : (b + c) : (c + a) = 6 : 7 : 8 and a + b + c = 14, then the value of c is :-
(a) 8 (b) 7 (c) 6 (d) 12

Q.52. If the ratio of boys to girls in a class is B, the ratio of girls to boys is G, then B + G is :-
(a) Equal to 1 (b) Greater than 1 (c) Less than 1 (d) None of these

Answers

INDICES

1	b	11	c	21	c	31	a	41	b	51	a	61	c
2	b	12	b	22	a	32	c	42	c	52	b	62	b
3	c	13	d	23	b	33	b	43	b	53	a	63	c
4	a	14	a	24	b	34	a	44	a	54	b	64	b
5	b	15	b	25	b	35	b	45	c	55	b	65	b
6	c	16	a	26	a	36	a	46	b	56	c	66	c
7	a	17	b	27	c	37	a	47	a	57	a	67	a
8	d	18	d	28	b	38	b	48	a	58	a	68	d
9	d	19	d	29	a	39	c	49	b	59	b	69	c
10	b	20	a	30	b	40	a	50	c	60	a	70	c

MISCELLANEOUS QUESTIONS OF *INDICES*

1	b	11	d	21	d	31	a
2	b	12	a	22	b		
3	c	13	a	23	a		
4	b	14	c	24	c		
5	b	15	b	25	c		
6	b	16	d	26	a		
7	c	17	a	27	c		
8	c	18	d	28	a		
9	b	19	d	29	c		
10	b	20	d	30	b		

LOGARITHM

1	a	11	b	21	a	31	c
2	b	12	a	22	b	32	a
3	d	13	c	23	c	33	b
4	b	14	b	24	b	34	c
5	c	15	c	25	a	35	a
6	b	16	a	26	c	36	b
7	a	17	b	27	a		
8	c	18	b	28	c		
9	b	19	a	29	b		
10	c	20	c	30	c		

MISCELLANEOUS QUESTIONS OF *LOGARITHM*

1	c	11	d	21	b	31	a	41	c
2	d	12	b	22	c	32	b	42	c
3	c	13	c	23	b	33	a	43	a
4	d	14	b	24	a	34	b	44	c
5	c	15	a	25	c	35	a	45	b
6	b	16	b	26	b	36	a	46	a
7	b	17	a	27	b	37	c	47	b
8	c	18	a	28	b	38	a		
9	b	19	a	29	d	39	b		
10	b	20	b	30	c	40	c		

RATIO

1	b	11	b	21	a
2	a	12	c	22	c
3	c	13	a	23	c
4	b	14	b		
5	a	15	a		
6	c	16	b		
7	b	17	c		
8	c	18	b		
9	b	19	c		
10	a	20	b		

PROPORTION

1	a	11	b	21	a
2	b	12	a		
3	c	13	c		
4	b	14	a		
5	c	15	b		
6	a	16	c		
7	c	17	a		
8	b	18	b		
9	a	19	c		
10	c	20	b		

MISCELLANEOUS QUESTIONS OF *RATIO & PROPORTION*

1	a	11	a	21	c	31	c	41	c	51	c
2	a	12	b	22	b	32	c	42	c	52	b
3	c	13	b	23	b	33	c	43	c		
4	d	14	d	24	c	34	b	44	d		
5	b	15	a	25	b	35	a	45	b		
6	b	16	a	26	d	36	b	46	d		
7	b	17	a	27	a	37	b	47	d		
8	a	18	d	28	a	38	b	48	c		
9	c	19	c	29	a	39	d	49	c		
10	d	20	b	30	a	40	b	50	d		

CHAPTER – 2 EQUATIONS

One / Two / Three Variables EQUATIONS

Q.01. Find the value of x, if $-7x + 1 = 5 - 3x$
(a) –1 (b) 1 (c) 0 (d) None

Q.02. Find the value of x, if $\frac{x+4}{4} + \frac{x-5}{3} = 11$
(a) 11 (b) 20 (c) 15 (d) None

Q.03. Find the value of x, if $\frac{x+24}{5} = 4 + \frac{x}{4}$
(a) 21 (b) 18 (c) 16 (d) None

Q.04. Find the value of x, if $\frac{x+11}{6} - \frac{x+1}{9} = \frac{x+7}{4}$
(a) $\frac{1}{7}$ (b) $\frac{-1}{7}$ (c) $\frac{4}{7}$ (d) None

Q.05. Find the value of x, if $\frac{12x+1}{4} = \frac{15x-1}{5} + \frac{2x-5}{3x-1}$
(a) 12 (b) 5 (c) 7 (d) None

Q.06. Find the value of x, if $\frac{x}{0.5} - \frac{1}{0.05} + \frac{x}{0.005} - \frac{1}{0.0005} = 0$
(a) 10 (b) –10 (c) 0 (d) None

Q.07. Find the value of x and y, if $3x + 4y = 7$ & $4x - y = 3$
(a) x = 2, y = 1 (b) x = 1, y = 1 (c) x = 1, y = 2 (d) None

Q.08. Find the value of x and y, if $\frac{1}{16x} + \frac{1}{15y} = \frac{9}{20}$ & $\frac{1}{20x} - \frac{1}{27y} = \frac{4}{45}$
(a) $\left(\frac{1}{4}, \frac{1}{3}\right)$ (b) $\left(\frac{1}{3}, \frac{1}{4}\right)$ (c) (3, 4) (d) (4, 3)

Q.09. Find the value of x and y, if $\frac{4}{x} - \frac{5}{y} = \frac{x+y}{xy} + \frac{3}{10}$ & $3xy = 10(y-x)$
(a) (5, 2) (b) $(-2, -5)$ (c) $(2, -5)$ (d) (2, 5)

Q.10. Find the value of x and y, if $(1.5)x + (2.4)y = 1.8$ & $(2.5).(x + 1) = 7y$
(a) $-0.5, -0.4$ (b) 0.4, 0.5 (c) -0.5, 0.4 (d) None

Q.11. Find the value of x and y, if $\frac{3}{x+y} + \frac{2}{x-y} = 3$ & $\frac{2}{x+y} + \frac{3}{x-y} = 11/3$
(a) $(1, 2)$ (b) $(-1, -2)$ (c) $\left(1, \frac{1}{2}\right)$ (d) $(2, 1)$

Q.12. Find the value of x, y and z, if $\frac{x}{4} = \frac{y}{3} = \frac{z}{2}$ & $7x + 8y + 5z = 62$
(a) (2, 3, 4) (b) (16, 8, 5) (c) (4, 3, 2) (d) None of these

Q.13. $\frac{xy}{y+x} = 20$, $\frac{yz}{y+z} = 40$, $\frac{zx}{z+x} = 24$
(a) (120, 60, 30) (b) (60, 30, 120) (c) (30, 60, 120) (d) None

Q.14. $\frac{xy}{y-x} = 110, \quad \frac{yz}{z-y} = 132, \quad \frac{zx}{z+x} = \frac{60}{11}$
(a) 12, 10, 11 (b) 12, 11, 10 (c) 11, 10, 12 (d) None

Q.15. $\frac{1}{3}(x + y) + 2z = 21, \; 3x - \frac{1}{2}(y + z) = 65, \; x + \frac{1}{2}(x + y - z) = 38$
(a) 4, 9, 5 (b) 2, 9, 5 (c) 24, 9, 5 (d) None

Q.16. $\frac{x}{0.01} + \frac{y + 0.03}{0.05} = \frac{y}{0.02} + \frac{x + 0.03}{0.04} = 2$
(a) (1, 2) (b) (0.1, 0.2) (c) (0.01, 0.02) (d) (0.02, 0.01)

Q.17. $3x - 4y + 70z = 0, \; 2x + 3y - 10z = 0, \; x + 2y + 3z = 13$
(a) (1, 3, 7) (b) (1, 7, 3) (c) (2, 7, 3) (d) $(-10, 10, 1)$

Q.18. The sum of the two numbers is 52 and their difference is 2, then find the numbers.
(a) 24, 26 (b) 25, 27 (c) 27, 29 (d) 30, 22

Q.19. Divide 56 into two parts such that the three times of first part exceeds 1/3 of the second part by 48.
(a) (36, 20) (b) (20, 36) (c) (26, 26) (d) (30, 26)

Q.20. The sum of the digits of a two digit number is 10. If 18 be subtracted from it then the digits of the resulting number will be equal. Find the number.
(a) 73 (b) 55 (c) 28 (d) 64

Q.21. $\frac{1}{4}$ of a number exceed by 4 with its $\frac{1}{6}$ part, find the number.
(a) 50 (b) 48 (c) 51 (d) None

Q.22. Ten years ago, age of a father is 4 times the age of his son. 10 years hence the age of father will be twice of that of his son. Find the present age of the Father & Son.
(a) 60, 30 (b) 50, 20 (c) 60, 20 (d) None

Q.23. Product of the two numbers is 3200. When the larger number is divided by the smaller number, quotient is 2. Find the numbers.
(a) 40, 80 (b) 40, 20 (c) 32, 100 (d) None

Q.24. The denominator of a fraction is two more than its numerator, if 5 be added to the numerator, the fraction increases by unity. Find the fraction.
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{5}{7}$ (d) None

Q.25. The three persons A, B and C together have Rs.51. B has 4/5 times of A and C has 3/4 time of A, then they have the money as :-
(a) (20, 16, 15) (b) (15, 16, 20) (c) (20, 22, 15) (d) None

Q.26. In a two digit number the tens place digit is three times of the unit's place digit, if 54 is subtracted from it the digit of the resulting number will be reverse.
(a) 93 (b) 39 (c) 62 (d) None of these

Q.27. A student is asked to divide half of a given number by 6 and to divide the other half by 4 and then add the two quantities. Instead of doing so he divides the given number by 5. The answer he obtain was 4 short of the correct answer, then find the number give to him.
(a) 820 (b) 480 (c) 240 (d) None

Q.28. The half of a number is 15 more than 1/5 of the number. Find the number.
(a) 45 (b) 50 (c) 51 (d) None

Q.29. A number lies between 10 and 100, such that the number is 5 times of the sum of its digits. If 9 be added to it, the digit of the resulting number would be reversed.
(a) 54 (b) 96 (c) 45 (d) 105

Q.30. A two digit number is such that it is 4 times of the sum of its digits, if 27 be added to it the digits reverse. Find the number.
(a) 63 (b) 60 (c) 36 (d) 50

Q.31. The sum of two numbers is 16. If $\frac{1}{5}$ of the greater number is equal to $\frac{1}{3}$ of the smaller number, then find the number.
(a) 2, 12 (b) 6, 10 (c) 8, 8 (d) None

Q.32. The income of 8 men and 6 boys is Rs 33. If the income of 4 men is Rs 4.5 more than that of the 5 boys, then find the income of each men and each boy.
(a) Rs 3, Rs 1.5 (b) Rs 5, Rs. 3 (c) Rs 3.5, Rs 2 (d) None

Q.33. The sum of the digits of a three digit number is 12, if the digits are reversed the number increases by 495. If only units and tens places digits are reversed, the number increases by 36. Find the number.
(a) 327 (b) 372 (c) 237 (d) 273

Q.34. The two numbers are such that the twice of the greater number is 18 more than the twice of smaller number, if the sum of the 1/3 of the smaller and 1/5 of the greater is 21 then the numbers are
(a) 50, 41 (b) 55, 46 (c) 45, 36 (d) None of these

Q.35. The relation between price (P) and the Quantity (Q) of a certain commodity are given by the equations P = $\frac{Q}{3} + \frac{7}{4}$ and 4Q + 7P = 17, find the value of P and Q.
(a) 3/4 , 2 (b) 2 , 3/4 (c) 4 , 3/2 (d) None

MISCELLANEOUS QUESTIONS Of One / Two / Three Variables EQUATIONS

Q.01. If out of three numbers, the sum of first and second is 24, sum of second and third is 30 and sum of first and third is 26, then the smallest number is :-
(a) 18 (b) 14 (c) 16 (d) 10

Q.02. The difference between the ages of two men is 10 years. 15 years ago, the age of the older was twice the age of younger. What are their present ages?
(a) 10, 15 (b) 35, 40 (c) 25, 35 (d) 15, 25

Q.03. 10 years ago, the age of father was 5 times the age of son. If after 20 years, the father's age is twice that of his son, find the present age of the father.
(a) 40 years (b) 45 years (c) 60 years (d) 90 years

Q.04. A man sells 6 radios and 4 televisions for Rs.18480. If 14 radios and 2 televisions are also sold for the same amount, what is the price of a television?
(a) Rs.1848 (b) Rs.840 (c) Rs.1680 (d) Rs.3360

Q.05. Rs.630 were distributed among A, B and C, so that the shares of A and B were as 2 : 3 and the shares of B and C were as 4 : 5. What is the shares of C?
(a) Rs.270 (b) Rs.144 (c) Rs.216 (d) None of these

Q.06. If 2x + y = 5 and 3x – 4y = 2, then the value of 2xy is:-
(a) 4 (b) 6 (c) 8 (d) 10

Q.07. If the cost of 3 chairs and 1 table is Rs.900 and that of 5 chairs and 3 tables is Rs.2100, then the cost of 4 chairs and one table is :-
(a) Rs.1000 (b) Rs.1050 (c) Rs.1100 (d) Rs.1150

Q.08. The sum of two numbers is 80. If three times of one number is equal to five times of the other number, then the numbers are
(a) 20, 60 (b) 50, 30 (c) 10, 70 (d) 25, 55

Q.09. Which of the following equation is not a linear equation?
(a) 2x + 3 = 7 (b) (4/3) x + (7/2) = 6 + (1/2) x (c) $x + \frac{1}{2x} = 3$ (d) None of these

Q.10. A man leaves half his property to his wife. One third to his son and remaining to his daughter. If the daughter's share is Rs.15000, how much money did his wife get?
(a) ` 45000 (b) ` 30000 (c) ` 9000 (d) None of these

Q.11. "A" and "B" can do a piece of work in 8 days, which "A" alone can do in 12 days. In how many days can "B" alone do the same work?
(a) 20 days (b) 24 days (c) 18 days (d) 16 days

Q.12. 5 years ago, the age of a man was 7 times the age of his son. The age of the man will be 3 times the age of his son in five years from now. How old is he now?
(a) 40 years (b) 30 years (c) 35 years (d) None of these

Q.13. If a scooterist drives at the rate of 24 km per hour, he reaches his destination 5 minutes late. If he drives at the rate of 30 km per hour, he reaches his destination 4 minutes soon. How far is his destination?
(a) 18 km (b) 16 km (c) 20 km (d) 15 km

Q.14. The largest angle of triangle is twice the sum of the other two angles. The smallest angle is one-fourth of the largest angle. The smallest angle in degrees is:-
(a) 30° (b) 60° (c) 45° (d) None of these

Q.15. The sum of A's age 10 years ago and twice his age 15 years ago, is equal to his present age. Find his present age.
(a) 25 (b) 30 (c) 20 (d) 35

Q.16. If the numerator of a fraction is increased by 1 and the denominator also by 1, it becomes 2 and if the numerator is decreased by 4 and the denominator by 1, it becomes 1. Find the fraction.
(a) 5/2 (b) 3/5 (c) 5/9 (d) 4/5

Q.17. The sum of the digits of a two digit number is 7. The difference between the number and the number formed by reversing the digits is 27. Find the number.
(a) 52 (b) 55 (c) 53 (d) 54

Q.18. Five years ago the age of a father was three times of his son. Five years hence the age of the father will be twice that of his son. The present ages of the father and the son are ?
(a) 35, 15 (b) 40, 20 (c) 55, 25 (d) 36, 12

Q.19. A is older than B by 5 years. 10 years earlier A's age was 3/2 times of B's age. Their present ages are :-
(a) (25, 20) (b) (30, 25) (c) (45, 40) (d) (35, 30)

Q.20. Two numbers are such that thrice the greater number exceeds the smaller number by 15 and twice the difference of the numbers is 6. The numbers are :-
(a) (6, 9) (b) (6, 3) (c) (7, 10) (d) (12, 9)

Q.21. The system of equations $2x + 3y = 7$ and $6x + 5y = 11$ have :-
(a) No solution (b) Unique solution
(c) Infinite number of solutions (d) None of these

Q.22. The system of equations $6x + 5y = 11$ and $9x + \left(\frac{15}{2}\right) y = 21$ have :-
(a) Unique solution (b) Infinite number of solution
(c) No solution (d) None of these

Q.23. The system of equations $4x + 7y = 10$ and $10x + \left(\frac{35}{2}\right) y = 25$ have :-
(a) Infinite number of solution (b) Unique solution
(c) No solution (d) None of these

Q.24. The value of k for which the system of equations $kx + 2y = 5$ and $3x + y = 1$ have unique solution, is :-
(a) $K \neq 6$ (b) $k = 4$ (c) $k = 3$ (d) $k = 1$

Q.25. The value of k for which the system of equations $kx + 2y = 5$ and $3x + y = 1$ have no solution, is :-
(a) $k \neq 5$ (b) $k = 6$ (c) $k \neq 3$ (d) $k \neq 7$

Q.26. For what value of k the equations $9x + 4y = 9$ and $7x + ky = 5$ has no solution.
(a) k = 28/9 (b) k = 36/7 (c) k = 23/9 (d) k = 7

QUADRATIC EQUATIONS

Q.01. If $ax^2 + bx + c = 0$ then x = ?

(a) $x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$ (b) $x = \frac{-b \mp \sqrt{b^2 + 4ac}}{2a}$ (c) $x = \frac{2c}{-b \mp \sqrt{b^2 - 4ac}}$ (d) None

Q.02. The roots of the equation x.(x + 1) = 6 are :-

(a) 1, –6 (b) 1, 6 (c) –3, –2 (d) 2, –3

Q.03. If $\frac{x}{27} = \frac{3}{x}$ then the value of x is :-

(a) $\pm$ 10 (b) $\pm$ 9 (c) $\pm$ 8 (d) $\pm$ 7

Q.04. The roots of the equations $x^2 + 2x - 48 = 0$ are

(a) 8, – 6 (b) 8, 6 (c) – 8, 6 (d) – 8, – 6

Q.05. Quadratic equation whose roots are 3 and 5 is :-

(a) $x^2 - 8x + 15 = 0$ (b) $x^2 + 8x + 15 = 0$

(c) $x^2 + 8x - 15 = 0$ (d) None

Q.06. The common root of the equations $x^2 - 5x + 6 = 0$ and $x^2 - 6x + 8 = 0$ is :-

(a) 3 (b) 2 (c) 5 (d) 4

Q.07. If $\sqrt{x-7} + \sqrt{x-3} = 2$ then what is the value of x ?

(a) 9 (b) 7 (c) 19 (d) 3

Q.08. What is the sum of the roots of the equation $2x^2 - 11x + 5 = 0$?

(a) 11/2 (b) $-11/2$ (c) 2/11 (d) 10

Q.09. $\left(\frac{x^2 - 7x + 12}{x^2 - 9x + 20}\right)$ in lowest form is

(a) x–3/x–5 (b) x–5/x–3 (c) x–3/x– 4 (d) None of these

Q.10. $\left(\frac{x^3 - 64}{x^2 + 4x + 16}\right)$ is equal to

(a) (x + 4) (b) (x – 4) (c) (x – 3) (d) (x – 5)

Q.11. $3\left(\frac{3x^2 + 15}{6}\right) + 2x^2 + 9 = \frac{2x^2 + 96}{7} + 6$ find the value of x

(a) (1, 1) (b) $(-1, -1)$ (c) $(1, -1)$ (d) $(2, -1)$

Q.12. $(a + b - 2c)\,x^2 + (2a - b - c)\,x + (c + a - 2b) = 0$ find the value of x

(a) 1 (b) – 1 (c) 0 (d) None

Q.13. Find the value of $4 + \cfrac{1}{4 + \cfrac{1}{4 + \cfrac{1}{4 + \cfrac{1}{4 + \cdots \infty}}}}$

(a) $2+\sqrt{3}$ (b) $2+\sqrt{2}$ (c) $2+\sqrt{5}$ (d) None

Q.14. Find the value of $\sqrt{2-\sqrt{2-\sqrt{2\ldots\ldots\ldots\ldots\ldots\infty}}}$

(a) -1 (b) 1 (c) 2 (d) 0

Q.15. Find the value of x , if $2^{2x+3}-3^2.2^x+1=0$

(a) 0, 1 (b) 0, 3 (c) 0, -3 (d) 1, 2

Q.16. If the roots of the Quadratic Equation $2x^2 + 8x - k^3 = 0$ are equal, than find k

(a) 1 (b) – 2 (c) 0 (d) None

Q.17. If the roots of the Quadratic Equation $x^2 + (2p-1)x + p^2 = 0$ are real
Then find the value of p.

(a) $p \geq 1$ (b) $p \leq 4$ (c) $p \geq \frac{1}{4}$ (d) $p \leq \frac{1}{4}$

Q.18. If the roots of the Equation $x^2 - (p + 4)x + (2p + 5) = 0$ are equal then p = ?

(a) $p = 4$ (b) $p = \pm 2$ (c) $p = 0$ (d) None

Q.19. Roots of the Equation $5x^2 + 13x + k = 0$ are reciprocal to each other, then k = ?

(a) 5 (b) 6 (c) 8 (d) None

Q.20. Roots of the Quadratic Equation $(k + 2)x^2 - 7x + 5k = 0$ are reciprocal to each other, then find the value of k.

(a) 1/5 (b) 1/3 (c) 1/2 (d) None

Q.21. If one root of the Quadratic Equation $x^2 - 8x + k = 0$ is 4 more than the other root, then find the value of k.

(a) 12 (b) 14 (c) 18 (d) None

Q.22. Find the Quadratic Equation whose roots are 2 and 3

(a) $x^2 - 8x + 10 = 0$ (b) $x^2 - 8x + 1 = 0$ (c) $x^2 - 5x + 6 = 0$ (d) None

Q.23. Find the Quadratic Equation whose roots are 1/2 and 7

(a) $x^2 - 7x + 15 = 0$ (b) $2x^2 - 15x + 7 = 0$ (c) $2x^2 + 15x - 7 = 0$ (d) None

Q.24. Find the Quadratic Equation whose one of the root is $(2 + \sqrt{3})$

(a) $x^2 - 4x + 1 = 0$ (b) $x^2 + 4x + 7 = 0$ (c) $x^2 - 4x + 7 = 0$ (d) None

Q.25. Find the quadratic equation whose one of the root is $(\sqrt{5}-3)$

(a) $x^2 - 6x + 4 = 0$ (b) $x^2 + 6x + 4 = 0$ (c) $x^2 - 4x + 6 = 0$ (d) None

Q.26. Find the Quadratic Equation whose one of the root is 5 and the other root is -3

(a) $x^2 + 2x - 15 = 0$ (b) $x^2 - 2x + 15 = 0$ (c) $x^2 - 2x - 15 = 0$ (d) None

Q.27. If α and β are the roots of the Quadratic Equation $x^2 - 3x + 2 = 0$, then find the value of (i) $\alpha^2 + \beta^2$ (ii) $\frac{1}{\alpha}+\frac{1}{\beta}$ (iii) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ (iv) $\alpha^3 + \beta^3$ (v) $\alpha - \beta$

(a) $\frac{3}{2}, \frac{5}{2}, 9, -1, 5$ (b) $5, \frac{3}{2}, \frac{5}{2}, 9, 1$ (c) $5, \frac{3}{2}, \frac{5}{2}, 2, 1$ (d) None

Q.28. If α, β are the roots of the Quadratic Equation $x^2 - x - 1 = 0$ then $\frac{\alpha^2}{\beta} - \frac{\beta^2}{\alpha} = ?$

(a) $-2\sqrt{5}$ (b) $-3\sqrt{5}$ (c) $2\sqrt{5}$ (d) None

Q.29. If α, β are the roots of the Quadratic Equation $2x^2 - 4x - 3 = 0$, then $\alpha^2+\beta^2 = ?$

(a) 0 (b) –7 (c) 7 (d) None

Q.30. If α, β are the roots of the quadratic equation $x^2 - 3x + 5 = 0$, then find the quadratic equation whose roots are $1/\alpha$ and $1/\beta$

(a) $5x^2 - 3x - 1 = 0$ (b) $5x^2 - 3x + 1 = 0$

(c) $5x^2 + 3x + 1 = 0$ (d) None

Q.31. If α, β are the roots of the Quadratic Equation $5x^2 - 7x + 2 = 0$ then find the quadratic equation whose roots are $1/\alpha$ and $1/\beta$

(a) $2x^2 + 7x + 5 = 0$ (b) $2x^2 - 7x - 5 = 0$

(c) $2x^2 - 7x + 5 = 0$ (d) None

Q.32. If α, β are the roots of the Quadratic Equation $2x^2 + 3x - 1 = 0$ then find the quadratic equation whose roots are reciprocal.

(a) $x^2 - 3x - 2 = 0$ (b) $x^2 + 3x - 2 = 0$

(c) $x^2 - 3x + 2 = 0$ (d) None

Q.33. If α, β are the roots of the Quadratic Equation $x^2 - 5x + 4 = 0$ then find the quadratic equation whose roots are α / β and β / α

(a) $4x^2 + 17x + 4 = 0$ (b) $4x^2 - 17x + 4 = 0$

(c) $4x^2 - 17x + 9 = 0$ (d) None

Q.34. If α and β are the roots of the Quadratic Equation $x^2 - 2x + 1 = 0$ then find the quadratic equation whose roots are $(\alpha + \beta)$ & $(\alpha - \beta)$

(a) $x^2 - 2x + 2 = 0$ (b) $x^2 + 2x = 0$

(c) $x^2 - 2x = 0$ (d) None

Q.35. If the two Quadratic Equations are $p^2 = 5p - 3$ and $q^2 = 5q - 3$ where $p \neq q$, Find the quadratic equation whose roots are $\frac{p}{q}$ and $\frac{q}{p}$

(a) $3x^2 - 19x + 3 = 0$ (b) $3x^2 + 19x + 3 = 0$

(c) $3x^2 - 19x - 3 = 0$ (d) None

Q.36. Sum of the two numbers is 8 and Sum of their square is 34. Find the numbers.

(a) 3, 2 (b) 1, 2 (c) 3, 5 (d) None

Q.37. Difference of two $+ve$ integers is 3 & sum of their square is 89. Find the numbers.

(a) 8, 4 (b) 8, 5 (c) 5, 2 (d) None

Q.38. 5 times of a $+ve$ number is three less than the double of its square. Find the number.

(a) 5 (b) 2 (c) 3 (d) None

Q.39. The area of a rectangular field is 2000 m^2 and its perimeter is 180 meter, then find the length and breadth of the rectangular field.

(a) 20 and 40 (b) 50 and 40 (c) 50 and 20 (d) None

Q.40. Divide number 50 into two parts such that the sum of their reciprocal is 1/12.
(a) 24, 26 (b) 27, 23 (c) 20, 30 (d) None

Q.41. Find the two consecutive numbers such that the difference of their reciprocal is $\frac{1}{240}$.
(a) 15 and 16 (b) 16 and 17 (c) 18 and 19 (d) None

Q.42. The sum of the two numbers is 45 and the mean proportional between them is 18, then find the numbers.
(a) 27, 36 (b) 9, 36 (c) 9, 27 (d) None

Q.43. The sides of an equilateral triangle are reduces by 12, 13 and 14 units respectively, then it will be a right angle triangle. Find the side of the equilateral triangle.
(a) 15 unit (b) 16 unit (c) 17 unit (d) 18 unit

Q.44. A distributor of cold-drink has 5000 bottles in stock. Find the value of P, if a relation between demand of the cold-drink (D) and price per bottle (P) is given by the equation $D = -2000 P^2 + 2000 P + 17000$
(a) 3 (b) 2 (c) 5 (d) 7

Q.45. The two irrational numbers are such that if their sum is multiplied by the greater number it will give 70 and it their difference is multiplied by smaller number then it will give 12, find the numbers.
(a) $5\sqrt{2}$ & $2\sqrt{2}$ (b) $2\sqrt{3}$ & $3\sqrt{2}$ (c) $2\sqrt{5}$ & $5\sqrt{2}$ (d) None

Q.46. The sum of the area of two squares is $625\ \text{meter}^2$. If their sides are k meter and (k + 5) meter, then find the sides of the squares.
(a) 12 meter, 18 meter (b) 15 meter, 20 meter
(c) 14 meter, 16 meter (d) None

MISCELLANEOUS QUESTIONS of QUADRATIC EQUATIONS

Q.01. The roots of the equation $x^2 + 6x - 5 = 0$ are
(a) Real and equal (b) Imaginary
(c) Real and unequal (d) Rational and equal

Q.02. If the roots of the equation $px^2 + qx + 3 = 0$ are reciprocal to each other, then
(a) q = 3 (b) p = 3 (c) p – q = 0 (d) p + q = 0

Q.03. Sum of the squares of two numbers is 100. If the ratio in the numbers be 3 : 4, what is their sum ?
(a) 4 (b) 5 (c) 14 (d) 2

Q.04. If the sum of two positive numbers is 5 and the sum of their squares is 17, what is the product of the numbers ?
(a) 22 (b) 8 (c) 4 (d) 12

Q.05. The value of the expression $5x^2 + 6x + 7$ for x = 1 is :-
(a) 17 (b) 1 (c) 18 (d) 19

Q.06. If x = 2 is a solution of $x^2 + Kx + 4 = 0$, then the value of K is :-
(a) 2 (b) –2 (c) 4 (d) – 4

Q.07. The equations $x^2 - 6x + k = 0$ has real roots, if
(a) $K \leq 9$ (b) $K \geq 9$ (c) $K = 9$ (d) $K = 0$

Q.08. If α, β are the roots of $x^2 - px + q = 0$, the value of $\alpha^2 + \beta^2$ is
(a) $p^2 + 2q$ (b) $p^2 - 2q$ (c) $p(p^2 - 3q)$ (d) $p^2 - 4q$

Q.09. The perimeter of rectangle is 14 and area is 12. The breadth of rectangle is :-
(a) 4 (b) 3 (c) 5 (d) 6

Q.10. What must be subtracted from $3 + 2x - x^3$, to get $3 + 2x$?
(a) x^3 (b) $-x^3$ (c) $3 + 2x$ (d) None of these

Q.11. If the number increased by its reciprocal is equal to 17/4, then the number is :-
(a) 1/4 (b) 8 (c) 3 (d) 2

Q.12. The solution of the equation $x^{2/3} + x^{1/3} - 2 = 0$ is
(a) x = 2 or x = 4 (b) x = 1 or x = – 8
(c) x = 8 or x = –1 (d) x = 1 or x = 8

Q.13. If quadratic equation $2x^2 + 3x + p = 0$ has equal roots, then the value of p will be
(a) 4/3 (b) 5/4 (c) 6/5 (d) 9/8

Q.14. If $A = \frac{x+1}{x-1}$ and $B = \frac{x-1}{x+1}$, then the value of (A – B) is :-
(a) $\frac{2(x^2+1)}{x^2-1}$ (b) $\frac{4x}{x^2-1}$ (c) $\frac{x}{x^2-1}$ (d) None of these

Q.15. The roots of the equation (x + 3) (x – 3) = 25 are
(a) 5 & – 5 (b) 3 & – 3 (c) $\sqrt{34}$ & $-\sqrt{34}$ (d) 8 & 2

Q.16. If one roots of a quadratic equation is $2 + \sqrt{5}$, then the quadratic equation is :-
(a) $x^2 - 4x - 1 = 0$ (b) $x^2 + 4x - 1 = 0$
(c) $x^2 - 4x + 1 = 0$ (d) $x^2 + 4x + 1 = 0$

Q.17. If the sum of the roots of the equations $qx^2 + 2x + 3q = 0$, is equal to their product, then the value of q is equal to
(a) –2/3 (b) 3/2 (c) 3 (d) – 6

Q.18. A train travel a distance of 300 km at a constant speed. If the speed of the train is increased by 5 km an hour, the journey would have taken 2 hours less. The original speed of the train is :-
(a) 25 km/hr (b) 28 km/hr (c) 27 km/hr (d) None of these

Q.19. The value of P for which the equation $px^2 + 4x + 1 = 0$ has real root is :-
(a) $p \geq 4$ (b) $p \leq 4$ (c) p = 2 (d) $p \leq 2$

Q.20. If p and q are the roots of $x^2 + x + 1 = 0$, then value of $p^3 + q^3$ becomes
(a) 2 (b) –2 (c) 4 (d) – 4

Q.21. If α and β are the roots of $x^2 = x + 1$, then value of $\frac{\alpha^2}{\beta} - \frac{\beta^2}{\alpha}$ is :-
(a) $2\sqrt{5}$ (b) $\sqrt{5}$ (c) $3\sqrt{5}$ (d) $-2\sqrt{5}$

Q.22. The condition that one of the roots of $ax^2 + bx + c = 0$ is twice the other, is :-
(a) $b^2 = 4ac$ (b) $2b^2 = 9(c + a)$ (c) $2b^2 = 9ac$ (d) $2b^2 = 9(c – a)$

Q.23. If the roots of $x^2 – px + q = 0$ differ by unity, then $1 + 4q$ = ?
(a) p (b) p^2 (c) pq (d) p^3

Q.24. If α, β are the roots of the equation $2x^2 – 5x + 7 = 0$, then find the quadratic equation whose root are ($2\alpha + 3\beta$) and ($3\alpha + 2\beta$)
(a) $2x^2 – 25x + 82 = 0$, (b) $x^2 + 25x - 82 = 0$,
(c) $2x^2 + 25x + 82 = 0$, (d) None of these

Q.25. If α, β are the roots of $3x^2 – 4x + 1 = 0$ then the equation whose roots are $\frac{\alpha^2}{\beta}$ & $\frac{\beta^2}{\alpha}$
(a) $9x^2 –28x + 3 = 0$ (b) $9x^2 + 28x –3 = 0$
(c) $9x^2 –28x –3 = 0$ (d) None of these

Q.26. If α, β are the roots of equation $x^2 –5x + 6 = 0$ then the equation with roots $(\alpha\beta + \alpha + \beta)$ and $(\alpha\beta – \alpha – \beta)$ is :-
(a) $x^2 –12x + 11 = 0$ (b) $2x^2 – 6x + 12 = 0$
(c) $x^2 – 12x + 12 = 0$ (d) None of these

Q.27. $2 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \ldots\ldots\infty}}}$ = ?
(a) $1+\sqrt{2}$ (b) $1-\sqrt{2}$ (c) $1+\sqrt{3}$ (d) None

Q.28. $\sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2} \ldots\ldots\infty}}}}$ = ?
(a) 2 (b) –2 (c) 1 (d) –1

Q.29. The hypotenuse of a triangle is 5 cm and its base is 4 cm., its perimeter is :-
(a) 20 cm. (b) 12 cm. (c) 10 cm. (d) 6 cm

Q.30. The equation $kx^2 –2kx + k + 1 = 0$ has equal roots, the value of k is :-
(a) 1 (b) 2 (c) 0 (d) –2

Q.31. If α and β are the roots of the equation $x^2 – 5x + 3 = 0$, then the equation having roots as $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is :-
(a) $x^2 – 19x + 3 = 0$ (b) $3x^2 – 19x –3 = 0$
(c) $3x^2 – 19x + 3 = 0$ (d) $3x^2 + 19x + 3 = 0$

Q.32. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is Equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to :-
(a) 2 (b) –2 (c) 1 (d) –1

Sol:- Let the roots be α and β
We have $\alpha + \beta$ = –b/a and $\alpha\beta$ = c/a
Given that the sum of the roots of the quadratic equation
$ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals.

Therefore

$\Rightarrow \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2} \Rightarrow \alpha + \beta = \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2} \Rightarrow \alpha + \beta = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$

$\Rightarrow \frac{-b}{a} = \frac{\left(\frac{-b}{a}\right)^2 - 2\frac{c}{a}}{\left(\frac{c}{a}\right)^2} \Rightarrow \frac{-b}{a} \cdot \frac{c^2}{a^2} = \frac{b^2 - 2ac}{a^2} \Rightarrow \frac{-bc^2}{a} = b^2 - 2ac$

$\Rightarrow$ Divide both side by "ac" $\Rightarrow \frac{-bc^2}{a^2 c} = \frac{b^2}{ac} - 2 \Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$

Q.33. If a, b and c are in G.P., then the roots of the quadratic equation, $ax^2 + 2bx + c = 0$ are

(a) Real and Unequal (b) Real and Equal (c) Imaginary (d) Complex

CUBICAL EQUATION

Solve the given cubical equation *(Q.01 to Q.06)* and find the value of x

Q.01. $x^3 - 6x^2 + 11x - 6 = 0$

(a) $(-1, 1, -2)$ (b) $(1, 2, 3)$ (c) $(-2, 2, 3)$ (d) $(0, 4, -5)$

Q.02. $x^3 + 2x^2 - x - 2 = 0$

(a) $1, -1, 2$ (b) $-1, 1, -2$ (c) $-1, 2, -2$ (d) $1, 2, 2$

Q.03. $x^3 + 7x^2 - 21x - 27 = 0$

(a) $(-3, -9, -1)$ (b) $(3, -9, -1)$ (c) $(3, 9, 1)$ (d) $(-3, 9, 1)$

Q.04. $x^3 + x^2 - 20x = 0$

(a) $(1, 4, -5)$ (b) $(2, 4, -5)$ (c) $(0, -4, 5)$ (d) $(0, 4, -5)$

Q.05. $x^3 + x^2 - x - 1 = 0$

(a) $(-1, -1, 1)$ (b) $(1, 1, -1)$ (c) $(-1, -1, -1)$ (d) $(1, 1, 1)$

Q.06. $2x^3 - x^2 - 4x + 2 = 0$

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) -2 (d) 2

Q.07. The factors of the cubic equation $3x^3 + 5x^2 - 3x - 5 = 0$

(a) $(x-1), (x), (x+1)$ (b) $(x-1), (x-2), (x-\frac{5}{3})$

(c) $(x+1), (x-1), (x-1)$ (d) None of these

Q.08. If $4x^3 + 8x^2 - x - 2 = 0$, then find the value of $2x + 3$

(a) $(-4, -1, 2)$ (b) $(-4, 2, 1)$ (c) $(4, -1, 2)$ (d) None of these

Q.09. If $(x - 1)$ and $(x - 2)$ are the factors of $x^3 + ax^2 + bx - 2$ then the value of a and b respectively, are

(a) 4 and 5 (b) –4 and –5 (c) –4 and 5 (d) 4 and –5

Answers

One / Two / Three Variables EQUATIONS

1	a	11	d	21	b	31	b
2	b	12	c	22	b	32	a
3	c	13	c	23	a	33	c
4	b	14	d	24	b	34	c
5	c	15	c	25	a	35	b
6	a	16	c	26	a		
7	b	17	d	27	b		
8	a	18	b	28	b		
9	d	19	b	29	c		
10	b	20	a	30	c		

MISCELLANEOUS QUESTIONS Of One / Two / Three Variables EQUATIONS

1	d	11	b	21	b
2	c	12	a	22	c
3	c	13	a	23	a
4	d	14	a	24	a
5	a	15	c	25	b
6	a	16	a	26	a
7	b	17	a		
8	b	18	a		
9	d	19	a		
10	a	20	b		

QUADRATIC EQUATIONS

1	c	11	c	21	a	31	c	41	a
2	d	12	b	22	c	32	a	42	b
3	b	13	c	23	b	33	b	43	c
4	c	14	b	24	a	34	c	44	a
5	a	15	c	25	b	35	a	45	a
6	b	16	b	26	c	36	c	46	b
7	b	17	d	27	b	37	b		
8	a	18	b	28	a	38	c		
9	a	19	a	29	c	39	b		
10	b	20	c	30	b	40	c		

MISCELLANEOUS QUESTIONS of QUADRATIC EQUATIONS

1	c	11	a	21	d	31	c
2	b	12	b	22	c	32	a
3	c	13	d	23	b	33	b
4	c	14	b	24	a		
5	c	15	c	25	a		
6	d	16	a	26	a		
7	a	17	a	27	a		
8	b	18	a	28	a		
9	b	19	b	29	b		
10	b	20	a	30	c		

1	b
2	b
3	b
4	d
5	a
6	a
7	d
8	c
9	c



CHAPTER 3

LINER INEQUALITIES

3.1 INEQUALITIES

Inequalities are statements where two quantities are unequal but a relationship exists between them. These type of inequalities occur in business whenever there is a limit on supply, demand, sales etc. For example, if a producer requires a certain type of raw material for his factory and there is an upper limit in the availability of that raw material, then any decision which he takes about production should involve this constraint also. We will see in this chapter more about such situations.

3.2LINEAR INEQUALITIES IN ONE VARIABLE AND THE SOLUTION SPACE

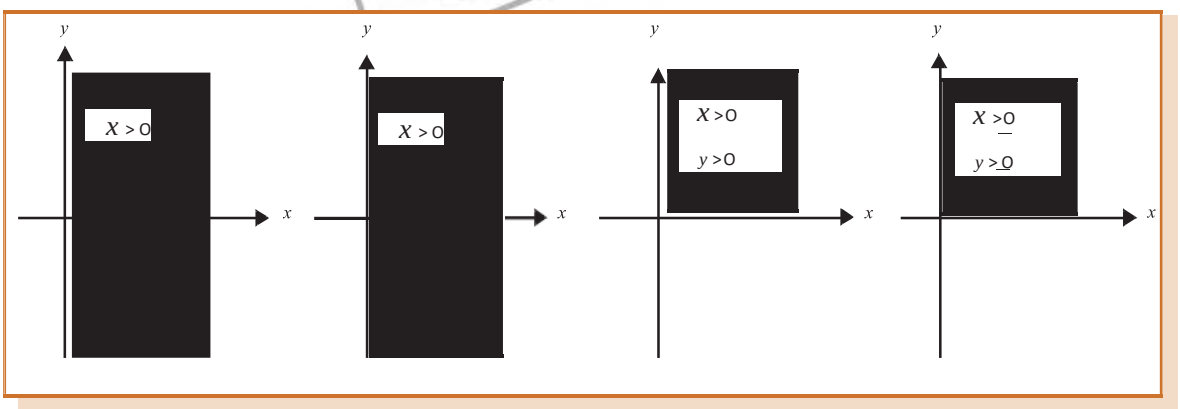
Any linear function that involves an inequality sign is a linear inequality. It may be of one variable, or, of more than one variable. Simple example of linear inequalities are those of one variable only; viz., x > 0, x < 0 etc.



The values of the variables that satisfy an inequality are called the solution space, and is abbreviated as S.S. The solution spaces for (i) x > 0, (ii) x 0 are shaded in the above diagrams, by using deep lines.

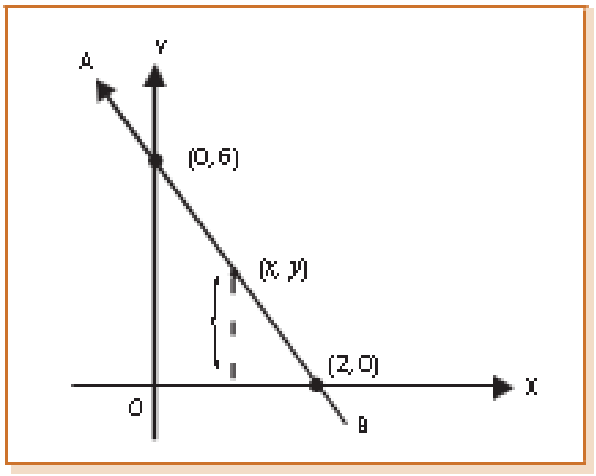
Linear inequalities in two variables: Now we turn to linear inequalities in two variables x and y and shade a few S.S.

Let us now consider a linear inequality in two variables given by 3x + y < 6



The inequality mentioned above is true for certain pairs of numbers (x, y) that satisfy 3x + y < 6. By trial, we may arbitrarily find such a pair to be (1,1) because 3 x 1 + 1 = 4, and 4 < 6.

Linear inequalities in two variables may be solved easily by extending our knowledge of straight lines.



For this purpose, we replace the inequality by an equality and seek the pairs of number that satisfy 3x + y = 6. We may write 3x + y = 6 as 6 – 3x, and draw the graph of this linear function.

Let x = 0 so that y = 6. Let y = 0, so that x = 2.

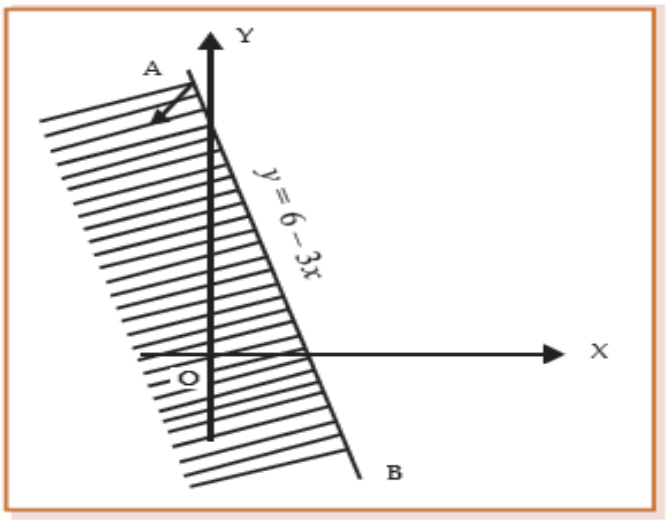
Any pair of numbers (x, y) that satisfies the equation y = 6 – 3x falls on the line AB.

Note: The pair of inequalities $x \geq 0$, $y \geq 0$ play an important role in linear programming problems.

Therefore, if y is to be less than 6 – 3x for the same value of x, it must assume a value that is less than the ordinate of length 6 – 3x.

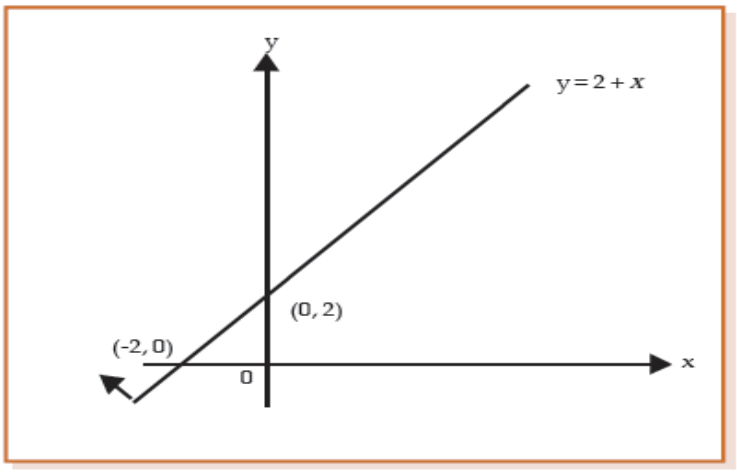
All such points (x, y) for which the ordinate is less than 6 – 3x lie below the line AB.

The region where these points fall is indicated by an arrow and is shaded too in the adjoining diagram. Now we consider two inequalities $3x + y \leq 6$ and $x – y \leq - 2$ being satisfied simultaneously by x and y. The pairs of numbers (x, y) that satisfy both the inequalities may be found b drawing the graphs of the two lines y = 6 – 3x and y = 2 + x, and determining the region where each equality with y on the left-side and the remaining terms in the right side. The first inequality $3x + y \leq 6$ is equivalent to $y \leq 6 – 3x$ and it requires the value of y for each x to be less than or equal to that of and on 6 – 3x. The inequality is therefore satisfied by all points lying below the line y = 6 – 3x. The region where these points fall has been shaded in the adjoining diagram.

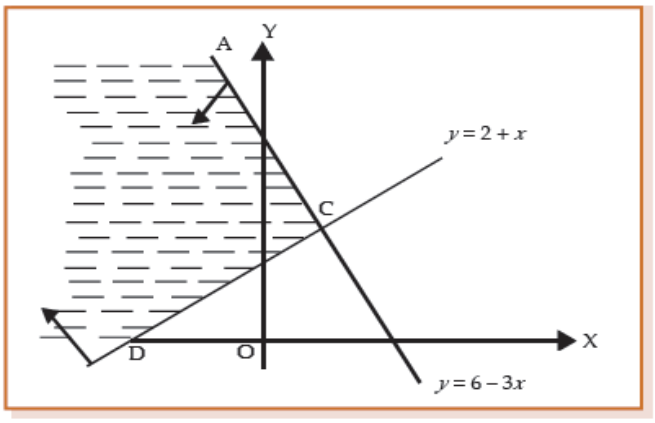


We consider the second inequality $x - y \leq -2$, and note that this is equivalent to $y \geq 2 + x$. It requires the value of y for each x to be larger than or equal to that of $2 + x$. The inequality is, therefore, satisfied by all points lying on and above the line $y = 2 + x$.

The region of interest is indicated by an arrow on the line $y = 2 + x$ in the diagram below.
For $x = 0$, $y = 2 + 0 = 2$;
For $y = 0$, $0 = 2.+ x$ i.e, $x = -2$



By superimposing the above two graphs we determine the common region ACD in which the pairs (x, y) satisfy both inequalities.



We now consider the problem of drawing graphs of the following inequalities

$x \geq 0, y \geq 0, x \leq 6, y \leq 7, x + y \leq 12$

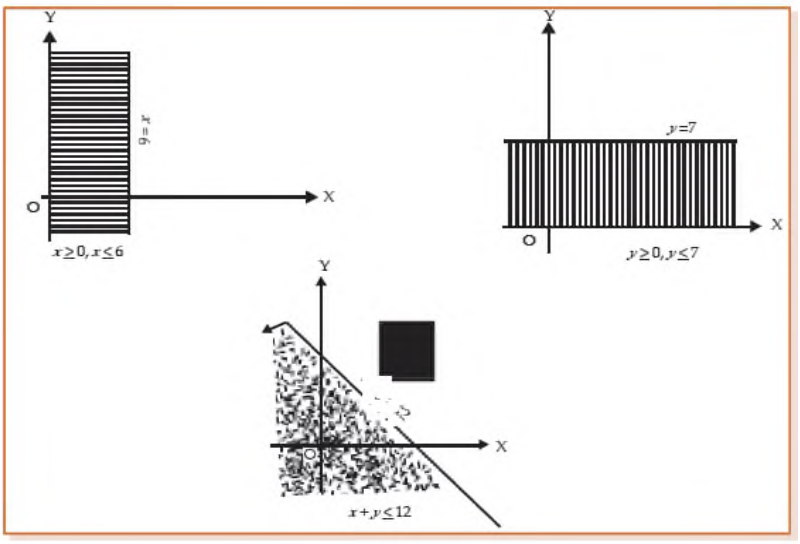
and shading the common region.

Note: 1. The inequalities $3x + y \leq 6$ and $x - y \leq 2$ differ from the preceding ones in that these also include equality signs. It means that the points lying on the corresponding lines are also included in the region.
2. The procedure may be extended to any number of inequalities.

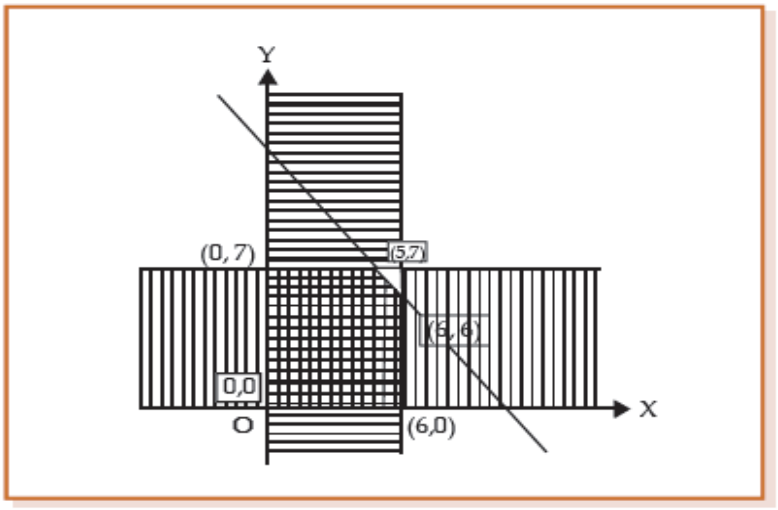
We note that given inequalities may be grouped as follows:

$x \geq 0$, $y \geq 0$

$x \leq 6,\ \ y \leq 7,\ \ x + y \leq 12$



By superimposing the above three graphs, we determine the common region in the xy plane where all the five inequalities are simultaneously satisfied.



This common region is known as feasible region or the solution set (or the polygonal convex sets).

A region is said to be bounded if it can be totally included within a (very large) circle. The shaded region enclosed by deep lines in the previous diagram is bounded, since it can be included within a circle.

The objective function attains a maximum or a minimum value at one of the corner points of the feasible solution known as extreme points of the solution set. Once these extreme points (the points of intersection of lines bounding the region) are known, a compact matrix representation of these points is possible. We shall denote the matrix of the extreme points by E.

The coefficients of the objective function may also be represented by a column vector. We shall represent this column vector by C.

The elements in the product matrix EC shows different values, which the objective function attains at the various extreme points. The largest and the smallest elements in matrix EC are respectively the maximum and the minimum values of the objective function. The row in matrix EC in which this happens is noted and the elements in that row indicate the appropriate pairing and is known as the optimal solution.

In the context of the problem under consideration.

$$E = \begin{matrix} X & Y \end{matrix} \begin{bmatrix} 0 & 0 \\ 0 & 7 \\ 5 & 7 \\ 6 & 0 \\ 6 & 6 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{matrix} x \\ y \end{matrix}$$

$$EC = \begin{bmatrix} 0 & 0 \\ 0 & 7 \\ 5 & 7 \\ 6 & 0 \\ 6 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0\times1+0\times2 \\ 0\times1+7\times2 \\ 5\times1+7\times2 \\ 6\times1+0\times2 \\ 6\times1+6\times2 \end{bmatrix} = \begin{bmatrix} 0 \\ 14 \\ 19 \\ 6 \\ 18 \end{bmatrix}$$

The given objective function viz. Z = x + 2y is maximum at the points (5, 7) present in the third row of the matrix E. Thus the optimal solution is x = 5, y = 7, and the maximum value of the objective function is 19.

We now list the steps to be followed under graphical solution to a linear programming problem.

Step 1 Determine the region that satisfies the set of given inequalities.

Step 2 Ensure that the region is bounded. If the region is not bounded, either there are additional hidden conditions which can be used to bound the region or there is no solution to the problem.

Step 3 Construct the matrix E of the extreme points, and the column vector C of the objective function.

Step 4 Find the matrix product EC. For maximization, determine the row in EC where the largest element appears; while for minimization, determine the row in EC where the smallest element appears.

Step 5 The objective function is optimized corresponding to the same row elements of the extreme point matrix E.

Note: If the slope of the objective function be same as that of one side of feasible region, there are multiple solutions to the problem. However, the optimized value of the objective function remains the same.

Example:

A manufacturer produces two products A and B, and has his machines in operation for 24 hours a day. Production of A requires 2 hours of processing in machine M1 and 6 hours in machine M2. Production of B requires 6 hours of processing in machine M1 and 2 hours in machine M2. The manufacturer earns a profit of ` 5 on each unit of A and ` 2 on each unit of B. How many units of each product should be produced in a day in order to achieve maximum profit?

Solution: Let X_2 be the number of units of type A product to be produced, and X_2 is that of type B product to be produced. The formulation of the L.P.P. in this case is as below:

Maximize Z = $5X_1 + 2X_2$

It is inconceivable for a practical problem to have an unbounded solution.

Subject to the constraints.

$2X_1 1 + 6X_2 < 24$

$6X_1 + 2X_2 < 24$

$X_1 \geq 0, X_2 \geq 0$

For the line $2X_1 + 6X_2 = 24$

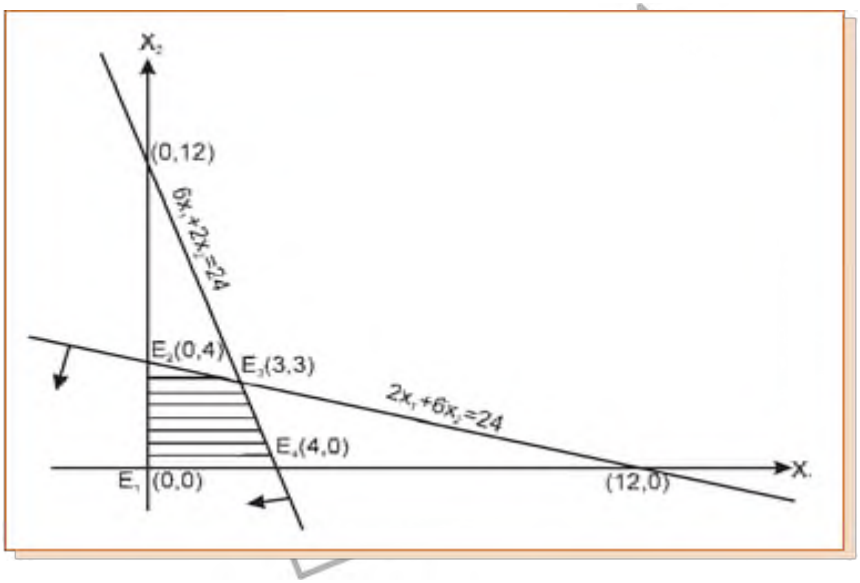
Let $X_1 = 0$, so that $X_2 = 4$

Let $X_2 = 0$, so that $X_1 = 12$

For the line $6X_1 + 2X_2 = 24$

Let $X_1 = 0$, so that $X_2 = 12$

Let $X_2 = 0$, so that $X_1 = 4$



The shaded portion in the diagram is the feasible region and the matrix of the extreme points E_1, E_2, E_3 and E_4 is

$$E = \begin{matrix} X_1 & X_2 \\ \end{matrix} \begin{bmatrix} 0 & 0 \\ 0 & 4 \\ 3 & 3 \\ 4 & 0 \end{bmatrix} \begin{matrix} E_1 \\ E_2 \\ E_3 \\ E_4 \end{matrix}$$

The column vector for the objective function is $C = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \end{matrix}$

The column vector the values of the objective function is given by

$$E = \begin{bmatrix} 0 & 0 \\ 0 & 4 \\ 3 & 3 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \times 5 + 0 & \times 2 \\ 0 \times 5 + 4 & \times 2 \\ 3 \times 5 + 3 & \times 2 \\ 4 \times 5 + 0 & \times 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 21 \\ 20 \end{bmatrix} \begin{matrix} E_1 \\ E_2 \\ E_3 \\ E_4 \end{matrix}$$

Since 21 is the largest element in matrix EC, therefore the maximum value is reached at the extreme point E3 whose coordinates are (3, 3).

Thus, to achieve maximum profit the manufacturer should produce 3 units each of both the products A and B.

Summary of Graphical Method

It involves:

(i) Formulating the linear programming problem, i.e. expressing the objective function and constraints in the standardised format.

(ii) Plotting the capacity constraints on the graph paper. For this purpose normally two terminal points are required. This is done by presuming simultaneously that one of the constraints is zero. When constraints concerns only one factor, then line will have only one origin point and it will run parallel to the other axis.

(iii) Identifying feasible region and coordinates of corner points. Mostly it is done by breading theto form a point on graph.

(iv) Testing the corner point which gives maximum profit. For this purpose the coordinates relating to the corner point should put in objectives function and the optimal point should be ascertained.

(v) For decision – making purpose, sometimes, it is required to know whether optimal point leaves some resources unutilized. For this purpose value of coordinates at the optimal point should be put with constraint to find out which constraints are not fully utilized.

Example: A company produces two products A and B, each of which requires processing in two machines. The first machine can be used at most for 60 hours, the second machine can be used at most for 40 hours. The product A requires 2 hours on machine one and one hour on machine two. The product B requires one hour on machine one and two hours on machine two. Express above situation using linear inequalities.

Solution: Let the company produce, x number of product A and y number of product B. As each of product A requires 2 hours in machine one and one hour in machine two, x number of product A requires 2x hours in machine one and x hours in machine two. Similarly, y number of product B requires y hours in machine one and 2y hours in machine two. But machine one can be used for 60 hours and machine two for 40 hours. Hence X_2 + y cannot exceed 60 and x + 2y cannot exceed 40. In other words,

$2x + y \leq 60$ and $x + 2y \leq 40$.

Thus, the conditions can be expressed using linear inequalities.

Example: A fertilizer company produces two types of fertilizers called grade I and grade II. Each of these types is processed through two critical chemical plant units. Plant A has maximum of 120 hours available in a week and plant B has maximum of 180 hours available in a week. Manufacturing one bag of grade I fertilizer requires 6 hours in plant A and 4 hours in plant B. Manufacturing one bag of grade II fertilizer requires 3 hours in plant A and 10 hours in plant B. Express this using linear inequalities.

Solution: Let us denote by x_1, the number of bags of fertilizers of grade I and by x_2, the number of bags of fertilizers of grade II produced in a week. We are given that grade I fertilizer requires 6 hours in plant A and grade

II fertilizer requires 3 hours in plant A and plant A has maximum of 120 hours available in a week. Thus $6x_1 + 3x_2 \leq 120$.

Similarly grade I fertilizer requires 4 hours in plant B and grade II fertilizer requires 10 hours in Plant B and Plant B has maximum of 180 hours available in a week. Hence, we get the inequality $4\ x_1 + 10x_2 \geq 180$.

Example: Graph the inequalities $5x_1 + 4x_2 \geq 9$, $x_1 + x_2 \geq 3$, $x_1 \geq 0$ and $x_2 \geq 0$ and mark the common region.

Solution: We draw the straight lines $5x_1 + 4x_2 = 9$ and $x_1 + x_2 = 3$.

Table for $5\ x_1 + 4x_2 = 9$

x_1	0	9/5
x_2	9/4	0

Table for $x_1 + x_2 = 3$.

x_1	0	3
x_2	3	0

Now, if we take the point (4, 4), we find

$5x_1 + 4x_2 \geq 9$

i.e., $5.4 + 4.4 \geq 9$

or, $36 \geq 9$ (True)

$x_1 + x_2 \geq 3$

i.e., $4 + 4 \geq 3$

$8 \geq 3$ (True)

Hence (4, 4) is in the region which satisfies the inequalities.
We mark the region being satisfied by the inequalities and note that the cross-hatched region is satisfied by all the inequalities.

Example: Draw the graph of the solution set of the following inequality and equality:

$x + 2y = 4$.

$x - y \leq 3$.

Mark the common region.

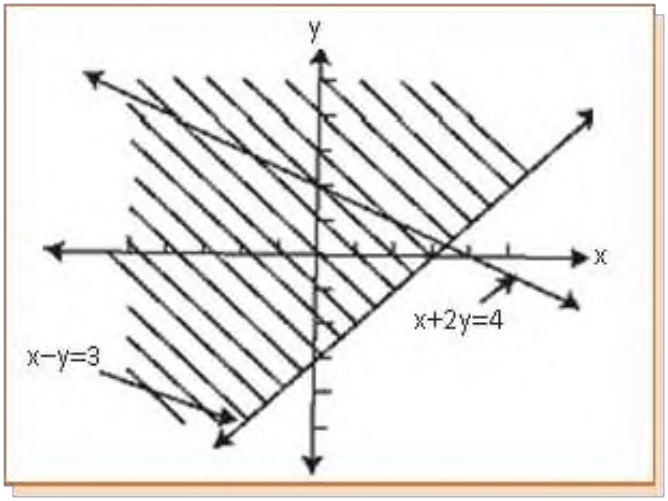
Solution: We draw the graph of both $x + 2y = 4$ and $x - y \leq 3$ in the same plane.
The solution set of system is that portion of the graph of $x + 2y = 4$ that lies within the half- plane representing the inequality $x - y \leq 3$.

For x + 2y = 4,

x	4	0
y	0	2

For x – y = 3,

x	3	0
y	0	–3



Example: Draw the graphs of the following inequalities:

$x + y \leq 4$,

$x – y \leq 4$,

$x \geq -2$.

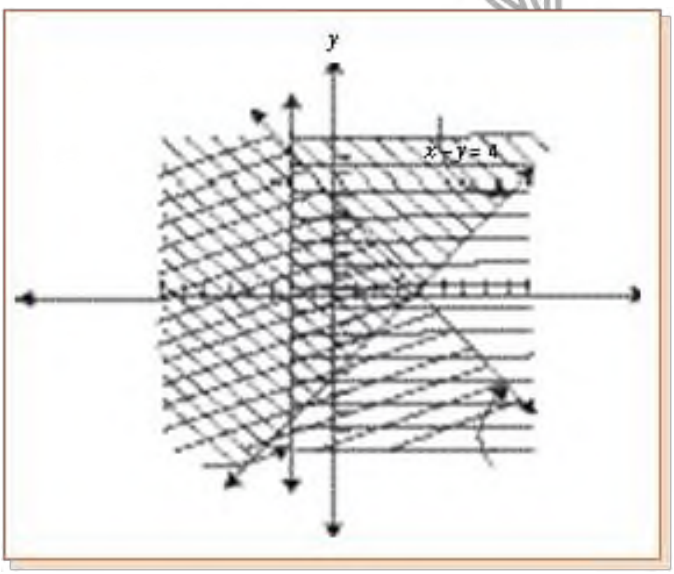
And mark the common region.

For x- y = 4,

x	4	0
y	0	- 4

For x-y = 4,

x	0	4
y	4	0



The common region is the on represented by overlapping of the shadings.

Example: Draw the graphs of the following linear inequalities:

$5x + 4y \leq 100$, $5x + y \geq 40$,

$3x + 5y \leq 75$, $x \geq 0, y \geq 0$.

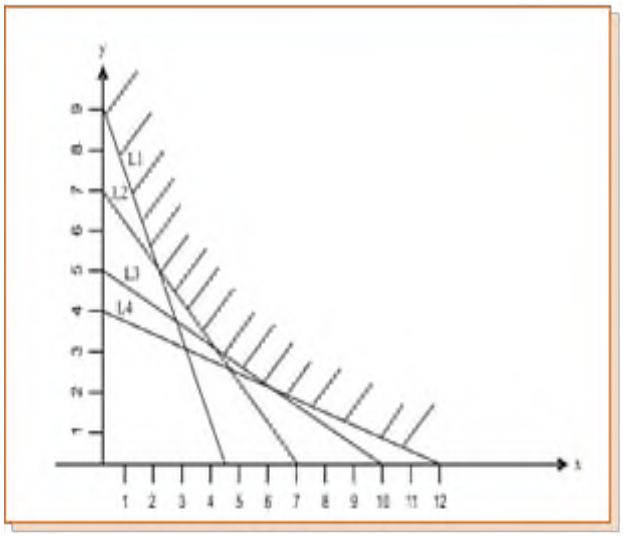
and mark the common region.

Solution:

5x + 4y = 100 or, $\frac{x}{20} + \frac{y}{25} = 1$

3x + 5y = 75 or, $\frac{x}{25} + \frac{y}{15} = 1$

5x + y = 100 or, $\frac{x}{8} + \frac{y}{40} = 1$



Example: Draw the graphs of the following linear inequalities:

5x + 8y ≤ 2000, x ≤ 175, x ≥ 0

7x + 4y ≤ 1400, y ≤ 225, y ≥ 0.

and mark the common region:

Solution: Let us plot the line AB (5x +8y = 2,000) by joining this points A (400, 0) and B (0. 250).

X	400	0
y	0	250

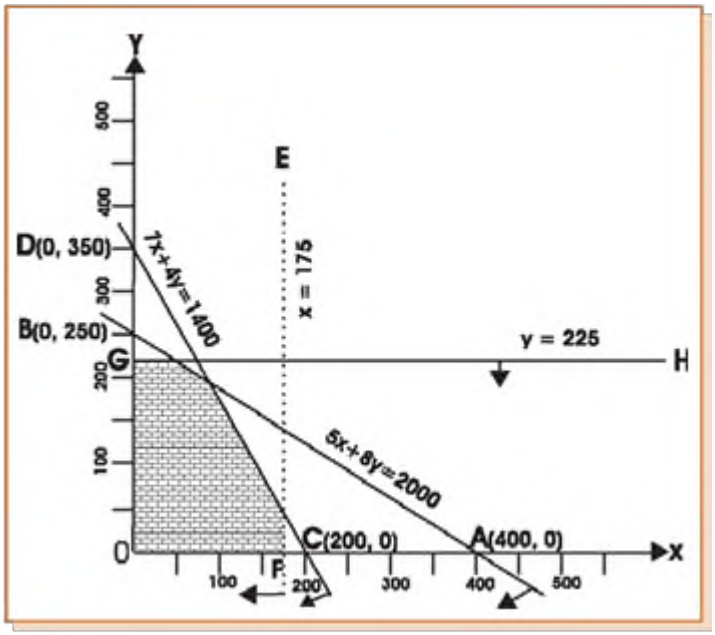
Similarly, we plot the line CD (7x + 4y = 1400) by

X	200	0
y	0	350

joining the points C(200, 0) and D(0, 350).

The required graph is shown alongside in which the common region is shaded.

Also, we draw the lines EF(x = 175) and GH (y = 225).



Example: Draw the graphs of the following linear inequalities:

$x + y \geq 1$, $7x + 9y \leq 63$,

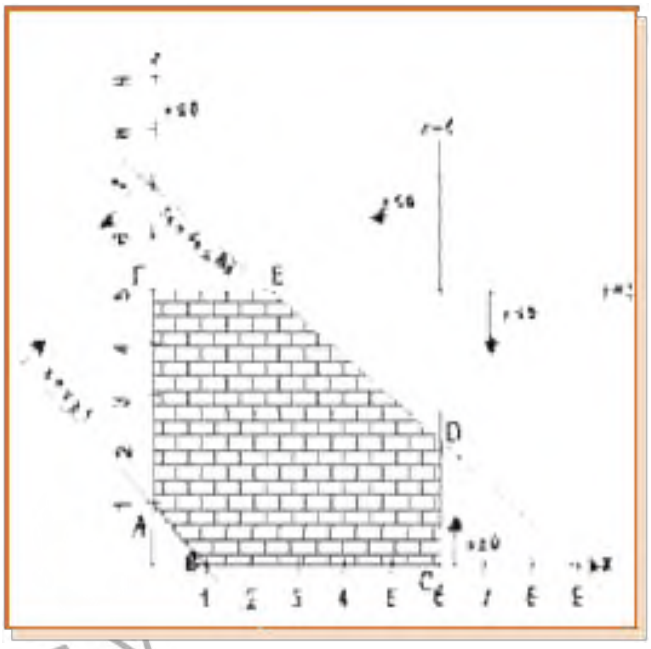
$y \leq 5$, $x \leq 6$, $x \geq 0, y \geq 0$.

and mark the common region.

Solution:

x-y = 1; $\frac{x}{y}\left|\frac{1}{0}\right|\frac{0}{1}$; 7x + 9y = 63, $\frac{x}{y}\left|\frac{9}{0}\right|\frac{0}{7}$.

We plot the line AB (x+ y = 1), CD 9(y = 5),
EF (x = 6), DE (7x +9y = 63).
Given inequalities are shown by arrows.
Common region ABCDEF is the shaded region.



Example: Two machines (I and II) produce two grades of plywood, grade A and grade B. In one hour off operation machine I produces two units of grade A and one unit of grade B, while machine II, in one hour of operation produces three units of grade A and four units of grade B. The machines are required to meet a production schedule of at least fourteen units of grade A and twelve units of grade B. Express this using linear inequalities and draw the graphs.

Solution: Let the number of hours required on machine I be x and that on machine II be y. Since in one hour, machine I can produce 2 units of grade A and one unit of grade B, in x hours it will produce 2x and x units of grade A and B respectively. Similarly, machine II, in one hour, can produce 3 units of grade A and 4 units of grade B. Hence, in y hours, it will produce 3y and 4y units Grade A & B respectively.

The given data can be expressed in the form of linear inequalities as follows: $2x + 3y \geq 14$ (Requirement of grade A)

$x + 4y \geq 12$ (Requirement of grade B)

Moreover x and y cannot be negative, thus $x \geq 0$ and $y \geq 0$

Let us now draw the graphs of above inequalities. Since both x and y are positive, it is enough to draw the graph only on the positive side.

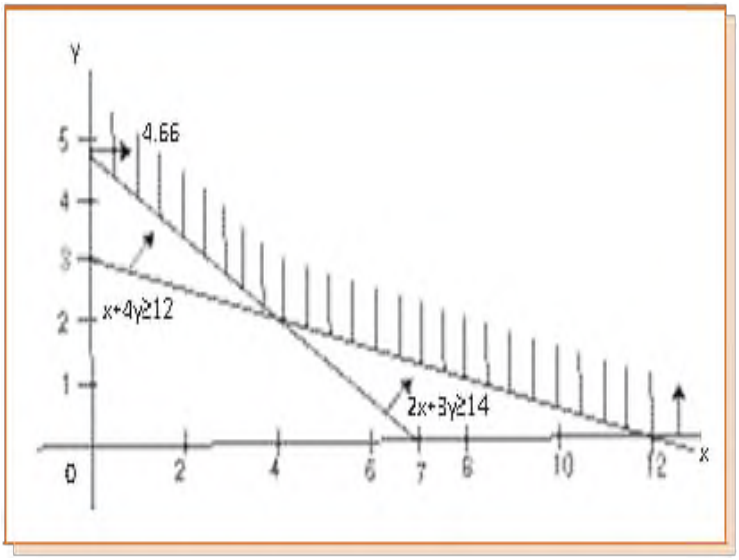
The inequalities are drawn in the following graph:

For 2x + 3y = 14,

x	7	0
y	0	4.66

For $x + 4y = 12$,

x	0	12
y	3	0



In the above graph we find that the shaded portion is moving towards infinity on the positive side. Thus the result of these inequalities is unbounded.

EXERCISE: 3 (A)

Choose the correct answer/ answers

(1) An employer recruits experienced (x) and fresh workmen (y) for his firm under the condition that he cannot employ more than 9 people. x and y can be related by the inequality

(a) $x + y \neq 9$ (b) $x + y \leq 9$ $x \geq 0, y \geq 0$

(c) $x + y \geq 9$ $x \geq 0, y \geq 0$ (d) none of these

(2) On the average experienced person does 5 units of work while a fresh one 3 units of work daily but the employer has to maintain an output of at least 30 units of work per day. This situation can be expressed as

(a) $5x + 3y \leq 30$ (b) $5x + 3y > 30$

(c) $5x + 3y \geq 30$ $x \geq 0, y \geq 0$ (d) none of these

(3) The rules and regulations demand that the employer should employ not more than 5 experienced hands to 1 fresh one and this fact can be expressed as

(a) $y \geq x/5$ (b) $5y \leq x$

(c) $5y \geq x$ (d) none of these

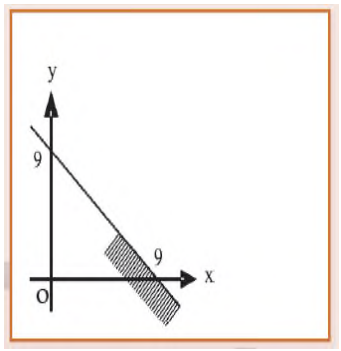
(4) The union however forbids him to employ less than 2 experienced person to each fresh person. This situation can be expressed as

(a) $x \leq y/2$ (b) $y \leq x/2$

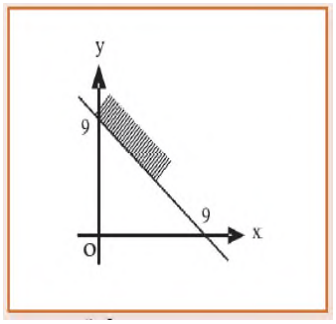
(c) $y \geq x/2$ (d) $x \geq 2y$

(5) The graph to express the inequality $x + y \leq 9$ is

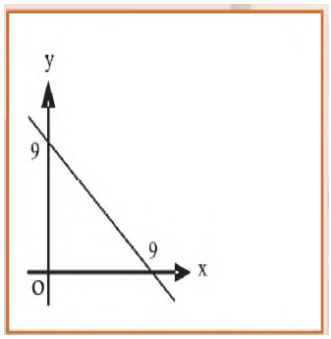
(a)



(b)



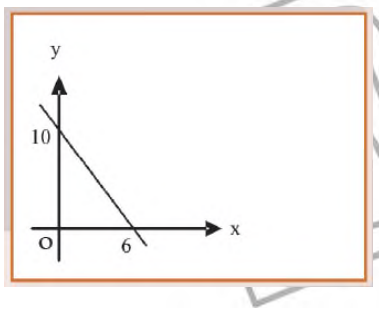
(c)



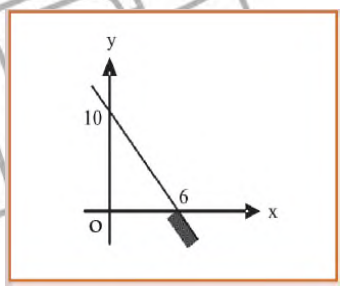
(d) none of these

(6) The graph to express the inequality $5x + 3y \geq 30$ is

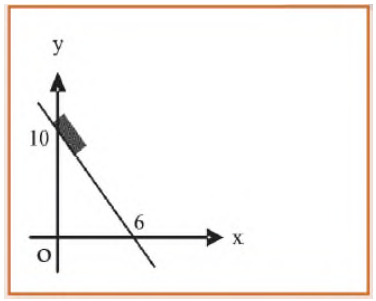
(a)



(b)

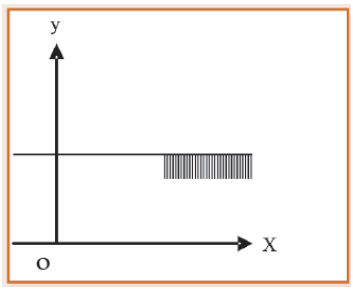


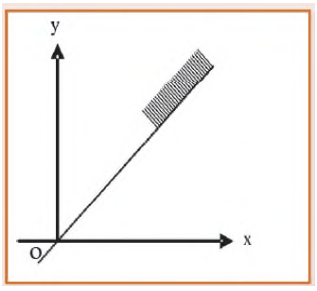
(c)

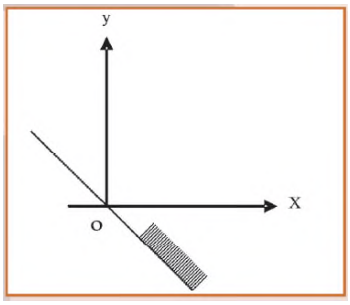


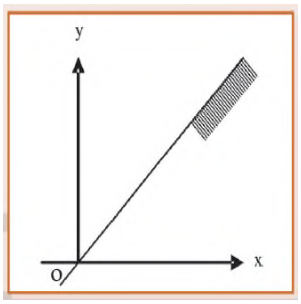
(d) none of these

(7) The graph to express the inequality $y \leq (1/2)\, x$ is indicated by

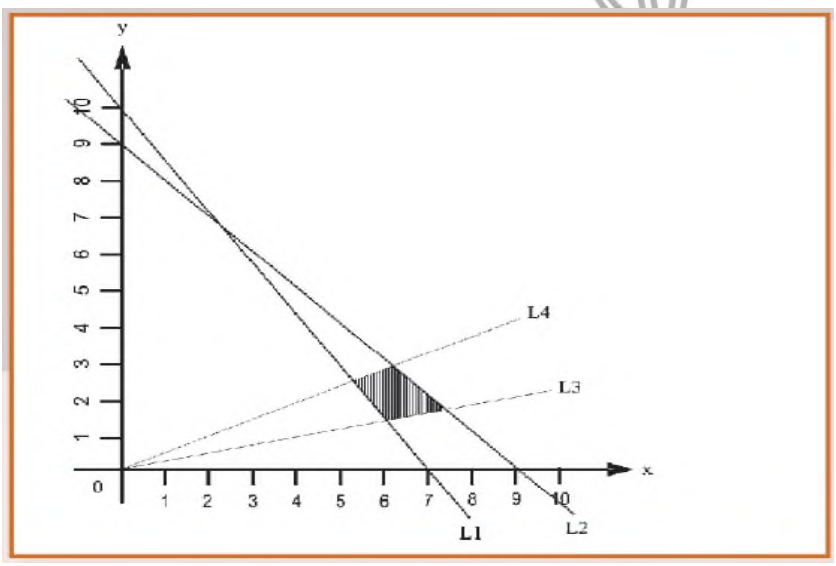
(a) 

(b) 

(c) 

(d) 

(8) L1 : 5x + 3y = 30 L2 : x + y = 9 L3 : y = x/3 L4 : y = x /2
The common region (shaded part) shown in the diagram refers to



(a) 5x + 3y ≤ 30
x + y ≤ 9
y ≤ 1/5 x
y ≤ x/2

(b) 5x + 3y ≥ 30
x + y ≤ 9
y ≥ x/3
y ≤ x/2
x ≥ 0, y ≥ 0

(c) 5x + 3y ≥ 30
x + y ≥ 9
y ≤ x/3
y ≥ x/2
x ≥ 0, y ≥ 0

(d) None of these

(9) A dietitian wishes to mix together two kinds of food so that the vitamin content of the mixture is at least 9 units of vitamin A, 7 units of vitamin B, 10 units of vitamin C and 12 units of vitamin D. The vitamin content per Kg. of each food is shown below:

	A	B	C	D
Food I :	2	1	1	2
Food II:	1	1	2	3

Assuming x units of food I is to be mixed with y units of food II the situation can be expressed as

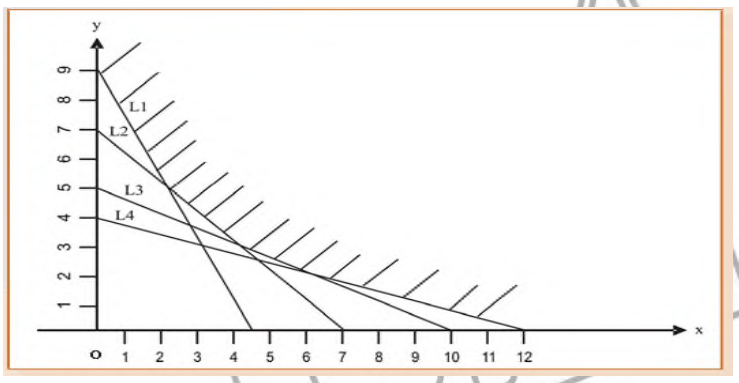
(a) $2x + y = 9$, $x + y \geq 7$, $x + 2y \geq 10$, $2x + 3y \geq 12$, $x > 0, y > 0$

(b) $2x + y = 30$, $x + y \geq 7$, $x + 2y \geq 10$, $x + 3y \geq 12$

(c) $2x + y = 9$, $x + y \geq 7$, $x + y \geq 10$, $x + 3y \geq 12$

(d) $2x + y = 9$, $x + y \geq 7$, $x + 2y \geq 10$, $2x + 3y \geq 12$, $x \geq 0, y \geq 0$,

(10) Graphs of the in equations are drawn below:



L1 : $2x + y = 9$ L2 : $x + y = 7$

L3 : $x + 2y = 10$ L4 : $x + 3y = 12$

The common region (shaded part) indicated on the diagram is expressed by the set of inequalities

(a) $2x + y = 9$, $x + y = 7$, $x + 2y = 10$, $x + 3y = 12$

(b) $2x + y = 9$, $x + y = 7$, $x + 2y = 10$, $x + 3y = 12$

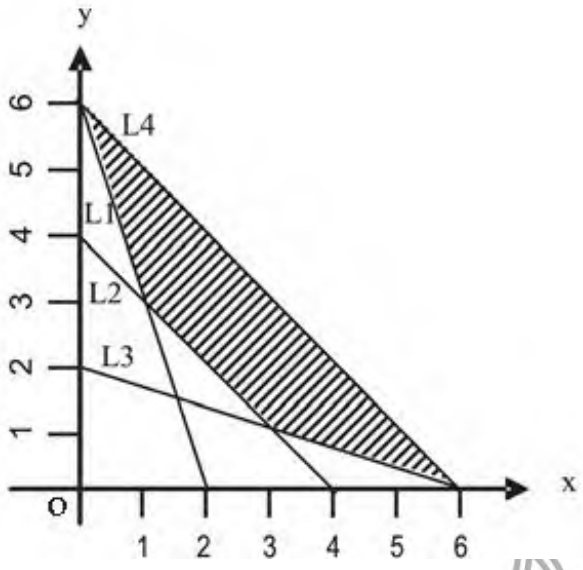
(c) $2x + y = 9$, $x + y = 7$, $x + 2y = 10$, $x + 3y = 12$, $x \geq 0, y = 0$

(d) none

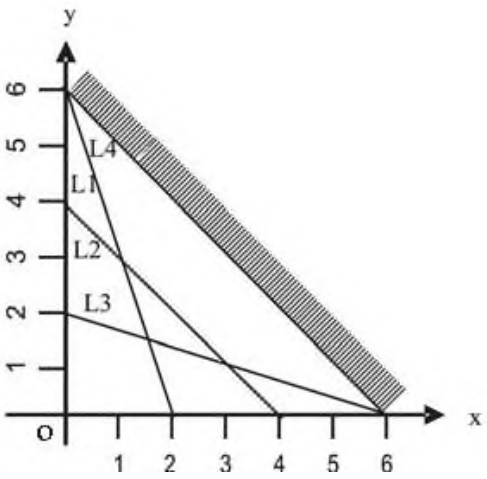
(11) The common region satisfied by the inequalities L1: $3x + y \geq 6$, L2: $x + y = 4$, L3: $x + 3y = 6$, and L4: $x +$

y = 6 is indicated by

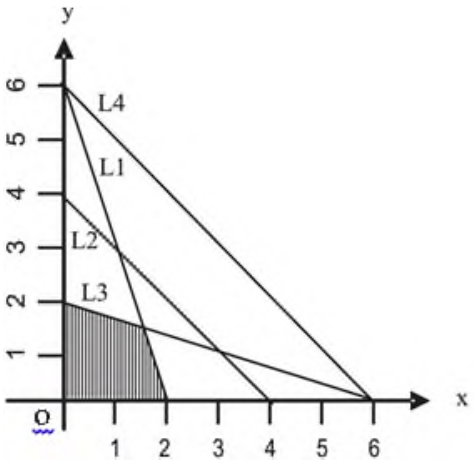
(a)



(b)

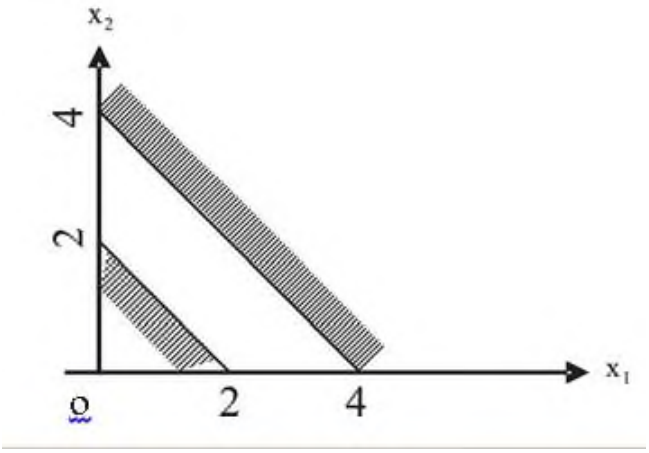


(c)



(d) None of these

(12) The region indicated by the shading in the graph is expressed by inequalities



(a) $x_1 + x_2 = 2$

$2x_1 + 2x_2 = 8$

$x_1 \geq 0$, $x_2 = 0$,

(b) $x_1 + x_2 = 2$

$x_2\, x_1 + x_2 = 4$

(c) $x_1 + x_2 = 2$

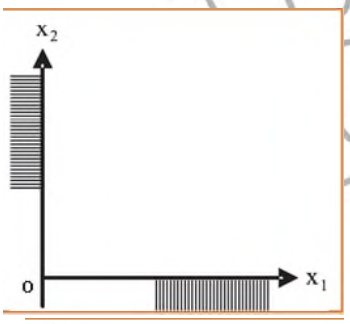
$2x_1 + 2x_2 = 8$

(d) $x_1 + x_2 = 2$

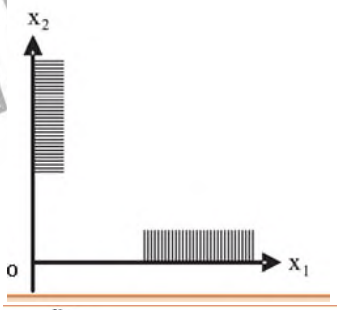
$2x_1 + 2x_2 > 8$

(13) The inequalities $x_1 = 0$, $x_2 = 0$, are represented by one of the graphs shown below:

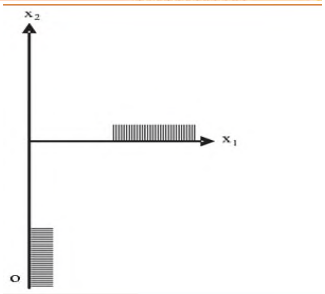
(a)



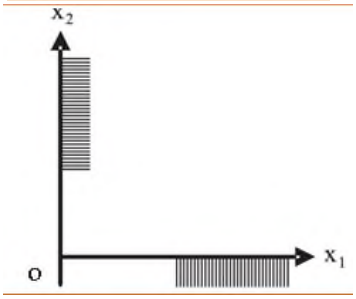
(b)

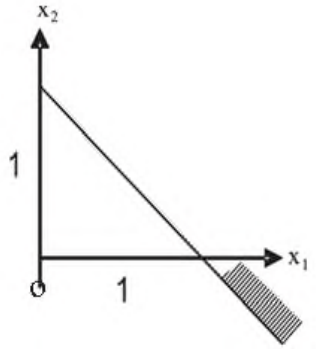


(c)



(d)



(14) 

The region is expressed as

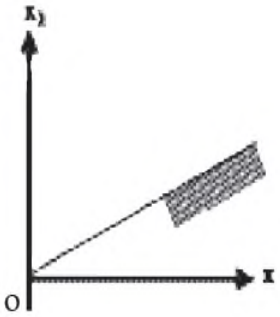
(a) $x_1 - x_2 = 1$

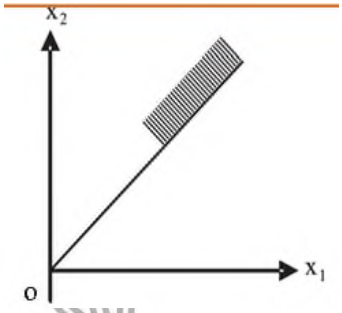
(b) $x_1 + x_2 = 1$

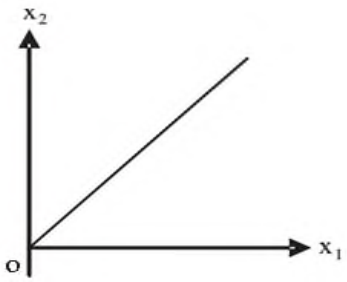
(c) $x_1 + x_2 = 1$

(d) none of these

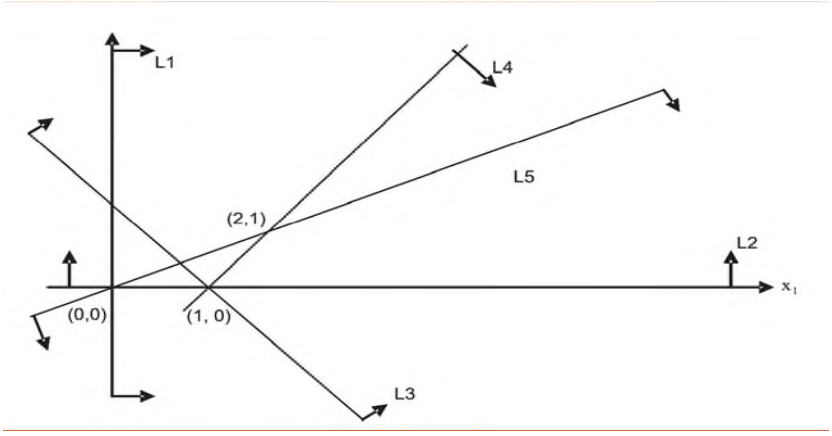
(15) The inequality $-x_1 + 2x_2 = 0$ is indicated on the graph as

(a) 

(b) 

(c) 

(d) None of these

(16) 

The common region indicated on the graph is expressed by the set of five inequalities

(a) L1 : $x_1 = 0$ (b) L1 : $x_1 = 0$

L2 : $x_2 = 0$ L2 : $x_2 = 0$

$L3 : x_1 + x_2 = 1$ $L3 : x_1+x_2 = 1$

$L4 : x_1 - x_2 = 1$ $L4 : x_1-x_2 = 1$

$L5 : -x_1 + 2x_2 = 0$ $L5 : -x_1+2x_2 = 0$

(c) $L1 : x_1 = 0$ (d) None of these

$L2 : x_2 = 0$

$L3 : x_1+ x_2 = 1$

$L4 : x_1-x_2 = 1$

$L5 : -x_1+2x_2 = 0$

(17) A firm makes two types of products: Type A and Type B. The profit on product A is Nu. 20 each and that on product B is Nu. 30 each. Both types are processed on three machines M1, M2 and M3. The time required in hours by each product and total time available in hours per week on each machine are as follows:

Machine	Product A	Product B	Available Time
M1	3	3	36
M2	5	2	50
M3	2	6	60

The constraints can be formulated taking x_1 = number of units A and x_2 = number of B as

(a) $x_1 + x_2 = 12$ (b) $3x_1 + 3x_2 = 36$

$5x_1 + 2x_2 = 50$ $5x_1 + 2x_2 = 50$

$2x_1 + 6x_2 = 60$ $2x_1 + 6x_2 = 60$

$x_1 = 0, x_2 = 0$

(c) $3x_1 + 3x_2 = 36$ (d) none of these

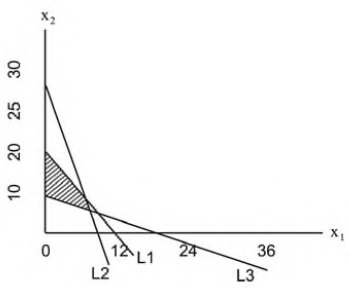
$5x_1 + 2x_2 = 50$

$2x_1 + 6x_2 = 60$

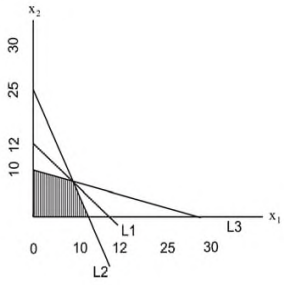
$x_1 = 0, x_2 = 0$

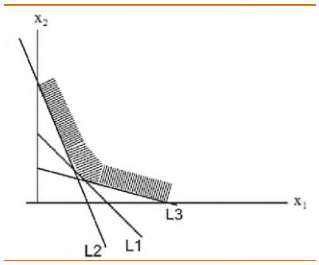
(18) The set of inequalities L1: $x_1 + x_2 = 12$, L2: $5x_1 + 2x_2 = 50$, L3: $x_1 + 3x_2 = 30$, $x_1 = 0$, and $x_2 = 0$ is represented by

(a)



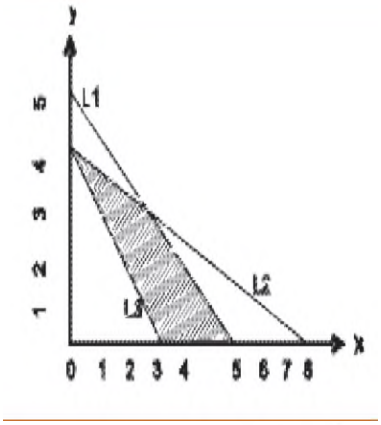
(b)

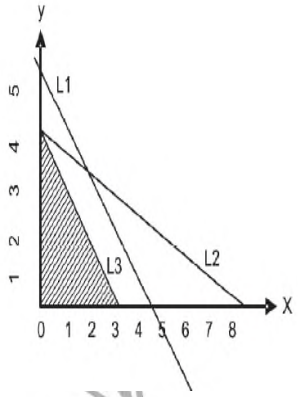


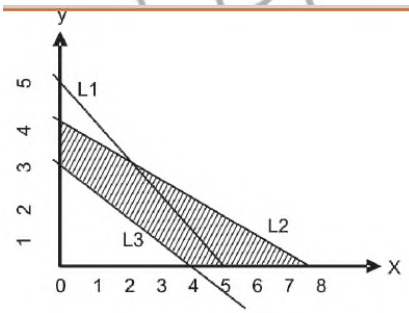
(c) 

(d) none of these

(19) The common region satisfying the set of inequalities x = 0, y = 0, L1: x + y = 5, L2: x +2y = 8 and L3: 4x +3y = 12 is indicated by

(a) 

(b) 

(c) 

(d) None of these

Answers:

1	b	6	c	11	a	16	b
2	C	7	d	12	a	17	c
3	A	8	d	13	b	18	b
4	b	9	b	14	c	19	a
5	A	10	c	15	a		

PAST EXAMS QUESTIONS

1. If $\left|x+\frac{1}{4}\right| > \frac{7}{4}$, then:

(a) $x < \frac{-3}{2}$ or $x > 2$ (b) $x < -2$ or $x > \frac{3}{2}$ (c) $-2 < x < \frac{3}{2}$ (d) None of these

2. If $\left|\frac{3x-4}{4}\right| \leq \frac{5}{12}$, the solution set is:

(a) $\{x : \frac{19}{18} \leq x \leq \frac{29}{18}\}$ (b) $\{x : \frac{7}{9} \leq x \leq \frac{17}{9}\}$ (c) $\{x : \frac{29}{18} \leq x \leq \frac{-19}{18}\}$ (d) None of these.

3. On solving the inequalities $6x + y \geq 18$, $x + 4y \geq 12$, $2x + y \geq 10$, we get the following situation:

(a) (0, 18), (12, 0), (4, 2), & (2, 6) (b) (3, 0), (0, 3), (4, 2), & (7, 6)

(c) (5, 0), (0, 10), (4, 2) & (7, 6) (d) (0, 18), (12, 0), (4, 2), (0, 0) and (7, 6)

4. A car manufacturing company manufactures cars of two types A and B. Model A requires 150 man-hours for assembling. 50 man-hours for painting and 10 man-hours for checking and testing. Model B requires 60 man-hours for assembling, 40 man-hours for painting and 20 man-hours for checking and testing. There are available 30 thousand man-hours for assembling, 13 thousand man-hours for painting and 5 thousand man-hours for checking and testing. Express the above situation using linear inequalities. Let the company manufacture x units of type of type A model of car and y units of type B model of car. Then, the inequalities are:

(a) $5x + 2y \geq 1000$; $5x + 4y \geq 1300$,
$x + 2y \leq 500$; $x \geq 0$, $y \geq 0$.

(b) $5x + 2y \leq 1000$, $5x + 4y \leq 1300$.
$x + 2y \geq 5000$; $x \geq 0$, $y \geq 0$.

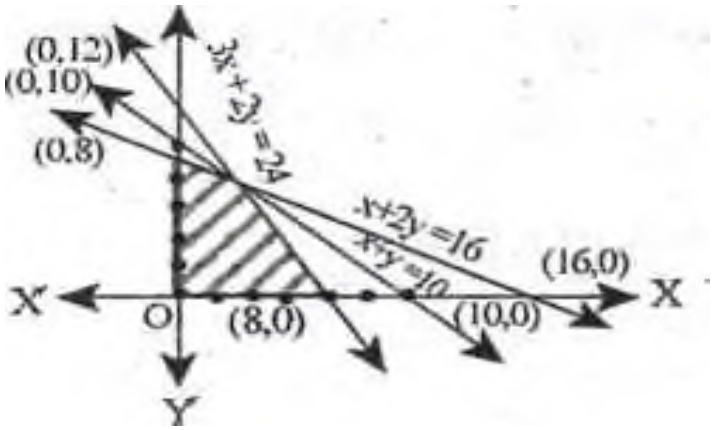
(c) $5x + 2y \leq 1{,}000$, $5x + 4y \leq 1300$,
$x + 2y \leq 500$; $x \geq 0$, $y \geq 0$.

(d) $5x + 2y = 1000$, $5x + 4y \geq 1300$,
$x + 2y = 500$; $x \geq 0$, $y \geq 0$.

5. The rules and regulations demand that the employer should employ not more than 5 experienced hands to 1 fresh one and this fact is represented by: (Taking experienced person as x and fresh person as y)

(a) $y \geq \frac{x}{5}$| (b) $5y \leq x$ (c) $5y \geq x$ (d) None.

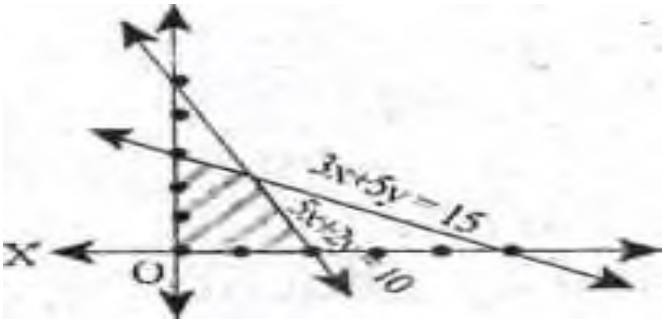
6. The shaded region represents:



(a) $3x + 2y \leq 24$, $x + 2y \geq 16$, $x + y \leq 10$, $x \geq 0$, $y \geq 0$

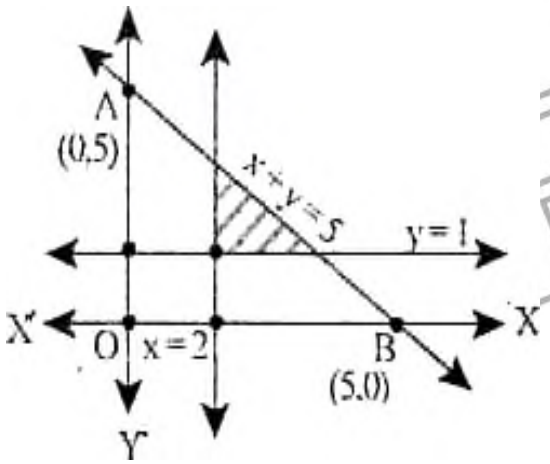
(b) $3x + 2y \leq 24, x + 2y \leq 16, x + y \geq 10, x \geq 0, y \geq 0$
(c) $3x + 2y \leq 24, x + 2y \leq 16, x + y \leq 10, x \geq 0, y \geq 0$
(d) None of these.

7. The shaded region represents:



(a) $3x + 5y \leq 15, 5x + 2y \geq 10, x, y \geq 0$
(b) $3x + 5y \leq 15, 5x + 2y \leq 10, x, y \geq 0$
(c) $3x + 5y \geq 15, 5x + 2y \geq 10, x, y, \geq 0$
(d) None of these.

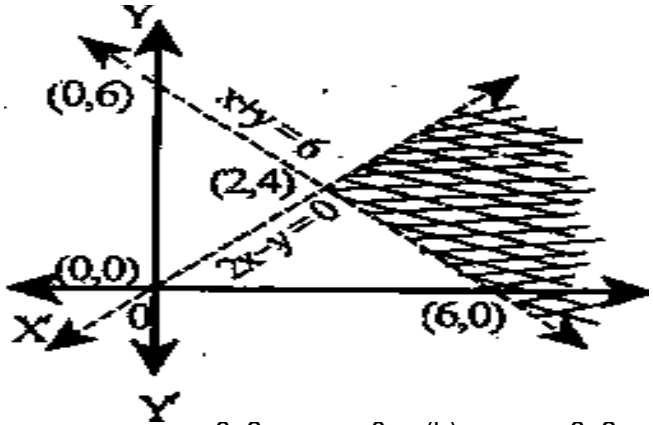
8. The shaded region represents:



(a) $x + y \leq 5, x \geq 2, y \leq 1$ (b) x + y *cc* 5, $x \geq 2, y \geq 1$

(c) $x + y \geq 5, x \geq 2, y \geq 1$ (d) None of these.

9. The shaded region represent:



(a) $x + y > 6, 2x - y > 0$ (b) $x + y < 6, 2x - y > 0$

(c) $x + y > 6, 2x - y < 0$ (d) None of these.

10. If $a > 0$ and $b < 0$, it follows that:

(a) $\frac{1}{a} > \frac{1}{b}$ (b) $\frac{1}{a} < \frac{1}{b}$

(c) $\frac{1}{a} = \frac{1}{b}$ (d) None of these.

11. The Linear relationship between two variables in an inequality:

(a) $ax + by \leq c$ (b) $ax + by \leq c$

(c) $axy + by \leq c$ (d) $ax + bxy \leq c$.

12. The solution of the inequality $\frac{(5-2x)}{3} \leq \frac{x}{6} - 5$ is

(a) $x - \geq 8$ (b) $x \leq 8$

(c) $x = 8$ (d) none of these.

13. On the average an experienced person does 7 units of work while a fresh one work 5 units of work daily but the employer has to maintain an output of at least 35 units of work per day. The situation can be expressed as:

(a) $7x + 5v < 35$ (b) $7x + 5v \leq 35$

(c) $7x + 5y > 35$ (d) $7x + 5y \geq 35$

14. Solution space of the inequalities $2x + y < 10$ and $x - y < 5$:

(i) Includes the origin.

(ii) includes the point (4, 3)

(iii) Which one is correct?

(a) Only (i)

(b) Only (ii)

(c) Both (ii) and (ii)

(d) None of the above.

15. on the average an experienced person does 7 units of work while a fresh one work 5 units of work daily but the employer has to maintain an output of at least 35 units of work per day. The situation can be expressed as:

(a) $5x + 3y \leq 30$ (b) $5x + 3y \geq 30$

(c) $5x + 3y > 30$ (d) $5x + 3y = 30$

16. Find the range of real values of x satisfying the inequalities $3x - 2 > 7$ and $4x - 13 > 15$

(a) $x > 3$ (b) $x > 7$ (c) $x < 7$ (d) $x < 3$

17. On the average, experienced person does 5 units of work while a fresh one 3 unit's work daily but the employer have to maintain the output of at least 30 units of work per day. The situation can be expressed as.

(a) $5x + 3y \leq 30$ (b) $5x + 3y \geq 30$

(c) $5x + 3y = 30$ (d) None of these.

18. The union forbids employer to employ less than two experienced person (x) to each fresh person (y). This situation can be expressed as:

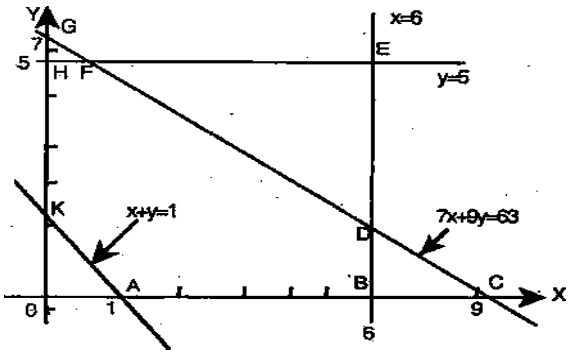
(a) $x < v/2$ (b) $y \leq x/2$

(c) $y \geq x/2$ (d) None of these.

19. The solution of the inequality $8x + 6 < 12x + 14$ is:

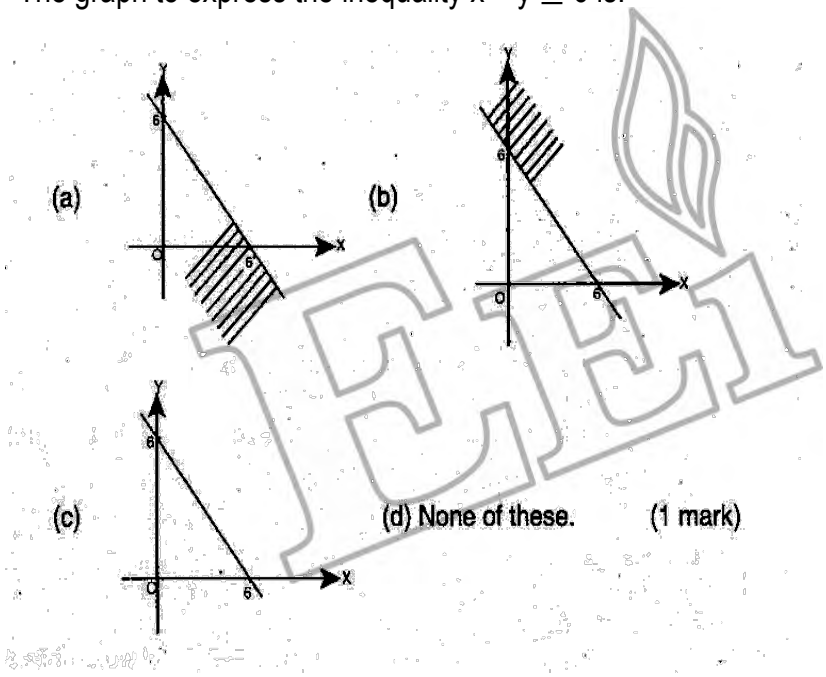
(a) $(-2, 2)$ (b) $(0, -2)$ (c) $(2, \infty)$ (d) $(-2, \infty)$

20. The graph of linear inequalities $7x + 9y \leq 63$, $x + y \geq 1$, $0 \leq x \leq 6$ and

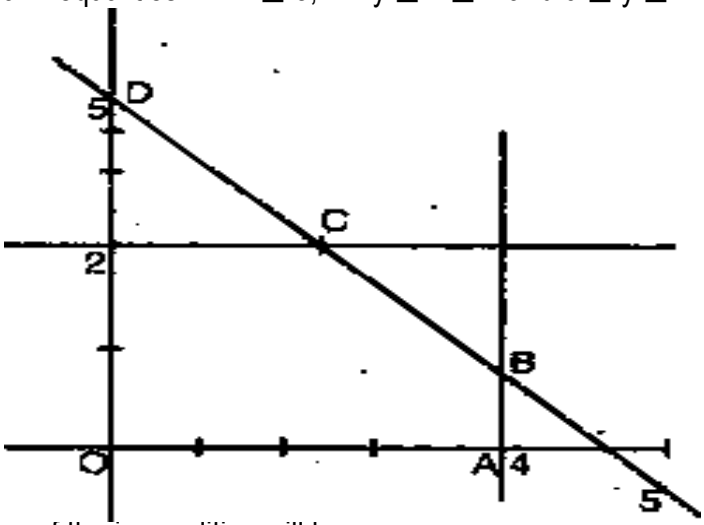


Common region of the inequalities is:
(a) BCDB and DEFD (b) Unbounded
(c) HFGH (d) ABDFHKA

21. The graph to express the inequality $x + y \leq 6$ is:



22. the graph of linear inequalities $x + Y \geq 5$; $x + y \leq x \leq 4$ and $0 \leq y \geq 2$ is given below:

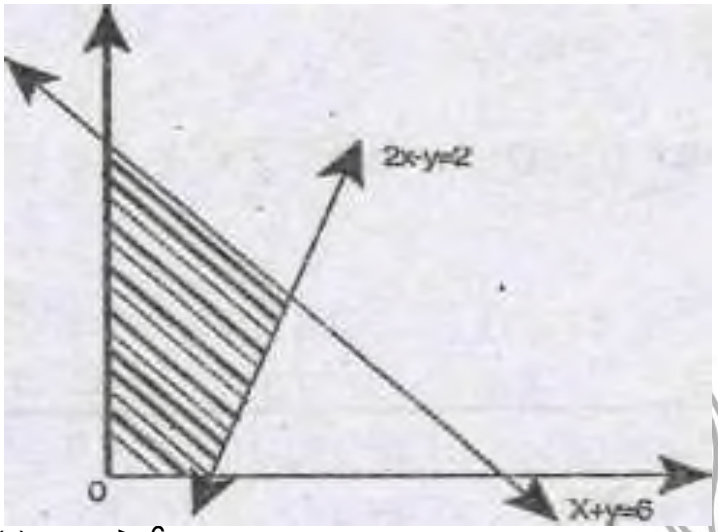


The common region of the inequalities will be:
(a) OABCEO (b) ECDE

(c) Line Segment DC (d) Line Segment BC

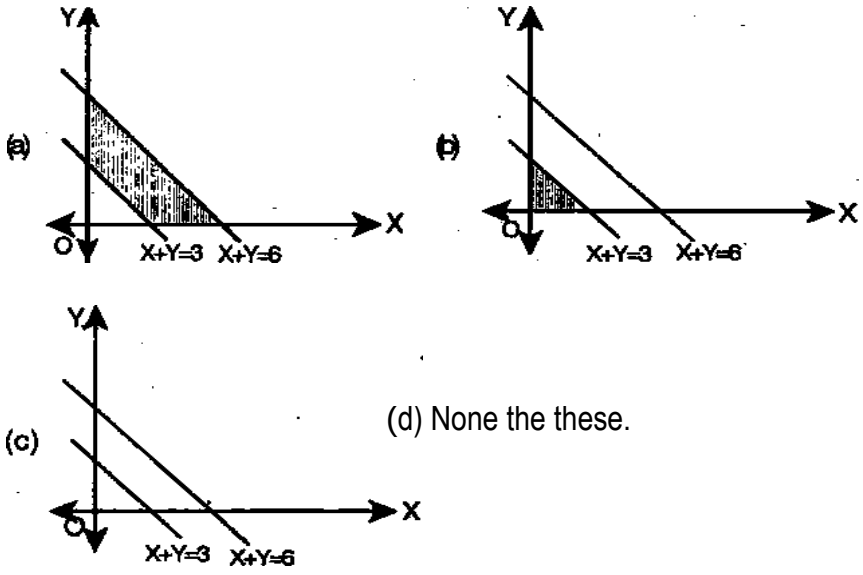
23. The common region in the graph of linear inequalities $2x + y \geq 18$, $x + y \geq 12$ and $3x + 2y \leq 34$ is:
(a) unbounded (b) infeasible
(c) feasible and bounded (d) feasible and unbounded

24. The common shaded region in the graph represents the linear inequalities as:



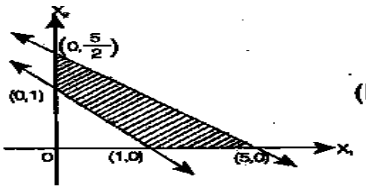
(a) $x + y \geq 6$
$2x - y - 2 \geq 0$
$x, y \geq 0$
(b) $x + y \geq 6$
$2x - y - 2 \leq 0$
$x, y \geq 0$
(c) $x + y \leq 6$
$2x - y - 2 \leq 0$
$x, y \geq 0$
(d) $x + y \leq 6$
$2x - y - 2 \geq 0$
$x, y \geq 0$

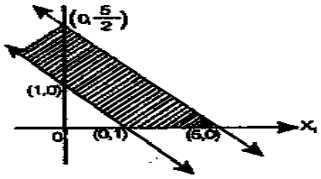
25. The common region of $x + y \leq 6$; $x + y \geq 3$; $x \geq 0$; $y \geq 0$, is (as shown by shaded region):

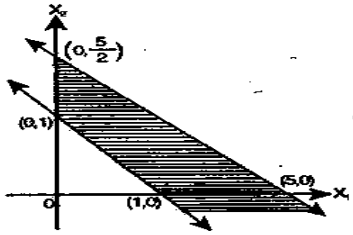


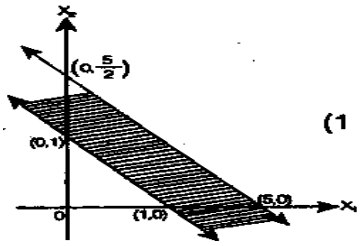
(d) None the these.

26. the common region by the inequalities $X_1 + + 2x_2 \leq 5$, $x, + x_2 \geq 1$, $x_1 \geq 0$, $x_2 \geq 0$ is given as shaded portion in:

(a) 

(b) 

(c) 

(d) 

(1 mark)

Answer:

27. A dietician wishes to mix together two kinds of food so that the vitamins content of the mixture is atleast 9 units of vitamin A, 7units of vitamin B, 10 units of vitamin C, 12 units of vitamin D. The vitamin content per kg. of each food is shown in table. Assuming 'x' units of food I is to be mixed with 'y' units of food II the situation can be expressed as:

	A	B	C	D
Food I	2	1	1	2
Food II	1	1	2	3

(a) $2x + y \leq 9$
$x + y \leq 7$
$x + 2y \leq 10$
$2x + 3y \leq 12$
$x > 0, y > 0$

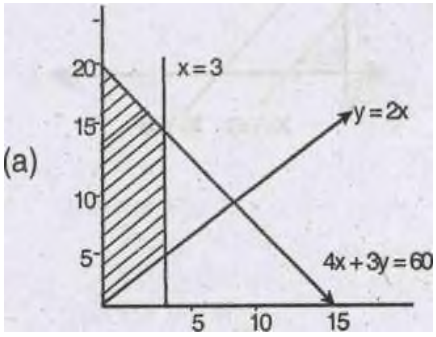
(b) $2x + y \geq 30$
$x + y \leq 7$
$x + 2y \geq 10$
$2x + 3y \geq 12$
$x > 0, y > 0$

(c) $2x + y \geq 9$
$x + y \geq 7$
$x + 2y \leq 10$
$x + 3v \leq 12$
$x \geq 0, y > 0$

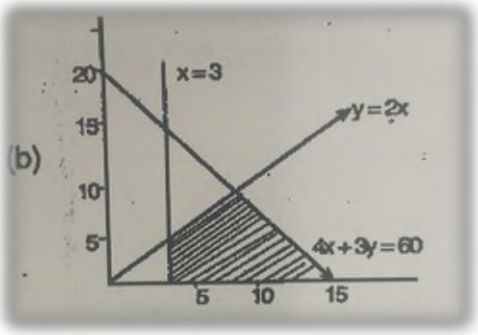
(d) $2x + v \geq 9$
$x + y \geq 7$
$x + 2v \geq 10$
$2x + 3v \geq 12$
$x \geq 0, y \geq 0$

28. The common regions by the inequalities $4x + 3y \leq 60$; $y \geq 2x$; $x \geq 3$, $x \geq 0$ and $y \geq 0$ is

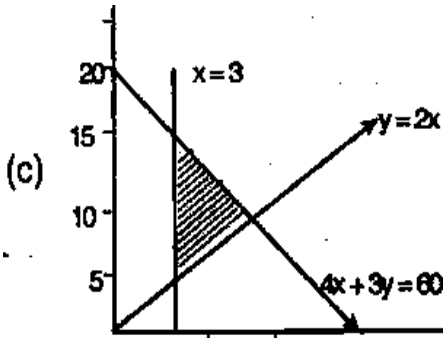
(a)



(b)



(c)



(d) None of these.

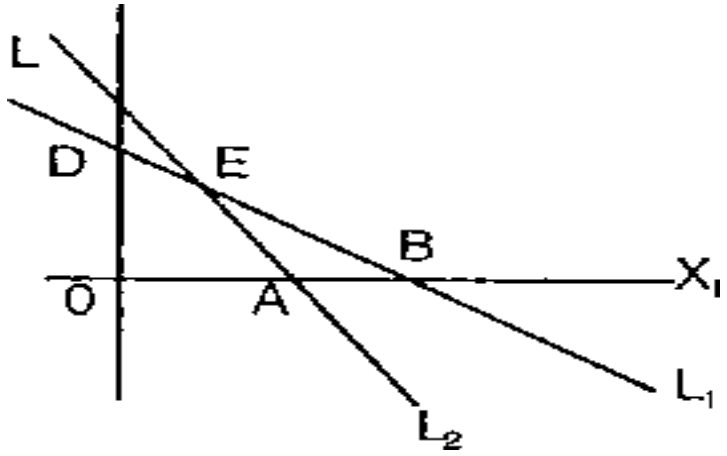
29. The linear relationship between two variables in an inequality:
(a) $ax + by \leq c$ (b) $ax.by \leq c$
(c) $axy + by \leq c$ (d) $ax + bxy \leq c$

30. On Solving the Inequalities $5x + y \leq 100$, $x + y \leq 60$, $x \geq 0$, $y \geq 0$, we get the following solution:
(a) (0, 0), (20,0), (10, 50) & (0,60)
(b) (0, 0), (60, 0), (10, 50) & (0,60)
(c) (0, 0), (20, 0), (0, 100) & (10,50)
(d) None of these

31. An employer recruits experienced (x) and fresh workmen (y) for his under the condition that he cannot employ more than 11 people, x and y can related by the inequality.
(a) $x + y \neq 11$
(b) $x + y \leq 11, x \geq 0, y \geq 0$
(c) $x + y \geq 11, x \geq 0, y \geq 0$
(d) None of these.

32. The solution set of the in equations $x + 2 > 0$ and $2x - 6 > 0$ is
(a) $(-2, \infty)$
(b) $(3, \infty)$
(c) $(-\infty, -2)$
(d) $(-\infty, -3)$

33. The common region represented by the following in equalities $L_1 = X_1 + X_2 < 4$; $L_2 = 2X_1 - X_2 > 6$



(a) OABC
(b) Outside of OAB
(c) Δ BCE
(d) Δ ABE

34. $6x + y \geq 18$, $x + 4y \geq 12$, $2x + y \geq 10$ on solving the inequalities; we get
(a) (0, 18), (12,0), (4,2), & (7, 6)
(b) (3, 0), (0, 3), (4. 2) & (7, 6)
(c) (5,0), (0,10), (4,2) & (7, 6)
(d) (0,18), (12, 0), (4,2), (0, 0), and (7, 6)

ANSWERS:

1	B	11	A	21	A	31	B
2	B	12	A	22	C	32	B
3	A	13	D	23	C	33	D
4	C	14	A	24	C	34	A
5	A	15	B	25	A		
6	C	16	B	26	A		
7	B	17	B	27	D		
8	B	18	B	28	C		
9	A	19	D	29	A		
10	A	20	D	30	A		

CHAPTER 4

TIME VALUE OF MONEY

SUMMARY OF TIME VALUE OF MONEY

A. Simple Interest	**B. Compound Interest**	**Future Value of annuity Regular**	**Present Value of annuity Regular**
SI = Pit	$A=P(1+i)^n$	FV = Ann. $[(1+i)^n-1]/i$	PV = Ann.$[(1+i)^n-1]/i(1+i)^n]$
A = P(1+it)	$CI = P[(1+i)^n-1]$		
	$ER = [(1+i)^n-1]$	**Future Value of Annuity due/immediate**	**Present Value of annuity Annuity due/ immediate**
	"WDV"	FV=Ann.$\left[\frac{(1+i)^n-1}{i}\right](1+i)$	Ann.$[(1+i)^{n-1}-1]/[i(1+i)^{n-1}]$+ one ann.
	$A= P(1-i)^n$		

Note:

Base of "i" and "n" should be same	"n"= number of "conversion period" in total time (कुल परिवर्तन अवधि)	$t \neq n$
t = n/conversion period in one year	PV factor = $1/(1+I)^n$	'i' = Rate/100
Rate of Interest for CI Formula = annual rete of interest / no. of conversion period per year	**conversion period** [परिवर्तन अवधि] "the period at the end of which the interest is compound is called conversion period."	
A = Accumulate amount (final value of investment) (निवेष का अंतिम मुल्य या संग्रहित राषि)		
P = principal (initial value of an investment) (मूलधन या निवेष का प्रारंभिक मूल्य)		
"i" = R/100		
I = amount of intrest (ब्याज की राषि)		
T = time in years (समय वर्षो में)		

Other concept:

Annuity – Sequence of Perodic Payments (or receipts) in same Period of time (if receipt/ payment takes place for ever then it is called perpetuity).

1. Annuity Regular (वार्षिकी नियमित)- First payment or receipt at the end of the period
2. Annuity due/ Immediate (वार्षिकी देय/वार्षिकी तुरत)- First payment receipt at the beginning of the annuity.

Practice for Calculation 'i' & 'n'

S. No	**Principle**	**Total time**	**Rate**	**Compounded Y/HY/Q/M/D**	**i**	**n**	**$(1+i)^n$**	**Amount**
1		5 years	12%p.a.	Yearly				
2		7 years	9% p.a.	Quarterly				
3		12 years	2% p.a.	Half Yearly				
4		20 years	6% H.Y.	Quarterly				
5		7 years	8% p.a.	Monthly				
6		10 years	9 % p.a.	Monthly				
7		9 years	3% Q.	Yearly				

Note : If nothing is given then interest rate will be 'p.a.' and compounding is done yearly.

Practice for calculating missing 'i' and 'n'

	Situation	n		Situation	i
1	$2.1436=(1.1)^n$	8	1	$2.6289 = (1+i)^{10}$	.102
2	$2.0789 = (1.05)^n$	15	2	$4.1772 = (1+i)^{15}$	0.1
3	$2 = (1.14)^n$	5.29	3	$2 = (1+i)^{10}$	0.72
4	$3 = (1.09)^n$	12.75	4	$2 = (1+i\}^8$	
5	$3 = (1.05)^n$	22.50	5	$2.55880 = (1+i)^9$	0..11
6	$2 = (1.15)^n$	4.95	6	$3 = (1+i)^7$	
7	$4 = (1.09)^n$	16.09	7	$4 = (1+i)^5$	

Special use of Business Calculator

1. For Log

Ex. log 25

25 √ (19 times) – 1 × 227695

2. For Antilog.

Ex. **0.3010 ÷ 227695 + 1 (× =)** → (19 times)

3. For Cube Root

$\sqrt[3]{27} = 3$

27 √ - 1 ÷ 3 + 1 [× =] → (12 times)

(12 times)

4. Multiplication with single amount and use of GT function

Year	P	$(1.1)^n$	$A = P(1.1)^n$	Use of Calculator
1	25000	1.21	30250	25000 × 1.21 =
2	25000	1.331	33275	1.331 =
3	25000	1.4641	36602.5	1.4641 =
4	25000	1.61051	40262.75	1.61051=
		Total	140390.25	GT

1.1 × (= = ==) GT × 25000 = 140390.25

(4 times)

5.

Year	Principle	$1/(1+i)^n$	$P \times 1/(1+i)^n$
1	25000	.9090	22725
2	25000	.8264	20660
3	25000	.7513	10782.5
4	25000	.6830	17075
		Total	**79242.5**

1 ÷ 1.1 (= = = =) GT × 25000 = 79246.64

(4 times)

or use of formula PV=Ann. $\left(\frac{(1+i)^n-1}{i(1+i)^n}\right)$

6. **If denominator come first in calculator**

25 ÷ 3 = 8.33

or

3 ÷ = 25 = 8.33

4.1 INTRODUCTION

People earn money for spending it on housing, food, clothing, education, entertainment etc. Sometimes extra expenditures have also to be met with. For example there might be a marriage in the family; one may want to buy house, one may want to set up his or her business, one may want to buy a car and so on. Some people can manage to put aside some money for such expected and unexpected expenditures. But most people have to borrow money for such contingencies. From where they can borrow money?

Money can be borrowed from friends or money lenders or Banks. If you can arrange a loan from your friend it might be interest free but if you borrow money from lenders or Banks you will have to pay some charge periodically for using money of money lenders or Banks. This charge is called interest.

Let us take another view. People earn money for satisfying their various needs as discussed above. After satisfying those needs some people may have some savings. People may invest their savings in debentures or lend to other person or simply deposit it into bank. In this way they can earn interest on their investment.

Most of you are very much aware of the term interest. Interest can be defined as the price paid by a borrower for the use of a lender's money.

We will know more about interest and other related terms later.

4.2 WHY IS INTEREST PAID?

Now question arises why lenders charge interest for the use of their money. There are a variety of reasons. We will now discuss those reasons.

1. **Time value of money:** Time value of money means that the value of a unity of money is different in different time periods. The sum of money received in future is less valuable than it is today. In other words the present worth of money received after some time will be less than a money received today. Since a money received today has more value rational investors would prefer current receipts to future receipts. If they postpone their receipts they will certainly charge some money i.e. interest.
2. **Opportunity Cost:** The lender has a choice between using his money in different investments. If he chooses one he forgoes the return from all others. In other words lending incurs an opportunity cost due to the possible alternative uses of the lent money.
3. **Inflation:** Most economies generally exhibit inflation. Inflation is a fall in the purchasing power of money. Due to inflation a given amount of money buys fewer goods in the future than it will now. The borrower needs to compensate the lender for this.
4. **Liquidity Preference:** People prefer to have their resources available in a form that can immediately be converted into cash rather than a form that takes time or money to realize.
5. **Risk Factor:** There is always a risk that the borrower will go bankrupt or otherwise default on the loan. Risk is a determinable factor in fixing rate of interest.

A lender generally charges more interest rate (risk premium) for taking more risk.

4.3 DEFINITION OF INTEREST AND SOME OTHER RELATED TERMS

Now we can define interest and some other related terms.

4.3.1 Interest

Interest is the price paid by a borrower for the use of a lender's money. If you borrow (or lend) some money from (or to) a person for a particular period you would pay (or receive) more money than your initial borrowing (or lending). This excess money paid (or received) is called interest. Suppose you borrow (or lend) ` 50,000 for a year and you pay (or receive) ` 55,000 after one year the difference between initial borrowing (or lending) ` 50,000 and end payment (or receipts) ` 55,000 i.e. ` 5,000 is the amount of interest you paid (or earned).

4.3.2 Principal

Principal is initial value of lending (or borrowing). If you invest your money the value of initial investment is also called principal. Suppose you borrow (or lend) ` 50,000 from a person for one year. ` 50,000 in this example is the 'Principal.' Take another example suppose you deposit ` 20,000 in your bank account for one year. In this example ` 20,000 is the principal.

4.3.3 Rate of Interest

The rate at which the interest is charged for a defined length of time for use of principal generally on a yearly basis is known to be the rate of interest. Rate of interest is usually expressed as percentages. Suppose you invest ` 20,000 in your bank account for one year with the interest rate of 5% per annum. It means you would earn ` 5 as interest every ` 100 of principal amount in a year.

Per annum means for a year.

4.3.4 Accumulated amount (or Balance)

Accumulated amount is the final value of an investment. It is the sum total of principal and interest earned. Suppose you deposit ` 50,000 in your bank for one year with a interest rate of 5% p.a. you would earn interest of ` 2,500 after one year. (Method of computing interest will be illustrated later). After one year you will get ` 52,500 (principal+ interest), ` 52,500 is amount here.

Amount is also known as the balance.

4.4 SIMPLE INTEREST AND COMPOUND INTEREST

Now we can discuss the method of computing interest. Interest accrues as either simple interest or compound interest. We will discuss simple interest and compound interest in the following paragraphs:

4.4.1 Simple Interest

Now we would know what is simple interest and the methodology of computing simple interest and accumulated amount for an investment (principal) with a simple rate over a period of time. As you already know the money that you borrow is known as principal and the additional money that you pay for using somebody else's money is known as interest. The interest paid for keeping ` 100 for one year is known as the

rate percent per annum. Thus if money is borrowed at the rate of 8% per annum then the interest paid for keeping ` 100 for one year is ` 8. The sum of principal and interest is known as the amount.

Clearly the interest you pay is proportionate to the money that you borrow and also to the period of time for which you keep the money; the more the money and the time, the more the interest. Interest is also proportionate to the rate of interest agreed upon by the lending and the borrowing parties. Thus interest varies directly with principal, time and rate.

Simple interest is the interest computed on the principal for the entire period of borrowing. It is calculated on the outstanding principal balance and not on interest previously earned. It means no interest is paid on interest earned during the term of loan.

Simple interest can be computed by applying following formulas:

I = Pit

A = P + I

= P + Pit

= P(1 + it) I = A – P

Here,

A = Accumulated amount (final value of an investment)

P = Principal (initial value of an investment)

i = Annual interest rate in decimal.

I = Amount of Interest t = Time in years

Let us consider the following examples in order to see how exactly are these quantities related.

Example 1: How much interest will be earned on ` 2000 at 6% simple interest for 2 years?

Answer: ` 240.

Example 2: Sania deposited ` 50,000 in a bank for two years with the interest rate of 5.5% p.a. How much interest would she earn?

Answer: ` 5,500.

Example 3: In example 2 what will be the final value of investment?

Answer: ` 55,500.

Example 4: Sachin deposited ` 1,00,000 in his bank for 2 years at simple interest rate of 6%. How much interest would he earn? How much would be the final value of deposit? **Answer: ` 1,12,000**

Example 5: Find the rate of interest if the amount owed after 6 months is ` 1050, borrowed amount being ` 1000.

Answer: 10%

Example 6: Rahul invested ` 70,000 in a bank at the rate of 6.5% p.a. simple interest rate. He received ` 85,925 after the end of term. Find out the period for which sum was invested by Rahul.

Answer: 3.5 years

Example 7: Kapil deposited some amount in a bank for 7 ½ years at the rate of 6% p.a. simple interest. Kapil received ` 1,01,500 at the end of the term. Compute initial deposit of Kapil.

Answer: ` 70,000.

Example 8: A sum of ` 46,875 was lent out at simple interest and at the end of 1 year 8 months the total amount was ` 50,000. Find the rate of interest percent per annum.

Answer: 4%.

Example 9: What sum of money will produce ` 28,600 as an interest in 3 years and 3 months at 2.5% p.a. simple interest?

Answer: ` 3,52,000.

Example 10: In what time will ` 85,000 amount to ` 1,57,675 at 4.5 % p.a. ?

Answer:19 Years.

EXERCISE 4 (A)

Choose the most appropriate option (a) (b) (c) or (d).

1. S.I on ` 3,500 for 3 years at 12% per annum is
   (a) ` 1,200 (b) ` 1,260 (c) ` 2,260 (d) none of these
2. P = 5,000, R = 15, T = 4 ½ using I = PRT/100, I will be
   (a) ` 3,375 (b) ` 3,300 (c) ` 3,735 (d) none of these
3. If P = 5,000, T = 1, I = ` 300, R will be
   (A) 5% (b) 4% (c) 6% (d) none of these
4. If P = ` 4,500, A = ` 7,200, than Simple interest i.e. I will be
   (a) ` 2,000 (b) ` 3,000 (c) ` 2,500 (d) ` 2,700
5. P = ` 12,000, A = ` 16,500, T= 2 ½ years. Rate percent per annum simple interest will be-
   (a) 15% (b) 12% (c) 10% (d) none of these
6. P = ` 10,000, I = ` 2,500, R = 12½% SI. The number of years T will be
   (a) 1 ½ years (b) 2 years (c) 3 years (d) none of these
7. P = ` 8,500, A = ` 10,200, R =2 ½ % SI, t will be.
   (a) 1 yr. 7 month. (b) 2 yrs. (c) ½ years (d) none of these
8. The sum required to earn a monthly interest of ` 1,200 at 18% per annum SI is
   (a) ` 50,000 (b) ` 60,000 (c) ` 80,000 (d) none of these

9. A sum of money amount to ` 6,200 in 2 years and ` 7,400 in 3 years. The principal and rate of interest are

(a) ` 3,800, 31.57% (b) ` 3,000, 20%

(c) ` 3,500, 15% (d) none of these

10. A sum of money doubles itself in 10 years. The number of years it would triple itself is

(a) 25 years. (b) 15 years. (c) 20 years (d) none of these

4.4.2 Compound Interest

We have learnt about the simple interest. We know that if the principal remains the same for the entire period or time then interest is called as simple interest. However in practice the method according to which banks, insurance corporations and other money lending and deposit taking companies calculate interest is different. To understand this method we consider an example:

Suppose you deposit ` 50,000 in ICICI bank for 2 years at 7% p.a. compounded annually. Interest will be calculated in the following way:

INTEREST FOR FIRST YEAR

I = Pit

$=50{,}000\ x\ \frac{7}{100}x1=$ ` 3,500

INTEREST FOR SECOND YEAR

For calculating interest for second year principal would not be the initial deposit. Principal for calculating interest for second year will be the initial deposit plus interest for the first year. Therefore principal for calculating interest for second year would be

= ` 50,000 + ` 3,500

= ` 53,500

Interest for the second year $=53{,}500\ x\ \frac{7}{100}x1$

= ` 3,745

Total interest = Interest for first year + Interest for second year

= ` (3,500 + 3,745)

= ` 7,245

This interest is ` 245 more than the simple interest on ` 50,000 for two years at 7% p.a. As you must have noticed this excess in interest is due to the fact that the principal for the second year was more than the principal for first year. The interest calculated in this manner is called compound interest.

Thus we can define the compound interest as the interest that accrues when earnings for each specified period of time added to the principal thus increasing the principal base on which subsequent interest is computed.

Example 11: Saina deposited ` 1,00,000 in a nationalized bank for three years. If the rate of interest is 7% p.a., calculate the interest that bank has to pay to Saina after three years if interest is compounded annually. Also calculate the amount at the end of third year.

Answer: C. I. ` 22,504.30 Amount at the end of third year ` 1,22,504.30

4.4.3 Conversion period

In the example discussed above the interest was calculated on yearly basis i.e. the interest was compounded annually. However in practice it is not necessary that the interest be compounded annually. For example in banks the interest is often compounded twice a year (half yearly or semiannually) i.e. interest is calculated and added to the principal after every six months. In some financial institutions interest is compounded quarterly i.e. four times a year. The period at the end of which the interest is compounded is called conversion period. When the interest is calculated and added to the principal every six months the conversion period is six months. In this case number of conversion periods per year would be two. If the loan or deposit was for five years then the number of conversion period would be ten.

Typical conversion periods are given below:

Conversion period	Description	Number of conversion period in a year
1 day	Compounded daily	3 6 5
1 month	Compounded monthly	1 2
3 months	Compounded quarterly	4
6 months	Compounded semi annually	2
12 months	Compounded annually	1

4.4.4 Formula for compound interest

Taking the principal as P, the rate of interest per conversion period as i (in decimal), the number of conversion period as n, the accrued amount after n payment periods as An we have accrued amount at the end of first payment period

$A1 = P + P\,i = P\,(1 + i)$;

at the end of second payment period $A2 = A1 + A1\,i = A1\,(1 + i)$

$= P\,(1 + i)\,(1 + i)$

$= P\,(1 + i)2$;

at the end of third payment period $A3 = A2 + A2\,i$

$= A2\ (1 + i)$

$= P(1 + i)2\ (1 + i)$

$= P(1 + i)3$

$An = An\text{-}1 + An\text{-}1\ i$

$= An\text{-}1\ (1 + i)$

$= P\ (1 + i)\ n\text{-}1\ (1 + i)$

$= P(1+ i)n$

Thus the accrued amount An on a principal P after n conversion periods at i (in decimal) rate of interest per conversion period is given by

$$A_n = P(1+i)^n$$

where, $i = \frac{\text{Annual rate off}}{\text{Number of conversion periods per year}}$

Inter

$= P_{\square}(1+i)^{n}_{\square}$

n is total conversions i.e. t x no. of

Note : Computation of A shall be quite simple with a calculator. However compound interest table and tables for at various rates per annum with (a) annual compounding; (b) monthly compounding and (c) daily compounding are available.

Example 12: ` 2,000 is invested at annual rate of interest of 10%. What is the amount after two years if compounding is done (a) Annually (b) Semi-annually (c) Quarterly (d) monthly?

Answer: (a) ` 2,420 (b) ` 2,431 (c) ` 2,436.80 (d) ` 2,440.58

Example 13: Determine the compound amount and compound interest on ` 1000 at 6% compounded semi-annually for 6 years. Given that (1 + i)n = 1.42576 for i = 3% and n = 12.

Answer: ` 1,425 and ` 425.76.

Example 14: Compute the compound interest on ` 4,000 for 1½ years at 10% per annum compounded half-yearly.

Answer: ` 630.50

To find the Principal/ Time/ Rate

The Formula $A_n = P(1+i)^n$ connects four variables A_n, P, i and n.

Similarly, C.I. (Compound Interest) = $P[(1+i)^n - 1]$ connects C.I., P, i and n. Whenever three out of these four variables are given the fourth can be found out by simple calculations.

Example 15: On what sum will the compound interest at 5% per annum for two years compounded annually be ` 1,640?

Answer: ` 16,000

Example 16: What annual rate of interest compounded annually doubles an investment in 7 years? Given that $2^{1/7} = 1.104090$

Answer: 10.41% pa.

Example 17: In what time will ` 8,000 amount to ` 8,820 at 10% per annum interest compounded half-yearly?

Answer: 1Year.

Example 18: Find the rate percent per annum if ` 2,00,000 amount to ` 2,31,525 in 1 ½ year interest being compounded half-yearly.

Answer: 10%.

Example 19: A certain sum invested at 4% per annum compounded semi-annually amounts to ` 78,030 at the end of one year. Find the sum.

Answer: ` 75,000.

Example 20: ` 16,000 invested at 10% p.a. compounded semi-annually amounts to ` 18,522. Find the time period of investment.

Answer: 1½ years.

Example 21: A person opened an account on April, 2011 with a deposit of ` 800. The account paid 6% interest compounded quarterly. On October 1, 2011 he closed the account and added enough additional money to invest in a 6 month time-deposit for `1,000, earning 6% compounded monthly.

(a) How much additional amount did the person invest on October1?
(b) What was the maturity value of his time deposit on April 1 2022?
(c) How much total interest was earned?

Given that $(1 + i)^n$ is 1.03022500 for i = 1 ½ % n = 2 and $(1 + i)^n$ is 1.03037751 for i = ½ % and n = 6.

Answer: (a) ` 175.82 (b) ` 1,030.38 (c) ` 54.56.

4.5 EFFECTIVE RATE OF INTEREST

If interest is compounded more than once a year the effective interest rate for a year exceeds the per annum interest rate. Suppose you invest ` 10,000 for a year at the rate of 6% per annum compounded semi annually. Effective interest rate for a year will be more than 6% per annum since interest is being compounded more than once in a year. For computing effective rate of interest first we have to compute the interest. Let out us compute the interest.

Principal for calculation of interest for first six months = ` 10,000 x 6/100 x 6/12

= ` 300

Principal for calculation of next six months =

Principal for first period one + Interest for first period one + Interest for first period

= ` (10,300 + 300)

= ` 10,300

Interest for next six months = `10,300 x 6/100 x 6/12 = ` 309

Total interest earned during the current year =

Interest for first six months + Interest for next six months

= ` (300 + 309) = ` 609

Interest of ` 609 can also be computed directly from the formula of compound interest.

We can compute effective rate of interest by following formula

I = Pet

Where I = Amount of interest
E = Effective rate of interest in decimal
t = Time period
P = Principal amount

Putting the values we have

609 = 10,000 x E x 1
E = 609/10,000
= 0.0609 or 6.09%

This if we compound the interest more than once a year effective interest rate for the year will be more than actual interest rate per annum. But if interest is compounded annually effective interest rate for the year will be equal to actual interest rate per annum.
So effective interest rate can be defined as the equivalent annual rate of interest compounded annually if interest is compounded more than once a year.

The effective interest rate can be computed directly by following formula;

$$E = (1 + i)^n - 1$$

Where E is the effective interest rate
i = actual interest rate in decimal
n = number of conversion period

Example 22: ` 5,000 is invested in a Term Deposit Scheme that fetches interest 6% per annum compounded quarterly. What will be the interest after one year? What is effective rate of interest?

Answer: I = ` 306.82 R = 6.13%

Example 23: Find the amount of compound interest and effective rate of interest if an amount of ` 20,000 is deposited in a bank for one year at the rate of 8% per annum compounded semiannually.

Answer: I = ` 1,632 R = 8.16%.

Example 24: Which is a better investment 3% per year compounded monthly or 3.2% per year simple interest? Given that $(1 + 0.0025)^{12} = 1,0304$.

Answer: 3.2%

EXERCISE 4 (B)

Choose the most appropriate option (a) (b) (c) or (d).

1. If P = ` 1,000, R = 5% p.a.; what is Amount and C.I. is

(a) ` 1,215.50, ` 215.50 (b) `1,125, ` 125

(c) ` 2,115, ` 115 (d) none of these

2. ` 100 will become after 20 years at 5% p.a. compound interest amount of

(a) ` 250 (b) ` 205 (c) ` 265.50 (d) none of these

3. The effective rate of invest corresponding to a nominal rate 3% p.a. payable half yearly is

(a) 3.2% p.a. (b) 3.25% p.a. (c) 3.0225% p.a. (d) None of these

4. A machine is depreciated at the rate of 20% on reducing balance. The original cost of the machine was ` 1,00,000 and its ultimate scrap value was ` 30,000. The effective life of the machine is

(a) 4.5 years (appx.) (b) 5.4 years (appx.)

(c) 5 years (appx.) (d) None of these

5. If A = ` 1,000, n = 2 years, R= 6% p.a. compound interest payable half-yearly, then principal (P) is

(a) ` 888.80 (b) ` 885 (c) 800 (d) none of these

6. The population of a town increases every year by 2% of the population at the beginning of that year. The number of years by which the total increase of population be 40% is
(a) 7 years (b) 10 years (c) 17 years (app) (d) none of these

7. The difference between C.I and S.I on a certain sum of money invested for 3 years at 6% p.a is ` 110.16. The sum is
(a) ` 3,000 (b) ` 3,700 (c) ` 12,000 (d) ` 10,000

8. The useful life of a machine is estimated to be 10 years and cost ` 10,000. Rate of depreciation is 10% p.a. The scrap value at the end of its life is
(a) ` 3,486.78 (b) ` 4,383 (c) ` 3,400 (d) none of these

9. The effective rate of interest corresponding a nominal rate of 7% p.a convertible quarterly is
(a) 7% (b) 7.5% (c) 5% (d) 7.18%

10. The C.I on ` 16000 for 1 ½ years at 10% p.a payable half -yearly is
(a) ` 2,222 (b) ` 2,522 (c) ` 2,500 (d) none of these

11. The C.I on ` 40000 at 10% p.a for 1 year when the interest is payable quarterly is
(a) ` 4,000 (b) ` 4,100 (c) ` 4,152.51 (d) none of these

12. The difference between the S.I and the C.I on ` 2,400 for 2 years at 5% p.a is
(a) ` 5 (b) ` 10 (c) ` 16 (d) ` 6

13. The annual birth and death rates per 1,000 are 39.4 and 19.4 respectively. The number of years in which the population will be doubled assuming there is no immigration or emigration is
(a) 35 years. (b) 30 years. (c) 25 years (d) none of these

14. The C.I on ` 4,000 for 6 months at 12% p.a payable quarterly is
(a) ` 243.60 (b) ` 240 (c) ` 243 (d) none of these

4.6 ANNUITY

In many cases you must have noted that your parents have to pay an equal amount of money regularly like every month or every year. For example payment of life insurance premium, rent of your house (if you stay in a rented house), payment of housing loan, vehicle loan etc. In all these cases they pay a constant amount of money regularly. Time period between two consecutive payments may be one month, one quarter or one year. Sometimes some people received a fixed amount of money regularly like pension rent of house etc. In all these cases annuity comes into the picture. When we pay (or receive) a fixed amount of money periodically over a specified time period we create an annuity.

Thus annuity can be defined as a sequence of periodic payments (or receipts) regularly over a specified period of time.

There is a special kind of annuity also that is called Perpetuity. It is one where the receipt or payment takes place forever. Since the payment is forever we cannot compute a future value of perpetuity. However we can compute the present value of the perpetuity. We will discuss later about future value and present value of annuity.

To be called annuity a series of payments (or receipts) must have following features:
(1) Amount paid (or received) must be constant over the period of annuity and
(2) Time interval between two consecutive payments (or receipts) must be the same.

Consider following tables. Can payments/ receipts shown in the table for five years be called annuity?

Table – 4.1

Year end	Payments/Receipts (`)
I	5,000
II	6,000
III	4,000
IV	5,000
V	7,000

Table – 4.2

Year end	Payments/Receipts (`)
I	5,000
II	5,000
III	–
IV	5,000
V	5,000

Table – 4.3

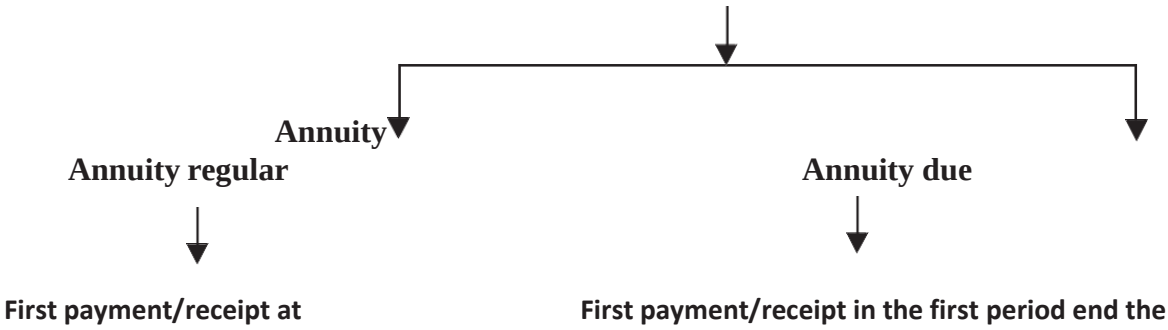
Year end	Payments/Receipts (`)
I	5,000
II	5,000
III	5,000
IV	5,000
V	5,000

Payments/ Receipts shown in table 4.1 cannot be called annuity. Payments/ Receipts though have been made at regular intervals but amount paid are not constant over the period of five years.

Payments/ Receipts shown in table 4.2 cannot also be called annuity. Though amounts paid/ received are same in every year but time interval between different payments/ receipts is not equal. You may note that time interval between second and third payment/ receipt is two year and time interval between other consecutive payments/ receipts (first and second third and fourth and (fourth and fifth) is only one year. You may also note that for first two year the payments/ receipts can be called annuity.

Now consider table 4.3. You may note that all payments/ receipts over the period of 5 years are constant and time interval between two consecutive payments/ receipts is also same i.e. one year. Therefore payments/ receipts as shown in table- 4.3 can be called annuity.

4.6.1 Annuity regular and Annuity due/ immediate



Annuity may be of two types:

(1) Annuity regular: In annuity regular first payment/ receipt takes place at the end of first period. Consider following table:

Table – 4.4

Year end	Payments/Receipts (`)
I	5,000
II	5,000
III	5,000
IV	5,000
V	5,000

We can see that first payment/receipts takes place at the end of first year therefore it is an annuity regular.

(2) Annuity Due or Annuity Immediate: When the first receipt or payment is made today (at the beginning of the annuity) it is called annuity due or annuity immediate. Consider following table:

In the beginning of	Payment/Receipt (`)
I year	5,000
II year	5,000
III year	5,000
IV year	5,000
V year	5,000

We can see that first receipt or payment is made in the beginning of the first year. this type of annuity is called annuity due or annuity immediate.

4.7 FUTURE VALUE

Future value is the cash value of an investment at some time in the future. It is tomorrow's value of today's money compounded at the rate of interest. Suppose you invest ` 1,000 in a fixed deposit that pays you 7% per annum as interest. At the end of first year you will have ` 1,070. This consist of the original principal of ` 1,000 and the interest earned of ` 70. ` 1,070 is the future value of ` 1,000 invested for one year at 7%. We can say that ` 1000 today is worth ` 1070 in one year's time if the interest rate is 7%.

Now suppose you invested ` 1,000 for two years. How much would you have at the end of the second year. You had ` 1,070 at the end of the first year. If you reinvest it you end up having ` 1,070(1+0.07)= ` 1144.90 at the end of the second year. Thus ` 1,144.90 is the future value of ` 1,000 invested for two years at 7%. We can compute the future value of a single cash flow by applying the formula of compound interest.

We know that

$$A_n = P(1+i)^n$$

Where A = Accumulated amount

n = number of conversion period

i = rate of interest per conversion period in decimal

P = principal

Future value of a single cash flow can be computed by above formula. Replace A by future value (F) and P by single cash flow (C.F.) therefore

$$F = C.F.\ (1 + i)^n$$

Example 25: You invest ` 3000 in a two year investment that pays you 12% per annum. Calculate the future value of the investment.

Answer: ` 3,763.20

4.7.1 Future value of an annuity regular

Now we can discuss how do we calculate future value of an annuity.

Suppose a constant sum of ` 1 is deposited in a savings account at the end of each year for four years at 6% interest. This implies that ` 1 deposited at the end of the first year will grow for three years, ` 1 at the end of second year for 2 years, ` 1 at the end of the third year for one year and ` 1 at the end of the fourth year will not yield any interest. Using the concept of compound interest we can compute the future value of annuity. The compound value (compound amount) of ` 1 deposited in the first year will be

A_3 = ` $1\ (1 + 0.06)^3$

= ` 1.191

The compound value of ` 1 deposited in the second year will be

A_2 = ` $1\ (1 + 0.06)^2$

= ` 1.124

The compound value of ` 1 deposited in the third year will be A1 = ` 1 (1 + 0.06)1

= ` 1.06

and the compound value of ` 1 deposited at the end of fourth year will remain ` 1.

The aggregate compound value of ` 1 deposited at the end of each year for four years would be:

` (1.191 + 1.124 + 1.060 + 1.00) = ` 4.375

This is the compound value of an annuity of ` 1 for four years at 6% rate of interest.

The above computation is summarized in the following table:

End of year	Amount Deposit (`)	Future value at the end of fourth year (`)
0	–	–
1	1	$1\ (1 + 0.06)^3 = 1.191$
2	1	$1\ (1 + 0.06)^2 = 1.124$
3	1	$1\ (1 + 0.06)^1 = 1.060$
4	1	$1\ (1 + 0.06)^0 = 1$
	Future Value	4.375

The computation shown in the table can be expressed as follows:

$A(4, i) = A(1 + i)^0 + A(1 + i) + A(1 + i)^2 + A(1 + i)^3$

i.e. $A(4, i) = A[1 + (1 + i) + (1 + i)^2 + (1 + i)^3]$

In above equation A is annuity, A (4, i) is future value at the end of year four, i is the rate of interest shown in decimal.

We can extend above equation for n periods and rewrite as follows:

$A(n, i) = A(1 + i)^0 + A(1 + i)^1 + \ldots\ldots\ldots\ldots + A(1 + i)^{n-2} + A(1 + i)^{n-1}$

Here A = Re.1

Therefore

$A(n, i) = 1(1 + i)^0 + 1(1 + i)^1 + \ldots\ldots\ldots\ldots + 1(1 + i)^{n-2} + 1(1 + i)^{n-1}$

$= 1 + (1 + i)^1 + \ldots\ldots\ldots\ldots + (1 + i)^{n-2} + (1 + i)^{n-1}$

[a geometric series with first term 1 and common ratio (1+ i)]

$$= \frac{1.[1 - (1 + i)^n]}{1 - (1 + i)}$$

$$= \frac{1 - (1 + i)^n}{-i}$$

$$= \frac{(1 + i)^n - 1}{i}$$

If A be the periodic payments, the future value A(n, i) of the annuity is given by

$$A(n, i) = \left[\frac{(1+i)^n - 1}{i}\right]$$

Example 26: Find the future value of an annuity of ` 500 made annually for 7 years at interest rate of 14% compounded annually. Given that $(1.14)^7 = 2.5023$.

Answer: ` 5,365.35

Example 27: ` 200 is invested at the end of each month in an account paying interest 6% per year compounded monthly. What is the future value of this annuity after 10th payment? Given that $(1.005)^{10} = 1.0511$.

Answer: ` 2,044

4.7.2 Future value of Annuity due or Annuity Immediate

As we know that in Annuity due or Annuity immediate first receipt or payment is made today. Annuity regular assumes that the first receipt or the first payment is made at the end of first period. The relationship between the value of an annuity due and an ordinary annuity in case of future value is:

Future value of an Annuity due/ Annuity immediate = Future value of annuity regular x (1 + i) where i is the interest rate in decimal.

Calculating the future value of the annuity due involves two steps.

Step-1 Calculate the future value as though it is an ordinary annuity.

Step-2 Multiply the result by (1+i)

Example 28: Z invests ` 10,000 every year starting from today for next 10 years. Suppose interest rate is 8% per annum compounded annually. Calculate future value of the annuity. Given that $(1 + 0.08)^{10} = 2.15892500$.

Answer: ` 1,56,454.875

4.8 PRESENT VALUE

We have read that future value is tomorrow's value of today's money compounded at some interest rate. We can say present value is today's value of tomorrow's money discounted at the interest rate. Future value and present value are related to each other in fact they are the reciprocal of each other. Let's go back to our fixed deposit example. You invested ` 1000 at 7% and get ` 1,070 at the end of the year. If ` 1,070 is the future value of today's ` 1000 at 7% then ` 1,000 is present value of tomorrow's ` 1,070 at 7%. We have also seen that if we invest ` 1,000 for two years at 7% per annum we will get ` 1,144.90 after two years. It means ` 1,144.90 is the future value of today's ` 1,000 at 7% and ` 1,000 is the present value of ` 1,144.90 where time period is two years and rate of interest is 7% per annum. We can get the present value of a cash flow (inflow or outflow) by applying compound interest formula.

The present value P of the amount A_n due at the end of n period at the rate of i per interest period may be obtained by solving for P the below given equation

$A_n = P(1 + i)^n$

i.e. $P = \frac{A_n}{(1+i)^n}$

- Computation of P may be simple if we make use of either the calculator or the present value table showing values of $\frac{1}{(1+i)^n}$ for various time periods/ per annum interest rates.
- For positive i the factor $\frac{1}{(1+i)^n}$ is always less than 1 indicating thereby future amount has smaller present value.

Example 29: What is the present value of ` 1 to be received after two years compounded annually at 10% interest rate?

Answer: ` 0.83

Example 30: Find the present value of ` 10,000 to be required after 5 years if the interest rate be 9%. Given that $(1.09)^5 = 1.5386$.

Answer: ` 6,499.42

4.8.1 Present value of an Annuity regular: We have seen how compound interest technique can be used for computing the future value of an Annuity. We will now see how we compute present value of an annuity. We take an example, Suppose your mom promise you to give you ` 1,000 on every 31st December for the next five years. Suppose today is 1st January. How much money will you have after five years from now if you invest this gift of the next five years at 10%? For getting answer we will have to compute future value of this annuity.

But you don't want ` 1,000 to be given to you each year. You instead want a lump sum figure today. Will you get ` 5,000. The answer is no. The amount that she will give you today will be less than ` 5,000. For getting the answer we will have to compute the present value of this annuity. For getting present value of this annuity we will compute the present value of these amounts and then aggregate them. Consider following table:

Year End	Gift Amount (`)	Present Value [$A_n / (1 + i)^n$]
I	1,000	1,000/(1 + 0.1) = 9.091
II	1,000	1,000/(1 + 0.1) = 6.446
III	1,000	1,000/(1 + 0.1) = 1.315
IV	1,000	1,000/(1 + 0.1) = 3.013
V	1,000	1,000/(1 + 0.1) = 0.921
Present Value		**= 3,790.86**

Thus the present value of annuity of ` 1,000 for 5 years at 10% is ` 3,790.79

It means if you want lump sum payment today instead of ` 1,000 every year you will get ` 3,790.79.

The above computation can be written in formula form as below.

The present value (V) of an annuity (A) is the sum of the present values of the payments.

$$\therefore \quad V = \frac{A}{(1+i)^1} + \frac{A}{(1+i)^2} + \frac{A}{(1+i)^3} + \frac{A}{(1+i)^4} + \frac{A}{(1+i)^5}$$

We can extend above equation for n periods and rewrite as follows:

$$V = \frac{A}{(1+i)^1} + \frac{A}{(1+i)^2} + \ldots\ldots\ldots\ldots \frac{A}{(1+i)^{n-1}} + \frac{A}{(1+i)^n} \quad \ldots\ldots\ldots(1)$$

Multiplying throughout by $\frac{1}{(1+i)}$ we get

$$\frac{V}{(1+i)} = \frac{A}{(1+i)^2} + \frac{A}{(1+i)^3} + \ldots\ldots\ldots\ldots \frac{A}{(1+i)^{n+1}} + \frac{A}{(1+i)^n} \quad \ldots\ldots\ldots(2)$$

subtracting (2) from (1) we get

$$V - \frac{V}{(1+i)} = \frac{A}{(1+i)^1} - \frac{A}{(1+i)^{n+1}}$$

Or $V(1+i) - V = A - \frac{A}{(1+i)^n}$

Or $Vi = A[1 - \frac{A}{(1+i)^n}]$

Where, $P(n, i) = \frac{(1+i)^n - 1}{i(1+i)^n}$

Consequently $A = \frac{V}{(n,i)}$

A loan with fixed rate of interest is said to be amortized if entire principal and interest are paid over equal periods of time by way of sequence of equal payment.

$A = \frac{V}{(n,i)}$ can be used to compute the amount of annuity if we have present value (V), n the number of time period and the rate of interest in decimal.

Example 31: S borrows ` 5,00,000 to buy a house. If he pays equal instalments for 20 years and 10% interest on outstanding balance what will be the equal annual instalment?

Answer: ` 45,130.43

Example 32: ` 5,000 is paid every year for ten years to pay off a loan. What is the loan amount if interest rate be 14% per annum compounded annually?

Answer: ` 26,080.55

Example 33: Y bought a TV costing ` 13,000 by making a down payment of ` 3000 and agreeing to make equal annual payment for four years. How much would be each payment if the interest on unpaid amount be 14% compounded annually?

Answer: 3,432.05

4.8.2 Present value of annuity due or annuity immediate

Present value of annuity due/ immediate for n years is the same as an annuity regular for (n-1) years plus an initial receipt or payment in beginning of the period. Calculating the present value of annuity due involves two steps.

Step 1: Compute the present value of annuity as if it were a annuity regular for one period short.

Step 2: Add initial cash payment/ receipt to the step 1 value.

Example 34: Suppose your mon decides to gift you ` 10,000 every year starting from today for the next five years. You deposit this amount in a bank as and when you receive and get 10% per annum interest rate compounded annually. What is the present value of this annuity?

Answer: ` 41,698.70

4.9 SINKING FUND

It is the fund credited for a specified purpose by way of sequence of periodic payments over a time period at a specified interest rate. Interest is compounded at the end of every period. Size of the sinking fund deposit is computed from A = P.A (n, i) where A is the amount to be saved, P the periodic payment, n the payment period.

Example 35: How much amount is required to be invested every year so as to accumulate ` 300000 at the end of 10 years if interest is compounded annually at 10%?

Answer: ` 18,823.62

4.10 APPLICATIONS

4.10.1 Leasing

Leasing is a financial arrangement under which the owner of the asset (lessor) allows the user of the asset (lessee) to use the asset for a defined period of time (lease period) for a consideration (lease rental) payable over a given period of time. This is a kind of taking an asset on rent. How can we decide whether a lease agreement is favorable to lessor or lessee, it can be seen by following example.

Example 36: ABC Ltd. wants to lease out an asset costing ` 3,60,000 for a five year period. It has fixed a rental of ` 1,05,000 per annum payable annually starting from the end of first year. Suppose rate of interest is 14% per annum compounded annually on which money can be invested by the company. Is this agreement favourable to the company?

Answer: ` 3,60,473.40

Example 37: A company is considering proposal of purchasing a machine either by making full payment of ` 4,000 or by leasing it for four years at an annual rate of ` 1,250. Which course of action is preferable if the company can borrow money at 14% compounded annually?

4.10.2 Capital Expenditure (investment decision)

Capital expenditure means purchasing an asset (which results in outflows of money) today in anticipation of benefits (cash inflow) which would flow across the life of the investment. For taking investment decision we compare the present value of cash outflow and present value of cash inflows. If present value of cash inflows is greater than present value of cash outflows decision should be in the favour of investment. Let us see how do we take capital expenditure (investment) decision.

Example 38: A machine can be purchased for ` 5,000. Machine will contribute ` 12000 per year for the next five years. Assume borrowing cost is 10% per annum compounded annually. Determine whether machine should be purchased or not.

Example 39: A machine with useful life of seven years costs ` 10,000 while another machine with useful life of five years costs ` 8,000. The first machine saves labour expenses of ` 1,900 annually and the second one saves labour expenses of ` 2,200 annually. Determine the preferred course of action. Assume cost of borrowing as 10% compounded per annum.

4.10.3 Valuation of Bond

A bond is a debt security in which the issuer owes the holder a debt and is obliged to repay the principal and interest. Bonds are generally issued for a fixed term longer than one year.

Example 40: An investor intends purchasing a three year ` 1,000 par value bond having nominal interest rate of 10%. At what price the bond may be purchased now if it matures at par and the investor requires a rate of return of 14%?

4.11 PERPETUITY

Perpetuity is an annuity in which the periodic payments or receipts begin on a fixed date and continue indefinitely or perpetually. Fixed coupon payments on permanently invested (irredeemable) sums of money are prime examples of perpetuities.

The formula for evaluating perpetuity is relatively straight forward. Two points which are important to understand in this regard are:

(a) The value of the perpetuity is finite because receipts that are anticipated far in the future have extremely low present value (today's value of the future cash flows).

(b) Additionally, because the principal is never repaid, there is no present value for the principal.

Therefore, the price of perpetuity is simply the coupon amount over the appropriate discount rate or yield.

4.11.1 Calculation of multi period perpetuity:

The formula for determining the present value of multi-period perpetuity is as follows:

$$PV\infty = \frac{R}{(1+i)^1} + \frac{R}{(1+i)^2} + \frac{R}{(1+i)^3} + \ldots\ldots\ldots + \frac{R}{(1+i)^\infty} = \sum_{n=1}^{\infty} \frac{R}{(1+i)^n} = \frac{R}{i}$$

Where

R = the payment or receipt each period

i = the interest rate per payment or receipt period

Example 41: Ramesh wants to retire and receive ` 3,000 a month. He wants to pass this monthly payment to future generations after his death. He can earn an interest of 8% compounded annually. How much will he need to set aside to achieve his perpetuity goal?

Solution:

R = ` 3,000

i = 0.08/12 or 0.00667

Substituting these values in the above formula, we get

$$PVA = \frac{3,000}{0.00667} = ` 4,49,775$$

4.11.2 Calculation of Growing Perpetuity:

A stream of cash flows that grows at a constant rate forever is known as growing perpetuity.
The formula for determining the present value of growing perpetuity is as follows:

$$\text{PVA} = \frac{R}{(1+i)^1} + \frac{R}{(1+i)^2} + \frac{R}{(1+i)^3} + \ldots\ldots\ldots\ldots + \frac{R(1+g)^{\infty}}{(1+i)^{\infty}}$$

$$\sum_{n=1}^{\infty} \frac{R(1+g)^{n-1}}{(1+i)^n} = \frac{R}{i-g}$$

Example 42: Assuming that the discount rate is 7% per annum, how much would you pay to receive ` 50, growing at 5%, annually, forever?

Solution:

$$\text{PVA} = \frac{R}{i-g} = \frac{50}{0.07-0.05} = 2{,}500$$

Calculating Rate of Return:

1. Calculating the rate of return provides important information that can be used for future investments. For example, if you invested in a stock that showed a substantial gain after several months of performance, you may decide to purchase more of that stock. If the stock showed a continual loss, it may be wise to conduct research to find a better- performing stock.
2. Calculating the rate of return is that it allows you to gauge your investment and decision-making skills. Investments that create a gain or profit are great. However, if you continually make investments at a loss, then you may want to change your investment strategies. A great attribute of successful business people is knowing how and when to make investments, as is knowing when to change strategies. With a firm grasp of calculating the rate of return, you can manage and monitor your investments at various stages to determine the outcome of your investments. This leads to a higher level of confidence and the skills necessary to be a savvy investor.

Net Present Value Technique (NPV): The net present value technique is a discounted cash flow method that considers the time value of money in evaluating capital investments. An investment has cash flows throughout its life, and it is assumed that a rupee of cash flow in the early years of an investment is worth more than a rupee of cash flow in a later year.

The net present value method uses a specified discount rate to bring all subsequent net cash inflows after the initial investment to their present values (the time of the initial investment is year 0).

Determining Discount Rate

Theoretically, the discount rate or desired rate of return on an investment is the rate of return the firm would have earned by investing the same funds in the best available alternative investment that has the same risk. Determining the best alternative opportunity available is difficult in practical terms so rather that using the true opportunity cost, organizations often use an alternative measure for the desired rate of return. An organization may establish a minimum rate of return that all capital projects must meet; this minimum could be based on an industry average or the cost o other investment opportunities. Many organizations choose to

use the overall cost of capital or Weighted Average Cost of Capital (WACC) that an organization has incurred in raising funds or expects to incur in raising the funds needed for an investment.

The net present value of a project is the amount, in current value of rupees, the investment earns after paying cost of capital in each period.

4.12 NET PRESENT VALUE

Net present value = Present value of net cash inflow – Total initial investment

Since it might be possible that some additional investment may also be required during the life time of the project then appropriate formula shall be:

Net present value = Present value of cash inflow – Present value of cash outflow

The steps to calculating net present value are:-

1. Determine the net cash inflow in each year of the investment.
2. Select the desired rate of return or discounting rate or Weighted Average Cost of Capital.
3. Find the discount factor for each year based on the desired rate of return selected.
4. Determine the present values of the net cash flows by multiplying the cash flows by respective the discount factors of respective period called Present Value (PV) of Cash flows.
5. Total the amounts of all PVs of Cash Flows

Decision Rule:

If NPV ≥ 0 Accept the Proposal

If NPV ≤ 0 Reject the Proposal

Example 43: Compute the net present value for a project with a net investment of ` 1,00,000 and net cash flows year one is ` 55,000; for year two is ` 80,000 and for year three is ` 15,000. Further, the company's cost of capital is 10%?

Solution:

Year	Net Cash Flows	PVIF @ 10%	Discounted Cash Flows
0	(1,00,000)	1.000	(1,00,000)
1	55,000	0.909	49,995
2	80,000	0.826	66,080
3	15,000	0.751	11,265
Net Present Value			27,340

Recommendation: Since the net present value of the project is positive, the company should accept the project.

4.13 NOMINAL RATE OF RETURN

The nominal rate is the stated interest rate. If a bank pays 5% annually on a savings account, then 5% is the nominal interest rate. So if you deposit ` 100 for 1 year, your will receive ` 5 in interest. However, that $5 will probably be worth less at the end of the year than it would have been at the beginning. This is because inflation lowers the value of money. As goods, services, and assets, such as real estate, rise in price.

The nominal interest rate is conceptually the simplest type of interest rate. It is quite simply the stated interest rate of a given or loan. It is also defined as a stated interest rate. This interest works according to the simple interest and does not take into account the compounding periods.

Real Rate of Return: The real interest rate is so named because it states the "real" rate that the lender or investor receives after inflation is factored in; that is, the interest rate that exceeds the inflation rate.
A comparison of real and nominal interest rates can therefore be summed up in this equation:

Nominal Rate of Return – Inflation = Real Rate of Return
Nominal Interest Rate = Real Interest Rate + Inflation
Effective Rate:
It is the actual equivalent annual rate of interest at which an investment grows in value when interest is credited more often than once a year. If interest is paid m times in a year is can be found by calculating:

$E_i = (1 + \frac{i}{m})^m$

The chief advantage to knowing the difference between nominal, real and effective rates is that it allows consumers to make better decisions about their loans and investments. A loan with frequent compounding periods will be more expensive than one that compounds annually.
A bond that only pays a 1% real interest rate may not be worth investors' time if they seek to grow their assets over time. These rates effectively reveal the true return that will be posted by a fixed-income investment and the true cost of borrowing for an individual or business.
Effective and nominal interest rates allow banks to use the number that looks most advantageous to the consumer. When banks are charging interest, they advertise the nominal rate, which is lower and does not reflect how much interest the consumer would owe on the balance after a full year of compounding. On the other hand, with deposit accounts where banks are paying interest, they generally advertise the effective rate because it is higher than the nominal rate.

4.14 COMPOUND ANNUAL GROWTH RATE (CAGR)

Compounded Annual Growth Rate (CAGR) is a business and investing specific term for the smoothed annualized gain of an investment over a given time periodic is not an accounting term, but remains widely used, particularly in growth industries or to compare the growth rates of two investments because CAGR dampens the effect of volatility of periodic returns that can render arithmetic means irrelevant. CAGR is often used to describe the growth over a period of time of some element of the business, for example revenue, units delivered, registered users, etc.

$$\mathrm{CAGR}(t_{0,}t_n) = \left[\frac{V(t_n)}{V(t_0)}\right]^{\frac{1}{t_n - t_0}} - 1$$

Note: It is nothing but calculating 'i' in the formula of $A = P(1 + i)^n$
Where $V(t_0)$ = Beginning Period ; V(tn) = End Period

Example 44: Suppose the revenues of a company for four years, V(t) in the above formula, have been

Year	2013	2014	2015	2016
Value of investment	100	120	160	210

Calculate Compound annual Growth Rate.

Solution:

$t_n - t_0$ = 2016 – 2013 = 3

The CAGR revenues over the three-year period from the end of 2013 to the end of 20016 is

$$\text{CAGR}(0,3) = \left(\frac{210}{100}\right)^{1/3} - 1 = 1.2806 - 1 = 28.06\%$$

Note: using formula $A = P(1 + i)^n$

$210 = 100(1 + i)^3$

$\frac{210}{100} = (1 + i)^3$

Log 2.10 = 3 log 1+ i

$\frac{.3222}{3}$ = log 1 + i

(1 + i) = AL 0.1074

(1 +i) = 1.2806

i = .2806 (28.06%)

Applications: These are some of the common CAGR applications:

a. Calculating average returns of investment funds.
b. Demonstrating and comparing the performance of investment advisors.
c. Comparing the historical returns of stocks with bonds or with a savings account.
d. Forecasting future values based on the CAGR of a data series.
e. Analyzing and communicating the behavior, over a series of years, of different business measures such as sales, market share, costs, customer satisfaction, and performance.

EXERCISE 4 (C)

Choose the most appropriate option (a) (b) (c) or (d).

1. The present value of an annuity of ` 3000 for 15 years at 4.5% p.a CI is

   (a) ` 23,809.41 (b) ` 32,218.63

   (c) ` 32,908.41 (d) none of these

2. The amount of an annuity certain of ` 150 for 12 years at 3.5% p.a C.I is

   (a) ` 2,190.28 (b) ` 1,290.28

   (c) ` 2,180.28 (d) none of these

3. A loan of ` 10,000 is to be paid back in 30 equal instalments. The amount of each installment to cover the principal and at 4% p.a CI is

   (a) ` 587.87 (b) ` 587 (c) ` 578.87 (d) none of these

4. A = ` 1,200 n = 12 years i = 0.08, V = ?

Using the formula $V = \frac{A}{i}[1-\frac{A}{(1+i)^n}]$ Value of v will be:

(a) ` 3,039 (b) ` 3,990

(c) ` 9930 (d) none of these

5. a = ` 100 n = 10, i = 5% find the FV of annuity

Using the formula FV = a / {1 + i)^n - 1}, FV is equal to

(a) ` 1,258 (b) ` 2,581 (c) ` 1,528 (d) none of these

6. If the amount of an annuity after 25 years at 5% p.a. C.I. is ` 50,000 the annuity will be

(a) `1,406.90 (b) ` 1,046.90 (c) ` 1,146.90 (d) none of these

7. Given annuity of ` 100 amounts to ` 3137.12 at 4.5% p.a C. I. The number of years will be

(a) 25 years (appx.) (b) 20 years (appx.)
(c) 22 years (d) none of these

8. A company borrows ` 10,000 on condition to repay it with compound interest at 5% p.a by annual installments of ` 1000 each. The number of years by which the debt will be clear is

(a) 14.2 years (b) 10 years
(c) 12 years (d) none of these

9. Mr. X borrowed ` 5,120 at 12 ½ % p.a C.I. At the end of 3 yrs, the money was repaid along with the interest accrued. The amount of interest paid by him is

(a) ` 2,100 (b) ` 2,170 (c) ` 2,000 (d) none of these

10. Mr. Paul borrows ` 20,000 on condition to repay it with C.I. at 5% p.a in annual installments of ` 2000 each. The number of years for the debt to be paid off is

(a) 10 years (b) 12 years (c) 11 years (d) none of these

11. A person invests ` 500 at the end of each year with a bank which pays interest at 10% p. a C.I. annually. The amount standing to his credit one year after he has made his yearly investment for the 12th time is.

(a) ` 11,764.50 (b) ` 10,000 (c) ` 12,000 (d) none of these

12. The present value of annuity of ` 5,000 per annum for 12 years at 4% p.a. C.I. annually is

(a) ` 46,000 (b) ` 46,850 (c) ` 15,000 (d) none of these

13. A person desires to create a fund to be invested at 10% CI per annum to provide for a prize of ` 300 every year. Using V = a/I find V and V will be

(a) ` 2,000 (b) ` 2,500 (c) ` 3,000 (d) none of these

MISCELLANEOUS PROBLEMS

EXERCISE 4 (D)

Choose the most appropriate option (a), (b), (c) or (d).

1. A = ` 5,200, R = 5% p.a., T = 6 years, P will be

(a) ` 2,000 (b) ` 3,880 (c) ` 3,000 (d) none of these

2 If P = 1,000, n = 4 years., R = 5% p.a then C. I will be

(a) ` 215.50 (b) ` 210 (c) ` 220 (d) none of these

3 The time in which a sum of money will be double at 5% p.a C.I is

(a) ` 10 years (b) 12 years (c) 14.2 years (d) none of these

4. If A = ` 10,000, n = 18yrs., R = 4% p.a C.I, P will be

(a) ` 4,000 (b) ` 4,900 (c) ` 4,500 (d) none of these

5. The time by which a sum of money would treble it self at 8% p. a C. I. is

(a) 14.28 years (b) 14 years (c) 12 years (d) none of these

6. The present value of an annuity of ` 80 a years for 20 years at 5% p.a is

(a) ` 997 (appx.) (b) ` 900 (c) ` 1,000 (d) none of these

7. A person bought a house paying ` 20,000 cash down and ` 4,000 at the end of each year for 25 yrs. at 5% p.a. C.I. The cash down price is

(a) ` 75,000 (b) ` 76,000

(c) ` 76,392 (d) none of these.

8. A man purchased a house valued at ` 3,00,000. He paid ` 2,00,000 at the time of purchase and agreed to pay the balance with interest at 12% per annum compounded half yearly in 20 equal half yearly instalments. If the first instalment is paid after six months from the date of purchase then the amount of each instalment is

[Given log 10.6 = 1.0253 and log 31.19 = 1.494]

(a) ` 8,719.66 (b) ` 8,769.21

(c) ` 7,893.13 (d) none of these.

ANSWERS									
Exercise 4(a)									
1.	B	2	A	3.	C	4.	D	5.	A
6.	B	7.	A	8.	C	9.	A	10.	C
Exercise 4(b)									
1.	A	2.	C	3.	C	4.	B	5.	A
6.	C	7.	D	8.	A	9.	D	10.	B
11.	C	12.	D	13.	A	14.	A		
Exercise 4(c)									
1.	B	2.	A	3.	C	4.	D	5.	A
6.	B	7.	B	8.	A	9.	B	10.	D
11.	A	12.	D	13.	C				
Exercise 4(d)									
1.	B	2.	A	3.	C	4.	D	5.	A
6.	A	7.	C	8.	A				

ADDITIONAL QUESTION BANK

1. The difference between compound and simple interest at 5% per annum for 4 years on ` 20,000 is `
   (a) 250 (b) 277 (c) 300 (d) 310

2. The compound interest on half-yearly rests on ` 10,000 the rate for the first and second years being 6% and for the third year 9% p.a. is `. .
   (a) 2,200 (b) 2,287 (c) 2,285 (d) None

3. The present value of ` 10,000 due in 2 years at 5% p.a. compound interest when the interest is paid on yearly basis is ` .
   (a) 9,070 (b) 9,000 (c) 9,061 (d) None

4. The present value of ` 10,000 due in 2 years at 5% p.a. compound interest when the interest is paid on half-yearly basis is ` .
   (a) 9,070 (b) 9,069 (c) 9,061 (d) None

5. Johnson left ` 1,00,000 with the direction that it should be divided in such a way that his minor sons Tom, Dick and Harry aged 9, 12 and 15 years should each receive equally after attaining the age 25 years. The rate of interest being 3.5%, how much each son receive after getting 25 years old?
   (a) 50,000 (b) 51,994 (c) 52,000 (d) None

6. In how many years will a sum of money double at 5% p.a. compound interest?
   (a) 15 years 3 months (b) 14 years 2 months
   (c) 14 years 3 months (d) 15 years 2 months

7. In how many years a sum of money trebles at 5% p.a. compound interest payable on half- yearly basis?
(a) 18 years 7 months (b) 18 years 6 months
(c) 18 years 8 months (d) 22 years 3 months

8. A machine depreciates at 10% of its value at the beginning of a year. The cost and scrap value realized at the time of sale being ` 23,240 and ` 9,000 respectively. For how many years the machine was put to use?
(a) 7 years (b) 8 years (c) 9 years (d) 10 years

9. A machine worth ` 4,90,740 is depreciated at 15% on its opening value each year. When its value would reduce to ` 2,00,000?
(a) 4 years 6 months (b) 4 years 7 months
(c) 4 years 5 months (d) 5 years 7 months approximately

10. A machine worth ` 4,90,740 is depreciated at 15% of its opening value each year. When its value would reduce by 90%?
(a) 11 years 6 months (b) 11 years 7 months
(c) 11 years 8 months (d) 14 years 2 months approximately

11. Alibaba borrows ` 6 lakhs Housing Loan at 6% repayable in 20 annual installments commencing at the end of the first year. How much annual payment is necessary.
(a) 52,420 (b) 52,419 (c) 52,310 (d) 52,320

12. A sinking fund is created for redeming debentures worth ` 5 lakhs at the end of 25 years. How much provision needs to be made out of profits each year provided sinking fund investments can earn interest at 4% p.a.?
(a) 12,006 (b) 12,040 (c) 12,039 (d) 12,035

13. A machine costs ` 5,20,000 with an estimated life of 25 years. A sinking fund is created to replace it by a new model at 25% higher cost after 25 years with a scrap value realization of ` 25000. what amount should be set aside every year if the sinking fund investments accumulate at 3.5% compound interest p.a.?
(a) 16,000 (b) 16,500 (c) 16,050 (d) 16,005

14. Raja aged 40 wishes his wife Rani to have ` 40 lakhs at his death. If his expectation of life is another 30 years and he starts making equal annual investments commencing now at 3% compound interest p.a. how much should he invest annually?
(a) 84,448 (b) 84,450 (c) 84,449 (d) 84,077

15. Appu retires at 60 years receiving a pension of 14,400 a year paid in half-yearly installments for rest of his life after reckoning his life expectation to be 13 years and that interest at 4% p.a. is payable half-yearly. What single sum is equivalent to his pension?
(a) 1,45,000 (b) 1,44,900 (c) 1,44,800 (d) 1,44,700

ANSWERS

1.	(d)	**2.**	(d)	**3.**	(a)	**4.**	(c)	**5.**	(d)	**6.**	(b)
7.	(d)	**8.**	(c)	**9.**	(d)	**10.**	(d)	**11.**	(c)	**12.**	(a)
13.	(c)	**14.**	(d)	**15.**	(b)						

PAST EXAMS QUESTIONS

1. ` 87,000 becomes ` 10,000 in two years at simple interest. The amount that will become ` 6,875 in 3 years at the same rate of interest is:
   (a) ` 4,850 (b) ` 5,000 (c) ` 5,500 (d) ` 5,275

2. The difference between the simple and compound interest on a certain sum for 3 year at 5% p.a. is ` 228.75. The compound interest on the sum for 2 years at 5% p.a. is:
   (a) ` 3,175 (b) ` 3,075 (c) ` 3,275 (d) ` 2,975

3. Mr. X invests ` 10,000 every year starting from today for next 10 years suppose interest rate is 8% per annum compounded annually. Calculate future value of the annuity:
   (a) ` 1,56,454.88 (b) ` 1,44,865.625
   (c) ` 1,56,554.88 (d) None of these

4. The present value of an annuity of ` 3,000 for 15 years at 4.5% p.a. C.I. is:[Given that $(1.045)^{15} = 1.935282$]
   (a) ` 23,809.67 (b) ` 32,218.67
   (c) ` 32,908.67 (d) None of these

5. The rate of simple interest on a sum of money is 6% p.a. for first 3 years, 8% p.a. for the next five years and 10% p.a. for the period beyond 8 years. If the simple interest accrued by the sum for a period for 10 years is ` 1,560. The sum is:
   (a) ` 1,500 (b) ` 2,000 (c) ` 3,000 (d) ` 5,000

6. In what time will ` 3,90,625 amount to ` 4,56,976 at 8% per annum, when the interest in compounded semi-annually? [Given : $(1.04)^4 = 1.16986$]
   (a) 2 years (b) 4 years (c) 5 years (d) 7 years

7. A machine can be purchased for ` 50,000. Machine will contribute ` 12,000 per year for the next five years. Assume borrowing cost is 10% per annum. Determine whether machine should be purchased or not:
   (a) Should be purchased (b) Should not be purchased
   (c) Can't say about purchase (d) None of the above

8. How much amount is required to be invested every year so as to accumulate ` 3,00,000 at the end to 10 years, if interest is compounded annually at 10%? **[Give $(1.1)^{10} = 2.5937$]**
   (a) ` 18,823.65 (b) ` 18,828.65
   (c) ` 18,832.65 (d) ` 18,882.65

9. A certain sum of money amounts to ` 6,300 in two years and ` 7,875 in three years nine months at simple interest. Find the rate of interest per annum:
(a) 20% (b) 18% (c) 15% (d) 10%

10. How long will ` 12,000 take to amount to ` 14,000 at 5% p.a. converted quarterly?
[Given: $(1.0125)^{12.4} = 1.1666$]
(a) 3 years (b) 3.1 years (c) 13.5 years (d) 12.4 years

11. A company is considering proposal of purchasing a machine either by making full payment of ` 4,000 or by leasing it for four years at an annual rate of ` 1,250. Which course of action is preferable, if the company can borrow money at 14% compounded annually? **Given: $(1.14)^4 = 1.68896$]**
(a) Leasing is preferable (b) Should be purchased
(c) No difference (d) None of the above

12. Vipul purchases a car for ` 5,50,000. He gets a loan of ` 5,00,000 at 15% p.a. from a Bank and balance ` 50,000 he pays at the time of purchases. He has to pay the whole amount of loan in 12 equal monthly instalments with interest starting from the end of the first month. The money he has to pay at the end of every months is: **[Given $(1.0125)^{12} = 1.16075452$]**
(a) ` 45,130.43 (b) ` 45,230.43
(c) ` 45,330.43 (d) None of these

13. If ` 1,000 be invested at interest rate of 5% and the interest be added to the principal every 10 years, then the number of years in which it will amount to ` 2,000 is:
(a) $16\frac{2}{3}$ years (b) $6\frac{1}{4}$ years (c) 16 years (d) $6\frac{2}{3}$ years

14. The annual birth and death rates per 1000 are 39.4 and 19.4 respectively. The number of years in which the population will be doubled assuming there is no immigration or emigration is:
(a) 35 years (b) 30 years (c) 25 years (d) None of these

15. The effective rate equivalent to nominal rate of 6% compounded monthly is:
(a) 6.05 (b) 6.16 (c) 6.26 (d) 6.07

16. A company establishers a sinking fund to provide for the payment of ` 2,00,000 debt maturing in 20 years. Contributions to the fund are to be made at the end of every year. Find the amount of each annual deposit if interest is 5% per annum:
(a) ` 6,142 (b) ` 6.049 (c) ` 6,052 (d) ` 6,159

17. A person borrows ` 5,000 for 2 years at 4% p.a. simple interest. He immediately lends to another person at $6\frac{1}{4}$ % p.a. for 2 years. Find his gain in the transaction per years:
(a) ` 112.50 (b) ` 125 (c) ` 225 (d) ` 167.50

18. Raja aged 40 wishes his wife Rani to have ` 40 lakhs at his death. If his expectation of life is another 30 years and he starts making equal annual investments commencing now at 3% compound interest p.a. How much should he invest annually?

(a) ` 84,077 (b) ` 81,628 (c) ` 84,449 (d) ` 84,247

19. Two equal sums of money were lent at simple interest at 11% p.a. for $3\frac{1}{2}$ years and $4\frac{1}{2}$ years respectively. If the difference in interests for two periods was ` 412.50, then each sum is:

(a) ` 3,250 (b) ` 3,500 (c) ` 3,750 (d) ` 4,350

20. Anshul's father wishes to have ` 75,000 in bank account when his first college expenses begin. How much amount his father should deposit at 6.5% compounded annually if Anshul is to start college in 8 years hence from now?

(a) ` 45,360 (b) ` 46,360 (c) ` 55,360 (d) ` 48,360.

21. A Company may obtain a machine either by leasing it for 5 years (useful life) at an annual rent of ` 2,000 or by purchasing the machine for ` 8,100. If the company can borrow money at 18% per annum, which alternative is preferable?

(a) Leasing (b) Purchasing (c) Can't say (d) None of these

22. In how much time would the simple interest on a certain sum be 0.125 times the principal at 10% per annum?

(a) $1\frac{1}{4}$ years (b) $1\frac{3}{4}$ years (c) $2\frac{1}{4}$ years (d) $2\frac{3}{4}$ years

23. The difference between compounded interest and simple interest on a certain sum of 2 years @ 10% p.a. is ` 10. Find the sum:

(a) ` 1,010 (b) ` 1,095 (c) ` 1,000 (d) ` 990

24. A machine worth ` 4,90,740 is depreciated at 15% on its opening value each year. When its value would reduce to ` 2,00,000:

(a) 5 years 6 months (b) 5 years 7 months
(c) 5 years 5 months (d) None

25. A sinking fund is created for redeeming debentures worth ` 5 lacs at the end of 25 years. how provision needs to be made out of profits each year provided sinking fund investments can earn interest at 4% p.a.?

(a) ` 12,006 (b) ` 12,040 (c) ` 12,039 (d) ` 12,035

26. If the difference between simple interest and compound interest is ` 11 at rate of 10% for two years, then find the sum.

(a) ` 1,200 (b) ` 1,100 (c) ` 1,000 (d) None of these

27. Future value of an ordinary annuity:

(a) $A(n, i) = A\left[\frac{(1+i)^n-1}{i}\right]$ (b) $A(n, i) = A\left[\frac{(1+i)^n+1}{i}\right]$

(c) $A(n, i) = A\left[\frac{(1+i)^n}{i}\right]$ (d) $A(n, i) = A\left[\frac{(1+i)^n-1}{i(1+i)^n}\right]$

28. Find the numbers of years in which a sum doubles itself at the rate of 8% per annum.

(a) $11\frac{1}{2}$ (b) $12\frac{1}{2}$ (c) $9\frac{1}{2}$ (d) $13\frac{1}{2}$

29. In how many years, a sum will become double at 5% p.a. compound interest.

(a) 14.0 years (b) 14.1 years (c) 14.2 years (d) 14.3 years

30. The time by which a sum of money if 8 time of itself if it doubles itself in 15 years.

(a) 42 years (b) 43 years (c) 45 years (d) 46 years

31. What is the rate of simple interest if a sum of money amounts to ` 2,784 in 4 years and ` 2,688 in 3 years?

(a) 1% p.a. (b) 4% p.a. (c) 5% p.a. (d) 8% p.a.

32. A sum amount to ` 1,331 at a principal of ` 1,000 at 10% compounded annually. Find the time.

(a) 3.31 years (b) 4 years (c) 3 years (d) 2 years

33. Paul borrows ` 20,000 on condition to repay it with compound interest at 5% p.a. in annual instalment of ` 2,000 each. Find the number of years in which the debt would be paid off.

(a) 10 years (b) 12 years (c) 14 years (d) 15 years

34. In how many years, a sum of ` 1,000 compounded annually @ 10% will amount to ` 1,331?

(a) 6 years (b) 5 years (c) 4 years (d) 3 years

35. The compound interest for a certain sum @ 5% p.a. for fist year us ` 25. The S-I for the same money @ 5% p.a. for 2 years will be.

(a) ` 40 (b) ` 50 (c) ` 60 (d) ` 70

36. At what % rate of compound interest (C.I.) will a sum of money become 16 times in four years, if interest is being calculated compounding annually:

(a) r = 100% (b) r = 10% (c) r = 200% (d) r = 20%

37. Find the preset value of an annuity of ` 1,000 payable at the end of each year for 10 years. If rate of interest is 6% compounding per annum (given $(1.06)^{-10} = 0.5584$):

(a) ` 7,360 (b) ` 8,360 (c) ` 12,000 (d) None of these.

38. If the simple interest on a sum of money at 12% p.a. for two years is ` 3,600. The compound interest on the same sum for two years at the same rate is:

(a) ` 3,816 (b) ` 3,806 (c) ` 3,861 (d) ` 3,860

39. The future value of an annuity of ` 5,000 is made annually for 8 years at interest rate of 9% compounded annually [Given that $(1.09)^8 = 1.99256$] is-

(a) ` 55, 142.22 (b) ` 65,142.22

(c) ` 65,532.22 (d) ` 57,425.22

40. The effective annual rate of interest corresponding to nominal rate 6% p.a. payable half yearly if

(a) 6.06% (b) 6.07% (c) 6.08% (d) 6.09%

41. The cost of Machinery is ` 1,25,00/- if its useful life is estimated to be 20 years and the rate of depreciation of its cost is 10% p.a., than the scrap value of the Machinery is

(a) ` 15,187 (b) ` 15,400 (c) ` 15,300 (d) ` 15,250

42. ` 8,000/- at 10% per annum interest compounded yearly will become at the end of one year

(a) ` 8,800 (b) ` 8,820 (c) ` 8,900 (d) ` 9,600

43. The value of furniture depreciates by 10% a year, it the present value of the furniture in an office ` 25,515 calculate the value of furniture 3 years ago

(a) ` 30,000 (b) ` 35,000 (c) ` 40,000 (d) ` 50,000

44. Mr. X invests 'P' amount at Simple Interest rate 10% and Mr. Y invests 'Q' amount at Compound interest rate 5% compounded annually. At the end of two years both get the same amount of interest, then the relation between two amounts P and Q is given by:

(a) $P = \frac{41Q}{80}$ (b) $P = \frac{41Q}{40}$ (c) $P = \frac{41Q}{100}$ (d) $P = \frac{41Q}{200}$

45. If the difference of S.I. and C.I. is ` 72 at 12% for 2 years. Calculate the amount.

(a) ` 8,000 (b) ` 6,000 (c) ` 5,000 (d) ` 7,750

46. If a simple interest on a sum of money at 6% p.a. for 7 years is equal to twice of simple interest on another sum for 9 years at 5% p.a. The ratio will be:

(a) 2 : 15 (b) 7 : 15 (c) 15 : 7 (d) 1 : 7

47. By mistake a clerk, calculated the simple interest on principal for 5 months at 6.5% p.a. instead of 6 months at 5.5% p.a. if the error in calculation was ` 25.40. The original sum of principal was ___.

(a) ` 60,690 (b) ` 60,960 (c) ` 90,660 (d) ` 90,690

48. If the Simple interest on ` 1,400 for 3 years is less than the simple interest on ` 1,800 for the same period by ` 80, then the rate of interest is

(a) 5.67% (b) 6.67% (c) 7.20% (d) 5.00%

49. Nominal rate of interest is 9.9% p.a. if interest is compounded monthly. What will be the effective rate of interest?

(a) 10.36% (b) 9.36% (c) 11.36% (d) 9.9%

50. The S.I. on a sum of money is $\frac{4}{9}$ of the principal and the no. of years is equal to the rate of interest per annum. Find the rate of interest per annum?
(a) 5% (b) 20 % (c) 22/7% (d) 6%

51. Simple interest on ` 2,000 for 5 months at 16% p.a. is _____
(a) ` 133.33 (b) ` 133.26 (c) ` 134.00 (d) ` 132.09

52. How much investment is required to yield an Annual income of ` 420 at 7% p.a. Simple interest.
(a) ` 6,000 (b) ` 6,420 (c) ` 5,580 (d) ` 5,000

53. Mr. X invests ` 90,500 in post office at 7.5% p.a. simple interest. While calculating the rate was wrongly taken as 5.7% p.a.
The difference in amounts at maturity is ` 9,774. Find the period for which the sum was invested:
(a) 7 years (b) 5.8 years (c) 6 years (d) 8 years

54. The difference between compound and simple interest on a certain sum of money for 2 years at 4% p.a. is ` 1. The sum (in `) is:
(a) 625 (b) 630 (c) 640 (d) 635

55. A sum of money compounded annually becomes ` 1,140 in two years and ` 1,710 in three years.
Find the rate of interest per annum.
(a) 30% (b) 40% (c) 50% (d) 60%

56. On what sum difference between compound interest and simple interest for two years at 7% p.a. interest is ` 29.4
(a) ` 5,000 (b) ` 5,5000 (c) ` 6,000 (d) ` 6,500

57. In what time will a sum of money double its y at 6.25% p.a. simple interest?
(a) 5 years (b) 8 years (c) 12 years (d) 16 years

58. What principal will amount to ` 370 in 6 years at 8% p.a. simple interest?
(a) ` 210 (b) ` 250 (c) ` 310 (d) ` 350

59. The partners A and B together lent ` 3,903 at 4% per annum interest compounded annually. After a span of 7 years. A gets the same amount as B gets after 9 years. The share of A in the sum of ` 3,903 would have been:
(a) ` 1,875 (b) ` 2,280 (c) ` 2,028 (d) ` 2,820

60. If a sum triples in 15 years at simple rate of interest, the rate of interest per annum will be:
(a) 13.0% (b) 13.3% (c) 13.5% (d) 18.0%

61. How much amount is required to be invested every year as to accumulate ` 6,00,000 at the end of 10 years, if interest is compounded annually at 10% rate of interest

(a) ` 37,467 (b) ` 37, 476 (c) 37,647 (d) ` 37,674

62. The future value of an annuity of ` 1,000 made annually for 5 years at the interest of 14% compounded annually is:

(a) ` 5,610 (b) ` 6,610 (c) ` 6,160 (d) ` 5,160

63. A sum of money invested of compound interest doubles itself in four years. It becomes 32 times of itself at the same rate of compound interest in

(a) 12 years (b) 16 years (c) 20 years (d) 24 years

64. A certain sum of money was invested at simple rate of interest for three years. If the same has been invested at a rate that was seven percent higher, the interest amount would have been ` 882 more. The amount of sum invested is:

(a) ` 12,600 (b) ` 6,800 (c) ` 4,200 (d) ` 2,800

65. A sum of money doubles itself in 8 years at simple interest. The number of years it would triple itself is _____.

(a) 20 years (b) 12 years (c) 16 years (d) None of these

66. A sum of ` 44,000 is divided into three parts such that the corresponding interest earned after 2 years, 3 years and 6 years may be equal. If the rates of simple interest are 6% p.a., 8% p.a. and 6% p.a. respectively, then the smallest part of the sum will be.

(a) ` 4,000 (b) ` 8,000 (c) ` 10,000 (d) ` 12,000

67. Suppose your parent decides to open a PPF (Public Provident Fund) account in a bank towards your name with ` 10,000 every year starting from today for next 15 years. When you receive and get 8.5% per annum interest rate compounded annually. What is the present value of this annuity?

(a) 83,042 (b) 1, 66,084 (c) 93,042 (d) 8,30,423

68. In how many years will a sum of money become four times at 12% p.a. simple interest?

(a) 18 years (b) 21 years (c) 25 years (d) 28 years

69. The simple interest for a certain sum for 2 years at 10% per annum is ` 90. The corresponding compound interest is (in `):

(a) 99 (b) 95.60 (c) 94.50 (d) 108

70. Mr. X bought an electronic item for ` 1,000. What would be the future value of the same item after 2 years, if the value is compounded semi annually at 22% per annum?

(a) ` 1,488.40 (b) ` 1,518.07 (c) ` 2,008.07 (d) ` 2,200.00

71. If an amount is kept at simple interest, it earns an interest of ` 600 in first two years but when kept at compound interest it earns an interest of ` 660 for the same period, then the rate of interest and principal amount respectively are:

(a) 20%, ` 1,200 (b) 10%, ` 1,200

(c) 20%, ` 1,500 (d) 10%, ` 1,500

72. The sum invested at 4% per annum compounded semi-annually amounts to ` 7,803 at the end of one year, is:

(a) ` 7,000 (b) ` 7,500 (c) ` 7,225 (d) ` 8,000

73. A compound interest on a sum for 2 years is ` 30 more than the simple interest at the rate of 5% per annum then the sum is

(a) ` 11,000 (b) ` 13,000 (c) ` 12,000 (d) ` 15,000

74. A person lends ` 6,000 for 4 years and ` 8,000 for 3 years at simple interest. If he gets ` 2,400 as total interest, the rate of interest is:

(a) 5% (b) 4% (c) 6% (d) 7%

75. The future value of an annuity of ` 1,500 made annually for five years at interest rate 10% compounded annually is (Given that $(1.1)^5 = 1.61051$):

(a) ` 9,517.56 (b) ` 9,157.65 (c) ` 9,715.56 (d) ` 9,175.65

76. The Difference between the compound interest and simple interest at 10% per annum for 4 years on ` 10,000 is `_____

(a) 650 (b) 640 (c) 641 (d) 600

77. How much amount is required to be invested every year as to accumulate ` 7,96,870 at the end of 10 years, if interest compounded annually at 10% given that A(10, 0.1)= 15.9374?

(a) ` 40,000 (b) ` 4,50,000 (c) ` 48,000 (d) ` 50,000

78. If compound interest on any sum at the rate of 5% for two years is ` 512.50 then the sum would be:

(a) ` 3,000 (b) ` 4,000 (c) ` 5,000 (d) ` 6,000

79. The effective rate of interest equivalent to the nominal rate of 7% converted monthly:

(a) 7.26% (b) 7.22% (c) 7.02% (d) 7.20%

80. Mr. X invest ` 10,000 every year starting from today for next: 10 years suppose interest rate is 8% per annual compounded annually. Calculate future value of the annuity.

(a) ` 1,56,454.88 (b) ` 1,56,554.88

(c) ` 1,44,865.625 (d) None of the above

81. How much amount is required to be invested every year so as to accumulate ` 3,00,000 at the end of 10 years, if interest is compounded annually at 10%?
(a) ` 18,823.65 (b) ` 18 (c) ` 18,828.65 (d) ` 18,882.65

82. If ` 1,000 be invested at interest at rate of 5% and the interest be added to the principal every 10 years, than the number of years in which it will amount to ` 2,000 is:
(a) $16\frac{2}{3}$ years (b) $6\frac{1}{4}$ years (c) 16 years (d) $6\frac{2}{3}$ years

83. A person borrows ` 5,000 for 2 years at 4% per annual simple interest. He immediately lends to another person at $6\frac{1}{4}$ % per annual for 2 years find his gain in the transaction for year:
(a) ` 112. 50 (b) ` 225 (c) ` 125 (d) ` 107.50

84. If an amount is kept at S.I. it earns an interest of ` 600 in first two years but when kept at compound interest it earns an interest of ` 660 for the same period, then the rate of interest and principal amount respectively are:
(a) 20%., ` 1,200 (b) 20%., ` 1,500
(c) 10%., ` 1,200 (d) 10%., ` 1,00

85. The future value of an annuity of ` 1,000 made annually for 5 years at the interest of 14% compounded annually is:
(a) ` 5,610 (b) ` 6,610 (c) ` 6,160 (d) ` 5,160

86. If ` 10,000 is invested at 8% per year compound quarterly, then the value of the investment after 2 years is
(a) ` 11,716.59 (b) ` 10,716.59 (c) ` 117.1659 (d) None

87. A bank pays 10% rate of interest, interest being calculated half yearly. A sum of ` 400 is deposited in the bank. The amount at the end of 1 year will be:
(a) ` 440 (b) ` 439 (c) ` 441 (d) ` 442

88. A certain money doubles itself in 15 years when deposited on simple interest. It would triple itself in
(a) 20 years (b) 15 years (c) 25 years (d) 30 years

89. A man deposited ` 8,000 in a bank for 3 years at 5% per annum compound interest, after 3 years he will get
(a) ` 8,800 (b) ` 9,261 (c) ` 9,200 (d) ` 9,000

90. If in two years time a principal of ` 100 amounts to ` 121 when the interest at the rate of r% is compounded annually, then the value of r will be:
(a) 10.5(b) 10 (c) 15 (d) 14

91. A certain sum of money Q was deposited for 5 years and 4 months at 4.5 % simple interest and amounted to ` 248, then the value of Q is:

(a) ` 200 (b) 210 (c) ` 220 (d) ` 240

92. If compound interest on a sum for 2 years at 4% per annum is ` 102, then the simple interest on the same sum for the same period at the same rate will be:

(a) ` 99 (b) ` 101 (c) ` 100 (d) ` 95

93. A man invests an amount of ` 15,860 in the names of his three sons A, B and C in such a way that they get the same amount after 2,3 and 4 years respectively. If the rate of interest is 5%, then ratio of amount invested in the name of A, B and C is:

(a) 6:4:3 (b) 3:4:6 (c) 30:12:5 (d) None of these

94. If the difference between the compound interest compounded annually and simple interest on a certain amount at 10 % per annum for two years is ` 372, then the principal amount is:

(a) ` 37,200 (b) ` 37,000 (c) ` 37,500 (d) None of these

95. What is the net present value of piece of property which would be valued at ` 2 lakh at the end of 2 years? (Annual rate of increase = 5%)

(a) ` 1. 81 lakh (b) ` 2.01 lakh (c) ` 2.00 lakh (d) 2.81 lakh

96. The effective rate of interest for one year deposit corresponding to a nominal 7% rate of interest per annum convertible monthly is:

(a) 7% (b) 7.5% (c) 7.23% (d) 7.18%

ANSWERS:

1	B	16	B	31	B	46	C	61	C	76	C	91	A
2	B	17	A	32	C	47	B	62	B	77	D	92	C
3	A	18	B	33	D	48	B	63	C	78	C	93	A
4	B	19	C	34	D	49	A	64	C	79	B	94	A
5	B	20	A	35	B	50	B	65	C	80	A	95	A
6	A	21	A	36	A	51	A	66	B	81	A	96	C
7	B	22	A	37	A	52	A	67	C	82	A		
8	A	23	C	38	A	53	C	68	C	83	B		
9	A	24	A	39	A	54	A	69	C	84	B		
10	B	25	A	40	D	55	C	70	B	85	B		
11	A	26	B	41	A	56	C	71	C	86	A		
12	A	27	A	42	A	57	D	72	B	87	A		
13	A	28	B	43	B	58	B	73	C	88	A		
14	A	29	D	44	A	59	C	74	A	89	B		
15	B	30	C	45	C	60	B	75	B	90	B		

CHAPTER – 5 PERMUTATION AND COMBINATION

PERMUTATION

Q.1. If $\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$ then value of x
(a) 121 (b) 144 (c) 132 (d) None

Q.2. (n + 1) ! = 30 × (n – 1) ! Find the value of n.
(a) 6 (b) 8 (c) 5 (d) None

Q.3. How many Different words can be formed with the letters of the word PEN
(a) 6 (b) 3 (c) 9 (d) None

Q.4. How many Different words can be formed with the letters of the word MOBILE.
(a) 730 (b) 720 (c) 740 (d) None

Q.5. How many various words can be formed with the letters of the word COMPUTER
(a) 40380 (b) 40360 (c) 40320 (d) None

Q.6. How many Different words can be formed with the letters of the word FAILURE
(a) 5040 (b) 5020 (c) 5080 (d) None

Q.7. How many Different words can be formed with the letters of the word EYE
(a) 9 (b) 3 (c) 12 (d) None

Q.8. How many Different words can be formed with the letters of the word RAIPUR
(a) 360 (b) 390 (c) 340 (d) None

Q.9. How many Different words can be formed with the letters of the word CALCUTTA
(a) 5020 (b) 5060 (c) 5040 (d) None

Q.10. Find the value of ${}^{7}P_{2}$
(a) 42 (b) 45 (c) 48 (d) None

Q.11. In the ${}^{n}P_{r} = n(n-1)(n-2)$ …….. The no. of factors is :-
(a) n (b) r (c) n – r (d) r + n

Q.12. If ${}^{n}P_{4} = 12 \times {}^{n}P_{2}$ then find the value of n
(a) – 1 (b) 6 (c) 5 (d) Both a and b

Q.13. ${}^{n}P_{3} : {}^{n}P_{2} = 3 : 1$ then find the value of n
(a) n = 7 (b) n = 4 (c) n = 5 (d) n = 6

Q.14. If ${}^{m+n}P_{2} = 56$ & ${}^{m-n}P_{2} = 30$ then find the value of m and n.
(a) (6, 2) (b) (7, 1) (c) (4, 4) (d) (8, 3)

Q.15. Mr. x and Mr. y enter into a railway compartment having 6 vacant seats, then in how many ways then can occupy the seats ?
(a) 20 (b) 30 (c) 40 (d) None

Q.16. 6 persons enter into a railway compartment having 2 vacant seats, then in how many ways they can occupy the seats?
(a) 60 (b) 80 (c) 30 (d) None

Q.17. In how many ways two photo frames can be arranged on a wall having 6 nails.
(a) 30 (b) 20 (c) 10 (d) None

Q.18. In how many ways 2 letters can be dropped in the 6 letter boxes.
(a) 30 (b) 36 (c) 40 (d) None

Q.19. In how many ways 6 letters can be dropped in the 2 letter boxes.
(a) 64 (b) 81 (c) 96 (d) None

Q.20. ${}^{(m+n)}P_2 = 132$ & ${}^{m-n}P_2 = 30$ then find the value of m and n
(a) $m = 6, n = 6$ (b) $m = 10, n = 2$ (c) $m = 9, n = 3$ (d) $m = 7, n = 3$

Q.21. In a group of boys, the number of arrangement of 4 boys is 12 times the number of arrangement of 2 boys, find the number of boys in the group.
(a) 10 (b) 8 (c) 6 (d) 7

Q.22. There are 10 trains running between Raipur and Delhi, in how many ways a person can go from Raipur to Delhi and return by any other trains.
(a) 120 (b) 90 (c) 60 (d) None

Q.23. If 12 students are participating in a competition, in how many ways the 1st, 2nd and 3rd prizes can be won?
(a) 1250 (b) 1380 (c) 1320 (d) None

Q.24. 12 students are participating in three different competition, in how many ways the winner can be declared.
(a) 1728 (b) 1751 (c) 1740 (d) None

Q.25. How many two digit no. can be formed using the digits 5, 3, 7, 9,
(i) When repetition not allowed (ii) when repetition is allowed.
(a) 15, 31 (b) 13, 21 (c) 12, 16 (d) None

Q.26. How many numbers of 3 digits can be formed using the digits 2, 5, 6, 8 only (i) Repetition not allowed (ii) repetition allowed.
(a) 24, 64 (b) 45, 125 (c) 60, 120 (d) None

Q.27. How many numbers of 3 digit can be formed using digits 2, 3, 5, 0, 8, 9.
(i) Repetition not allowed (ii) repetition allowed.
(a) 280, 100 (b) 160, 100 (c) 100, 180 (d) None

Q.28. In how many ways the letter of the word MOBILE can be arranged so that the Consonants always occupy the ODD places.
(a) 38 (b) 36 (c) 51 (d) None

Q.29. How many number of four digit numbers can be formed which are greater than 5000, using only the digits 3, 4, 5, 6, 7. (i) Repetition not allowed (ii) repetition allowed
(a) 72, 375 (b) 45, 375 (c) 72, 320 (d) None

Q.30. How many different words can be formed with the letter of the word MONDAY So that the each words so formed always begins with M but do not end with N.
(a) 64 (b) 100 (c) 96 (d) None

Q.31. In how many ways, 8 Sweets of different sizes can be distributed among 8 children's so that the biggest sweet always goes to the youngest children.
(a) 5040 (b) 4050 (c) 4550 (d) None

Q.32. How many different words can be formed with the letter of the word PEN.
(a) 9 (b) 6 (c) 12 (d) None

Q.33. How many different words can be formed with the letters of the word PEN so that E and N always come together.
(a) 4 (b) 8 (c) 1 (d) None

Q.34. How many different words can be formed with the letters of the word PEN so that E and N never come together.
(a) 6 (b) 2 (c) 12 (d) None

Q.35. How many different words can be formed with the letters of the word FAILURE so that all the vowels come together.
(a) 569 (b) 567 (c) 576 (d) None

Q.36. How many different words can be formed with the letters of the word FAILURE so that all the vowels do not come together
(a) 4356 (b) 4464 (c) 4246 (d) None

Q.37. How many different words can be formed with the letter of the word TRIANGLE (i) So that the letters A,N,G,L,E always come together (ii) So that each word so formed always contains the word ANGLE.
(a) 40320, 45 (b) 3040, 24 (c) 2880, 24 (d) None

Q.38. How many numbers of 9 digits can be formed using different digits.
(a) $9 \times 9!$ (b) $10 \times 9!$ (c) $9 \times 10!$ (d) None

Q.39. How many numbers lying between 10 and 1000 can be formed using only the digits 2, 3, 4, 0, 8, 9
(a) 120 (b) 125 (c) 100 (d) None

Q.40. There are Five speakers A, B, C, D and E, in how many ways they can speak if speaker A will speak Just before speaker B always.
(a) 28 (b) 36 (c) 24 (d) None

CIRCULAR PERMUTATION

Q.41. In how many ways 7 girls can form a ring.
(a) 720 (b) 730 (c) 710 (d) None

Q.42. In how many ways a Necklace can be formed with 50 different jewels.
(a) $\frac{39!}{2}$ (b) $\frac{49!}{2}$ (c) $\frac{59!}{2}$ (d) None

Q.43. In how many ways 4 men and 4 women can be seated in a round table so that any two women never be seated side by side.
(a) 144 (b) 145 (c) 414 (d) None

Q.44. In how many ways 7 boys can be seated in a round table so that the two particulars boys always sit together.
(a) 260 (b) 364 (c) 240 (d) None

Q.45. In how many ways five persons can be seated in a round table so that the tallest person is always on the right side of the shortest person.
(a) 6 (b) 18 (c) 9 (d) None

Q.46. In how many ways the 7 members of a committee can be seated in a round table so that the President, Vice President and the Secretary always sit together.
(a) 121 (b) 132 (c) 144 (d) None

Q.47. In how many ways 7 members of a committee can be seated in a round table so that the Vice President and the Secretary are always be the neighbor of the President.
(a) 48 (b) 58 (c) 46 (d) None

COMBINATION

Q.48. Find the value of ${}^{12}C_4$
(a) 496 (b) 495 (c) 469 (d) None

Q.49. Find the value of ${}^{52}C_2$
(a) 1326 (b) 1339 (c) 1352 (d) None

Q.50. If ${}^{18}C_r = {}^{18}C_{r+2}$, then find the value of r
(a) r = 2 (b) r = 18 (c) r = 8 (d) None

Q.51. If ${}^{18}C_r = {}^{18}C_{r+2}$, Then Find the value of ${}^{r}C_5$.
(a) 48 (b) 56 (c) 69 (d) None

Q.52. If ${}^{n}C_{10} = {}^{n}C_{14}$, Find the value of ${}^{25}C_n$
(a) 25 (b) 21 (c) 51 (d) None

Q.53. If ${}^{n}P_r = 336$ & ${}^{n}C_r = 56$ then find the value of n and r.
(a) (3, 2) (b) (8, 3) (c) (9, 3) (d) None

Q.54. Every two persons shake hands with each other in a party and the total number of handshakes is 66. Find the number of guests in the party.
(a) 12 & – 11 (b) 24 (c) 12 (d) None

Q.55. 8 points are marked on the circumference of a circle, how many chords can be drawn by them.
(a) 48 (b) 28 (c) 21 (d) None

Q.56. There are 12 points in a plane & not more than two of them are collinear, How many straight lines can be formed by them?
(a) 66 (b) 69 (c) 96 (d) None

Q.57. In the previous question, how many different Triangles can be formed?
(a) 240 (b) 220 (c) 260 (d) None

Q.58. There are 12 points are marked in a plane of which five points are collinear, how many Triangles can be formed by them?
(a) 210 (b) 232 (c) 100 (d) None

Q.59. In the previous question, how many different straight lines can be form?
(a) 75 (b) 55 (c) 57 (d) None

Q.60. How many Diagonals can be formed in a Decagon.
(a) 32 (b) 35 (c) 51 (d) None

Q.61. How many Parallelogram can be drawn from a set of four parallel lines and an another set of three parallel lines intersecting each other?
(a) 14 (b) 16 (c) 18 (d) None

Q.62. A person has 2 friends, in how many ways he may invite one or more of them to a dinner.
(a) 6 (b) 3 (c) 4 (d) None

Q.63. A person has 8 friends, in how many ways he may invite one or more of them to a dinner.
(a) 255 (b) 245 (c) 232 (d) None

Q.64. In how many ways a person can select one or more devices from the four devices AC, TV, Mobile and Washing Machine.
(a) 14 (b) 16 (c) 15 (d) None

Q.65. In how many ways a person can contribute to a fund out of one note of `10, one note of `5, one note of `2 and one note of `1.
(a) 15 (b) 25 (c) 18 (d) None

Q.66. A question paper contains 6 questions and each question having an alternative question, in how many ways an examinee can solve one or more questions.
(a) 730 (b) 732 (c) 728 (d) None

Q.67. In an election there are five candidates and three member are to be elected. A voter is allowed to vote for any number of candidates but not greater than the number to be elected. In how many ways a voter can vote.
(a) 40 (b) 25 (c) 30 (d) None

Q.68. The Supreme Court has given 6 to 3 decision uploading the lower court. The number of ways it can give a majority decision reversing the lower court is?
(a) 256 (b) 251 (c) 255 (d) None

Q.69. ${}^{n}C_1 + {}^{n}C_2 + {}^{n}C_3 + \ldots\ldots\ldots\ldots {}^{n}C_n = ?$
(a) 2^n (b) 2^{n-1} (c) $2^n - 1$ (d) $2^n + 1$

Q.70. $(2n)! = ?$
(a) $2^n . n!$ (b) $1 \times 3 \times 5 \times \ldots . (2n-1)$
(c) 2^n (d) $2^n[1 \times 3 \times 5 \ldots (2n-1)] \times n!$

Q.71. A committee of 5 members is to be form from a group of 7 boys and 4 girls. Then, in how many way such committee can be formed.
(a) 462 (b) 465 (c) 469 (d) None

Q.72. Form a group of 7 boys and 4 girls a committee of 5 members is to be formed such that:-
(i) Committee includes exactly 3 boys and 2 girls (ii) At least 3 girls (iii) At most 2 girls.
(a) 220, 95, 321 (b) 210, 91, 371 (c) 210, 91, 317 (d) None

Q.73. A student is required to solve only 6 questions out of the 12 question. The question paper is divided into 2 parts, containing 6 questions in each part. A students is not allowed to solve more than 4 questions from any part. In how many ways a student can solve the question paper?
(a) 840 (b) 850 (c) 805 (d) None

Q.74. A committee of 7 members is to be formed from a group of 6 CA, 4 CS and 5 CMA. In how many ways can this be done if, in the committee there must be at least one member from each group and at least 3 CA.
(a) 3705 (b) 3507 (c) 3570 (d) None

Q.75. In how many ways 4 consonants and 3 vowels can be selected from 12 consonants and 5 vowels.
(a) 4950 (b) 4960 (c) 4955 (d) None

Q.76. How many different words can be formed with 12 consonants and 5 vowels by taking 4 consonants and 3 vowels in each word.
(a) $8! \times 4950$ (b) $4650 \times 7!$ (c) $4950 \times 7!$ (d) None

Q.77. In how many a person can invite 7 male friends and 3 female friends out of 10 male friends and 5 female friends to a dinner.
(a) 1200 (b) 1400 (c) 1800 (d) None

Q.78. In how many ways a person can invite 7 male friends and three female friends out of 10 male friends and 5 female friends and celebrate a dinner party with various sitting arrangement.
(a) $1100 \times 10!$ (b) $1200 \times 10!$ (c) $15! \times 1200$ (d) None

Q.79. A committee of three ladies and four gents is to be formed out of eight ladies and seven gents. Mrs. X refuses to serve in a committee in which Mr. Y is a member. The number of such committee that can be formed, is
(a) 1540 (b) 1450 (c) 1560 (d) None

Q.80. Five Bulbs of which three are defective are to be tried in the two bulb points in a dark room, find the number of trials the room shall be lighted is :-
(a) 4 (b) 7 (c) 9 (d) None

Q.81. In how many ways 4 letters can be selected from the letters of the word EXAMINATION.
(a) 136 (b) 132 (c) 139 (d) None

Q.82. 8 guests have to be seated, four on each side of a long rectangular table. Two particular guests desire to sit on the one side of the table and the three particular guest desire to sit on the particular/other side of the table, then the number of ways in which the sitting arrangement can be made is.
(a) 7218 (b) 2178 (c) 1728 (d) None

Q.83. In how many ways six plus (+) sign and four negative (−) sign can be arranged in a line such that no two negative sign occur together.
(a) 35 (b) 38 (c) 51 (d) None

Q.84. Three ladies and three gents can be seated in round table so that any two and only two of the ladies sit together, the number of ways is:-
(a) 81 (b) 72 (c) 45 (d) None

Q.85. If ${}^{12}C_5 + 2 \times {}^{12}C_4 + {}^{12}C_3 = {}^{14}C_x$ then find the value of x
(a) 3 (b) 6 (c) 9 (d) None

Q.86. ${}^{500}C_{92} = {}^{499}C_{92} + {}^{n}C_{91}$ then find the value of n
(a) 499 (b) 500 (c) 498 (d) None

Q.87. The number of ways in which 9 objects can be divided into three group containing 2, 3 and 4 objects respectively?
(a) 1250 (b) 1260 (c) 1245 (d) None

Q.88. What is the Rank of the word PEN, when all the letters are arranged in dictionary format.
(a) 3 (b) 6 (c) 5 (d) None

Q.89. A boy has 3 library tickets and 8 books of his interest in the library. Of these 8, he does not want to borrow mathematics part-II unless mathematics part-I is also borrowed. In how many ways can he choose the three books to be borrowed?
(a) 51 (b) 41 (c) 61 (d) 71

Q.90. An examination paper with 10 questions consists of 6 questions in mathematics and 4 questions in statistic part. At least one question from each part is to be attempted in how many ways can this be done?
(a) 945 (b) 1024 (c) 1005 (d) None

MISCELLANEOUS QUESTIONS 0f *Permutation & Combination*

Q.91. The ways of arranging smaller or equal number of persons or objects from a group of persons or collection of objects with due regard being paid to the order of arrangement is/are known as
(a) Permutations (b) Combinations lap;
(c) (a) or (b) (d) (a) and (b)

Q.92. The number of ways in which smaller or equal number of things are arranged or selected from a collection of things where the order of selection or arrangement is not important, is/are known as
(a) Permutations (b) Combinations
(c) (a) or (b) (d) (a) and (b)

Q.93. If ${}^{n}p_r = 720\ {}^{n}C_r$ then the value of r is :-
(a) 6 (b) 5 (c) 4 (d) None

Q.94. If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, then n is equal to
(a) 4 (b) 9 (c) 3 (d) None

Q.95. If ${}^{35}C_{n+7} = {}^{35}C_{4n-2}$ then value of n is :-
(a) 5 (b) 6 (c) 7 (d) None

Q.96. If ${}^{n}C_{r-1} = 36$, ${}^{n}C_r = 84$ and ${}^{n}C_{r+1} = 126$, then
(a) n = 8, r = 4 (b) n = 9, r = 3 (c) n = 7, r = 5 (d) None

Q.97. In how many ways can a student choose a programme of 5 courses, if 9 courses are available and 2 courses are compulsory for every student?
(a) 45 ways (b) 35 ways (c) 55 ways (d) 65 ways

Q.98. If ${}^{n+2}C_8 : {}^{n-2}P_4 = 57 : 16$ then the value of n is :-
(a) 20 (b) 19 (c) 18 (d) 17

Q.99. If 7 points out of 12 are in the same straight line, then the number of triangles that can be formed, is :-
(a) 19 (b) 158 (c) 185 (d) 201

Q.100. A polygon has 44 diagonals, the number of its sides is:-
(a) 9 (b) 10 (c) 11 (d) 12

Q.101. The number of different permutations of the word "BANANA" is:-
(a) 270 (b) 60 (c) 120 (d) 360

Q.102. Number of straight lines that can be formed out of 10 points of which 7 are collinear
(a) 26 (b) 21 (c) 25 (d) None

Q.103. The number of ways in which 5 boys and 5 girls can sit in a ring are
(a) 10 ! (b) 9 ! (c) 5 ! (d) 6 !

Q.104. There are 6 candidates for 3 posts, in how many ways can the posts be filled?
(a) 20 (b) 30 (c) 10 (d) 11

Q.105. A code word is to consist of 2 English alphabets followed by 2 distinct numbers from 1 to 9. For example CA23 is a code word, how many such code words are there?
(a) 615800 (b) 46800 (c) 719500 (d) 410800

Q.106. A number lock on a suitcase has 3 wheels, each labelled with 10 digits from 0 to 9. If opening of the lock is a particular sequence of three digits with no repeats, how many such sequence of three digits with no repeats will be possible?
(a) 720 (b) 760 (c) 680 (d) 780

Q.107. A customer forgets a four-digit code for an Automatic teller machine (A.T.M.) in a bank. However, he remembers that code consists of digits 3, 5, 6 and 9. Find the largest possible number of trials necessary to obtain the correct code.
(a) 12 (b) 24 (c) 48 (d) 36

Q.108. If $(n+1)\,! = 6\,[(n-1)\,!]$, then find n
(a) 6 (b) 4 (c) 8 (d) 2

Q.109. Find n , if ${}^{n}P_4 = 18\ {}^{n-1}P_2$.
(a) 4 (b) 8 (c) 6 (d) 12

Q.110. 10 students are participating in a race. In how many ways first three prizes be won?
(a) 920 (b) 680 (c) 820 (d) 720

Q.111. ${}^{n}P_r : {}^{n}C_r = ?$
(a) n (b) n ! (c) r ! (d) r

Q.112. The Chief Ministers of 11 states of India meet to discuss the language problem. In how many ways can they seat themselves at a round table so that the Punjab and Maharashtra Chief Minister sit together?
(a) 725760 (b) 625760 (c) 925760 (d) 825760

Q.113. In how many ways can a cricket team of 11 players be selected out of 16 players, if the two particular player are always to be included?
(a) 2006 (b) 2004 (c) 2008 (d) 2002

Q.114. In how many ways can a cricket team of 11 players be selected of 16 players, if one particular player is to be excluded?
(a) 1565 (b) 1365 (c) 1165 (d) 1265

Q.115. In how many ways can a cricket team of 11 players be selected out of 16 players, if two particular players are to be included and one particular player is to be rejected?
(a) 715 (b) 615 (c) 915 (d) 515

Answers

PERMUTATION

1	a	11	b	21	c	31	a
2	c	12	b	22	b	32	b
3	a	13	c	23	c	33	a
4	b	14	b	24	a	34	b
5	c	15	b	25	c	35	c
6	a	16	c	26	a	36	b
7	b	17	a	27	c	37	c
8	a	18	b	28	b	38	a
9	c	19	a	29	a	39	b
10	a	20	c	30	c	40	c

CIRCULAR PERMUTATION

41	a
42	b
43	a
44	c
45	a
46	c
47	a

COMBINATION

48	b	58	a	68	a	78	b	88	c
49	a	59	c	69	c	79	a	89	b
50	c	60	b	70	d	80	b	90	a
51	b	61	c	71	a	81	a		
52	a	62	b	72	b	82	c		
53	b	63	a	73	b	83	a		
54	c	64	c	74	c	84	b		
55	b	65	a	75	a	85	c		
56	a	66	c	76	c	86	a		
57	b	67	b	77	a	87	b		

MISCELLANEOUS QUESTIONS 0f *Permutation & Combination*

91	a	101	b	111	c
92	b	102	c	112	a
93	a	103	b	113	d
94	a	104	a	114	b
95	b	105	b	115	a
96	b	106	a		
97	b	107	b		
98	b	108	d		
99	c	109	c		
100	c	110	d		

CHAPTER - 6 ARITHMETIC & GEOMETRIC PROGRESSION

ARITHMETIC PROGRESSION

Q.1. Find the 20th term of the series 1, 4, 7, 10 ……
(a) 58 (b) 45 (c) 51 (d) None

Q.2. Find the last term of the series, 0.6, 1.2, 1.8, ……upto 13 terms.
(a) 7.6 (b) 7.8 (c) 7.9 (d) None

Q.3. Which term of the series –1, –3, –5..... is – 39
(a) 20 (b) 15 (c) 21 (d) None

Q.4. If a = 5 and d = 3, then find the value of its 47th term
(a) 149 (b) 145 (c) 143 (d) None

Q.5. 25th term of an AP is 350, and if the first term is 62 then find the value of d
(a) 14 (b) 12 (c) 18 (d) None

Q.6. In an A.P. a = 7 and d = 3, T_n = 145, then find the number of terms –
(a) 49 (b) 57 (c) 47 (d) None

Q.7. In an A.P. a = 5, a_n = 270 & d = 6 then find the number of terms.
(a) 2 (b) 4 (c) 8 (d) None

Q.8. If Tn = n^2 –2n, then find the first three term of Arithmetic series.
(a) 3, 0 , -1 (b) 1, 0, 3 (c) -1, 0, 3 (d) None

Q.9. If the 3 terms of an A.P. are (8x + 4) , (6x-2) , (2x + 7) then find the value of x
(a) 2/15 (b) – 15/2 (c) 15/2 (d) None

Q.10. If T_5 of AP is 12, and the twelfth term is 5, then find value of its 17th term.
(a) 0 (b) 1 (c) 7 (d) None

Q.11. In an AP, T_7 = 10 and T_{10} = 7 then find the value of it's 25th term
(a) 8 (b) – 8 (c) 2 (d) None

Q.12. $\sum_{i=4}^{7} \sqrt{2i-1} = ?$
(a) $\sqrt{7}+\sqrt{9}+\sqrt{11}+\sqrt{13}+\ldots\ldots$ (b) $2\sqrt{7}+3+2\sqrt{11}+\sqrt{13}+\ldots\ldots$
(c) $\sqrt{7}+3+\sqrt{11}+\sqrt{13}$ (d) None of these

Q.13. Find the nth term of the series –1, 2, – 4, 8……
(a) $(-1)^n\ 2^{n-1}$ (b) -2^{n-1} (c) 2^{n-1} (d) 2^n

Q.14. 3 numbers are in AP and their sum is 15 and product is 80, then find the numbers.
(a) 2, 5, 8 (b) 8, 5, 2 (c) 5, 8, 2 (d) both A and B

Q.15. Find the number of terms between 74 and 25556 which are exactly divisible by 5.
(a) 5097 (b) 5079 (c) 5197 (d) None

Q.16. Find the sum of 100 terms of the series 9, 5, 1……..
(a) – 18900 (b) 18900 (c) 16900 (d) None

Q.17. Find the sum of the series $\frac{7}{2}$, 7 $\frac{21}{2}$, 14…….. upto 17 terms
(a) $553\ \frac{1}{2}$ (b) $535\ \frac{-1}{2}$ (c) $535\ \frac{1}{2}$ (d) None

Q.18. If $a = -4$ and T_n = 146 & if the sum of n terms of the series is 7171. Find n
(a) 101 (b) 105 (c) 108 (d) None

Q.19. If S_n is 155 then find the number of terms in the series $10,\ 9\ \frac{2}{3}\ ,\ 9\ \frac{1}{3}$, ……...
(a) 31 (b) 30 (c) $31 \text{ and } 30$ (d) None

Q.20. In an A.P. a = 14 & if S_5 and S_{10} are equal in magnitude but apposite in sign. Then find the third term of the series.
(a) $\frac{75}{11}$ (b) $\frac{70}{11}$ (c) $\frac{-70}{11}$ (d) None

Q.21. A person saves Rs.16500 in 10 years. He saved Rs.100 more every years in comparison to last year, then find the amount that he saved in the first year.
(a) 1200 (b) 1800 (c) 1600 (d) None

Q.22. A person pays Rs. 975 by monthly installment such that each installment is Rs. 5 less than the previous installment. If the first installment was of Rs. 100 then find the time by which entire amount will be paid.
(a) 15 (b) 26 (c) Both 15 & 26 (d) None of these

Q.23. A person pay Rs. 6240 in 30 installment, if each instalment is Rs. 10 more than the previous installment, than find the amount of first installment.
(a) 68 (b) 63 (c) 69 (d) None

Q.24. Find the sum of all the add numbers between 200 and 300.
(a) 12,500 (b) 12,600 (c) 12,800 (d) None

Q.25. Sum of all numbers between 500 & 1000 which are exactly divisible by 13 is :-
(a) 28540 (b) 28402 (c) 28405 (d) None

Q.26. Find the sum of all the natural numbers from 100 to 300 which are exactly Divisible by 4 *and* 5.
(a) 2250 (b) 2200 (c) 2280 (d) None

Q.27. Find the sum of all the natural number from 100 to 300 which are exactly Divisible by 4 *or* 5.
(a) 16,200 (b) 16,400 (c) 16,800 (d) None

Q.28. If unity is added to the sum of any number of terms of the AP 3, 5, 7, 9…. Then, the resulting number will be:-
(a) A Prime number (b) A Perfect square number
(c) A Perfect cube number (d) None of these

Q.29. The nth term of an AP is $\frac{3n-1}{6}$, then find the sum of its n term.
(a) $n(3n+1)$ (b) $\frac{n}{12}(3n+1)$ (c) $\frac{n}{2}(3n-1)$ (d) None

Q.30. The sum of n terms of an AP is $5n^2 + 2n$, then find its nth term.
(a) $3n-10$ (b) $5n+2$ (c) $10-3n$ (d) $10n-3$

Q.31. Find the Arithmetic Mean between 33 and 77.
(a) 45 (b) 55 (c) 57 (d) None

Q.32. Find two Arithmetic Mean between 5 and 23.
(a) 17 , 21 (b) 12 , 17 (c) 11 , 17 (d) None

Q.33. Find 4 Arithmetic Mean between -2 and 23
(a) 18, 3, 8, 13 (b) 3, 8, 13, 18 (c) 18, 13, 8, 3 (d) None

Q.34. The three numbers 2x, x + 10, 3x + 2 are in AP , find the value of x.
(a) 9 (b) 3 (c) 6 (d) None

MISCELLANEOUS QUESTIONS OF *ARITHMETIC PROGRESSION*

Q.01. In an A.P., terms are increased or decreased by a fixed number is called…
(a) Fixed difference (b) Sequential difference
(c) Common difference (d) None.

Q.02. If a be the first term and d be the common difference, then the nth term of A.P. is
(a) $t_n = a + nd$ (b) $t_n = a + (n-1)d$
(c) $t_n = a + (n+1)d$ (d) None

Q.03. Determine k so that 2/3, k and (5/8).k are the three consecutive terms of an A.P.
(a) 16/33 (b) 14/33 (c) 12/33 (d) 18/33

Q.04. Determine k, so that k + 2, 4k – 6 & 3k – 2 are the 3 consecutive terms of an A.P.
(a) 5 (b) 7 (c) 9 (d) 3

Q.05. The sum of n terms of an A. P. with first terms 'a' and common difference 'd' is:-
(a) $S_n = \frac{n}{2}[a + d]$ (b) $S_n = \frac{n}{2}[2a - (n+1)d]$
(c) $S_n = \frac{n}{2}[a + (n+1)d]$ (d) $S_n = \frac{n}{2}[2a + (n-1)d]$

Q.06. Sum of n terms of an A.P., whose first terms is 'a' and last term is L, is:-
(a) $S_n = \frac{n}{2}[a + L]$ (b) $S_n = a + (n-1)d$
(c) $S_n = \frac{n}{2}[2L - (n-1)d]$ (d) None

Q.07. If m times the mth term of an A.P. is equal to the n times of its nth term, then its $(m+n)^{th}$ term is equal to :-
(a) n/m (b) mn (c) m/n (d) Zero

Q.08. In an A.P. if the sum of m terms is equal to the sum of n terms, then the sum of (m + n) terms of the same A.P. is:-
(a) (m – n) (b) mn (c) 0 (d) – 2 (m + n)

Q.09. If the pth term of an A.P. is q and the qth term is p, then its nth terms is :-
(a) pq + n (b) pqn (c) p + q – n (d) p – q – n

Q.10. If the mth term of an A.P. is 1/n and n^{th} term is 1/m , then sum of mn terms is :-
(a) $\frac{1}{2}$ (mn + 1) (b) $\frac{1}{2}$ (mn – 1) (c) $\frac{1}{2}$ (m + n + 1) (d) $\frac{1}{2}$ (m – n + 1)

Q.11. Determine 25th term of an A.P.,whose 9th term is – 6 and common difference is 5/4
(a) 16 (b) 18 (c) 12 (d) 14

Q.12. Which term of the A.P. 5, 13, 21, ……. is 181
(a) 21st (b) 22nd (c) 23rd (d) 24th

Q.13. Find the nth term of the series $\frac{1}{n} + \frac{n+1}{n} + \frac{2n+1}{n}$ + ……
(a) $\frac{3+n^2+n}{n}$ (b) $\frac{1+n^2-n}{n}$ (c) $\frac{2+n^2-n}{n}$ (d) None of these.

Q.14. The ratio of the 7th term to the 3rd term of an A.P. is 12 : 5. Find the ratio of 13th term to the 4th term.
(a) 8 : 5 (b) 9 : 4 (c) 7 : 3 (d) 10 : 3

Q.15. If the 9th term of an A.P. is 99 and 99th term is 9, find 108th term
(a) 0 (b) 2 (c) 4 (d) 6

Q.16. Determine the sum of first 35 terms of an A.P., if $T_2 = 2$ and $T_7 = 22$
(a) 2,510 (b) 2,310 (c) 2,710 (d) 2,910

Q.17. The sum of a series in A.P. is 525. Its first term is 3 and last term is 39. Find the common difference.
(a) 3/2 (b) 3/3 (c) 2/3 (d) 1/3

Q.18. Sum the series 51 + 50 + 49 + …….. + 21.
(a) 1,116 (b) 1,112 (c) 1,128 (d) 1,124

Q.19. The sum of p term of an A.P. is $3p^2 + 4p$. Find the nth term
(a) 5n + 2 (b) 6n + 1 (c) 8n + 3 (d) 7n + 3.

Q.20. How many terms of the A.P. 1, 4, 7, …….. are needed to give the sum 715
(a) 33 (b) 22 (c) 24 (d) 27.

Q.21. Find the sum of the numbers of three digits, divisible by 7
(a) 70,334 (b) 70,338 (c) 70,336 (d) 70,332

Q.22. If the 9th term of an A.P. is zero, then $T_{29}/T_{19} = k$, where k =
(a) 1 (b) 2 (c) 3 (d) 4

Q.23. The fourth term of an A.P. is equal to 3 times of the first term and the seventh term exceeds twice the third term by 1. Find the common difference.
(a) 2 (b) 3 (c) 1 (d) 4

Q.24. How many numbers of two digits are divisible by 7
(a) 13 (b) 15 (c) 11 (d) 17

Q.25. A club consists of members whose ages are in A.P., the common difference being 3 months. The youngest member of the club is just 7 years old and the sum of the ages of all members is 250 years. The numbers of members in the club is:-
(a) 15 (b) 20 (c) 25 (d) 30

Q.26. The sum of the digits of a 3 digit number is 12. The digits are in A.P. If the digits are reversed, then the number is diminished by 396. The number is :-
(a) 6 4 2 (b) 1 3 8 (c) 2 3 7 (d) None of these

Q.27. The Seventh term of the sequence $\sqrt{3},\ \sqrt{12},\ \sqrt{27},$ ……. is:-
(a) $\sqrt{147}$ (b) 147 (c) $\sqrt{243}$ (d) 243

Q.28. 300 plants are planted in a regular pattern in rows in the shape of an isosceles triangle. The numbers in the successive rows diminished by 1 from the base to the apex. How many plants are there in the row which forms the base of the triangle?
(a) 22 (b) 20 (c) 24 (d) 36

Q.29. Which is the first positive term of the sequence $-111, -107, -103 -99$ ……
(a) 20 (b) 29 (c) 30 (d) 35

Q.30. Which is the first negative term of the sequence 252, 230, 208, ……
(a) 11 (b) 12 (c) 13 (d) 14

Q.31. The sum of all natural numbers between 200 & 400 exactly divisible by 3 and 7 is:-
(a) 3045 (b) 3405 (c) 3540 (d) 3504

Q.32. The sum of a series in AP is 72. The first term being 17 and common difference -2. The number of terms is:-
(a) 6 (b) 12 (c) 6 or 12 (d) None of these

Q.33. If the sum of n terms of an A.P. is $3n^2 + 5n$, then which of its terms is 164
(a) 26th (b) 25th (c) 28th (d) 27th

Q.34. If the 6th term of an AP is 121, then the sum of its first 11 terms is:-
(a) 1210 (b) 1331 (c) 1330 (d) 1333

Q.35. If the sum of n terms of an A.P. is $(3n^2 - n)$ and its common difference is 6,then its first term is :-
(a) 3 (b) 2 (c) 4 (d) 1

Q.36. How many terms of the A.P. 80, 75, 70 …… are required to make their sum equal to zero ?
(a) 25 (b) 30 (c) 32 (d) 33

GEOMETRIC PROGRESSION

Q.01. Find the 17th term of the series, 6, 12, 24,……..
(a) 393216 (b) 393217 (c) 393219 (d) None

Q.02. Find the 12th term of the series, $-128, 64, -32$,……...
(a) $\frac{-1}{16}$ (b) $\frac{1}{16}$ (c) $\frac{2}{16}$ (d) None

Q.03. Find the 4th term of the series 0.04, 0.2, 1,……..
(a) 4 (b) 3 (c) 5 (d) None

Q.04. Find the last term of the series 1, 2, 4……up to 10 term.
(a) 518 (b) 512 (c) 521 (d) None

Q.05. Find the last term of the series, 1, −3, 9, −27……..up to 7 terms.
(a) 719 (b) 729 (c) 739 (d) None

Q.06. Find the last term of the series $x^2,\ x,\ 1,$ …….. up to 31 terms.
(a) x^{-28} (b) x^{28} (c) x^{-31} (d) None

Q.07. Three numbers are in GP. Their sum is 39 & product is 729 , find the numbers.
(a) 3, 27, 9 (b) 27, 9, 3 (c) 3, 9, 27 (d) Both (b)&(c)

Q.08. Three numbers are in GP and their product is 27/8, then find the middle term.
(a) 3/2 (b) 2/3 (c) −3/2 (d) None

Q.09. Find the sum of ∞ terms of GP 1, $\frac{-1}{3}$, $\frac{1}{9}$, $\frac{-1}{27}$, ……..
(a) 3/4 (b) 3/2 (c) −3/4 (d) None

Q.10. Find the sum of infinite terms of GP. 14, −2, $\frac{2}{7}$, ……..
(a) 12.26 (b) 12.29 (c) 12.25 (d) None

Q.11. Find the sum of 7 terms of the series −2, 6, −18……
(a) −1094 (b) 1094 (c) 1099 (d) None

Q.12. Find the sum of 8 terms of the series, 243, 81, 27,……
(a) 363.44 (b) 364.44 (c) 365.44 (d) None

Q.13. Find the sum of 18 terms of the series $\frac{1}{\sqrt{3}}$, 1, $\frac{3}{\sqrt{3}}$ …….
(a) $\frac{9841\ (\sqrt{3}-1)}{\sqrt{3}}$ (b) $\frac{9841\ (\sqrt{3}+1)}{\sqrt{2}}$ (c) $\frac{9841\ (\sqrt{3}+1)}{\sqrt{3}}$ (d) None

Q.14. If you save 1 paisa today, 2 paise tomorrow, 4 paise in the succeeding day & so on, then find your total saving in the two weeks.
(a) Rs. 163.83 (b) Rs. 164.84 (c) Rs. 169.84 (d) None

Q.15. Sum of first 20 terms of a G.P. is 244 times of the sum of its first 10 term then find the common ratio.
(a) $-\sqrt{3}$ (b) $\pm\sqrt{3}$ (c) $\sqrt{3}$ (d) None

Q.16. Three numbers are in GP, their product is 729, and the sum of their squares is 819. Find the numbers.
(a) 3, 9, 27 (b) 27, 9, 3 (c) 9, 27, 3 (d) Both (a) & (b)

Q.17. 4 + 44 + 444 + 4444 + ……. up to n terms =
(a) $\frac{4}{9}\left\{\frac{10}{9}\ [10^n-1]-n\right\}$ (b) $\frac{10}{9}\{\ [10^n-1]-n\ \}$
(c) $\frac{4}{9}[\ 10^n-1\ ]-n$ (d) None of these

Q.18. 0.1 + 0.11 + 0.111 + ……..up to n terms =

(a) $\frac{1}{9}\{n - [1-(0.1)^n]\}$ (b) $\frac{1}{9}\left\{n - 1 - \frac{(0.1)^n}{9}\right\}$

(c) $n - 1 - \frac{(0.1)^n}{9}$ (d) None of these

Q.19. If p, q & r are in AP and x, y, z are in GP, then find the value of $x^{q-r} \times y^{r-p} \times z^{p-q}$

(a) – 1 (b) 1 (c) 0 (d) None

Q.20. Find the Geometric Mean of 5 and 13.

(a) 8.06225 (b) 8.06226 (c) 8.06228 (d) None

Q.21. Find the 4 Geometric Mean between 4 and 972.

(a) 324, 108, 36, 12 (b) 12, 36, 108, 324

(c) Both (a) & (b) (d) None

Q.22. Find the sum of the series $6^2 + 7^2 + 8^2$……$+ 11^2$

(a) 471 (b) 457 (c) 451 (d) None

Q.23. In a GP, $T_4 = x$, $T_{10} = y$, $T_{16} = z$

(a) $x^2 = yz$ (b) $y^2 = xz$ (c) $z^2 = xy$ (d) None of these

Q.24. If x, y, z are in GP, then

(a) $y(z^2 + x^2) = x(z^2 + y^2)$ (b) $2y = x + z$

(c) $y^2 = xz$ (d) None of these

Q.25. The first term of a GP is 1 and its common ratio is 1/2, if the sum of its n terms is $1\frac{127}{128}$ then find the value of n.

(a) 6 (b) 8 (c) 4 (d) None

Q.26. x, 8, y are in GP, and x, y, -8 are in AP , then find the value of x and y

(a) $-8, -8$ (b) $8, 8$ (c) $16, 4$ (d) Both (a) & (c)

Q.27. x, y, z are in AP and x, y, $(z+1)$ are in GP, then

(a) $(x-z)^2 = 4x$ (b) $z^2\ (x-y)$ (c) $z = x - y$(d) None of these

Q.28. If x, y, z are in GP and $x^p = y^q = z^r$ then $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}$ will be in

(a) AP (b) GP (c) HP (d) None of these

Q.29. If a, b, c are in GP then log a, log b, log c will be in

(a) GP (b) AP (c) HP (d) None of these

Q.30. If a, b, c are in GP then, $\log_2 a$, $\log_2 b$, $\log_2 c$ will be in

(a) GP (b) HP (c) AP (d) None of these

Q.31. The sum of the four numbers which are in GP is 60 and the mean of first and last term is 18, find the numbers.

(a) 4, 8, 16, 32 (b) 32, 16, 8, 4 (c) 16, 8, 4, 2 (d) both A and B

Q.32. Three numbers are in AP, and their sum is 15. If 8, 6, and 4 are added to them respectively then they will form a GP. find the numbers.
(a) 2, 6, 7 (b) 3, 5, 7 (c) 5, 3, 7 (d) None of these

Q.33. If AM of the two numbers is 40 and their GM is 24. Then find the numbers.
(a) 72, 8 (b) 70, 10 (c) 60, 20 (d) None of these

Q.34. Find the sum of the series $1+\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots+\frac{1}{3^{n-1}}+\cdots\infty$
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{5}{2}$ (d) None

Q.35. Three numbers are in GP and their sum is 70. If the two extreme numbers are multiplied each by 4 and the middle numbers is multiplied by 5 then the resulting number will form an AP. Find the Numbers.
(a) 10, 20, 40 (b) 40, 20, 10 (c) 12, 18, 40 (d) Both A and B

Q.36. If a, b, c are in AP, a, x, b are in GP, b, y, c are in GP, x^2, b^2, y^2 will be in = ?
(a) AP (b) GP (c) HP (d) None of these

Q.37. If a, b, c are in AP, then $(b+c,\ c+a,\ a+b)$ will be in
(a) AP (b) GP (c) HP (d) None of these

Q.38. If a^2, b^2, c^2 are in AP, then (b + c), (c + a), (a + b) will be in
(a) AP (b) GP (c) HP (d) None of these

Q.39. If a^2, b^2, c^2, are in AP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$, will be in
(a) AP (b) GP (c) HP (d) None of these

Q.40. 1.2 + 2.3 + 3.4 +……up to n term.
(a) $S_n = \frac{n}{3}(n+1).(n+2)$ (b) $S_n = \frac{n}{2}(n+1).(n+2)$
(c) $S_n = \frac{n}{3}(n+1).(n-2)$ (d) $S_n = \frac{n}{3}(n-3).(n+5)$

Q.41. Find the sum of n terms of series 1.2.3 + 2.3.4 + 3.4.5 ……
(a) $\frac{n}{4}(n+1)(n+2)(n+3)$ (b) $\frac{n}{3}(n+1)(n+2)(n+3)$
(c) $\frac{n}{2}(n+1)(n+2)(n+3)$ (d) None of these

Q.42. $1 + a + a^2 + a^3 + \ldots\ldots\infty = x$, $1 + b + b^2 + b^3 + \ldots\ldots\infty = y$
$1 + ab + a^2b^2 + a^3b^3 + \ldots\ldots\infty = ?$
(a) xy (b) $\frac{xy}{x+y-1}$ (c) $\frac{xy}{x-y-1}$ (d) $\frac{xy}{x+y+1}$

Q.43. Find the sum of ∞ terms of the series $\frac{1}{2}+\frac{1}{3^2}+\frac{1}{2^3}+\frac{1}{3^4}+\frac{1}{2^5}+\frac{1}{3^6}+\ldots\ldots\ldots$
(a) $\frac{18}{24}$ (b) $\frac{19}{24}$ (c) $\frac{15}{24}$ (d) None

Q.44. The Sum of the first two terms of terms of GP is $\frac{5}{3}$ and the sum of its infinite terms is 3. Find the value of r.
(a) $\frac{-2}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) None

MISCELLANEOUS QUESTIONS OF *GEOMETRIC PROGRESSION*

Q.01. How many terms of the GP 3 + 9 + 27 + 81 ……. Will add up to 363.
(a) 5 (b) 6 (c) 8 (d) 7

Q.02. The next term of the sequence $-1, \frac{1}{4}, \frac{-1}{16}, \frac{1}{64}$
(a) $\frac{1}{128}$ (b) $\frac{-1}{128}$ (c) $\frac{1}{256}$ (d) $\frac{-1}{256}$

Q.03. The third term of a GP is 7. The product of the first five terms is :-
(a) 7^5 (b) 7^3 (c) 7^4 (d) 7^6

Q.04. The 4th term of a G.P. is x, the 10th term is y and the 16th term is z, then xz =
(a) x^2 (b) y^2 (c) z^2 (d) None of these

Q.05. If p^{th}, q^{th}, r^{th} terms of a G.P. are a, b, c respectively, then (q – r) log a + (r – p) log b + (p – q) log c =
(a) 1 (b) 2 (c) 0 (d) 4

Q.06. The amount of a certain sum of money at compound interest of r % per year for n years, will form a:-
(a) H.P. (b) A.P. (c) G.P. (d) None of these

Q.07. It the continued product of three numbers in G.P. is 27 and the sum of their products in pairs is 39, the numbers are:-
(a) 1, 3, 9 (b) 9, 3, 1 (c) Both (a)&(b) (d) None

Q.08. four number are in G.P. in which the third term is greater than the first by 9 and the second term is greater than the fourth by 18, then the common ratio is :-
(a) 2 (b) −2 (c) 3 (d) None

Q.09. If a, G, b are in G.P. then
(a) 2G = ab (b) G^2 = ab (c) G = 1/2 ab (d) G = 1/2 (a + b)

Q.10. Which term of the geometric sequence $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots\ldots\ldots\ldots\ldots$ is $\frac{1}{19683}$?
(a) 9 (b) 7 (c) 11 (d) 13

Q.11. Find the 10th term of the geometric series 5 + 25 + 125 +…….
(a) 5^{10} (b) 5^9 (c) 5^{11} (d) 5^8

Q.12. Write down the 20th term of the G.P. 1, −1, 1, −1, ……
(a) 1 (b) −1 (c) + 1 (d) None of these

Q.13. Write down the 5th term of the series $\frac{1}{4} - \frac{1}{2} + 1 \ldots\ldots\ldots\ldots$
(a) 6 (b) 8 (c) 4 (d) 10

Q.14. For any GP, a = 729, and T_7 = 64, then S_7 = ?
(a) 2259 (b) 2059 (c) 3059 (d) 2459

Q.15. For what value of x, the numbers $\frac{-2}{7}$, x, $\frac{-7}{2}$ are in G.P.?
(a) 1, −2 (b) 1, −3 (c) 1, −5 (d) 1, −1

Q.16. Evaluate $\sum_{j=1}^{11} (2 + 3^j)$
(a) $22 + \frac{3}{2}(3^{11} - 1)$ (b) $11 + \frac{3}{2}(3^{11} - 1)$
(c) $22 + \frac{3}{2}(3^{10} - 1)$ (d) None of these

Q.17. The common ratio of a G.P. is −4/5 and the sum to infinity is 80/9. Find the first term of the G.P.
(a) 14 (b) 16 (c) 14 (d) 10

Q.18. Sum the series to infinity $\frac{3}{4} - \frac{5}{4^2} + \frac{3}{4^3} - \frac{5}{4^4} + \frac{3}{4^5} - \frac{5}{4^6} + \ldots\ldots\ldots$
(a) 8/15 (b) 7/17 (c) 7/15 (d) 8/17

Q.19. A progression 2, 8, 32 ……..……..is given the 10th number will be
(a) 2^{18} (b) 2^{19} (c) 2^{20} (d) None of these

Q.20. Find the sum of series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} +$ ……..…… ∞
(a) 3 (b) 4 (c) 2 (d) 5

Q.21. 6, 3, $\frac{3}{2}$, X, $\frac{3}{8}$ …….. is a given progression, the value of X is :-
(a) 3/4 (b) 3/10 (c) 3/5 (d) None of these

Q.22. The 19th term of the series −4, 2, 8, …….. is ?
(a) 104 (b) 112 (c) −40 (d) 140

Q.23. The nth term of the series 3, $\sqrt{3}$, 1, ……..... is 1/243 , then n is :-
(a) 12 (b) 13 (c) 14 (d) 15

Q.24. Which term is 729 for the G.P. – 1/27, 1/9, – 1/3 ……..
(a) 10 (b) 20 (c) 30 (d) None of these

Q.25. The value of $9^{1/3} \times 9^{1/9} \times 9^{1/27} \times$ …… ∞ is :-
(a) 9 (b) 1 (c) 3 (d) None of these

Q.26. For any GP. $T_4 = 24$ and $T_8 = 384$ then r = ?
(a) 2 (b) 6 (c) 4 (d) 8

Q.27. $1 - \frac{1}{3} + \frac{1}{3^2} - \frac{1}{3^3} +$ …… ∞ = ?
(a) 1/4 (b) 1/6 (c) 3/4 (d) 4/9

Q.28. If a, b, c are in A.P. then 3^a, 3^b, 3^c are in
(a) A.P. (b) G.P. (c) H.P. (d) None

Q.29. If the geometric mean of 3 observation 40, 50 and x is 10, then the value of x is :-
(a) 1/2 (b) 1 (c) 2 (d) 4

Q.30. The value of $0.2\overline{34}$ is :-
(a) 232/990 (b) 232/9990 (c) 232/9900 (d) 232/9909

Q.31. The A.M. of two numbers is 34 and G.M. is 16 , the numbers are
(a) 2 and 64 (b) 64 and 3 (c) 64 and 4 (d) None of these

Q.32. 8 + 88 + 888 + ……sum upto n terms
(a) $\frac{80}{81}(10^n - 1) - \frac{8n}{9}$ (b) $\frac{10}{81}(10^n - 1)$
(c) $\frac{80}{81}(10^n - 1) + \frac{8n}{9}$ (d) None of these

Q.33. The value of $15^2 + 16^2 + 17^2 + \ldots\ldots\ldots\ldots + 70^2$ is
(a) 117580 (b) 115780 (c) 118750 (d) None of these

Q.34. If a, b, c are in A.P. and (b – a), (c – b), (a) are in G.P, then a : b : c =
(a) 2 : 3 : 4 (b) 1 : 2 : 4 (c) 1 : 2 : 3 (4) 1 : 3 : 5

Q.35. The sum of n terms of the series 1 + (1 + 3) + (1 + 3 + 5) + ……. is
(a) $\frac{n\,(n+1)(2n+1)}{6}$ (b) $\frac{n(n+1)(n+2)}{6}$ (c) $\frac{n\,(n+1)(2n+2)}{3}$ (d) None of these

Q.36. If $\sum n$ = 55, then $\sum n^2$ is equal to :-
(a) 3025 (b) 506 (c) 385 (d) 1015

Q.37. Let x be the A.M. and y, z be the two G.Ms between the two positive numbers, Then $\frac{y^3 + z^3}{xyz}$ is equal to:-
(a) 2 (b) 1 (c) 1/2 (d) 1/3

Q.38. The next term of the sequence $\frac{3}{2}, \frac{5}{4}, \frac{9}{8}, \frac{17}{16}$ …….. is :-
(a) $\frac{33}{32}$ (b) $\frac{29}{32}$ (c) $\frac{43}{32}$ (d) $\frac{35}{32}$

Q.39. The sum of the series $11^2 + 12^2 + 13^2 + 14^2 + \ldots\ldots\ldots\ldots 20^2$ is :-
(a) 2485 (b) 4285 (c) 8524 (d) 5428

Q.40. The sum of the series $11^3 + 12^3 + 13^3 + 14^3 + \ldots\ldots\ldots\ldots 20^3$ is :-
(a) 47015 (b) 47105 (c) 40175 (d) 41075

Q.41. If x, y, z are in A.P. and x, y – x, z – x are in G.P., Then x : y : z is :-
(a) 1 : 3 : 5 (b) 1 : 2 : 3 (c) 1 : 3 : 7 (d) 1 : 4 : 6

Q.42. The sum of three numbers in a geometric progression is 28. When 7, 2 and 1 are subtracted from the first, second and the third numbers respectively, then the resulting numbers are in arithmetic progression. What is the sum of squares of the original three numbers?
(a) 336 (b) 510 (c) 456 (d) 400

Q.43. Relation between A.M. and G.M. is :-
(a) A.M. $\leq$ 4/5 G.M. (b) A.M. $\geq$ G.M.
(c) A.M. = 3/4 G.M. (d) A.M. < G.M.

Answer

ARITHMETIC PROGRESSION

1	a	11	b	21	a	31	b
2	b	12	c	22	a	32	c
3	a	13	c	23	b	33	b
4	c	14	d	24	a	34	c
5	b	15	a	25	c		
6	c	16	a	26	b		
7	d	17	c	27	a		
8	c	18	a	28	b		
9	c	19	c	29	b		
10	a	20	b	30	d		

MISCELLANEOUS QUESTIONS OF *ARITHMETIC PROGRESSION*

1	c	11	d	21	c	31	a
2	b	12	c	22	b	32	c
3	a	13	b	23	a	33	d
4	d	14	d	24	a	34	b
5	d	15	a	25	c	35	b
6	a	16	b	26	a	36	d
7	d	17	a	27	a		
8	c	18	a	28	c		
9	c	19	b	29	b		
10	a	20	b	30	c		

GEOMETRIC PROGRESSION

1	a	11	a	21	b	31	d	41	a
2	b	12	b	22	c	32	b	42	b
3	c	13	c	23	b	33	a	43	b
4	b	14	a	24	c	34	b	44	b
5	b	15	b	25	b	35	d		
6	a	16	d	26	d	36	a		
7	d	17	a	27	a	37	a		
8	a	18	d	28	a	38	c		
9	a	19	b	29	b	39	a		
10	c	20	a	30	c	40	a		

MISCELLANEOUS QUESTIONS OF *GEOMETRIC PROGRESSION*

1	a	11	a	21	a	31	c	41	a
2	b	12	b	22	a	32	a	42	a
3	a	13	c	23	b	33	b	43	b
4	b	14	b	24	a	34	c		
5	c	15	d	25	c	35	a		
6	c	16	a	26	a	36	c		
7	c	17	b	27	c	37	a		
8	b	18	c	28	b	38	a		
9	b	19	b	29	a	39	a		
10	a	20	c	30	a	40	d		

7.7 CONCEPT OF LIMIT

I) We consider a function f(x) = 2 x. If x is a number approaching to the number 2 then f(x) is a number approaching to the value 2x2= 4

The following table shows f(x) for different values of x approaching 2

x	f(x)
1.90	3.8
1.99	3.98
1.999	3.998
1.9999	3.9998
2	4

Here x approaches 2 from values of x<2 and for x being very close to 2 f(x) is very close 4. This situation is defined as left-hand limit of f(x) as x approaches 2 and is written as $\lim f(x) = 4,\ x \to 2-$

x	f(x)
2.0001	4.0002
2.001	4.002
2.01	1.02
2.0	4

Here x approaches 2 from values of x greater than 2 and for x being very close 2f(x) is very close to 4. This situation is defined as right-hand limit of f(x) as x approaches 2 and is written as lim f(x)= 4 as $x \to {}_{2+}$

So we write

$$\lim_{x\to 2-} f(x) = \lim_{x\to 2-+} f(x) = 4$$

Thus $\lim_{x\to a} f(x)$ is said to exist when both left-hand and right-hand limits exists and they are equal.

We write as

$$\lim_{x\to a-} f(x) = \lim_{x\to a-+} f(x) = \lim_{x\to a} f(x)$$

Thus $\lim_{x\to a} f(x)$ is said to exist when both left-hand and right-hand limits exists and they are equal.

We write as

$$\lim_{x\to a} f(x) = \lim_{x\to a+} f(x) = \lim_{x\to a} f(x)$$

Thus, if $\lim_{h\to o} f(a+h) = \lim_{h\to o} f(a-h), \quad (h > o)$

then $\lim_{x \to a} f(x)$ exists

We now consider a function defined by

$$fx = \begin{cases} 2x-2 & \text{for } x < 0 \\ 1 & \text{for } x = 0 \\ 2x+2 & \text{for } x > 0 \end{cases}$$

We calculate limit of f(x) as x tend to zero. At x =0 , f(x) =1 (given). If x tends to zero from left-hand side for the value of x<0, f(x) is approaching (2x0) –2 + –2 which is defined as left-hand limit of f(x) as $x \to 0$. We can write this as

$$\lim_{x \to 0-} = -2$$

Similarly if x approaches zero from right-hand side for values of x>0, f(x) is approaching 2 x 0 + 2 = 2. We can write this as $\lim_{x \to 0+} f(x) = -2$.

In this case both left-hand and right-hand exist, but they are not equal. So we may conclude that $\lim_{x \to 0} f(x)$ does not exist.

7.8. USEFUL RULES (THEOREMS) ON LIMITS

Let $\lim_{x \to a} f(x) = l$ and $\lim_{x \to a} g(x) = m$

where l and m are finite quantities

i) $\lim_{x \to a} \{f(x) + g(x)\} = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = l + m$

That is limit of the sum of the difference of two functions is equal to the sum of their limits

ii) $\lim_{x \to a} \{f(x) - g(x)\} = \lim_{x \to a} f(x) - \lim_{x \to a} g(x) = l - m$

That is limit of the difference of two functions is equal to difference of their limits.

iii) $\lim_{x \to a} \{f(x) . g(x)\} = \lim_{x \to a} f(x) . \lim_{x \to a} g(x) = lm$

That is limit of the product of two functions is equal to the product of their limits

iv) $\lim_{x \to a} \{f(x) / g(x)\} = \left\{\lim_{x \to a} f(x)\right\} / \left\{\lim_{x \to a} g(x)\right\} = l / m$ if $m \neq 0$

That is limit of the quotient of two functions is equal to the quotient of their limits.

v) $\lim_{x \to a} c = c$ where c is constant

That is limit of a constant is the constant

vi) $\lim_{x \to a} cf(x) = c \lim_{x \to a} f(x)$

vii) $\lim_{x \to a} F\{f(x)\} = F\{\lim_{x \to} F(x)\} = F(l)$

viii) $\lim_{x\to 0+}\frac{1}{x}=\lim_{h\to 0}\frac{1}{h}\to+\infty$ (h>0)

$\lim_{x\to 0+}\frac{1}{x}=\lim_{h\to 0}\frac{1}{-h}\to-\infty$ (h>0)

∞ is a very-every large number called infinity

Thus $\lim_{x\to 0+}\frac{1}{x}$ does not exist.

Example 1: Evaluate (i) $\lim_{x\to 2}(3x+9)$; (ii) $\lim_{x\to 5}\frac{1}{x-1}$ (iii) $\lim_{x\to a}\frac{1}{x-a}$

Solution:

(i) $\lim_{x\to 2}(3x+9)=3.2+9=(6+9)=15$

(ii) $\lim_{x\to 5}\frac{1}{x-1}=\frac{1}{5-1}=\frac{1}{4}$

(iii) $\lim_{x\to a}\frac{1}{x-a}$ does not exist, since $\lim_{x\to a+}\frac{1}{x-a}\to+\infty$ and $\lim_{x\to a-}\frac{1}{x-a}\to-\infty$

Example-2 Evaluate $\lim_{x\to 2}\frac{x^2-5x+6}{x-2}$

Solution: At x = 2 the function becomes undefined as 2-2 =0 and division by zero is not mathematically defined.

So $\lim_{x\to 2}\{x^2-5x+6/(x-2)\}=\lim_{x\to 2}\{(x-3)/(x-2)\}=\lim_{x\to 2}(x-=3)\ (\because x-2\neq 0)$

= 2-3 = -1

Example-2 Evaluate $\lim_{x\to 2}\frac{x^2+2x-1}{\sqrt{x^2+2}}$

$$\lim_{x\to 2}\frac{x^2+2x-1}{\sqrt{x^2+2}}=\frac{\lim_{x\to 2}(x^2+2x-1)}{\lim_{x\to 2}\sqrt{x^2+2}}=\frac{\lim_{x\to 2}(x^2+\lim_{x\to 2}2x-1)}{\sqrt{\lim_{x\to 2}x^2+2}}$$

$$\frac{(2)^2+2x\ 2-1}{\sqrt{(2)^2+2}}=\frac{7}{\sqrt{6}}$$

SOME IMPORTANT LIMITS

We now state some important limits

a) $\lim_{x \to 0} \frac{(e^x - 1)}{x} = 1$

b) $\lim_{x \to 0} \frac{a^x - 1}{x} = \log_e \quad a(a > 0)$

c) $\lim_{x \to 0} \frac{\log(1+x)}{x} = 1$

d) $\lim_{x \to x} \left(1 + \frac{1}{x}\right)^x = e$ or $\lim_{x \to 0} \frac{(1+x)^{\frac{1}{x}}}{x} = e$

e) $\lim_{x \to 0} \frac{x^n - a^n}{x} = na^{n-1}$

f) $\lim_{x \to 0} \frac{(1+x)^n - 1}{x} = n$

(A) The number e called exponential number is given by e = 2.718281828_____=2.7183. This number e is one of the useful constants in mathematics.

(B) In calculus all logarithms are taken with respect to base 'that is $\log x = \log_e x$.

ILLUSTRATIVE EXAMPLES

Example 1: Evaluate : $\lim_{x \to 3} \frac{x^2 - 6x+9}{x-3}$, where $f(x) = \frac{x^2 - 6x+9}{x-3}$. Aso find f (3)

Solution: At x = 3 the function is undefined as division by zero is meaningless. While taking the limit as $x \to 3$, x cannot be exactly equal to 3 i.e. x-3 $\neq 0$ and consequently division by x-3 is permissible.

Now $\lim_{x \to 3} \frac{x^2 - 6x+9}{x-3} = \lim_{x \to 3} \frac{(x-3)^2}{x-3} = \lim_{x-3} (x-3) = 3-3 = 0$ $\quad f(3) \frac{0}{0}$ is undefined

The reader may compute the left-hand and the right-hand limits as an exercise.

Example 2: A function is defined as follows:

$$fx = \begin{cases} -3x \text{ when } x < 0 \\ 2x \text{ when } x > 0 \end{cases}$$

Test the existence of $\lim_{x \to 0} f(x)$.

Solution: For x approaching 0 from the left $x < 0$.

Left-hand limit $= \lim_{x\to 0-} f(x) = \lim_{x\to 0-} (-3x) = 0$

When x approaches 0 from the right $x > 0$

Right-hand limit $= \lim_{x\to 0+} f(x) = \lim_{x\to 0+} 2x = 0$

Since L.H. limit = R.H. Limit, the limit exists. Thus, $\lim_{x\to 0} f(x) = 0$.

Example 3: $\lim_{x\to 3} \dfrac{x^2+4x+3}{x^2+6x+9}$.

Solution: $\dfrac{x^2+4x+3}{x^2+6x+9} = \dfrac{x^2+3x+x+3}{(x+3)^2} = \dfrac{x(x+3)+1(x+3)}{(x+3)^2} = \dfrac{(x+3)(x+1)}{(x+3)^2} = \dfrac{x+1}{x+3}$

$\therefore \lim_{x\to 3} \dfrac{x^2+4x+3}{x^2+6x+9} = \lim_{x\to 3} \dfrac{x+1}{x+3} = \dfrac{4}{3} = \dfrac{2}{3}$.

Example 4: Find the following limits:

(i) $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$ (ii) $\lim_{h\to 0} \dfrac{\sqrt{x+h}-\sqrt{x}}{h}$ if $h > 0$.

Solution:

(i) $\dfrac{\sqrt{x}-3}{x-9} = \dfrac{\sqrt{x}-3}{(\sqrt{x}+3)(\sqrt{x}-3)} = \dfrac{1}{\sqrt{x}+3}$. $\therefore \lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9} = \lim_{x\to 9} \dfrac{1}{\sqrt{x}+3} = \dfrac{1}{6}$.

(ii) $\dfrac{\sqrt{x+h}-\sqrt{x}}{h} = \dfrac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})} = \dfrac{1}{\sqrt{x+h}+\sqrt{x}}$ $\therefore \lim_{h\to 0} \dfrac{\sqrt{x+h}-\sqrt{x}}{h} = \lim_{h\to 0} \dfrac{1}{\sqrt{x+h}+\sqrt{x}}$

$= \dfrac{1}{\lim_{h\to 0}\sqrt{x+h} + \lim_{h\to 0}\sqrt{x}} = \dfrac{1}{\sqrt{x}+\sqrt{x}} = \dfrac{1}{2\sqrt{x}}$.

Example 5: Find $\lim_{x\to 0} \dfrac{3x+|x|}{7x-5|x|}$.

Solution: Right-hand limit $= \lim_{x\to 0+} \dfrac{3x+|x|}{7x-5|x|} = \lim_{x\to 0+} \dfrac{3x+x}{7x-5x} = \lim_{x\to 0+} 2 = 2$

Left-hand limit $= \lim_{x\to 0-} \dfrac{3x+|x|}{7x-5|x|} = \lim_{x\to 0} \dfrac{3x-(x)}{7x-5(-x)} = \lim_{x\to 0-} \dfrac{1}{6} = \dfrac{1}{6}$.

Since Right-hand limit $\neq$ Left-hand limit the limit does not exist.

Example 6: Evaluate $= \lim_{x \to 0} \frac{e^x - e^{-x}}{x}$

Solution: $= \lim_{x \to 0} \frac{e^x - e^{-x}}{x} == \lim_{x \to 0} \frac{(e^x - 1) - (e^{-x} - 1)}{x} = \lim_{x \to 0} \frac{e^x - 1}{x} - = \lim_{x \to 0} \frac{e^x - 1}{x} = 1 - 1 = 0$

Example 7: Find $\lim_{x \to \hat{a}} \left(1 + \frac{9}{x}\right)^x$. (Form 1)

Solution: It may be noted that $\frac{x}{9}$ approaches α as x approaches ∞. i.e. $\lim_{x \to \infty} \frac{x}{9} \to \infty$

$$\lim_{x \to \infty} \left(1 + \frac{9}{x}\right)^x = \lim_{x/9 \to \alpha} \left\{ \left(1 + \frac{1}{\frac{x}{9}} \right)^{x/9} \right\}^9$$

Substitution x / 9 = z the above expression takes the form $\lim_{z \to \alpha} \left\{ \left(1 + \frac{1}{z}\right)^z \right\}^9$

$$= \left\{ \lim_{z \to \infty} \left(1 + \frac{1}{z}\right)^z \right\}^9 = e^9.$$

Example 8: Evaluate: $\lim_{z \to \infty} \frac{2x + 1}{x^3 + 1}$. $\left[\text{Form} \frac{\infty}{\infty}\right]$

Solution: As x approaches $\infty, 2x + 1$ and $x^3 + 1$ both approach ∞ and therefore the given function takes the form $\frac{\infty}{\infty}$ which is determinate. Therefore, instead of evaluating directly let us try for suitable algebraic transformation so that the indeterminate for is avoided.

$$\lim_{x \to \infty} \frac{x^{\frac{2}{2}} + x^{\frac{1}{3}}}{1 + x^{\frac{1}{3}}} = \frac{\lim_{x \to \infty} \left(x^{\frac{2}{2}} + x^{\frac{1}{3}} \right)}{\lim_{x \to \infty} \left(1 + x^{\frac{1}{3}} \right)} = \frac{\lim_{x \to \infty} x^{\frac{2}{2}} \lim_{x \to \infty} x^{\frac{1}{3}}}{\lim_{x \to \infty} 1 + \lim_{x \to \infty} x^{\frac{1}{3}}} = \frac{0 + 0}{1 + 0} = \frac{0}{1} = 0$$

Example 9: Find $\lim_{x \to \infty} \frac{1^2 + 2^2 + 3^2 + \ldots\ldots\ldots\ldots + x^2}{x^3}$

$$\lim_{x \to \infty} \frac{[x(x+)(2x+1)]}{6x^3} = \frac{1}{6} \lim_{x \to \infty} \left\{ \left(1 + \frac{1}{x}\right) \left(2 + \frac{1}{x}\right) \right\}$$

$$= \frac{1}{6} \text{ x } 1 \text{ x } 2 = \frac{1}{3}.$$

7.9 CONTINUITY:

By the term continuous we mean some thing which goes on without interruption and without abrupt changes. Here in mathematics the term continuous carries the same meaning. Thus, we define continuity of a function in the following way.

A function f(x) is said to be continuous at x= a if and only if

(i) f(x) is defined x= a

(ii) $\lim_{x\mapsto a-} f(x) = \lim_{x\mapsto a+} f(x)$

(iii) $\lim_{x\mapsto a} f(x) = f(a)$

In the second condition both left-hand and right-hand limits are equal. In the third condition limiting value of the function must be equal to its function value at x= a

Useful Information:

(i) The sum, difference and product of two continuous functions is a continuous function. This property holds good for any finite number of functions.

(ii) The quotient of two continuous functions is continuous function provided the denominator is not equal to zero.

Example:1 f(x) = [

$= \frac{3}{2} - x$ when $\frac{1}{2} < x < 1$

$= \frac{1}{2}$ when $x = \frac{1}{2}$

Discuss the continuity of f(x) at x= $\frac{1}{2}$

$$\lim_{x\mapsto \frac{1}{2}-} f(x) = \lim_{x\to \frac{1}{2}+}\left(\frac{1}{2} - x\right) = \frac{1}{2} - \frac{1}{2} = 0$$

$$\lim_{x\mapsto \frac{1}{2}+} f(x) = \lim_{x\to \frac{1}{2}+}\left(\frac{3}{2} - x\right) = \frac{3}{2} - \frac{1}{2} = 1$$

Solution: Since LHL ≠ RHL

$\lim_{x\mapsto \frac{1}{2}} f(x)$ does not exist

Moreover f(1/2)

Hence f(x) is not continuous at x=1/2 , ie f(x) is discontinuous at x= 1/2

Example 2: Find points of discontinuity of the function f(x) = $\frac{x^2+2x+5}{x^2-3x+2}$

Solution: f(x) = $\frac{x^2+2x+5}{x^2-3x+2}$

For x=1 and x=2 the denominator becomes zero and the function f(x) is undefined at x=1 and x= 2. Hence the points of discontinuity are at x=1 and x= 2

Example 3: A function g(x) is defined as follows

g(x) = x when 0<x<1

=2-x when x≥1

is g(x) is continuous at x= 1

Solution:

$\lim_{x\mapsto 1-} g(x) = \lim_{x\mapsto 1+} g(x)$

$\lim_{x\mapsto 1+} g(x) = \lim_{x\mapsto 1+} (2-x) = 2-1 = 1$

$\therefore \lim_{x\mapsto 1-} g(x) = \lim_{x\mapsto 1-} g(x) = 1$

Also g(1) =2-1=1

and $\lim_{x\mapsto 1} g(x) = g(1) = 1$

Hence f(x) is coninuous at x=1

Example 4: The function f(x) = $\frac{x^2-9}{x-3}$ is undefined at x=3 . What value must be assigned to f (3), if f(x) is to be continuous at x=3

Solution: When x approaches 3, x≠3 ie x-3≠0

$\lim_{x\mapsto 3} f(x) = \lim_{x\mapsto 3} \frac{(x-3)(x+3)}{(x-3)}$

$= \lim_{x\mapsto 3} (x+3) = 3+3 = 6$

Therefore if f(x) is to be continuous at x=3 =f(3) = $\lim_{x\mapsto 3} f(x) = 6$

Example 5: Is the function f(x) = $|x|$ continuous at x= 0

Solution: We know that $|x|$ =x when x>0

= 0 when x= 0

= -x when x<0

$\lim_{x\to 0-} f(x) = \lim_{x\mapsto 0-} (-x) = 0$ and $\lim_{x\to 0+} f(x) = \lim_{x\mapsto 0+} x = 0$

Hence $\lim_{x\mapsto 0} f(x) = 0 = f(0)$

So f(x) is continuous at x = 0

CHAPTER – 7 SETS, FUNCTIONS AND RELATIONS

SETS

"The collection of well-defined distinct elements"

***Examples*:–**

(i) A = {a, e, i, o, u} => Roster form or Braces form
A = The set of all the vowels of English alphabets. => Set builder form

(ii) B = {2, 4, 6, 8, 10} => Roster form or Braces form
B = The set of all the even numbers from 2 to 10 => Set builder form
B = {x : x = 2n, where $1 \leq n \leq 5$} => Algebraic form

(iii) C = {1, 2} => Roster form or Braces form
C = The set of the first two natural number. => Set builder form
C = {x : $x^2 - 3x + 2 = 0$} => Algebraic form

Q.01. A collection of well-defined distinct objects in called……
(a) A set (समुच्चय) (b) A sub set (उप समुच्चय)
(c) Power set (पॉवर समुच्चय) (d) Disjoint sets (डिस जॉइंन्ट)

Q.02. Roaster form of the set {x : x is a prime number less than 10} is:-
(a) {2, 4, 6, 8} (b) {1, 3, 5, 7, 9} (c) {1, 3, 5, 7} (d) {2, 3, 5, 7}

Q.03. Set in which the number of element is finite, is called
(a) Finite set (परिमीत) (b) Infinite set (अपरिमीत)
(c) Empty set (खाली) (d) Equal sets (समान)

CARDINAL NUMBER: – The number of elements in a set is called its "cardinal number".

EMPTY set or Null set or VOID set: - "The set having NO element"
It is generally represented as { } or Ø

***Note: -** Wrong representation of Null set are {Ø} or {0} or {{}}

Q.04. The cardinal number of Empty set is
(a) 0 (b) Ø (c) 1 (d) ∞

SINGLETON set: – "The set having only ONE element".

***Examples*:–**

(i) {{}} or {0} or {7} or {1} or {x} etc.
(ii) The set of even prime number.

Q.05. A set containing one element is called……
(a) Unit set (b) Single set
(c) Singleton set (एकाकी समुच्चय) (d) None

Q.06. Cardinal Number of Singleton set is
(a) 0 (b) Ø (c) 1 (d) ∞

EQUAL sets – The two sets are said to be equal, if they have the same elements and also they have equal cardinal number.

***Example*: -** A = {2, 4, 7} & B {4, 2, 7} are equal sets.

EQUIVALENT sets – The two sets are said to be equivalent, if their cardinal numbers are equal but not having the same elements.

***Example*: -** A = {2, 7, 8} & B = {2, 9, 4} because here n(A) = 3 and n(B) = 3

Q.07. If n(A) = n(B) for the two finite sets A & B, is known as
(a) Singleton set (एकाकी समुच्चय) (b) Equal sets (समान समुच्चय)
(c) Equivalent sets (समतुल्य समुच्चय) (d) Power sets (पॉवर समुच्चय)

***Note: -** Let a given set be A = {5, 8}
Then the Subsets of the set A are {}, {5}, {8}, {5, 8}
So, the total number of subsets = 4
And
The Proper Subset of the set A are {}, {5}, {8},
So, the number of proper subset = 3
Hence
"*If the cardinal number of a given set is **n**, then the number of Subsets will be **2^n***
*And the number of Proper Subsets will be **$2^n - 1$***"

Q.08. If every element of a set P is also an element of set Q and set P is a subset of set Q, then the set Q is known as :-
(a) Set of P (P का समुच्चय) (b) Subset of P (P का उप समुच्चय)
(c) Super set of P (P का सुपरसैट) (d) None

Q.09. The number of subsets (उपसमुच्चयो की संख्या) of the set {a, b, c} is :-
(a) 3 (b) 4 (c) 8 (d) 7

Q.10. Of the following sets, the one having exactly eight subsets is :-
(a) {1, 2} (b) {0, 1, 2}
(c) {8} (d) {1, 2, 3, 4, 5, 6, 7, 8}

***Note: -** If A = {1, 3, 8} and B = {2, 3, 8, 9} then A ∪ B = {1, 2, 3, 8, 9}
A ∪ B is also known as *(A + B)* or *(A OR B)* or (*At least One)*

Q.11. If A = {2, 5, 7} and B = {3, 5, 7, 8} the value of A ∪ B is:-
(a) {5, 7} (b) {3, 8} (c) {2, 3, 5, 7, 8} (d) {2}

Q.12. {1, 2, 3} ∪ {2, 3, 4} is equal to :-
(a) {1, 2} (b) {1} (c) {1, 2, 3} (d) {1, 2, 3, 4}

***Note: -** If A = {1, 2, 3} and B = {2, 3, 8, 9} then A ∩ B = {2, 3}
A ∩ B is also known as *(A X B)* or *(A AND B)* or *(Both)*

Q.13. Inter-sections of two sets say A & B is written as :-
(a) A ⊂ B (b) A ∩ B (c) A ∪ B (d) A ∈ B

Q.14. If A = {1, 2, 3} and B = {2, 4} the value of A ∩ B is :-
(a) {1, 2, 4} (b) {2} (c) {2, 3} (d) {1}

Q.15. If A = {1, 2, 3, 4} and B = {5, 6, 7}, then A ∩ B is :-
(a) {1, 2, 3} (b) {5, 6, 7} (c) {4} (d) Ø

Q.16. If P = {1, 2, 3, 5, 7} and Q = {1, 3, 6, 10, 15} then the cardinal number of P ∩ Q
(तो P ∩ Q का कार्डिनल नम्बर है ?)
(a) 3 (b) 2 (c) 0 (d) None

Q.17. If A = {1, 2, 3}, B = {3, 4} and C = {4, 5, 6} then A ∪ (B ∩ C) is equal to:-
(a) {1,2,5,6} (b) {3} (c) {1,2,3,4,5,6} (d) {1, 2, 3, 4}

***Note: - <u>POWER Set</u>** – "The Set of Subsets."
For Example, If A = {5, 8}, then the Subsets of the set A are {5}, {8}, {}, {5, 8}
So, the Power set of the set A will be P(A) = {{5}, {8}, {}, {5, 8}}

Q.18. What is the power set of A = {8, 9}
(a) {Ø,{8},{9},{8,9}} (b) {Ø,{8},{9}}
(c) Ø (d) None of these

Q.19. What is the power set of the set {0}
(a) Ø (b) {0} (c) {Ø} (d) {Ø, {0}}

Q.20. Total number of elements in the Power Set of a set A containing 10 elements, is:-
(a) 2^{10} (b) 10^2 (c) 2^{10-1} (d) None of these

***Note: - <u>PRODUCT Set</u>** – If A = {2, 5} and B = {3, 7}
Then A × B = {(2, 3), (2, 7), (5, 3), (5, 7)}
And B × A = {(3, 2), (7, 2), (3, 5), (7, 5)}
So, n(A × B) = n(B × A) But, A × B ≠ B × A

Q.21. Let A = {1, 2} and B = {2, 3} then Evaluate A × B
(a) {(2, 1), (3, 1), (2, 3)} (b) {(1, 2), (1, 3), (2, 3)}
(c) {(1, 2), (1, 3), (2, 2), (2, 3)} (d) None of these

Q.22. If A = {a, b}, B = {2, 3, 5, 6, 7} and C = {5, 6, 7, 8, 9} then find A × (B ∩ C)
(a) A (b) {(5, a), (6, a), (7, a), (5, b), (6, b), (7, b)}
(c) {(a, 5), (a, 6), (a, 7), (b, 5), (b, 6), (b, 7)} (d) Ø

Q.23. If A = {a, d}, B = {b, c, e} and C = {b, c, f} then A × (B ∩ C) =
(a) Ø (b) (A × B) ∩ (A × C)
(c) (A × B) ∪ (A × C) (d) None of these

Q.24. The Cartesian Product A × A has 9 elements and two of them are (−1, 0) and (0, 1) then Find the set A?
(a) A = {0, 1} (b) A = {−1, 0, 1} (c) A = {0, −1} (d) A = {1, −1}

Q.25. If A × B = {(3, 2), (3, 4), (5, 2), (5, 4)}, then find A
(a) {2, 3, 4} (b) {2, 3, 4, 5} (c) {3, 5} (d) {2, 4}

***Note: -** **UNIVERSAL Set** – The Set which contains all the elements under consideration in a particular problem is called the "Universal Set". It is generally denoted by 'S'.

The Complement Set of a given set A is represented by $\mathbf{A^c}$ or **A'** or **S–A** or **S∼A**

***Example: -* (i)** Let a Universal Set S = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
And Set A = {3, 5, 6, 8, 9}
Then, S – A or A' or A^c or S∼A = {0, 1, 2, 4, 7}

***Example: -* (ii)** Let a Universal Set S = {3, 5, 6, 7, 9} And Set A = {2, 5, 7}
Then, S – A or A' or A^c or S∼A = {3, 6, 9}

Q.26. If A = {1, 3, 5, 7} and B = {2, 3, 4, 5}, the B – A is equal to:-
(a) {0, 1} (b) {3, 5} (c) {2, 4} (d) {3, 4}

Q.27. If A = {1, 2, 3, 4}, B = {2, 4, 6, 8} and C ={3, 4, 5, 6}, the value of A –{B ∪ C}
(a) {4, 6, 8} (b) Ø (c) {1} (d) {2, 3, 4, 5, 6, 8}

Q.28. If A = {a, d}, B = {b, c, e} and C = {b, c, f}, then A × (B — C)
(a) (A × B) — (A × C) (b) (A × B)
(c) A × C (d) None

Q.29. Given that ∪ = {1, 2, 3, 4, 5}, A = {2, 4}, then Find the value of (A')
(a) {1, 3, 5} (b) {2, 4} (c) {1, 2, 3} (d) None

Q.30. Find the value of B – A, where A = {3, 4, 5, 6} and B = {5, 6, 7, 8}
(a) {3, 4} (b) {7, 8} (c) {3, 4, 7, 8} (d) None

Q.31. If A = {x : x^2 –3x + 2 = 0} and B = {x : x^2 +2x –8 = 0}, then A – B is :-
(a) {1, 2} (b) {2} (c) {1} (d) {4, 3}

Q.32. If U = {2, 3, 4, 5, 6, 7, 8, 9, 10, 11}, A = {3, 5, 7, 9, 11} and B = {7, 8, 9, 10, 11), compute the value of (A – B)'
(a) Ø (b) {2, 4, 6, 8, 10, 12}
(c) {2, 4, 6, 7, 8, 9, 10 11} (d) 0

Q.33. A = {1, 2, 3, 5, 7}, B = {1, 3, 6, 10, 15} then find the cardinal number of A–B
(a) 3 (b) 5 (c) 10 (d) None

Q.34. The …… of two set A and B denoted by A Δ B, which is defined as
A Δ B = (A – B) ∪ (B – A)
(a) Union (संघ) (b) Equivalent Elements (समतुल्य अवयव)
(c) Disjoint difference (डिस्जाइन्ट अन्तर) (d) Symmetric difference (सममीत अन्तर)

Q35. If set A = {1, 2, 3, 4} and set B = {3, 5, 7}, then find the value of A Δ B.
(a) {3, 4} (b) {1, 2, 4, 5, 7} (c) {3, 7, 8} (d) None

***Note: -** Let S = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}, P = {0, 2, 4, 6, 8} and Q = {1, 3, 5, 2, 4}
Then P' => S – P = {1, 3, 5, 7, 9} and Q' => S – Q = {0, 6, 7, 8, 9}
P' ∪ Q' = {0, 1, 3, 5, 6, 7, 8, 9} and P' ∩ Q' = {7, 9}

$P \cap Q = \{2, 4\}$ and $(P \cap Q)' = \{0, 1, 3, 5, 6, 7, 8, 9\}$
$P \cup Q = \{0, 1, 2, 3, 4, 5, 6, 8\}$ and $(P \cup Q)' = \{7, 9\}$
Hence **$(P \cup Q)' = P' \cap Q'$** and **$(P \cap Q)' = P' \cup Q'$** ***["De – Morgan's Law"]***

Q.36. Let (मान लो) X = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} be the universal set (सार्वभौमिक समुच्च;) and A = {2, 4, 6} , B = {1, 3, 7}, then A' ∩ B' is equal to :-
(a) {5, 8, 9, 10} (b) {2, 4, 5, 6, 8, 9, 10}
(c) {1, 3, 5, 7, 8, 9, 10} (d) {2, 4, 6, 8, 9, 10}

***Note: - (i)** $n(A \cup B) = n(A) + n(B) – n(A \cap B)$
(ii) $n(A \cup B \cup C) = n(A) + n(B) + n(C) – n(A \cap B) – n(A \cap C) – n(B \cap C) + n(A \cap B \cap C)$

Q.37. If n(A) = 3, n(B) = 4 and $n(A \cap B) = 2$, then $n(A \cup B)$ is :-
(a) 5 (b) 7 (c) 6 (d) 4

Q.38. Draw the **Venn diagram** for the sets:-
(i) A = {3, 5, 6, 7, 9} and B = {1, 2, 3, 4, 8, 9}
(ii) A = {5, 7, 6, 9, 8} and B = {2, 4, 7, 8}
(iii) A = {1, 2, 3, 5, 7}, B = {2, 4, 6, 8} and C = {1, 3, 9}

Q.39. In a class, 50 student's play cricket, 20 students play football and 10 play both cricket and football. How many play at least one of these two games?
(a) 60 (b) 45 (c) 55 (d) 65

Q.40. If X and Y are two sets such that $X \cup Y$ has 18 elements. X has 8 elements, and Y has 15 elements, how many elements does $X \cap Y$ have?
(a) 5 (b) 7 (c) 9 (d) 11

Q.41. If S and T are two sets such that S has 21 elements, T has 32 elements, and $S \cap T$ has 11 elements, how many elements does $S \cup T$ have?
(a) 52 (b) 32 (c) 42 (d) None of these

Q.42. In a group of 1,000 people, there are 750 people who can speak Hindi and 400 who can speak English. How many can speak Hindi only?
(a) 600 (b) 650 (c) 750 (d) 800

Q.43. In a town with a population of 5000, 3200 people are Egg-eaters, 2500 Meat-eaters and 1500 eat both Egg and Meat. How many are pure vegetarians?
(a) 600 (b) 800 (c) 900 (d) 850

Q.44. In a group of 75 persons, 20 take tea but not coffee and 30 take tea. How many take coffee but not tea?
(a) 45 (b) 35 (c) 40 (d) None of these

Q.45. In a committee, 50 people speak Hindi, 20 speak Bengali and 10 speak both Hindi and Bengali. How many speak at least one of these two languages?
(a) 50 (b) 40 (c) 60 (d) 70

Q.46. In a group of students 10 like Economic, 15 likes Accounts and 7 like both. Find the total number of Students.
(a) 25 (b) 18 (c) 32 (d) None

Q.47. A town has a total population of 50000 people. 28000 people read the Newspaper X and 23000 read Newspaper Y and 4000 read both the newspaper. Find the number of people NOT reading any of the Newspaper.
(a) 2500 (b) 3200 (c) 3000 (d) None of these

Q.48. In a class of 60 students, 40 like Math, 36 like Physics and 24 like both subjects. Find the number of students who likes –(i) Only Math (ii) Only Physics (iii) Either Math or Physics (iv) Neither Math nor Physics
(a) 24, 36, 40, 52 (b) 40, 24, 28, 8
(c) 16, 12, 52, 8 (d) None of these

Q.49. Out of 20,000 families, 40% families read the newspaper A, 20% families read the newspaper B and 10% families read the newspaper C, 5% families read both A and B, 3% families read both B and C, 4% families read both A and C and 2% families read all newspapers. Find the number of families read only newspaper A.
(a) 6600 (b) 2800 (c) 1000 (d) None of these

Q.50. In a group of 300 people, 200 people prefer newspaper, 40 people prefer television and 100 prefer radio. 50 prefer newspaper and radio both, 20 prefer radio and television both and 25 prefer newspaper and television both and 5 prefer all. Find the number of people not using any of the media.
(a) 75 (b) 50 (c) 0 (d) None

FUNCTIONS

Q.1. If $F(x) = 3x^2 + 7x - 5$ then find the value of f (0), f (2) and f (-3)
(a) -5, 21, 1 (b) 5, 21, -1 (c) -5, -21, 1 (d) None of these

Q.2. If $F(x) = 7x - 1$ when $x < 0$
$F(x) = 5x + 3$ when $0 \leq x \leq 5$
$F(x) = x^2 + 2$ when $x > 5$
Find the value of F (-3), F (0), F (2) and F (6)
(a) -22, 3, 13, 28 (b) 22, 3, 13, 28 **(c)** -22, 3, 13, 38 (d) None

Note: - (i)** If **f (– x) = f (x)** then, f (x) is an ***"Even Function"
(ii) If **f (– x) = – f (x)** then, f (x) is an ***"Odd Function"***

Q.3. If $F(x) = 4x^5 - 7x^3 + 2x$, then the function is:-
(a) An Even Function (b) An Odd Function
(c) Not a Function (d) None

Q.4. If $F(x) = 7x^4 + 2x^2 - 6$, then the function is:-
(a) An Even Function (b) An Odd Function
(c) Not a Function (d) None

Q.5. If F (x) = $\frac{4^x+1}{4^x-1}$, then the function is:-
(a) An Even Function (b) An Odd Function
(c) Not a Function (d) None

Q.6. If F (x) = $\sqrt{x^2-1}$, then the function is:-
(a) An Even Function (b) An Odd Function
(c) Not a Function (d) None

QUESTIONS OF INVERSE FUNCTION

Q.7. If f (x) = 2x then, find $f^{-1}(x)$
(a) 2/x (b) x (c) x/2 (d) None of these

Q.8. If f (x) = $\frac{1}{1-x}$ then, find $f^{-1}(x)$
(a) x / (x – 1) (b) (x – 1) / x (c) (x + 1) / x (d) None of these

Q.9. If f (x) = $\log_{10} x$ then, find $f^{-1}(x)$
(a) x (b) 10 (c) 10^x (d) None of these

Q.10. If f (x) = $\frac{2+x}{2-x}$ then, find $f^{-1}(x)$
(a) $\frac{2.(x-1)}{x+1}$ (b) $\frac{(x-1)}{x+1}$ (c) $\frac{2.(x+1)}{x-1}$ (d) None of these

QUESTIONS OF COMPOSITE FUNCTION

Q.11. If f (x) = x^2 and g (x) = x + 3, then find fog (x) and gof (x).
(a) $(x+3)^2$ & x^2+3+6x (b) $(x-3)^2$ & x^2+3
(c) $(x+3)^2$ & x^2+3 (d) None

Q.12. If f (x) = $\frac{1}{1-x}$ and g (x) = $\frac{x-1}{x}$, then find fog (x) and gof (x).
(a) x^2 & x (b) x & x (c) x & x^2 (d) None

Q.13. If f (x) = $[a - x^n]^{1/n}$, then find the value of f (f (x))
(a) x (b) x^2 (c) x^n (d) None

Q.14. If f (x-1) = $x^2 - 4x + 8$, then find the value of f (x+1)
(a) x^2 (b) $x^2 - 4$ (c) $x^2 + 4$ (d) None of these

Q.15. If f (x) = $\frac{x+1}{x+2}$, then find the value of $f\left(f\left(\frac{1}{x}\right)\right)$
(a) $\frac{3x+1}{5x+2}$ (b) $\frac{3x+2}{5x+3}$ (c) $\frac{x+3}{5x+2}$ (d) None

Q.16. If f (x) = kx + 1 & g (x) = 3x + 2 and if fog = gof , then the value of k is :-
(a) 1 (b) 2 (c) 3 (d) 4

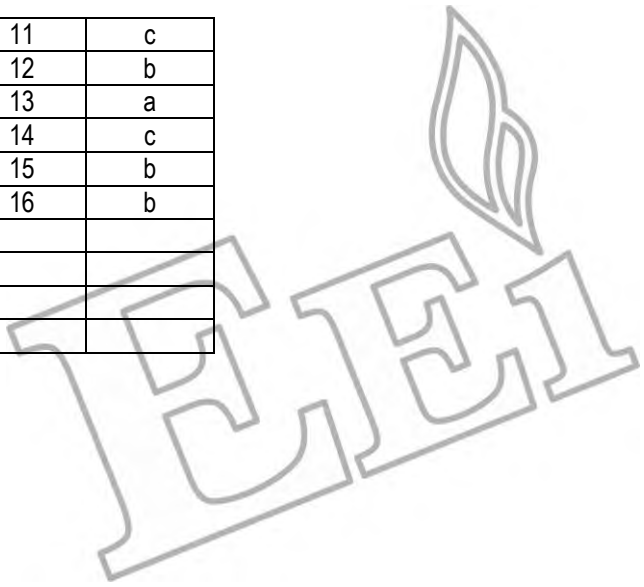
Answers

SETS

1	a	11	c	21	c	31	c	41	c
2	d	12	d	22	c	32	c	42	a
3	a	13	b	23	b	33	a	43	b
4	a	14	b	24	b	34	d	44	a
5	c	15	d	25	c	35	b	45	c
6	c	16	b	26	c	36	a	46	b
7	c	17	d	27	c	37	a	47	c
8	c	18	a	28	a	38		48	c
9	c	19	d	29	c	39	a	49	a
10	b	20	a	30	b	40	a	50	b

FUNCTIONS

1	a	11	c
2	c	12	b
3	b	13	a
4	a	14	c
5	b	15	b
6	b	16	b
7	c		
8	b		
9	c		
10	a		



CHAPTER-8 DIFFERENTIATION & INTEGRATION

DIFFERENTIATION

Q.1. $\frac{dx^5}{dx}$ (a) $5x^4$ (b) $4x^4$ (c) 1 (d) None

Q.2. $\frac{dx^{-7}}{dx}$ (a) $7x^{-8}$ (b) $-7x^{-8}$ (c) 0 (d) None

Q.3. $\frac{dx^{-3/7}}{dx}$ (a) $\frac{-3}{7}x^{\frac{-10}{7}}$ (b) $\frac{3}{7}x^{\frac{-10}{7}}$ (c) 1 (d) None

Q.4. $\frac{dx^{3/2}}{dx}$ (a) $\frac{1}{2}x^{\frac{1}{2}}$ (b) $\frac{3}{2}x^{\frac{1}{2}}$ (c) 3 (d) None

Q.5. $\frac{d\sqrt{x}}{dx}$ (a) $\frac{1}{\sqrt{x}}$ (b) $\frac{1}{2\sqrt{x}}$ (c) 2 (d) None

Q.6. $\frac{dx}{dx}$ (a) 0 (b) x (c) 1 (d) None

Q.7. $\frac{dk}{dx}$ (a) 0 (b) 1 (c) k (d) None

Q.8. $\frac{d(x^2+3)}{dx}$ (a) x (b) 3 (c) 2x (d) None

Q.9. $\frac{d\,(e^x+\sqrt{x}-logx)}{dx}$ (a) $e^x+\frac{1}{2\sqrt{x}}-\frac{1}{x}$ (b) $e^x+\frac{1}{2\sqrt{x}}+\frac{1}{x}$ (c) 1 (d) None

Q.10. $\frac{d(x-1)^2}{dx}$ (a) x – 2 (b) 2x – 2 (c) 0 (d) None

Q.11. $\frac{d[x(x-1)\,(x-2)]}{dx}$ (a) $3x^2-6x+2$ (b) $2x^2-6x+2$ (c) 1 (d) None

Q.12. $\frac{d}{dx}\left(\frac{1}{\sqrt[3]{x^2}}\right)$ (a) $\frac{-2}{3}x^{\frac{-5}{3}}$ (b) $\frac{2}{3}x^{\frac{-5}{3}}$ (c) $\frac{1}{3}$ (d) None

Q.13. If f(x) = x^3-5x, then find f ′(x) , f "(x) and f ′(2)
(a) $2x^2-5$, 6x, 7 (b) $3x^2-5$, 3x, 7 (c) $3x^2-5$, 6x, 7 (d) None

Q.14. f (x) = $\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2$ then find f ′(x) and f ′(2)
(a) $1+x^{-2}$, 3/4 (b) $1-x^{-2}$, 3/4 (c) 1 (d) None

Q.15. $\frac{d\,(x+2)^3}{dx}$ (a) $3x^2+12x+12$ (b) $3x^2+12x$ (c) 0 (d) None

Q.16. Find the Gradient of the curve y = $2x^3-3x^2-12x+8$ at the point x = 0.
(a) 8 (b) – 12 (c) 12 (d) None

Q.17. Find the Gradient of the curve y = $2x^3-5x^2-3x$ at the point x = 0
(a) – 3 (b) 3 (c) 5 (d) None

Q.18. $\frac{d\,(e^x \times \sqrt{x}\,)}{dx}$ (a) $e^x\left[\frac{3x+1}{\sqrt{x}}\right]$ (b) $e^x\left[\frac{2x+1}{2\sqrt{x}}\right]$ (c) 0 (d) None

Q.19. $\frac{d\,(x^2 \times 2^x\,)}{dx}$

(a) $2^x.x[x\log_e 2+2]$ (b) $2^x.x[\log_e 2+2]$

(c) $2^x.x[x\log_e 2]$ (d) None

Q.20. $\frac{d^{[logx \times 5^x]}}{dx}$

(a) $2^x.\left[\log_e x \times \log_e 5+\frac{1}{x}\right]$ (b) $5^x.\left[\log_e x \times \log_e 5+\frac{1}{x}\right]$

(c) $5^x.\left[\log_e x \times \log_e 5-\frac{1}{x}\right]$ (d) None

Q.21. $\frac{d\,(\sqrt{x} \times e^x \times logx)}{dx}$

(a) $e^x\left(\frac{\sqrt{x}}{x}+\sqrt{x}.logx+\frac{logx}{2\sqrt{x}}\right)$ (b) $e^x\left(\frac{\sqrt{x}}{x}-\sqrt{x}.logx+\frac{logx}{2\sqrt{x}}\right)$

(c) $e^x\left(\frac{\sqrt{x}}{x}+\sqrt{x}.logx-\frac{logx}{2\sqrt{x}}\right)$ (d) None

Q.22. $\frac{d\,(x^{\frac{3}{2}} \times \sqrt[5]{x})}{dx}$ (a) $\frac{11}{10}x^{\frac{7}{10}}$ (b) $\frac{17}{10}x^{\frac{7}{10}}$ (c) 0 (d) None

Q.23. $\frac{d(e^x/\sqrt{x})}{dx}$ (a) $\frac{e^x(2x-1)}{2x^{\frac{3}{2}}}$ (b) $\frac{\sqrt{x}.e^x(3x-1)}{2}$ (c) 1 (d) None

Q.24. $\frac{d\,(x^3/3^x)}{dx}$

(a) $\frac{3^x.x^2(3-x.\log_e 3)}{3^{2x}}$ (b) $\frac{2^x.x^2(3-x.\log_e 3)}{3^{2x}}$

(c) $\frac{3^x.x^2(3+x.\log_e 3)}{3^{2x}}$ (d) None

Q.25. If f (x) = $\frac{x^2+1}{x^2-1}$ (a) $\frac{-4x}{(x^2-1)^2}$ (b) $\frac{-2x}{(x^2-1)^2}$ (c) 0 (d) None

Q.26. If y = $\frac{e^x+1}{e^x-1}$ (a) $\frac{-2e^x}{(e^x+1)^2}$ (b) $\frac{-2e^x}{(e^x-1)^2}$ (c) 1 (d) None

Q.27. $\frac{d\left(\frac{x^2-1}{x}\right)}{dx}$ (a) 1+ $\frac{1}{x^2}$ (b) 1+ $\frac{2}{x^2}$ (c) 0 (d) None

Q.28. f(x) = $\frac{x^2}{e^x}$, then find the value of f ′(1)

(a) $\frac{-1}{e}$ (b) $\frac{1}{e}$ (c) $\frac{2}{e}$ (d) None

Q.29. f(x) = $\frac{4-2x}{2+3x+3x^2}$ & f ′(x) = 0, then find x

(a) $2\left[1+\sqrt{\frac{5}{3}}\right]$ (b) $2\left[1\pm\sqrt{\frac{2}{3}}\right]$ (c) $2\left[1\pm\sqrt{\frac{5}{3}}\right]$ (d) None

Q.30. f(x) = x^k and f ′(1) = 10 find the value of k.

(a) – 10 (b) 10 (c) 6 (d) None

Q.31. If f(x) = x^2 −6x +8 then, find the value of f ′(5) – f ′(8)
(a) – 6 (b) 6 (c) 8 (d) None

Q.32. If y = 1+ x + $\frac{x^2}{2!} + \frac{x^3}{3!}$ + ---------- ∞ , then find the value of $\frac{dy}{dx} - y = ?$
(a) 0 (b) – 1 (c) 2 (d) None

Q.33. $\frac{de^{\sqrt{x}}}{dx}$ (a) $\frac{e^{\sqrt{x}}}{3\sqrt{x}}$ (b) $\frac{e^{\sqrt{x}}}{2\sqrt{x}}$ (c) 1 (d) None

Q.34. $\frac{de^{x^3}}{dx}$ (a) $3e^{x^3}.x^2$ (b) $2e^{x^2}.x^2$ (c) 2 (d) None

Q.35. $\frac{d\sqrt{e^x}}{dx}$ (a) $\frac{\sqrt{e^x}}{2}$ (b) $\frac{\sqrt{e^x}}{3}$ (c) (d) None

Q.36. $\frac{d(\sqrt{logx})}{dx}$ (a) $\frac{1}{3x.\sqrt{logx}}$ (b) $\frac{1}{2x.\sqrt{logx}}$ (c) 0 (d) None

Q.37. $\frac{d(\sqrt{3^x})}{dx}$ (a) $\frac{\sqrt{3^x}.\log_e 3}{2}$ (b) $\frac{\sqrt{3^x}.log2}{2}$ (c) 1 (d) None

Q.38. $\frac{d\log(logx)}{dx}$ (a) $\frac{2}{x.logx}$ (b) $\frac{1}{x.logx}$ (c) 0 (d) None

Q.39. $\frac{d(x+3)}{dx}$ (a) –1 (b) 0 (c) 1 (d) None

Q.40. $\frac{d(x+3)^2}{dx}$ (a) 2(x–3) (b) 2(x+3) (c) 0 (d) None

Q.41. $\frac{d(x+3)^3}{dx}$ (a) $3(x+3)^2$ (b) $3(x-3)^2$ (c) 3 (d) None

Q.42. $\frac{d(x+3)^{11}}{dx}$ (a) $2(x+3)^{10}$ (b) $11(x+3)^{10}$ (c) 0 (d) None

Q.43. $\frac{d(7x^2+5)^7}{dx}$ (a) $98x(7x^2+5)^6$ (b) $98x(6x^2+5)^6$ (c) $98x(7x^2+5)^5$ (d) None

Q.44. $\frac{d(\sqrt{x+1})}{dx}$ (a) $\frac{1}{3\sqrt{x-1}}$ (b) $\frac{1}{2\sqrt{x+1}}$ (c) 0 (d) None

Q.45. $\frac{d(e^{ax^2+bx+c})}{dx}$ (a) $(2ax+b)(e^{ax^2+bx+c})$ (b)(e^{ax^2+bx+c})
(c) $(3ax+b)(e^{ax^2+bx+c})$ (d) None

Q.46. $\frac{d\sqrt{x+\sqrt{x}}}{dx}$ (a) $\frac{2\sqrt{x}-1}{4\sqrt{x(x+\sqrt{x})}}$ (b) $\frac{2\sqrt{x}+1}{4\sqrt{x(x+\sqrt{x})}}$ (c) 1 (d) None

Q.47. $\frac{d(e^{-x})}{dx}$ (a) $-e^{-x}$ (b) e^{-x} (c) 0 (d) None

Q.48. y = $\sqrt{2x} + 3^{2x}$, find $\frac{dy}{dx}$ (a) $\frac{1}{\sqrt{2x}} + 2.3^{2x}.\log_e 3$ (b) $\frac{1}{\sqrt{2x}}$ (c) $\frac{1}{\sqrt{2x}} + 3^{2x}.\log_e 3$ (d) None

Q.49. $\frac{d(e^{3x^2-6x+2})}{dx}$ (a) e^{3x^2-6x} (b) $e^{3x^2-6x+2}(6x-6)$ (c) $2e^{3x^2+6x+2}$ (d) None

Q.50. y = $e^{\sqrt{2x}}$ (a) $\frac{e^{\sqrt{2x}}}{\sqrt{2x}}$ (b) $\frac{2e^{\sqrt{2x}}}{\sqrt{2x}}$ (c) 1 (d) None

Q.51. $\frac{d\,[\log(\log(logx))]}{dx}$ (a) $\frac{1}{\text{x.logx.log(logx)}}$ (b) $\frac{1}{\text{logx.log(logx)}}$ (c) $\frac{1}{\text{logx}}$ (d) None

Q.52. If f(x) = ${}^{x}C_3$ then f '(1) = ?
(a) $\frac{1}{6}$ (b) $-\frac{1}{6}$ (c) $\frac{5}{6}$ (d) $-\frac{5}{6}$

Q.53. $\frac{d\,[e^{3x^2+6x-2}]}{dx}$
(a) $e^{2x^2+3x-2}.(6x+6)$ (b) e^{3x^2-4x+2}
(c) $e^{3x^2+6x-2}.(6x+6)$ (d) None

Q.54. If y = $\sqrt{x^2+m^2}$ then, find y. y_1
(a) x^2 (b) x (c) y^2 (d) None

Q.55. If y = $\left[x+\sqrt{x^2+m^2}\right]^n$, then find dy/dx
(a) $\frac{n\left[x+\sqrt{x^2+m^2}\right]^n}{\sqrt{x^2+m^2}}$ (b) $\frac{n\left[x-\sqrt{x^2-m^2}\right]^n}{\sqrt{x^2+m^2}}$
(c) $\frac{n\left[x-\sqrt{x^2+m^2}\right]^n}{\sqrt{x^2-m^2}}$ (d) None

Q.56. $x^3 + y^3 - 3axy = 0$, find $\frac{dy}{dx}$
(a) $\frac{xy+x^2}{y^2-ax}$ (b) $\frac{ay-x^2}{y^2+ax}$ (c) $\frac{ay-x^2}{y^2-ax}$ (d) None

Q.57. $\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$, find $\frac{dy}{dx}$ (a) $\frac{2x}{\sqrt{x^2-a^2}}$ (b) $\frac{x}{\sqrt{x^2-a^2}}$ (c) $\frac{x}{\sqrt{x^2+a^2}}$ (d) None

Q.58. $\log\left(\frac{x}{y}\right)$ = x + y, then find $\frac{dy}{dx}$
(a) $\frac{y(1-x)}{x(y+1)}$ (b) $\frac{x(1+x)}{y(y+1)}$ (c) $\frac{y(1-x)}{x(y-1)}$ (d) None

Q.59. If y = $\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}}\ \infty$, then find dy/dx
(a) 1 (b) $\frac{1}{2y+1}$ (c) $\frac{1}{2y-1}$ (d) None

Q.60. y = $\sqrt{e^x + \sqrt{e^x + \sqrt{e^x + }}}$ .....∞, then Find $\frac{dy}{dx}$

(a) $\frac{1}{2y+1}$ (b) $\frac{e^x}{2y-1}$ (c) $\frac{e^x}{3y+1}$ (d) None

Q.61. y = $\sqrt{log_x + \sqrt{log_x + \sqrt{log_x + }}}$ ..... ∞, $\frac{dy}{dx}$ =?

(a) $\frac{1}{(x+y)}$ (b) $\frac{1}{x(2y+1)}$ (c) $\frac{1}{x(2y-1)}$ (d) None

Q.62. If y = $\sqrt{\sqrt{x} + \sqrt{\sqrt{x} + \sqrt{\sqrt{x} + \sqrt{\sqrt{x} + \cdots}}}}$ ∞ , then find $\frac{dy}{dx}$

(a) $\frac{1}{2\sqrt{x}(2y-1)}$ (b) $\frac{1}{-2\sqrt{x}(2y-1)}$ (c) $\frac{1}{\sqrt{x}(2y-1)}$ (d) None

Q.63. y = $\sqrt{3^x + \sqrt{3^x + \sqrt{3^x + }}}$ ....∞, Find $\frac{dy}{dx}$

(a) $\frac{\log_e 3}{2y-1}$ (b) $\frac{2^x.\log_e 3}{2y+1}$ (c) $\frac{3^x.\log_e 3}{2y-1}$ (d) None

Q.64. y = $\sqrt{log_2 x + \sqrt{log_2 x + \sqrt{log_2 x + }}}$ ..... ∞, find $\frac{dy}{dx}$ =?

(a) $\frac{1}{x.\log_e 2(2y-1)}$ (b) $\frac{1}{\log_e 2(2y-1)}$ (c) $\frac{1}{x.\log_e 2(2y+1)}$ (d) None

Q.65. If x.y = 1, then $y^2 + \frac{dy}{dx}$ = ?

(a) 1 (b) 0 (c) –1 (d) None

Q.66. e^{-xy}– 4xy = 0 , find $\frac{dy}{dx}$

(a) $\frac{dy}{dx}$ (b) $\frac{x}{2y}$ (c) $-\frac{y}{x}$ (d) None

Q.67. Find the slope of the tangent to the curve y = $\sqrt{4 - x^2}$ at the point where "ordinate and abscissa are equal"

(a) 1 (b) –1 (c) 0 (d) None

Q.68. Find the slope of the tangent to the curve $y = x^2 - x$ at the point where the line y = 2 cuts the curve in the first quadrant.

(a) 3 (b) -3 (c) 2 (d) None

Q.69. If $x^2+y^2 + 2gx + 2hy = 0$, find $\left(\frac{dy}{dx}\right)_{(0,0)}$

(a) $\frac{g}{h}$ (b) $\frac{-g}{h}$ (c) $\frac{-g}{-h}$ (d) None

Q.70. If $x^3-2x^2 y^2 + 5x + y-5 = 0$, find $\left(\frac{dy}{dx}\right)$ at the point (1,1)

(a) $\frac{2}{3}$ (b) $\frac{4}{3}$ (c) $\frac{-4}{3}$ (d) None

Q.71. If y = $\log\left[e^x \times \left(\frac{x-2}{x+2}\right)^{3/4}\right]$

(a) $\frac{x^2-2}{x^2+4}$ (b) $\frac{x^2-1}{x^2-4}$ (c) $\frac{x^2}{x^2-4}$ (d) None

Q.72. If, y = x^x, find $\frac{dy}{dx}$

(a) $x^x(1-logx)$ (b) $(1+logx)$ (c) $x^x(1+logx)$ (d) None

Q.73. If y = x^y, find $\frac{dy}{dx}$

(a) $\frac{y}{x}\left(\frac{y}{1+logx}\right)$ (b) $\frac{y}{x}\left(\frac{y}{1-ylogx}\right)$ (c) $\frac{x}{y}\left(\frac{1}{1-logx}\right)$ (d) None

Q.74. If y = $x^{x^{x.....\infty}}$, find $\frac{dy}{dx}$

(a) $\frac{y}{x}\left(\frac{y}{1-ylogx}\right)$ (b) $\frac{y}{x}\left(\frac{y}{1+logx}\right)$ (c) $\frac{x}{y}\left(\frac{1}{1-logx}\right)$ (d) None

Q.75. If y = $\sqrt{x}^{\sqrt{x}^{\sqrt{x}.....\infty}}$ then find dy/dx

(a) $\frac{y^2}{x(2-ylogx)}$ (b) $\frac{y^2}{(3-ylogx)}$ (c) $\frac{x^2}{x(2+ylogx)}$ (d) None

Q.76. If $x^y \times y^x$ = m, find $\frac{dy}{dx}$

(a) $-\frac{y(xlogy+y)}{x(ylogx+x)}$ (b) $\frac{x(xlogy+y)}{y(ylogx-x)}$ (c) $\frac{y(xlogy-y)}{x(ylogx-x)}$ (d) None

Q.77. $\sqrt{x}+\sqrt{y}$ = 1, then $\frac{d^2y}{dx^2}$=?

(a) $\left(\frac{x}{2}^{-3/2}\right)$ (b) $\left(\frac{x}{2}^{3/2}\right)$ (c) $\left(\frac{x}{2}^{-1/2}\right)$ (d) None

Q.78. If y = $\frac{e^{3x}-e^{2x}}{e^{3x}+e^{2x}}$ find $\frac{dy}{dx}$

(a) $\frac{2e^x}{(e^x+1)^2}$ (b) $\frac{2e^x}{(e^x-1)^2}$ (c) $\frac{e^x}{(e^x+1)^2}$ (d) None

Q.79. If x = at^2 and y = 2at, then find $\frac{dy}{dx}$

(a) 1 (b) 1/t (c) t (d) None

Q.80. x = $3t^2$ -1 and y = t^3- t, find $\frac{dy}{dx}$

(a) $\frac{3t^2-1}{6t}$ (b) $\frac{3t^2+1}{6t}$ (c) $\frac{t^2-1}{6t}$ (d) None

Q.81. x = t + t^{-1} and y = t – t^{-1} find, $\frac{dy}{dx}$ at t = 2

(a) 1/2 (b) 2/3 (c) 5/3 (d) None

Q.82. x = $\frac{1-t^2}{1+t^2}$ and y = $\frac{2t}{1+t^2}$ then find dy/dx at t = 1

(a) 0 (b) 1 (c) t (d) None

Q.83. Gradient of the curve y – xy + 2 px + 3 qy = 0 at the point (3, 2) is (-2/3) find the value of p and q.
(a) $\left(\frac{1}{2},\frac{1}{2}\right)$ (b) $\left(\frac{-1}{2},\frac{-1}{2}\right)$ (c) $\left(\frac{1}{2},\frac{1}{6}\right)$ (d) None

Q.84. Curve $y^2 = ux^3 + v$ passes through the point (2, 3) and given that $\frac{dy}{dx} = 4$ then find the value of u and v
(a) 2, −7 (b) −2, 7 (c) 2, 7 (d) None

Q85. $f(x) = 2x^3-15x^2+36x+15$ find the value of x , so that the function will be minima.
(a) 3 (b) 2 (c) − 3 (d) None

Q.86. $f(x) = \left\{\frac{a+x}{1+x}\right\}^{a+1+2x}$ then find f ′(0)
(a) $a^{a+1}\left[\frac{1+a^2+aloga^2}{a}\right]$ (b) $a^{a+1}\left[\frac{1-a^2+aloga^2}{a}\right]$
(c) $a^{a+1}\left[\frac{2-a^3+aloga^2}{a}\right]$ (d) None

MISCELLANEOUS QUESTIONS OF DIFFERENTIATION

Q.87. If $y = 5x^5$ then dy/dx is :-
(a) $25x^6$ (b) 25x (c) $25x^4$ (d) None

Q.88. If y = 200 then dy /dx is :-
(a) 200x (b) 200 (c) 0 (d) None

Q.89. If $y = x^2 + x + 1$ then dy / dx is :-
(a) 2x + 2 (b) 2x + 1 (c) 2x + 2 (d) None

Q.90. If $y = (x + 1)^2$ then dy/dx is :-
(a) 2 (x + 1) (b) 2x + 1 (c) x + 1 (d) None

Q.91. If $y = \sqrt{x}$ then dy/dx is :-
(a) $\frac{1}{2\sqrt{x}}$ (b) $\frac{1}{\sqrt{x}}$ (c) $2\sqrt{x}$ (d) None

Q.92. If $f(x) = x^2 + 1$ then f '(x) at x = 2 is :-
(a) 1 (b) 2 (c) 3 (d) 4

Q.93. If $f(x) = (x – 1)^2$ then f '(x) at x = 1 is :-
(a) 0 (b) 1 (c) 2 (d) None

Q.94. If $f(x) = \frac{2}{x^2}$ then f '(2) is :-
(a) 1/2 (b) −1/2 (c) 0 (d) None

Q.95. If $f(x) = 4x^4 + 3x^3 + 2x^2 + 1$ then f '(1) is :-
(a) 29 (b) 30 (c) 25 (d) None

Q.96. If $f(x) = 2x^3 + 3x^2 – 12x$, for what value of x will f '(x) = 0.
(a) 1, 2 (b) −1, −2 (c) 1, −2 (d) 0, 2

Q.97. If $y = (x + 1).(2x^3 + 21)$ then dy/dx is :-
(a) $8x^3 – 6x^2 + 21$ (b) $8x^3 + 6x^2 + 21$ (c) $8x^3 – 6x^2 – 21$ (d) None

Q.98. If y = (x + 1).(x + 2) then dy/dx is :-
(a) 2x + 3 (b) 2(x + 3) (c) x + 3 (d) None

Q.99. If y = x.e^x then dy/dx is :-
(a) x^3 .e^x + e^x (b) x^2 e^x + e^x (c) x^2 e^x + 2e^x (d) None

Q.100. If y = e^x.log x , then dy/dx is :-
(a) $e^x\left(\frac{1}{x}+\log x\right)$ (b) e^x(x+logx) (c) $e^x\left(x+\frac{1}{logx}\right)$ (d) None

Q.101. If y = x^2.e^x.log x , then dy/dx is :-
(a) e^x (1 + 2log x + log x) (b) e^x.x.(1 + 2log x + x.log x)
(c) x^2 (1 + 2log x + log x) (d) None

Q.102. If y = $\frac{x}{1+x^2}$ then dy/dx is :-
(a) (1 – x^2)/(1 + x^2) (b) (1 – x^2)^2/(1 + x^2)
(c) (1 – x^2)/ (1 + x^2)^2 (d) None

Q.103. If f(x) = x^2 + $\log_e$x , then f '(1) is :-
(a) 5/2 (b) 2 (c) 3 + $\log_e$ 5 (d) None

Q.104. If y = $e^{\sqrt{x}}$ then dy/dx is :-
(a) $e^{\sqrt{x}}$ (b) $\frac{e^{\sqrt{x}}}{\sqrt{x}}$ (c) $\frac{e^{\sqrt{x}}}{2\sqrt{x}}$ (d) None

Q.105. x^3 + y^3 = 3axy , then dy/dx = ?
(a) $\frac{ay-x^2}{y^2-ax}$ (b) $\frac{ay-a^2x^2}{y^2-ax}$ (c) $\frac{ax-y^2}{x^2-ay}$ (d) None

Q.106. If x = at^2 and y = 2at , then dy/dx = ?
(a) t (b) 1/t (c) t^2 (d) None

Q.107. If $\sqrt{x}+\sqrt{y}$ = 1 , then $\frac{d^2y}{dx^2}$ is :-
(a) $x^{-1/2}$ (b) $\frac{1}{2}x^{-3/2}$ (c) $x^{-3/2}$ (d) None

Q.108. If f (x) = 3e^{-3x} + 2e^{3x} , then f "(x) is:-
(a) 6 f(x) (b) 3 f(x) (c) 9 f(x) (d) None

Q.109. If y = $\frac{1+x}{1-x}$, then d^2y / dx^2 is :-
(a) $\frac{4}{(1-x)^2}$ (b) $\frac{2}{(1-x)^2}$ (c) $\frac{4}{(1-x)}$ (d) None

Q.110. If y = $\frac{x^2}{a+x}$, then d^2y/dx^2 is :-
(a) $\frac{2a^2}{(a+x)^3}$ (b) $\frac{2a}{(a+x)^2}$ (c) $\frac{2}{(a+x)}$ (d) None.

Q.111. If y = 5^x, then d^2y/dx^2 is :-
(a) ($\log_e$ 5). 5^x (b) ($\log_e$ 5)^2.5^x (c) ($\log_e$ 5)^2.5^{2x} (d) None

Q.112. If the cost functions is $C = x^3 - 15x^2 - 3600x + 700$, the value of x for which cost is minimum is :-
(a) 40 (b) 100 (c) 80 (d) None.

Q.113. If $y = x^x$, then dy/dx = ?
(a) 0 (b) $x^x.(1 + \log x)$ (c) x^x (d) None

Q.114. If $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots\ldots\infty$, then $\frac{dy}{dx}$ = ?
(a) y (b) 2y (c) y^2 (d) None

Q.115. If $x + y^2 = a^2$, then find dy/dx
(a) 1/y (b) -1/(2y) (c) a/y (d) None

Q.116. If $x = at$ and $y = at^2$, then find dy/dx
(a) 2t (b) 2a (c) 2at (d) None

Q.117. If $y = mn\,x^5$, then find dy/dx
(a) $5mnx^4$ (b) $20\,mnx^4$ (c) 0 (d) $5x^4$

INTEGRATION

Q.1. $\int x^3 dx$
(a) $\frac{x^4}{4}$ + c (b) $\frac{x^3}{4}$ + c (c) $\frac{x^4}{3} - c$ (d) None

Q.2. $\int x^{-2} dx$
(a) $\frac{1}{x}$ + c (b) $\frac{-1}{x}$ + c (c) $\frac{-1}{x}$ (d) None

Q.3. $\int x^{5/3} dx$
(a) $\frac{3}{8}x^{\frac{8}{3}}$ (b) $\frac{2}{8}x^{\frac{8}{3}}$ + c (c) $\frac{3}{8}x^{\frac{8}{3}}$ + c (d) None

Q.4. $\int x^{-2/5} dx$
(a) $\frac{x^2}{8}$ + c (b) $\frac{5}{3}x^{\frac{3}{5}}$ + c (c) $\frac{3x^8}{8} - c$ (d) None

Q.5. $\int \sqrt{x}\, dx$
(a) $\frac{1}{3}x^{\frac{3}{2}}$ + c (b) $\frac{2}{3}x^{\frac{3}{2}}$ (c) $\frac{2}{3}x^{\frac{3}{2}}$ + c (d) None

Q.6. $\int 5\, dx$
(a) 0 (b) $3x + c$ (c) $5x + c$ (d) None

Q.7. $\int \frac{1}{x}\, dx$
(a) $2\log x + c$ (b) $\log x$ (c) $\log x + c$ (d) None

Q.8. $\int \frac{1}{\sqrt[3]{x}}\, dx$
(a) $\frac{3}{2}x^{\frac{2}{3}}$ + c (b) $\frac{1}{2}x^{\frac{2}{3}}$ + c (c) $\frac{3}{2}x^{\frac{2}{3}}$ (d) None

Q.9. $\int \left(\frac{1}{x} + e^x - \sqrt{x}\right) dx$

(a) $logx - e^x - \frac{3}{2}x^{\frac{3}{2}} + c$ (b) $logx + e^x - \frac{2}{3}x^{\frac{3}{2}} + c$

(c) $logx + e^x + \frac{3}{2}x^{\frac{3}{2}} - c$ (d) None

Q.10. $\int \left(\frac{\sqrt{x}+x^2}{x} - 5\right) dx$

(a) $3x^{1/2} + \frac{x^2}{2} - 5x + c$ (b) $2x^{1/2} + \frac{x^2}{2} - 5x + c$

(c) $3x^{1/2} + \frac{x^2}{2} + 5x + c$ (d) None

Q.11. $\int 5x^2\, dx$

(a) $\frac{5}{3}x^3$ (b) $\frac{5}{3}x^3 + c$ (c) $\frac{2}{3}x^{\frac{3}{2}} + c$ (d) None

Q.12. $\int (3 - 2x - x^4) dx$

(a) $3x - x^2 - \frac{x^5}{5} + c$ (b) $3x + x^2 - \frac{x^5}{5} + c$

(c) $3x + x^2 + \frac{x^5}{5} - c$ (d) None

Q.13. $\int (4x^3 + 3x^2 - 2x + 5) dx$

(a) $x^4 + x^3 - x^2 + 5x + c$ (b) $x^4 + x^3 + x^2 + 3x - c$

(c) $x^4 - x^3 + x^2 + 5x + c$ (d) None

Q.14. $\int (1 - 3x)\,(1 + x) dx$

(a) $x - x^2 - x^3 + c$ (b) $x + x^2 + x^3 + c$

(c) $x - x^2 + x^3 - c$ (d) None

Q.15. $\int \frac{\sqrt{x}-1}{\sqrt{x}} dx$

(a) $x - 2\sqrt{x} + c$ (b) $x - 2\sqrt{x}$

(c) $x + 2\sqrt{x} + c$ (d) None

Q.16. $\int \left(px^3 + qx^2 + rx + \frac{w}{x}\right) dx$

(a) $\frac{p}{4}x^4 + \frac{q}{3}x^3 + \frac{r}{2}x^2 + wlogx + c$

(b) $\frac{p}{4}x^4 - \frac{q}{3}x^3 - \frac{r}{2}x^2 + wlogx + c$

(c) $\frac{p}{4}x^4 + \frac{q}{3}x^3 + \frac{r}{2}x^2 - wlogx - c$ (d) None

Q.17. $\int_0^1 (3x - 2) dx$

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) $\frac{-1}{2} + c$ (d) None

Q18. $\int_0^1 (2x^2 - x^3)dx$

(a) $\frac{1}{12}$ (b) $\frac{-5}{12}$ (c) $\frac{5}{12}$ (d) None

Q.19. $\int_2^4 (3x-2)^2 dx$

(a) 102 (b) 104 (c) 105 (d) None

Q.20. $\int_0^2 3x^2 dx$

(a) 0 (b) – 8 (c) 8 (d) None

Q.21. $\int e^x . x\, dx$

(a) $e^x.(x-1)+c$ (b) $e^x.(x+1)+c$

(c) $e^x.(x+1)-c$ (d) None

Q.22. $\int x^3 . logx\, dx$

(a) $\frac{x^4}{4}logx + \frac{x^4}{16} - c$ (b) $\frac{x^4}{4}logx - \frac{x^4}{16} + c$

(c) $\frac{x^4}{4}logx - \frac{x^4}{15} - c$ (d) None

Q.23. $\int \log x\, dx$

(a) $x(logx-1)+c$ (b) $x(logx-2)-c$

(c) $x(logx+1)+c$ (d) None

Q.24. $\int (logx)^2\, dx$

(a) $x.(logx)^2 - 2x.logx + 2x + c$ (b) $x.(logx)^2 + 2x.logx + 2x - c$

(c) $x.(logx)^2 + 3x.logx - 2x + c$ (d) None

Q.25. $\int x^2 . e^x\, dx$

(a) $[x^2 - 2x + 2]$ (b) $e^x[x^2 + 2x - 2]$

(c) $e^x[x^2 - 2x + 2]$ (d) None

Q.26. $\int_1^2 (x.logx)dx$

(a) $log4 - \frac{3}{4}$ (b) $log4 + \frac{3}{4}$ (c) $log4 - \frac{2}{4}$ (d) None

Q.27. $\int (2x.e^{x^2})\, dx$

(a) $2e^{x^2} + c$ (b) $e^{x^2} + c$ (c) e^{x^2} (d) None

Q.28. $\int x.e^{x^2}\, dx$

(a) $\frac{e^{x^2}}{2}$ (b) $\frac{e^{x^2}}{2} + c$ (c) $\frac{2e^{x^2}}{2} + c$ (d) None

Q.29. $\int \frac{e^{\sqrt{x}}}{2\sqrt{x}}\, dx$

(a) $e^{\sqrt{x}} + c$ (b) $2.e^{\sqrt{x}}$ (c) $e^{\sqrt{2x}} + c$ (d) None

Q.30. $\int \left(\frac{5+logx}{x}\right) dx$

(a) $\left(\frac{5+\log x}{2}\right)^2$ (b) $\frac{(5+\log x)^2}{2} + C$ (c) $\left(\frac{5-\log x}{2}\right)^2 + C$ (d) None

Q.31. $\int_1^2 \frac{2x}{1+x^2}\, dx$

(a) $log\, \frac{5}{2}$ (b) $log\, \frac{-5}{2}$ (c) $2log\, \frac{5}{2}$ (d) None

Q.32. $\int_1^{e^2} \frac{1}{(x)\,(1+logx)^2}\, dx$

(a) $\frac{-2}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) None

Q.33. $\int (4x+5)dx$

(a) $x^2 + 5x + c$ (b) $2x^2 - 5x + c$ (c) $2x^2 + 5x + c$ (d) None

Q.34. $\int (4x+5)^2 dx$

(a) $\frac{x^3}{3} + 25x + 20x^2 + c$ (b) $\frac{16x^3}{3} + 25x + 20x^2 + c$

(c) $\frac{16x^3}{3} + 2x - 20x^2 - c$ (d) None

Q.35. $\int (4x+5)^7 dx$

(a) $\frac{(4x+5)^8}{32}$ + c (b) $\frac{(4x-5)^8}{32}$ + c (c) $\frac{(2x+5)^8}{32}$ + c (d) None

Q.36. $\int (x+a)^n dx$

(a) $\frac{(x-a)^{n+1}}{n+1}$ + c (b) $\frac{(x+a)^{n+1}}{n+1}$ + c (c) $\frac{(x+a)^{n-1}}{n-1}$ + c (d) None

Q.37. $\int \frac{8x^2}{(x^3+2)^3}\, dx$

(a) $\frac{-4}{3(x^3+2)^2} + c$ (b) $\frac{-16}{3(x^2+3)^2} + c$ (c) $\frac{16}{3(x^3+2)^2} - c$ (d) None

Q.38. $\int x^2 \times e^{3x}\, dx$

(a) $\frac{e^{3x}}{3}\left[x^2 - \frac{2}{3}(x+1)\right]$ (b) $\frac{e^{3x}}{3}\left[x^2 - \frac{2x}{3} + \frac{2}{9}\right]$

(c) $\frac{e^{3x}}{3}\left[x^2 + \frac{2}{3}(x+1)\right]$ (d) None

Q.39. $\int (e^x + e^{-x})^2.(e^x - e^{-x})dx$

(a) $\frac{(e^x+e^{-x})^3}{3} + c$ (b) $\frac{(e^x+e^{-x})^3}{3}$ (c) $\frac{(e^x-e^{-x})^3}{2} + c$ (d) None

Q.40. $\int \frac{e^x - e^{-x}}{e^x + e^{-x}}\, dx$

(a) $log(e^x - e^{+x})$ (b) $log(e^x + e^{-x}) - c$ (c) $log(e^x + e^{-x})$ (d) None

Q.41. $\int \frac{(1-x)^3}{x} dx$

(a) $logx - \frac{x^3}{3} - 3x + \frac{3x^2}{2} + c$ (b) $logx - \frac{x^3}{3} + 3x - \frac{3x^2}{2} - c$

(c) $logx + \frac{x^3}{3} - 2x + \frac{3x^2}{2} + c$ (d) None

Q.42. $\int_0^4 \sqrt{3x+4}\, dx$

(a) $\frac{-112}{9}$ (b) $\frac{112}{9}$ (c) $\frac{110}{9}$ (d) None

Q.43. $\int_0^2 \frac{x+2}{x+1} dx$

(a) $2 - log3$ (b) $4 + log3$ (c) $2 + log3$ (d) None

Q.44. $\int \frac{x-3}{x+1} dx$

(a) $x - 4\log(x+1) + c$ (b) $x - 2\log(x+1) + c$

(c) $x + 4\log(x+1) - c$ (d) None

Q.45. $\int \frac{x+2}{x-3} dx$

(a) $x + 5log(x+2) + c$ (b) $x + 3log(x+3) + c$

(c) $x + 5log(x-3) + c$ (d) None

Q.46. $\int \frac{1}{(x-2)(x+3)} dx$

(a) $\frac{2}{5} log\left(\frac{x-2}{x+3}\right) - c$ (b) $\frac{1}{5} log\left(\frac{x-2}{x+3}\right) + c$ (c) $\frac{1}{5} log\left(\frac{x+2}{x-3}\right) + c$ (d) None

Q.47. $\int \frac{3}{(x-4)(x+5)} dx$

(a) $\frac{1}{3} log\left(\frac{x+4}{x-5}\right) + c$ (b) $\frac{1}{3} log\left(\frac{x-4}{x+5}\right) + c$ (c) $\frac{1}{2} log\left(\frac{x-4}{x+5}\right) - c$ (d) None

Q.48. $\int \frac{2x}{(x-2)(x-5)} dx$

(a) $\frac{1}{3}[log(x-5)^{10} - log(x-2)^4] + c$

(b) $\frac{1}{2}[log(x-5)^{10} - log(x-2)^4] + c$

(c) $\frac{1}{3}[log(x-5)^{10} + log(x-2)^4] + c$ (d) None

Q.49. $\int \frac{3x}{x^2-x-2} dx$

(a) $2log|x+2| + log|x+1| + c$ (b) $2log|x-2| + log|x-1| - c$

(c) $2log|x-2| + log|x+1| + c$ (d) None

Q.50. $\int_1^2 \frac{x^2-1}{x^2} \times e^{(x+1/x)} dx$

(a) $e^{\frac{5}{2}} - e^2$ (b) $e^{\frac{5}{2}} + e^2$ (c) $e^{\frac{1}{2}} - e^2$ (d) None

Q.51. $\int e^x \left(\frac{xlogx+1}{x}\right) dx$

(a) $e^x.logx$ (b) $e^x.logx + c$ (c) $2.logx + c$ (d) None

Q.52. $\int e^x.\left[\sqrt{x} + \frac{1}{2\sqrt{x}}\right] dx$

(a) $e^x.\sqrt{x} + c$ (b) $e^x.\sqrt{x}$ (c) $2e^x.\sqrt{x} + c$ (d) None

Q.53. $\int \frac{e^x}{(1+x)^3} dx - \int \frac{e^x}{2(1+x)^2} dx$

(a) $\frac{e^x}{2.(1+x)^2}$ (b) $\frac{-e^x}{2.(1-x)^2}$ (c) $\frac{-e^x}{2.(1+x)^2}$ (d) None

Q.54. $\int_0^2 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{2-x}} dx$

(a) – 1 (b) 0 (c) 1 (d) None

Q.55. $\int_0^5 \frac{x^2}{x^2+(5-x)^2} dx$

(a) $\frac{-5}{2}$ (b) $\frac{5}{2}$ (c) $\frac{4}{2}$ (d) None

Q.56. $\int_2^7 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{9-x}} dx$

(a) $\frac{5}{2}$ (b) $-\frac{5}{2}$ (c) $\frac{3}{2}$ (d) None

Q.57. $\int_3^{11} \frac{x^{5/2}}{x^{5/2}+(14-x)^{5/2}} dx$

(a) 0 (b) – 4 (c) 4 (d) None

Q.58. $\int_{-1}^{+1} (e^x - e^{-x}) dx$

(a) 1 (b) 0 (c) – 1 (d) None

Q.59. $\int_{-1}^{+1} \frac{|x|}{x} dx$

(a) 0 (b) 2 (c) – 1 (d) None

Q.60. $\int \frac{e^x(2-x)}{(1-x)^2} dx$

(a) $\frac{2e^x}{(1-x)}$ (b) $\frac{e^x}{(1+x)}$ (c) $\frac{e^x}{(1-x)}$ (d) None

MISCELLANEOUS QUESTIONS 0f INTEGRATION

Q.61. Integrate x^{-11} with respect to x.

(a) $\frac{x^{-9}}{-9}$ (b) $\frac{x^{-10}}{-10}$ (c) $\frac{x^{-11}}{-11}$ (d) $\frac{x^{-12}}{-12}$

Q.62. Integrate $\sqrt{x^5}$ with respect to x.

(a) $\frac{1}{7}x^{1/2}$ (b) $\frac{2}{7}x^{2/7}$ (c) $\frac{2}{7}x^{7/2}$ (d) $\frac{5}{7}x^{7/2}$

Q.63. Integrate $3\sqrt{x^2}$ with respect to x.

(a) $\frac{2}{5}x^{2/5}$ (b) $\frac{2}{5}x^{5/2}$ (c) $\frac{3}{5}x^{3/5}$ (d) $\frac{3}{5}x^{5/3}$

Q.64. Integrate $\int x\sqrt{x}$ dx

(a) $x^{2/5}$ (b) $\frac{2}{5}x^{2/5}$ (c) $\frac{2}{5}x^{5/2}$ (d) None

Q.65. Integrate $\int 3x^2\sqrt{x}$ dx

(a) $\frac{2}{7}x^{2/7}$ (b) $\frac{3}{7}x^{3/7}$ (c) $\frac{6}{7}x^{2/7}$ (d) $\frac{6}{7}x^{7/2}$

Q.66. Integrate $\int(a+x)^2$ dx

(a) $a^2x + \frac{x^2}{2}$ (b) $a^2x + \frac{x^3}{3}$ (c) $ax + (ax)^2$ (d) $a^2x + ax^2 + \frac{x^3}{3}$

Q.67. Integrate $\int e^{-4x}$ dx

(a) e^{-4x} (b) $4e^{-4x}$ (c) $\frac{e^{-4x}}{-4}$ (d) $\frac{e^{4x}}{4}$

Q.68. Integration of $\int e^{2\log x}$ dx is

(a) x/2 (b) x^2/x (c) $x^3/2$ (d) $x^3/3$

Q.69. Integration of $\int 5^{2x}$ dx is :-

(a) $\frac{5^x}{log_e 5}$ (b) $\frac{5^2}{log_e 5}$ (c) $\frac{5^{2x}}{2\ log_e 5}$ (d) $\frac{5^x}{2\ log_e 5}$

Q.70. Integration of $\int 2K\,x\,dx$ is :-

(a) Kx (b) Kx^2 (c) $(Kx)^2$ (d) $2Kx^2$

Q.71. Integration of $\int \frac{1}{x}\left(x+\frac{1}{x}\right)$ dx is :-

(a) $\frac{1}{x} - x$ (b) $x - \frac{1}{x}$ (c) $x + \frac{1}{x}$ (d) $x - \frac{1}{x^2}$

Q.72. $\int\left(x+\frac{1}{x}\right)^2$ dx = ?

(a) $x^3/3 + 1/x + 2x$ (b) $x^3/3 + 1/x - 2x$ (c) $x^3/3 - 1/x + 2x$ (d) 0

Q.73. $\int(a+b)$ dx = ?

(a) (a + b) x (b) (a + b) (c) ax (d) bx

Q.74. Value of $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}$ dx is :-

(a) $2e^{\sqrt{x}}$ (b) $e^{\sqrt{x}}$ (c) $\frac{2}{e^{\sqrt{x}}} + c$ (d) None

Q.75. $\int \frac{e^x}{e^x - 5}$ dx =

(a) log e^x – log 5 + c (b) log e^x – 5 + c (c) log (e^x – 5) + c (d) None

Q.76. $\int \sqrt{x^2 - a^2}$ dx is

(a) $\frac{x}{2}\sqrt{x^2 - a^2} + \frac{a^2}{2}\log\left(x + \sqrt{x^2 - a^2}\right)$ + c

(b) $\frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left(x + \sqrt{x^2 - a^2}\right)$ + c

(c) $\frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left(x - \sqrt{x^2 - a^2}\right)$ + c (d) None

Q.77. Value of $\int \frac{1}{(x-1)(x-2)}$ dx is :-

(a) $\log\left(\frac{x-2}{x-1}\right)$ + c (b) $\log\left(\frac{x-1}{x-2}\right)$ + c

(c) $\frac{1}{2}\log\left(\frac{x-2}{x-1}\right)$ + c (d) None

Q.78. Value of $\int \frac{x^2}{x+2}$ dx is :-

(a) $\frac{x^2}{2} + \frac{x^{-3}}{6}$ + c (b) $\frac{x^2}{2}$ – 2x + 4 log (x + 2) + c

(c) 2x log (x + 2) + c (d) None

Q.79. Integration of (1 – 3x) (1 + x) is :-

(a) $-x + x^2 + x^3$ + c (b) $x - x^2 - x^3$ + c

(c) $x - 2x^2 - 3x^3$ + c (d) None of them.

Q.80. If f '(x) = $3x^2$ + 2 then f (x) is :-

(a) x^3 (b) x^3 + 2x + c (c) 2x (d) 0

Q.81. Value of $\int \frac{1}{x^2 - a^2}$ dx is :-

(a) $\log\left(\frac{x-a}{x+a}\right)$ + c (b) $\frac{1}{a}\log\left(\frac{x-a}{x+a}\right)$ + c

(c) $\frac{1}{2a}\log\left(\frac{x-a}{x+a}\right)$ + c (d) None

Q.82. Value of $\int \frac{5x^4 + 3x^2}{x^5 + x^3 + 1}$ dx is :-

(a) $\log(x^5 + x^3 + 1)$ + c (b) $\log(5x^4 + 3x^2)$ + c

(c) $\log(x^5 + x^3)$ + c (d) None

Q.83. Value of $\int \frac{(1-x)^3}{x}$ dx is :-

(a) log x – 3x + $\frac{3x^2}{2} - \frac{x^3}{3}$ + c (b) $\frac{1}{x}$ – 3 + 3x – x^2 + c

(c) log x – 3x + $3x^2$ – x^3 + c (d) None of them.

Q.84. Value of $\int \log x \, dx$ is:-

(a) $x \log x - x + c$ (b) $1/x$ (c) $x \log x - 1 + c$ (d) None

Q.85. Value of $\int \frac{1}{x \log x} dx$ is :-

(a) log (log x) + c (b) log.x + c (c) $\frac{-1}{x^2}$ log x + c (d) None

Q.86. Value of $\int \left(\frac{x \log x + 1}{x}\right) e^x dx$ is :-

(a) $x \log x. e^x + c$ (b) $\log x. e^x + c$ (c) $e^x + c$ (d) None

Q.87. Value of $\int_1^4 \frac{1}{\sqrt{x}} dx$ is :-

(a) 1 (b) 4 (c) 2 (d) None

Q.88. Value of $\int_{-1}^{1} (x^5 - 3x^3 + 2x) dx$ is :-

(a) 0 (b) 5 (c) -2 (d) None

Q.89. Value of $\int_0^1 \frac{1}{(x+2)(x+3)} dx$ is :-

(a) log 9 (b) log (9/8) (c) log (8/9) (d) None

Q.90. Value of $\int_0^1 \frac{dx}{(1+x)(2+x)} dx$ is :-

(a) log 4/3 (b) log 2/3 (c) log 2/3 (d) None

Q.91. Value of $\int_2^5 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{7-x}} dx$ is :-

(a) 5/2 (b) 7/2 (c) 3/2 (d) None.

Q.92. Value of $\int_0^8 \frac{\sqrt{8-x}}{\sqrt{8-x} + \sqrt{x}} dx$ is

(a) 8 (b) 4 (c) 2 (d) None

Q.93. Value of $\int_1^2 \frac{2x+5}{x^2+5x+1} dx$ is :-

(a) log 15/7 (b) log 15 (c) log 7 (d) None

Answers

DIFFERENTIATION

1	a	11	a	21	a	31	a	41	a	51	a	61	c	71	b	81	c
2	b	12	a	22	b	32	a	42	b	52	b	62	a	72	c	82	a
3	a	13	c	23	a	33	b	43	a	53	c	63	c	73	b	83	c
4	b	14	b	24	a	34	a	44	b	54	b	64	a	74	a	84	a
5	b	15	a	25	a	35	a	45	a	55	a	65	b	75	a	85	a
6	c	16	b	26	b	36	b	46	b	56	c	66	c	76	a	86	b
7	a	17	a	27	a	37	a	47	a	57	b	67	b	77	a		
8	c	18	b	28	b	38	b	48	a	58	a	68	a	78	a		
9	a	19	a	29	c	39	c	49	b	59	c	69	b	79	b		
10	b	20	b	30	b	40	b	50	a	60	b	70	b	80	a		

MISCELLANEOUS QUESTIONS OF DIFFERENTIATION

87	c	97	b	107	b	117	a
88	c	98	a	108	c		
89	b	99	d	109	d		
90	a	100	a	110	a		
91	a	101	b	111	b		
92	d	102	c	112	a		
93	a	103	d	113	b		
94	b	104	c	114	a		
95	a	105	a	115	b		
96	c	106	b	116	a		

INTEGRATION

1	a	11	b	21	a	31	a	41	a	51	b
2	b	12	a	22	b	32	b	42	b	52	a
3	c	13	a	23	a	33	c	43	c	53	c
4	b	14	a	24	a	34	a	44	a	54	c
5	c	15	a	25	c	35	a	45	c	55	b
6	c	16	a	26	a	36	b	46	b	56	a
7	c	17	a	27	b	37	a	47	b	57	c
8	a	18	c	28	b	38	b	48	a	58	b
9	b	19	b	29	a	39	a	49	c	59	a
10	b	20	c	30	b	40	b	50	a	60	c

MISCELLANEOUS QUESTIONS 0f INTEGRATION

61	b	71	b	81	c	91	c
62	c	72	c	82	a	92	b
63	d	73	a	83	a	93	a
64	c	74	a	84	a		
65	d	75	c	85	a		
66	d	76	b	86	b		
67	c	77	a	87	c		
68	d	78	b	88	a		
69	c	79	b	89	b		
70	b	80	b	90	a		

Chapter – 9 Number Series, Coding – Decoding, Odd-One Out Series

Number Series

Q.1. 15, 20, 25, 30, 35, ?
(a) 50 (b) 40 (c) 60 (d) 30

Q.2. 5, 25, 125, 625, 3125, ?
(a) 15625 (b) 15525 (c) 15657 (d) 15565

Q.3. 2, 9, 28, 65, ?
(a) 169 (b) 179 (c) 126 (d) 144

Q.4. 2, 6, 10, 3, 9, 13, 4, 12, ?
(a) 18 (b) 24 (c) 49 (d) 16

Q.5. 3, 9, 27, 4, 16, 64, 5, 25, 125, ?
(a) 5, 25, 125 (b) 6, 36, 216 (c) 4, 64, 128 (d) 3, 81, 169

Q.6. 2, 5, 10, 17, 26, 37, ?, 65
(a) 49 (b) 64 (c) 50 (d) 81

Q.7. 21, 23, 46, 48, 96, 98, ?
(a) 196 (b) 169 (c) 144 (d) 961

Q.8. 8, 9, 125, 49, 1331, ?
(a) 196 (b) 961 (c) 169 (d) None

Q.9. 4, 5, 12, 39, 160, ?
(a) 805 (b) 508 (c) 850 (d) 580

Q.10. 8, 4, 12, 6, 18, ?, 27
(a) 81 (b) 27 (c) 9 (d) 127

Q.11. D, G, J, M, P, ?
(a) K (b) S (c) V (d) Z

Coding – Decoding

Q.12. If the word "PARTNER" is coded as "OZQSMDQ", Then what is the code for the word "SEGMENT" in that Language?
(a) SDFNDMS (b) RDFLDMS (c) RWFLDMC (d) BDFPDMS

Q.13. If the word SPHERE is coded as EREHPS, Then how is the word "EXHIBITION" Coded as:-
(a) MOITIBIHRE (b) NPITIBIQXE (c) UOITIBIHXB (d) NOITIBIHXE

Q.14. If LOVE is coded as 5782 and JOCKEY is coded as 371429, what do figure 4295714 stands for?
(a) KEYLOCK (b) MEYLOCK (c) WEYLOCK (d) QEYLOCK

Q.15. If "EXCEL" is coded as 93596 and PAINT is coded as 74128, then "ANCIENT" will be coded as ?
(a) 3251926 (b) 1251928 (c) 4251928 (d) 4251988

Odd-One Out

Choose Out the Odd One:-

Q.16. 4, 9, 16, 36, 64, 100.
(a) 9 (b) 36 (c) 64 (d) 4

Q.17. 11, 13, 15, 17, 19
(a) 17 (b) 11 (c) 15 (d) 13

Q.18. 121, 144, 169, 196, 200, 225
(a) 200 (b) 144 (c) 121 (d) 169

Q.19. 1, 3, 4, 7, 11, 18, 27, 47
(a) 11 (b) 1 (c) 4 (d) 27

Q.20. 23, 35, 57, 79
(a) 23 (b) 79 (c) 35 (d) 57

Q.21. 24, 46, 68, 810, 1011
(a) 24 (b) 1011 (c) 68 (d) 46

Q.22. 5, 8, 12, 16, 24, 30
(a) 16 (b) 30 (c) 8 (d) 5

Q.23. 263, 482, 551, 362, 285
(a) 482 (b) 263 (c) 285 (d) 551

Q.24. December, February, March, July.
(a) February (b) March (c) July (d) December

Q.25. Metre, Mile, Yard, Acre.
(a) Mile (b) Acre (c) Metre (d) Yard

Q.26. Mother, Friend, Sisters, Father.
(a) Mother (b) Sisters (c) Friend (d) Father

Q.27. Ear, Lung, Eye, Heart.
(a) Ear (b) Heart (c) Lung (d) Eye

Q.28. Ear, Nose, Tongue, Throat.
(a) Ear (b) Nose (c) Tongue (d) Throat

Q.29. Curd, Butter, Oil, Cheese.
(a) Oil (b) Curd (c) Butter (d) Cheese

Q.30. Carrot, Potato, Tomato, Ginger
(a) Carrot (b) Potato (c) Tomato (d) Ginger

Q.31. Gold, Silver, Brass, Platinum.
(a) Brass (b) Gold (c) Silver (d) Platinum

Q.32. Tiger, Cow, Cheetah, Leopard.
(a) Tiger (b) Cheetah (c) Cow (d) Leopard

Q.33. House, Cottage, School, Palace.
(a) House (b) Cottage (c) Palace (d) School

Q.34. Mercury, Moon, Jupiter, Saturn.
(a) Moon (b) Mercury (c) Jupiter (d) Saturn

Q.35. Manipur, Sikkim, MP, Lakshadweep.
(a) Sikkim (b) Lakshadweep (c) Manipur (d) MP

Q.36. V.V. Giri, Dr. Rajendra Prasad, Dr. S. Radha Krishanan, Lal Bahadur sastri.
(a) Lal Bahadur Sastri (b) Dr. S. Radha Krishanan
(c) VV. Giri (d) Dr. Rajendra Prasad

Q.37. Listen, Swim, Walk, Climb.
(a) Walk (b) Swim (c) Listen (d) Climb

Q.38. Green, Saffron, White, Orange.
(a) Orange (b) Saffron (c) White (d) Green

Q.39. Rose, Lotus, Marigold, Lily
(a) Rose (b) Marigold (c) Lily (d) Lotus

Q.40. X-Ray, T.V., Radio, Telephone,
(a) X-Ray (b) Radio (c) T.V. (d) Telephone

Q.41. Geometry, Algebra, Arithmetic, Mathematics,
(a) Geometry (b) Algebra (c) Mathematics (d) Arithmetic

Q.42. Brick, Heart, Spade, Club.
(a) Brick (b) Heart (c) Club (d) Spade

Q.43. Broker, Salesman, Customer, Hawker
(a) Customer (b) Broker (c) Salesman (d) Hawker

Q.44. Table, Cupboard, Chair, Paper Weight.
(a) Table (b) Chair (c) Paper Weight (d) Cupboard

Q.45. Cotton, Jute, Silk , Nylon
(a) Cotton (b) Nylon (c) Jute (d) Silk

Answers

Number Series

1	b	11	b
2	a		
3	c		
4	d		
5	b		
6	c		
7	a		
8	c		
9	a		
10	c		

Coding – Decoding

12	b
13	d
14	a
15	c

Odd-One Out

16	a	26	c	36	a
17	c	27	b	37	c
18	a	28	d	38	a
19	d	29	a	39	d
20	b	30	c	40	a
21	b	31	a	41	c
22	a	32	c	42	a
23	c	33	d	43	a
24	a	34	a	44	c
25	b	35	b	45	b

CHAPTER 10 DIRECTION SENSE TEST

Q.1. Ram starts from his house and walks 3 km towards North, turns towards his right and walks 2 km, turns right again and walks 3 km. What is the direction now he is facing?
(a) North (b) South (c) East (d) West

Q.2. In the preceding problem in which direction is he from the starting point?
(a) North (b) South (c) East (d) West

Q.3. Ram starts from his house and walks 3 km towards West, turns towards his right and walks 2 km, turns left and walks 3 km and finally turns left again and walks 2 km. What is the direction now he is facing?
(a) North (b) South (c) East (d) West

Q.4. Ram started walking 3 km straight from his school. Then he turned right and walked 2 km. Again he turned right and walked 1 km to reach his house. If his house is sourth-east from his school, then in which direction did Ram start walking from the school?
(a) East (b) West (c) South (d) North

Q.5. Ram starts from his house and walks 3 km towards West, turns towards his right and walks 2 km, thereafter he turns left and walks 3 km. Further he turns left and walks 2 km. Finally, he turns towards his right and walks 3 km. How far is Ram from his house?
(a) 4 km (b) 6 km (c) 8 km (d) 9 km

Q.6. Ram walks 10 km towards North. Then he walks 6 km towards South. From there he moves 3 km towards East. How far and in which direction is he with reference to his starting point?
(a) 4 km N-E (b) 5 km N-E (c) 6 km S-W (d) 8 km S-E

Q.7. On a playground Devang, Kunjal, Nilesh, Aakash and Pinku are standing as directed below facing the north.
(i) Kunjal is 60 m to the right of Aakash (ii) Devang is 50 m to the South of Kunjal (iii) Nilesh is 25 m to the West of Aakash (iv) Pinku is 90 m to the north of Devang. *Who is the N-E of the person who is to the left of Kunjal.*
(a) Nilesh (b) Devang (c) Aakash (d) Pinku

Q.8. A child is looking for his father. He went 90 m in the east before turning to his right. He went 20 m before turning to is right again to look for his father at his uncle's place 30 m from this point. His father was not there. From there, he went 100 m to his north before meeting his father in a street. How far did the son meet his father from starting point?
(a) 60 m (b) 90 m (c) 100 m (d) None of these

Q.9. Johnson left for his office in his car. He drove 15 km toward north and then 10 km toward west. He then turned to the south and covered 5 km. Further, he turned to the east and moved 8 km. lastly, he turned right and drove 10 km. How far and in which direction is he from his starting point?
(a) 2 km West (b) 5 km East (c) 10 km North (d) None of these

Q.10. A man is facing West. He turns 45 degree in the clockwise direction and then another 180 degree in the same direction and then 270 degree in the anticlockwise direction.
Find which direction he is facing now?
(a) South-East (b) West (c) South (d) South-West

Q.11. P, Q, R and S are playing cards, P and Q are partners. S faces toward North. If P faces toward West, then who faces toward South?
(a) Q (b) R (c) S (d) P

Answers

1.	b
2.	c
3.	b
4.	a
5.	d
6.	a
7.	d
8.	c
9.	a
10.	d
11.	b



CHAPTER – 11 SEATING ARRANGEMENTS

Q.01. Friends are sitting on a bench. A is to the left of B but on the right of C, D is to the right of B but one the left of E. Who are at the extremes?
(a) A, B (b) A, D (c) C, E (d) B, D

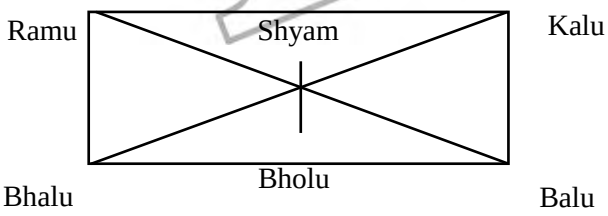
Q.02. Four Children are sitting in arrow. A is occupying seat next to B but not next to C. If C is not sitting next to D? Who is occupying seat next to adjacent to D.
(a) B (b) B and A (c) Impossible to tell (d) A

Q.03. Six persons M, N, O, P, Q and R are sitting in two row with three persons in each row. Both the row are in front of each other. Q is not at the end of any row. P is second the left of R. O is the neighbor of Q and diagonally opposite to P. N is the neighbor of R. Who are the neighbors of Q?
(a) M and O (b) M and N (c) P and R (d) None

Q.04. Four ladies & A, B, C and D and Four men E, F, G and H are sitting in a circle around a table facing each other.
I. No tow ladies or men are sitting side by side. II. C, who is sitting between G and E, is facing D.
III. F is between D and A and facing G. IV. H is to the right of B.
(a) (b) (c) (d)

Q.05. Five senior citizens are living in a multi-storeyed building. A lives in a flat above B, C in a flat below D, B lives in a flat below D and E lives in a flat below C. Who lives in the topmost flat?
(a) C (b) D (c) A (d) E

Q.06. Consider the following seating arrangement and answer the questions 1-5.
(1) Who is sitting opposite to Bhalu? (2) Who is sitting opposite to Bholu?
(3) Who is sitting opposite to Balu? (4) Who is sitting diagonally opposite to Kalu?
(5) Who is sitting diagonally opposite to Ramu?



Q.07. Eight books are kept one over the other. Counting from the top, the second fifth and sixth books are on Plays. Two books on Plays are between two books on Composition. One book of Plays is between two books on Poetry while the book at the top of the book of Literature is a book of Composition. Which books is fourth from the top?
(a) Plays (b) Composition (c) poetry (d) Literature

Q.08. Seven persons A, B, C, D, E, F and G are sitting around a circle and are facing the center, G is second to the left of C who is to the immediate left of F and A is third to the left of E finally B is between D and E.
On the basis of these information answer the following questions:
(1) Who is fourth to the right of B? (2) Who is immediate right of G?
(3) Which person is sitting between B and F? (4) Who is at the fourth position to the right of D?

Answers

1.	c
2.	d
3.	a
4.	
5.	c
6.	
7.	b
8.	

CHAPTER – 12 BLOOD RELATIONS

Q.01. A is the sister of B. B is the brother of C. C is the son of D. How is D related to A?
(a) Mother/Father (b) Daughter (c) Son (d) Uncle

Q.02. F is the brother of A. G is the daughter of A. K is the sister of F. G is the sister of C. Who is uncle of C ?
(a) A (b) C (c) K (d) F

Q.03. In the preceding problem, Who is the aunt of G?
(a) K (b) G (c) A (d) F

Q.04. E is the sister of B. A is the father of C. B is the son of C. How is A related of E?
(a) Grandfather (b) Grand-daughter (c) Father (d) Great-grandfather

Q.05. A is D' brother. D is B's father. B and C are sisters. How is A related to C?
(a) Son (b) Niece (c) Father (d) Uncle

Q.06. A is the mother of B. C is the son of A. D is the brother of E. E is the daughter of B. who is grandmother of D?
(a) A (b) B (c) C (d) D

Q.07. In the preceding problem, how E and D are related to C?
(a) (b) (c) (d)

Q.08. C is wife of B. E is the son of C. A is the brother of B and father of D. What is the relationship of E to D?
(a) Mother (b) Sister (c) Brother (d) Cousin

Q.09. In the preceding problem, how is C related of A?
(a) (b) (c) (d)

Q.10. A is father of C and D is son of B. E is brother of A. If C is sister of D, how is B related to E?
(a) Sister-in-law (b) Sister (c) Brother (d) Brother-in-law

Q.11. Annanya is mother of Satya and Shyam is the son of Bhima. Shiva is brother of Annanya. If Satya is sister of Shyam. How Bhima is related to Shiva?
(a)Son (b) Cousin (c) Brother-in-law (d) Son-in-law

Q.12. Anjali introduces Rohini as the daughter of the only brother of his father's wife. How is Rohini related to Anjali?
(a) (b) (c) (d)

Q.13. Pointing toward a person in a photograph, Roshni said, "He is the only son of the father of my sister's brother." How is that person related to Roshni?
(a) (b) (c) (d)

Q.14. A and B are brothers. C and D are sisters. A's son is D's brother. How is B related to C?
(a) (b) (c) (d)

Q.15. If X + Y means that X is the mother of Y, X ÷ Y means that X is the father of Y, X – Y means that X is the sister of Y. Then, which of the following relationship shows that Q is the – daughter of R?
(a) R ÷ Q + P (b) R + P ÷ Q (c) R – Q ÷ P (d) None of these

Answers

1.	A	11.	C
2.	D	12.	
3.	A	13.	
4.	A	14.	
5.	D	15.	
6.	A		
7.			
8.	d		
9.			
10.	a		

UNIT 2 SAMPLING

LEARNING OBJECTIVES

After reading this unit a student will learn -

- Different procedure of sampling which will be the best representative of the population;

13.2.1 INTRODUCTION

There are situations when we would like to know about a vast, infinite universe or population. But some important factors like time, cost, efficiency, vastness of the population make it almost impossible to go for a complete enumeration of all the units constituting the population. Instead, we take recourse to selecting a representative part of the population and infer about the unknown universe on the basis of our knowledge from the known sample. A somewhat clear picture would emerge out if we consider the following cases.

In the first example let us share the problem faced by Mr. Basu. Mr. Basu would like to put a big order for electrical lamps produced by Mr. Ahuja's company "General Electricals". But before putting the order, he must know whether the claim made by Mr. Ahuja that the lamps of General Electricals last for at least 1500 hours is justified.

Miss Manju Bedi is a well-known social activist. Of late, she has noticed that the incidence of a particular disease in her area is on the rise. She claims that twenty per cent of the people in her town have been suffering from the disease.

In both the situations, we are faced with three different types of problems. The first problem is how to draw a representative sample from the population of electrical lamps in the first case and from the population of human beings in her town in the second case. The second problem is to estimate the population parameters i.e., the average life of all the bulbs produced by General Electricals and the proportion of people suffering form the disease in the first and second examples respectively on the basis of sample observations. The third problem relates to decision making i.e., is there enough evidence, once again on the basis of sample observations, to suggest that the claims made by Mr. Ahuja or Miss Bedi are justifiable so that Mr. Basu can take a decision about buying the lamps from General Electricals in the first case and some effective steps can be taken in the second example with a view to reducing the outbreak of the disease. We consider tests of significance or tests of hypothesis before decision making.

13.2.2 BASIC PRINCIPLES OF SAMPLE SURVEY

Sample Survey is the study of the unknown population on the basis of a proper representative sample drawn from it. How can a part of the universe reveal the characteristics of the unknown universe? The answer to this question lies in the basic principles of sample survey comprising the following components:

(a) Law of Statistical regularity

(b) Principle of Inertia

(c) Principle of Optimization

(d) Principle of Validity

(a) According to the law of statistical regularity, if a sample of fairly large size is drawn from the population under discussion at random, then on an average the sample would posses the characteristics of that population.

Thus the sample, to be taken from the population, should be moderately large. In fact larger the sample size, the better in revealing the identity of the population. The reliability of a statistic in estimating a population characteristics varies as the square root of the sample size. However, it is not always possible to increase the sample size as it would put an extra burden on the available resource. We make a compromise on the sample size in accordance with some factors like cost, time, efficiency etc.

Apart from the sample size, the sample should be drawn at random from the population which means that each and every unit of the population should have a pre-assigned probability to belong to the sample.

(b) The results derived from a sample, according to the principle of inertia of large numbers, are likely to be more reliable, accurate and precise as the sample size increases, provided other factors are kept constant. This is a direct consequence of the first principle.

(c) The principle of optimization ensures that an optimum level of efficiency at a minimum cost or the maximum efficiency at a given level of cost can be achieved with the selection of an appropriate sampling design.

(d) The principle of validity states that a sampling design is valid only if it is possible to obtain valid estimates and valid tests about population parameters. Only a probability sampling ensures this validity.

13.2.3 COMPARISON BETWEEN SAMPLE SURVEY AND COMPLETE ENUMERATION

When complete information is collected for all the units belonging to a population, it is defined as complete enumeration or census. In most cases, we prefer sample survey to complete enumeration due to the following factors:

(a) **Speed:** As compared to census, a sample survey could be conducted, usually, much more quickly simply because in sample survey, only a part of the vast population is enumerated.

(b) **Cost:** The cost of collection of data on each unit in case of sample survey is likely to be more as compared to census because better trained personnel are employed for conducting a sample survey. But when it comes to total cost, sample survey is likely to be less expensive as only some selected units are considered in a sample survey.

(c) **Reliability:** The data collected in a sample survey are likely to be more reliable than that in a complete enumeration because of trained enumerators better supervision and application of modern technique.

(d) **Accuracy:** Every sampling is subjected to what is known as sampling fluctuation which is termed as sampling error. It is obvious that complete enumeration is totally free from this sampling error. However, errors due to recording observations, biases on the part of the enumerators, wrong and faulty interpretation of data etc. are prevalent in both sampling and census and this type of error is termed as non-sampling errors. It may be noted that in sample survey, the sampling error can be reduced to a great extent by taking several steps like increasing the sample size, adhering to a probability sampling design strictly and so on. The non-sampling errors also can be contained to a desirable degree by a proper planning which is not possible or feasible in case of complete enumeration.

(e) **Necessity:** Sometimes, sampling becomes necessity. When it comes to destructive sampling where the items get exhausted like testing the length of life of electrical bulbs or sampling from a hypothetical population like coin tossing, there is no alternative to sample survey.

However, when it is necessary to get detailed information about each and every item constituting the population, we go for complete enumeration. If the population size is not large, there is hardly any merit to take recourse to sampling. If the occurrence of just one defect may lead to a complete destruction of the process as in an aircraft, we must go for complete enumeration.

13.2.4 ERRORS IN SAMPLE SURVEY

Errors or biases in a survey may be defined as the deviation between the value of population parameter as obtained from a sample and its observed value. Errors are of two types.

I. Sampling Errors

II. Non-Sampling Errors

Sampling Errors : Since only a part of the population is investigated in a sampling, every sampling design is subjected to this type of errors. The factors contributing to sampling errors are listed below:

(a) Errors arising out due to defective sampling design: Selection of a proper sampling design plays a crucial role in sampling. If a non- probabilistic sampling design is followed, the bias or prejudice of the sampler affects the sampling technique thereby resulting some kind of error.

(b) Errors arising out due to substitution: A very common practice among the enumerators is to replace a sampling unit by a suitable unit in accordance with their convenience when difficulty arises in getting information from the originally selected unit. Since the sampling design is not strictly adhered to, this results in some type of bias.

(c) Errors owing to faulty demarcation of units: It has its origin in faulty demarcation of sampling units. In case of an agricultural survey, the sampler has, usually, a tendency to underestimate or overestimate the character under consideration.

(d) Errors owing to wrong choice of statistic: One must be careful in selecting the proper statistic while estimating a population characteristic.

(e) Variability in the population: Errors may occur due to variability among population units beyond a degree. This could be reduced by following somewhat complicated sampling design like stratified sampling, Multistage sampling etc.

Non-sampling Errors

As discussed earlier, this type of errors happen both in sampling and complete enumeration. Some factors responsible for this particular kind of biases are lapse of memory, preference for certain digits, ignorance, psychological factors like vanity, non- responses on the part of the interviewees wrong measurements of the sampling units, communication gap between the interviewers and the interviewees, incomplete coverage etc. on the part of the enumerators also lead to non-sampling errors.

13.2.5 SOME IMPORTANT TERMS ASSOCIATED WITH SAMPLING

Population or Universe

It may be defined as the aggregate of all the units under consideration. All the lamps produced by "General Electricals" in our first example in the past, present and future constitute the population. In the second example, all the people living in the town of Miss Manju form the population. The number of units belonging to a population is known as population size. If there are one lakh people in her town then the population size, to be denoted by N, is 1 lakh.

A population may be finite or infinite. If a population comprises only a finite number of units, then it is known as a finite population. The population in the second example is obviously, finite. If the population contains an infinite or uncountable number of units, then it is known as an infinite population. The population of electrical lamps of General Electricals is infinite. Similarly, the population of stars, the population of mosquitoes in Kolkata, the population of flowers in Mumbai, the population of insects in Delhi etc. are infinite population.

Population may also be regarded as existent or hypothetical. A population consisting of real objects is known as an existent population. The population of the lamps produced by General Electricals and the population of Miss Manju's town are example of existent populations. A population that exists just hypothetically like the population of heads when a coin is tossed infinitely is known as a hypothetical or an imaginary population.

Sample

A sample may be defined as a part of a population so selected with a view to representing the population in all its characteristics selection of a proper representative sample is pretty important because statistical inferences about the population are drawn only on the basis of the sample observations. If a sample contains n units, then n is known as sample size. If a sample of 500 electrical lamps is taken from the production process of General Electricals, then n = 500. The units forming the sample are known as "Sampling Units". In the first example, the sampling unit is electrical lamp and in the second example, it is a human. A detailed and complete list of all the sampling units is known as a "Sampling Frame". Before drawing sample, it is a must to have a updated sampling frame complete in all respects before the samples are actually drawn.

Parameter

A parameter may be defined as a characteristic of a population based on all the units of the population. Statistical inferences are drawn about population parameters based on the sample observations drawn from that population. In the first example, we are interested about the parameter "Population Mean". If x a denotes the a th member of the population, then population mean m, which represents the average length of life of all the lamps produced by General Electricals is given by

$$\mu = \frac{\sum_{a=1}^{n} x_a}{N} \qquad (13.2.1)$$

Where N denotes the population size i.e. the total number of lamps produced by the company. In the second example, we are concerned about the population proportion P, representing the ratio of the people suffering from the disease to the total number of people in the town. Thus if there are X people possessing this attribute i.e. suffering from the disease, then we have

$$P = \frac{X}{N} \qquad (13.2.2)$$

Another important parameter namely the population variance, to be denoted by s^2 is given by

$$\sigma^2 = \frac{\sum (X_a - \mu)^2}{N} \qquad (13.2.3)$$

$$\text{Also we have SD} = \sigma = \sqrt{\frac{\sum (X_a - \mu)^2}{N}} \qquad (13.2.4)$$

Statistics

A statistic may be defined as a statistical measure of sample observation and as such it is a function of sample observations. If the sample observations are denoted by $x_1, x_2, x_3, \ldots\ldots\ldots x_n$, then a statistic T may be expressed as $T = f(x_1, x_2, x_3, \ldots\ldots\ldots x_n)$

A statistic is used to estimate a particular population parameter. The estimates of population mean, variance and population proportion are given by

$$\bar{x} = \hat{\mu} = \frac{\sum x_i}{n} \qquad (13.2.5)$$

$$S_2 = \hat{\sigma}^2 = \frac{\sum \left(x_i - \bar{x}\right)^2}{n} \qquad (13.2.6)$$

$$\text{and } p = P = \hat{P} = \frac{x}{n} \qquad (13.2.7)$$

Where x, in the last case, denotes the number of units in the sample in possession of the attribute under discussion.

Sampling Distribution and Standard Error of a Statistic

Starting with a population of N units, we can draw many a sample of a fixed size n. In case of sampling with replacement, the total number of samples that can be drawn is and when it comes to sampling without replacement of the sampling units, the total number of samples that can be drawn is ${}^{N}c_{n}$.

If we compute the value of a statistic, say mean, it is quite natural that the value of the sample mean may vary from sample to sample as the sampling units of one sample may be different from that of another sample. The variation in the values of a statistic is termed as "Sampling Fluctuations".

If it is possible to obtain the values of a statistic (T) from all the possible samples of a fixed sample size along with the corresponding probabilities, then we can arrange the values of the statistic, which is to be treated as a random variable, in the form of a probability distribution. Such a probability distribution is known as the sampling distribution of the statistic. The sampling distribution, just like a theoretical probability distribution possesses different characteristics. The mean of the statistic, as obtained from its sampling distribution, is known as "Expectation" and the standard deviation of the statistic T is known as the "Standard Error (SE)" of T. SE can be regarded as a measure of precision achieved by sampling. SE is inversely proportional to the square root of sample size. It can be shown that

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} \text{ for SRS WR}$$

$$= \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}} \qquad \text{for SRS WOR} \qquad (13.2.8)$$

Standard Error for Proporation

$$SE(p) = \sqrt{\frac{Pq}{n}} \qquad \text{for SRS WR}$$

$$\sqrt{\frac{Pq}{n}} \sqrt{\frac{N-n}{N-1}} \qquad \text{for SRS WOR} \qquad (13.2.9)$$

SRSWR and SRSWOR stand for simple random sampling with replacement and simple random sampling without replacement.

The factor $\sqrt{\frac{N-n}{N-1}}$ is known as finite population correction (fpc) or finite population multiplier and may be ignored as it tends to 1 if the sample size (n) is very large or the population under consideration is infinite when the parameters are unknown, they may be replaced by the corresponding statistic.

Illustrations

Example 13.2.1: A population comprises the following units: a, b, c, d, e. Draw all possible samples of size three without replacement.

Solution: Since in this case, sample size (n) = 3 and population size (N) = 5. the total number of possible samples without replacement = ${}^5c_3 = 10$

These are abc, abd, abe, acd, ace, ade, bcd, bce,bde,cde.

Example 13.2.2: A population comprises 3 member 1, 5, 3. Draw all possible samples of size two

(i) with replacement

(ii) without replacement

Find the sampling distribution of sample mean in both cases.

Solution: (i) With replacement :- Since n = 2 and N = 3, the total number of possible samples of size 2 with replacement = $3^2 = 9$.

These are exhibited along with the corresponding sample mean in table 15.1. Table 15.2 shows the sampling distribution of sample mean i.e., the probability distribution of $\bar{x}$.

Table 13.2.1

All possible samples of size 2 from a population comprising 3 units under WR scheme

Serial No.	Sample of size 2 with replacement	Sample mean ($\bar{x}$)
1	1, 1	1
2	1, 5	3
3	1, 3	2
4	5, 1	3
5	5, 5	5
6	5, 3	4
7	3, 1	2
8	3, 5	4
9	3, 3	3

Table 13.2.2

Sampling distribution of sample mean

$\bar{x}$	1	2	3	4	5	Total
P	1 / 9	2 / 9	3 / 9	2 / 9	1 / 9	1

(ii) without replacement: As N = 3 and n = 2, the total number of possible samples without replacement = ${}^NC_2 = {}^3C_2 = 3$.

Table 13.2.3

Possible samples of size 2 from a population of 3 units under WOR scheme

Serial No	Sample of size 2 without replacement	Sample mean ($\bar{x}$)
1	1 , 3	2
2	1 , 5	3
3	3 , 5	4

Table 13.2.4

Sampling distribution of mean

:	2	3	4	Total
P:	1 / 3	1/3	1/3	1

Example 13.2.3: Compute the standard deviation of sample mean for the last problem. Obtain the SE of sample mean applying 15.8 and show that they are equal.

Solution: We consider the following cases:

(i) with replacement :

Let U = $\bar{x}$ The sampling distribution of U is given by

U:	1	2	3	4	5
P:	1/9	2/9	3/9	2/9	1/9

$E(U) = \Sigma P_i U_i$

$= 1/9\times1 + 2/9\times2 + 3/9\times3 + 2/9\times4 + 1/9\times5$

$= 3$

$E(U\,2) = \Sigma P_i U_i^2$

$= 1/9\times1^2 + 2/9\times2^2 + 3/9\times3^2 + 2/9\times4^2 + 1/9\times5^2$

$= 31/3$

$\therefore v(\bar{x}) = v(U) = E(U^2) - [E(U)]^2$

$= 31/3 - 3^2$

$= 4/3$

Hence $SE = \frac{2}{\sqrt{3}}$ (1)

Since the population comprises 3 units, namely 1, 5, and 3 we may take $X_1 = 1, X_2 = 5, X_3 = 3$

The population mean (m) is given by

$$\mu = \frac{\sum X_a}{N}$$

$$= \frac{1+5+3}{3} = 3$$

and the population variance $\sigma^2 = \frac{\sum (X_a - \mu)^2}{N}$

$$\frac{(1-3)^2 + (5-3)^2 + (3-3)^2}{3} = 8/3$$

Applying 15.8 we have, $SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{3}} \times \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{3}}$ (2)

Thus comparing (1) and (2), we are able to verify the validity of the formula.

(ii) without replacement :

In this case, the sampling distribution of V = is given by

V:	2	3	4
P:	1/3	1/3	1/3

$E(\bar{x}) = E(V) = 1/3 \times 2 + 1/3 \times 3 + 1/3 \times 4$

$= 3$

$V(\bar{x}) = Var(V) = E(v^2) - [E(v)]^2$

$= 1/3 \times 2^2 + 1/3 \times 3^2 + 1/3 \times 4^2 - 3^2$

$= 29/3 - 9$

$= 2/3$

$\therefore SE_{\bar{x}} = \frac{2}{\sqrt{3}}$

Applying 13.2.8, we have

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$

$$= \frac{8}{\sqrt{3}} \times \frac{1}{\sqrt{2}} \times \frac{8}{\sqrt{3}} \times \frac{1}{\sqrt{2}} \times \sqrt{\frac{3-2}{3-1}} = \frac{2}{\sqrt{3}}$$

and thereby, we make the same conclusion as in the previous case.

13.2.6 TYPES OF SAMPLING

There are three different types of sampling which are

I. Probability Sampling

II. Non – Probability Sampling

III. Mixed Sampling

In the first type of sampling there is always a fixed, pre assigned probability for each member of the population to be a part of the sample taken from that population . When each member of the population has an equal chance to belong to the sample, the sampling scheme is known as Simple Random Sampling. Some important probability sampling other than simple random sampling are stratified sampling, Multi Stage sampling, Multi Phase Sampling, Cluster Sampling and so on. In non- probability sampling , no probability attached to the member of the population and as such it is based entirely on the judgement of the sampler. Non-probability sampling is also known as Purposive or Judgement Sampling. Mixed sampling is based partly on some probabilistic law and partly on some pre decided rule. Systematic sampling belongs to this category. Some important and commonly used sampling process are described now.

Simple Random Sampling (SRS)

When the units are selected independent of each other in such a way that each unit belonging to the population has an equal chance of being a part of the sample, the sampling is known as Simple random sampling or just random sampling. If the units are drawn one by one and each unit after selection is returned to the population before the next unit is being drawn so that the composition of the original population remains unchanged at any stage of the sampling then the sampling procedure is known as Simple Random Sampling with replacement. If, however, once the units selected from the population one by one are never returned to the population before the next drawing is made, then the sampling is known as sampling without replacement. The two sampling methods become almost identical if the population is infinite i.e. vary large or a very large sample is taken from the population. The best method of drawing simple random sample is to use random sampling numbers.

Simple random sampling is a very simple and effective method of drawing samples provided (i) the population is not very large (ii) the sample size is not very small and (iii) the population under consideration is not heterogeneous i.e. there is not much variability among the members forming the population. Simple random sampling is completely free from Sampler's biases. All the tests of significance are based on the concept of simple random sampling.

Stratified Sampling

If the population is large and heterogeneous, then we consider a somewhat, complicated sampling design known as stratified sampling which comprises dividing the population into a number of strata or sub-populations in such a way that there should be very little variations among the units comprising a stratum and maximum variation should occur among the different strata. The stratified sample consists of a number of sub samples, one from each stratum. Different sampling scheme may be applied to different strata and , in particular, if simple random sampling

is applied for drawing units from all the strata, the sampling procedure is known as stratified random sampling. The purpose of stratified sampling are (i) to make representation of all the sub populations (ii) to provide an estimate of parameter not only for all the strata but also and overall estimate (iii) reduction of variability and thereby an increase in precision.

There are two types of allocation of sample size. When there is prior information that there is not much variation between the strata variances. We consider "Proportional allocation" or "Bowely's allocation where the sample sizes for different strata are taken as proportional to the population sizes. When the strata-variances differ significantly among themselves, we take recourse to "Neyman's allocation" where sample size vary jointly with population size and population standard deviation i.e. $n_i \mu N_i S_i$. Here n_i denotes the sample size for the ith stratum, N_i and S_i being the corresponding population size and population standard deviation. In case of Bowley's allocation, we have $n_i \mu N_i$.

Stratified sampling is not advisable if (i) the population is not large (ii) some prior information is not available and (iii) there is not much heterogeneity among the units of population.

Multi Stage Sampling

In this type of complicated sampling, the population is supposed to compose of first stage sampling units, each of which in its turn is supposed to compose of second stage sampling units, each of which again in its turn is supposed to compose of third stage sampling units and so on till we reach the ultimate sampling unit.

Sampling also, in this type of sampling design, is carried out through stages. Firstly, only a number of first stage units is selected. For each of the selected first stage sampling units, a number of second stage sampling units is selected. The process is carried out until we select the ultimate sampling units. As an example of multi stage sampling, in order to find the extent of unemployment in India, we may take state, district, police station and household as the first stage, second stage, third stage and ultimate sampling units respectively.

The coverage in case of multistage sampling is quite large. It also saves computational labour and is cost-effective. It adds flexibility into the sampling process which is lacking in other sampling schemes. However, compared to stratified sampling, multistage sampling is likely to be less accurate.

Systematic Sampling

It refers to a sampling scheme where the units constituting the sample are selected at regular interval after selecting the very first unit at random i.e., with equal probability. Systematic sampling is partly probability sampling in the sense that the first unit of the systematic sample is selected probabilistically and partly non- probability sampling in the sense that the remaining units of the sample are selected according to a fixed rule which is non-probabilistic in nature.

If the population size N is a multiple of the sample size n i.e. N = nk, for a positive integer k which must be less than n, then the systematic sampling comprises selecting one of the first k units at random, usually by using random sampling number and thereby selecting every kth unit till the complete, adequate and updated sampling frame comprising all the members of the population is exhausted. This type of systematic sampling is known as "linear systematic sampling ". K is known as "sample interval".

However, if N is not a multiple of n, then we may write $N = nk + p$, $p < k$ and as before, we select the first unit from 1 to k by using random sampling number and thereafter selecting every kth unit in a cyclic order till we get the sample of the required size n. This type of systematic sampling is known as "circular systematic sampling."

Systematic sampling is a very convenient method of sampling when a complete and updated sampling frame is available. It is less time consuming, less expensive and simple as compared to the other methods of sampling. However, systematic sampling has a severe drawback. If there is an unknown and undetected periodicity in the sampling frame and the sampling interval is a multiple of that period, then we are going to get a most biased sample, which, by no stretch of imagination, can represent the population under investigation. Furthermore, since it is not a probability sampling, no statistical inference can be drawn about population parameter.

Purposive or Judgement sampling

This type of sampling is dependent solely on the discretion of the sampler and he applies his own judgement based on his belief, prejudice, whims and interest to select the sample. Since this type of sampling is non-probabilistic, it is purely subjective and, as such, varies from person to person. No statistical hypothesis can be tested on the basis of a purposive sampling.

UNIT II EXERCISE

Set A

Answer the following questions. Each question carries one mark.

1. Sampling can be described as a statistical procedure
   (a) To infer about the unknown universe from a knowledge of any sample
   (b) To infer about the known universe from a knowledge of a sample drawn from it
   (c) To infer about the unknown universe from a knowledge of a random sample drawn from it
   (d) Both (a) and (b).
2. The Law of Statistical Regularity says that
   (a) Sample drawn from the population under discussion possesses the characteristics of the population
   (b) A large sample drawn at random from the population would posses the characteristics of the population
   (c) A large sample drawn at random from the population would possess the characteristics of the population on an average
   (d) An optimum level of efficiency can be attained at a minimum cost.
3. A sample survey is prone to
   (a) Sampling errors (b) Non-sampling errors
   (c) Either (a) or (b) (d) Both (a) and (b)

Latin ward → Status
Italian ward → Statista
German word → Statistik
French word → Statistique

Note-1 Chandragupta Maurya
↓
Chankaya (kautilya)
Record of Death and birth (Aarthshashtra)

Note-2 Akbar
↓
Abu fazal
↓
Agriculture record (Ain-e-Akbari)

Note-3 Egypt
↓
Pharaoh
↓
Census record

Statistics

(i) Collection of data Primary Data
Secondary Data
(ii) Scrutiny of Data
(iii) Classification of Data
(iv) Presentation / Tabulation of Data
(v) Calculation of Data
(vi) Conclusion of Data

I (A) Methods of Collection of Primary Data
(a) Interview Method
(i) Personal Interview method
(ii) Indirect interview method
(iii) Telephonic interview method
(b) Mailed Questionnaire Method
(c) Observation Method
(d) Questionnaires filled and sent by enumerators (population)

(B) Methods of the Collection of Secondary
(a) Magazine, Newspaper
(b) Social Media
(c) Govt. Agencies
(d) libraries etc.

(II) Scrutiny of Data

(III) Classification of Data

The data may be classified as –

(a) Chronological data (time series) OR Temporal data
(b) Geographical data
(c) Qualitative data / ordinal or attribute data
(d) Quantitative data / cardinal data

Note – The data may be further classified as frequency data and non-frequency data
The qualitative as well as quantitative data belong to the frequency group whereas time-series data and geographic data always belong to non-frequency data.

(IV) Presentation of Data

Methods of presentation of Data –

(i) Textual presentation
(ii) Tabulation / Tabular Presentation
(iii) Diagrammatic Presentation

Line diagram	→	1D
Histogram / Area Diagram	→	2D
Bar diagram	→	1D
Pie chart	→	2D
Square diagram	→	2D
Rectangle diagram	→	2D
Cube diagram	→	3D

Q.1. Find the central angel for the various components given in the table.

Commodity	Expenses (`)	Central Angle
A	80	$\frac{360}{540} \times 80 = 53°$
B	190	$\frac{360}{540} \times 190 = 127$
C	160	$\frac{360}{540} \times 160 = 107$
D	75	$\frac{360}{540} \times 75 = 50$
E	35	$\frac{360}{540} \times 35 = 23$
	540	

Q.2. The cost of sugar in a month under the heads row material labour charge direct production and other charges are 12, 20, 35, and 23 units respectively what is the difference between the central angle for the largest and the smallest component of the sugar.

Largest = $\frac{360}{90} \times 35 = 140°$

Smallest = $\frac{360}{90} \times 12 = 48$

Difference = 140 – 48 = 92

Graphical representation of a frequency distribution

(i) Histogram or Area diagram

→ We can find mode with the help of histogram

(ii) Frequency polygon

(iii) Cumulative frequency graph [ogives]

→ We can find, median with the help of cumulative frequency polygon

→ We can find, median, quartile, decile, percentile and "partition value" with ogive.

Types of cumulative frequency graph –

(a) More than

(b) Less than

"WE CANNOT FIND THE MEAN BY ANY GRAPHICAL METHOD"

Frequency curve

→ The four types of frequency curves are –

(a) Bell Shape curve

(b) U – Shape curve

(c) J – Shape curve

(d) Mixed curve

Note – (i) Range = number of class interval x class length

(ii)

Class Limit

UCL
(Lopper class Limit)

LCL
(Upper Class Limit)

Mid Value = $\frac{\text{LCL} + \text{UCL}}{2}$

Conversion of discontinuous / discrete into continuous

CI	F	CI	F
44-48	3	43.5-48.5	3
49-53	2	48.5-53.5	2
54-58	7	53.5-58.5	7
59-63	5	58.5-63.5	5
64-68	4	63.5-68.5	4
69-73	1	68.5-73.5	1

Class boundary (CB)

LCB
(Lower CB)

UCB
(Upper CB)

Mid Value = $\frac{\text{UCB} + \text{LCB}}{2}$

"The mid value is also called mid – point / class – mark"

Q.3.

No. of accidents	0	1	2	3	4	5	6
Frequency	15	19	22	31	9	3	2

What is the number of cases when three or less accidents occurred.

Sol:- 15 + 19 + 22 + 31 = 87

Q.4. The following data relates to the income of 86 persons.

Income	500-999	1000-1499	1500-1999	2000-2499
No. of persons	15	28	36	7

Find – (i) what is the no. of person earning more than 1500.
(ii) What is the % of person earning more than 1500.

Sol:- (i) 36 + 7 = 43 (ii) $\frac{43}{86} \times 100$ = 50%

Q.5.

Marks	Below 10	Below 20	Below 30	Below 40	Below 50
No. of students	15	38	65	84	100

How many students got the marks more than 30

Sol:- 100 – below 30
100 – 65 = 35

Q.6.

Marks	More than 200	>250	>300	>350
No. of students	56	38	15	0

Find the no. of students whose marks lies between 250 and 300.

Chapter – 14 Measure of central Tendency and Dispersion

MEAN

Q.1. Find the average of the data 7, 17, 25, 11, 2, 21, 33

(a) 16.57 (b) 17.58 (c) 26.57 (d) 17.27

Q.2. The score of 10 students are 15, 19, 38, 7, k, 11, 25, 6, 13, 17

If mean is $\bar{x} = 21$ Then find the value of k

(a) 69 (b) 79 (c) 59 (d) 89

Q.3. Find the mean of the given table-

X	5	3	7	2	9
F	4	6	1	8	2

(a) 3.29 (b) 3.59 (c) 4.89 (d) 3.76

Q.4. Find mean of the given table-

X	11	7	14	19	2	25
F	6	10	3	1	17	2

(a) 8.29 (b) 7.20 (c) 4.89 (d) 3.76

Q.5. If the mean of the given table is 15. Then find k.

X	7	10	2	K	19	22	13	30
F	6	4	12	7	3	1	5	2

(a) 87.29 (b) 41.20 (c) 41.42 (d) 43.76

Q.6. Find the mean of the given table. {Exclusive / continuous series}

C.	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
F	9	7	6	5	6	3	2	1

(a) 28.5 (b) 26.44 (c) 36.96 (d) 25.11

Q.7. Find the mean of the given table- {Inclusive / Discontinuous series.}

C.I.	1-9	10-19	20-29	30-39	40-49
F	11	7	6	2	1

(a) 17.32 (b) 15.44 (c) 16.96 (d) 15.11

Q.8. Find mean of the given table-

X marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
No. of students	5	9	17	20	26	35

(a) 35 (b) 39 (c) 33 (d) None

Q.9. Find the mean of the given table-

Marks	More than 0	More than 10	More than 20	More than 30	More than 40	More than 50
No. of students	50	43	32	27	12	6

(a) 36 (b) 34 (c) 29 (d) 30

Q.10. In a class of 20 boys and 10 girls, the average score of the boys and girls are 65 and 55. Find the average score of all the students.

(a) 62.64 (b) 63.66 (c) $61.6\overline{6}$ (d) 62.56

Q.11. The average score of a class is 70. If the average score of 30 boys is 60. Then find the average score of 40 girls.

(a) 81.5 (b) 77.5 (c) 86.5 (d) 89.6

Q.12. The average score of 19 students was calculated as 60, later on score of a new student was added to the list which 67. Find the new average score.

(a) 81.35 (b) 77.55 (c) 86.5 (d) 60.35

Q.13. Average score of 15 students was found to be 75. later on it was found that the score of a student was wrongly taken as 79 instead of 69. Then find the correct mean?

(a) 74.33 (b) 76.45 (c) 85.79 (d) 78.16

Q.14. The average score of 25 students of a class is 60. Later on it was found that the score of two students was wrongly taken as 65 and 53 instead of 59 and 67. Find the correct mean.

(a) 60.32 (b) 80.48 (c) 78.49 (d) 77.92

MEDIAN and MODE

Q.15. Find median of the given table. 10, 7, 15, 3, 21, 33, 25

(a) 25 (b) 35 (c) 15 (d) 45

Q.16. Find median of the given data- 2, 10, 15, 20, 17, 7, 12, 19

(a) 12.14 (b) 35.15 (c) 15.54 (d) 13.5

Q.17. In a class of 10 students, 7 of them clears CA-foundation. If the score of the students who clears the exam are 25, 37, 30, 48, 41, 29, 31 then find median score of all the 10 students.

(a) 32.14 (b) 29.5 (c) 55.54 (d) 43.5

Q.18. Find the median of the given table-

X	3	10	7	20	15	11
F	4	2	5	3	6	1

(a) 10 (b) 20 (c) 30 (d) 40

Q.19. Find the median of the given table-

X	3	7	11	16	20	23	27	30
F	6	3	4	6	2	6	3	10

(a) 26.55 (b) 20 (c) 55.99 (d) 40.9

Q.20. Find median of the given table-

C.I.	0-10	10-20	20-30	30-40	40-50	50-60
F	6	3	7	5	8	4

(a) 31 (b) 81 (c) 67 (d) 41

Q.21. Find the median of the given table-

C.I.	**0-5**	**5-15**	**15-20**	**20-30**	**30-40**	**40-45**
F	**7**	**3**	**4**	**6**	**8**	**5**

(a) 28.188 (b) 81.145 (c) 24.166 (d) 41.145

Q.22. Find the median of the given table-

C.I.	1-10	11-20	21-30	31-40	41-50
F	3	5	4	6	2

(a) 45 (b) 78 (c) 79 (d) 25.5

Q.23. Find median of the given table-

X marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
No. of students	7	11	15	20	29	35

(a) 35 (b) 29 (c) 32 (d) 34

Q.24. Find median of the given table-

X marks	More than 10	More than 20	More than 30	More than 40	More than 50	More than 60
No. of students	40	31	24	17	10	5

(a) 40.79 (b) 35.71 (c) 32.5 (d) 34.6

Q.25. Find mode of the given data- 3, 4, 5, 7, 8, 5, 9, 5

(a) 5 (b) 7 (c) 6 (d) 8

Q.26. Find mode of the given data- 2, 5, 4, 7, 6, 5, 9, 7, 8, 5, 7

(a) 5 (b) 7 (c) 6 (d) both A and B

Q.27. Find mode –

X	3	5	4	7	10	6
F	2	8	15	11	7	10

(a) 5 (b) 7 (c) 6 (d) 4

Q.28. Find mode

X	3	5	9	7	8	10
F	6	8	17	17	2	4

(a) 5 (b) 9 (c) 6 (d) 4

Q.29. Find mode-

CI	0-10	10-20	20-30	30-40	40-50	50-60
F	3	7	12	6	5	4

(a) 55.45 (b) 45.12 (c) 24.54 (d) 78.19

Q.30. Find mode-

CI	1-9	10-19	20-25	26-35	36-41	42-50
F	3	5	10	6	4	2

(a) 55.45 (b) 45.12 (c) 24.54 (d) 23.33

QUARTILE, DECILE, PERCENTILE

Q.31. Find Q_3, P_{17}

x	3	8	15	20	26	30	37

(a) 30, 5.4 (b) 30, 4.8 (c) 29, 4.7 (d) 28, 3.3

Q.32. Find Q_1, D_6, P_{87}.

X	11	17	21	5	19	14	8	27

(a) 7.75, 15.2, 25.9 (b) 8.75, 12.6, 23.7
(c) 8.6, 18.9, 20.5 (d) 8.75, 17.8, 25.98

Q.33. Find Q_3, D_2, P_{67} of the given data-

X	2	7	13	17	20	25	32
F	6	5	4	5	4	3	2

(a) 22, 2, 20 (b) 20, 2, 20
(c) 20, 2, 22 (d) 20, 3, 22

Q.34. Find Q_1, D_8, P_{49}

X	7	11	19	2	13	27	22
F	5	3	7	10	6	4	2

(a) 2, 19, 13 (b) 10, 2, 20 (c) 21, 2, 22 (d) 2, 3, 22

Q.35. Find Q_1, Q_2, D_9, P_{23}, P_{77}

X	6	10	17	20	27	32	40
F	10	6	8	7	5	4	3

(a) 22, 2, 20, 10, 27 (b) 10, 17, 32, 10, 27
(c) 21, 2, 51, 45, 10 (d) 20, 3, 22, 74, 16

Q.36. Find Q_1, D_7, P_{39},

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70
F	6	5	3	4	2	3	5

(a) 12, 48, 22 (b) 12, 50, 20 (c) 12, 48, 19.84 (d) None of these

Q.37. Find, Q_3, D_2, P_{39}, P_{89}, Q_2

C.I	0-10	10-25	25-35	35-40	40-50	50-70
F	6	4	5	7	3	2

(a) 38.75, 9, 26.06, 46.76, 32 (b) 38.75, 11, 26.86, 46.76, 45
(c) 39.75, 9, 26.06, 36.76, 12 (d) None of these

RANGE & COEFFICIENT OF RANGE

Q.38. Find the Range & Coefficient of range of the given data- 6, 10, 5, 19, 27, 2, 11, 15, 18, 25, 32
(a) 33, 82.23% (b) 30, 88.235% (c) 35, 78.245% (d) None

Q.39. Find range and coefficient of

X	3	7	2	2	5	8	10
F	2	5	6	2	3	9	7

range-

(a) 17, 46.48% (b) 23, 66.42% (c) 8, 66.66% (d) None

Q.40. Find range and the coefficient of range of given data-

C.I.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
F	5	6	7	8	10	9	8

(a) 70, 100% (b) 69, 100% (c) 60, 100% (d) None of these

Q.41. Find range and the coefficient of range of given data-

C.I.	1-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
F	3	5	6	8	4	2	7	9

(a) 79, 98.75% (b) 89, 58.65% (c) 49, 98.74% (d) None

QUARTILE DEVIATION or SEMI–INTER QUARTILE RANGE & COQD

Q.42. Find Q.D. and Coefficient Of Quartile Deviation (C.O.Q.D.)

X	2	11	19	7	15	23	30	26
F	15	10	7	11	4	3	2	1

(a) 9.5, 40.15% (b) 7.5, 81.92% (c) 8.5, 80.95% (d) None

Q.43. Find Q.D. and C.O.Q.D. of the given data-

C.I.	5-20	20-30	30-50	50-60	60-75	75-80
F	6	5	8	7	9	4

(a) 19.458, 80.8 (b) 18.958, 40.8 (c) 48.158, 20.8 (d) None

Q.44. Find Q.D. and Coefficient Of Quartile Deviation (C.O.Q.D.)

X	5	11	19	8	15	2	21	27	24

(a) 8, 55.17% (b) 9, 54.18% (c) 4, 45.13% (d) None

MEAN DEVIATION & COEFFICIENT OF MEAN DEVIATION

Q.45. Find $MD_{\overline{X}}$ and $COMD_{\overline{X}}$ of 3, 5, 4, 6
(a) 2, 32.12% (b) 1, 22.22% (c) 5, 26.44% (d) None

Q.46. Find $MD_{\overline{X}}$ and $COMD_{\overline{X}}$ of

X	3	2	5	4
F	2	4	1	3

(a) 1.52, 29.47% (b) 0.94, 29.87% (c) 0.92, 29.67% (d) None

Q.47. Find $MD_{\overline{X}}$ and $COMD_{\overline{X}}$ of

C.I.	0-10	10-20	20-30	30-40	40-50
F	7	5	3	2	1

(a) 10.001, 59.99% (b) 11.041, 59.99% (c) 15.001, 59.89% (d) None

Q.48. Find MD_{md} & $COMD_{md}$ of

C.I.	5-15	15-20	20-30	30-50	50-60	60-70
F	3	7	8	6	4	2

(a) 15.16, 51.55% (b) 13.16, 50.15% (c) 63.86, 70.15% (d) None

Q.49. Find MD_{md} & $COMD_{md}$ of

C.I.	0-5	5-10	10-15	15-20	20-25	25-30
F	2	3	2	1	1	2

(a) 7.38, 65.65% (b) 5.28, 65.45% (c) 9.38, 35.62% (d) None

Q.50. Find MD_{md} & $COMD_{md}$ of 5, 17, 28, 21, 3, 11, 25, 30
(a) 2.5, 74.73% (b) 8.7, 94.53% (c) 8.5, 44.73% (d) None

Q.51. Find MD_{md} & $COMD_{md}$ of

X	3	5	9	13	18	20
F	7	6	5	4	3	2

(a) 4.85, 53.90% (b) 5.85, 53.60% (c) 3.85, 33.90% (d) None

STANDARD DEVIATION (σ), VARIANCE & C.O.S.D.

Q.52. Find σ , σ^2 and Coefficient of Standard Deviation (C.O.S.D.) of the given table-

X	2	3	7	13	10	5	1

(a) 2.096, 86.772, 79.01% (b) 4.446, 16.112, 75.01%
(c) 4.096, 16.772, 70.01% (d) None

Q.53. Find σ, Variance (σ^2) and C.O.S.D. of the given table-

X	5	3	4	6	2	1

(a) 1.707, 2.9138, 48.77% (b) 1.607, 2.9318, 98.77%
(c) 2.307, 2.9358, 48.77% (d) None

Q.54. Find σ, Variance (σ^2) and C.O.S.D. of the given table-

X	5	3	4	6	2	1
F	2	4	3	2	1	3

(a) 2.624, 2.657, 49.76% (b) 1.224, 2.537, 47.86%
(c) 1.624, 2.637, 47.76% (d) None

Q.55. Find σ , Variance (σ^2) and C.O.S.D. of the given table-

X	3	2	5	9	1
F	5	3	1	2	9

(a) 1.540, 3.749, 92.83% (b) 2.37, 5.62, 89.5%
(c) 1.224, 2.637, 47.76% (d) None

Q.56. Find σ , Variance (σ^2) and C.O.S.D. of the given table-

C.I.	0-10	10-20	20-30	30-40	40-50
F	7	6	4	3	1

(a) 12.01, 144.355, 67.28% (b) 11.01, 144.355, 57.28%
(c) 15.01, 184.355, 67.28% (d) None

Q.57. The average score of three batsman A, B, C in the Excel Premier League was 20,40,50. If the SD of their runs are 2, 10, 20 respectively. Which batsman is most consistent?
(a) Batsman C (b) Batsman A (c) Batsman B (d) None

Q.58. The quarterly profits of the two companies are given, which company is more solid.

Profit of Company A	2	5	3	4
Profit of Company B	7	1	2	6

(a) Company A (b) Company B (c) Both A and B (d) None

Q.59. The score of the two students in the last 5 test are given, which student scores consistently.

Student A	3	7	4	5	2
Student B	8	5	6	2	3

(a) Both Student (b) Student B (c) Student A (d) None

COMBINED STANDARD DEVIATION:-

Q.60. The average score of 10 boys and 20 girls are 45 and 50 respectively and the SD of their scores are 5 and 10 respectively. Find the combined standard deviation of their score.
(a) 8.97 (b) 9.85 (c) 7.98 (d) None

Q.61. Calculate the combined standard deviation, given that - $n_1 = 15$, $n_2 = 25$
$\sigma_1 = 4$, $\sigma_2 = 3$, $\overline{x}_1 = 5$, $\overline{x}_2 = 7$
(a) 3.54 (b) 5.43 (c) 4.53 (d) None

Q.62. The average score 9 students was found to be 50 and the SD of their score was 5. Later on, the score of a new student was added to the list which was 56. Find the New Standard deviation.
(a) 6.07 (b) 5.07 (c) 8.56 (d) None

Q.63. The average score of 20 students of a class is 70 and the SD of their score is 5. Later on it was found that the score of a students was wrongly taken as 68 instead of 78. Find correct SD.
(a) 4.56 (b) 6.25 (c) 5.26 (d) None

Q.64. The average score 15 students of a class is 60, and the SD of their score is 7. Later on it was found that the score of 2 students were wrongly taken as 55 and 67 instead of 65 and 57. Find correct SD.
(a) 6.80 (b) 8.60 (c) 7.86 (d) None

GEOMETRIC MEAN

Q.65. Find the GM of 7 and 13.
(a) 5.54 (b) 9.53 (c) 5.39 (d) None

Q.66. Find the GM of 3, 9, and 27.
(a) 6.5 (b) 81 (c) 9 (d) None

Q.67. Find the GM of 2, 7, and 9
(a) 5.2645 (b) 5.0133 (c) 4.8156 (d) None

Q.68. Find the geometric mean of 10, 110, 120, 50, 52, 37, 60.
(a) 52.8 (b) 51.5 (c) 53.2 (d) None

Q.69. Find the GM of given table-

X	12	13	14	15	16	17
F	5	4	4	3	2	1

(a) 15.127 (b) 13.703 (c) 19.129 (d) None

Q.70. Find the GM of given table-

CI	0-10	10-20	20-30	30-40	40-50
F	4	8	10	6	7

(a) 22.05 (b) 21.5 (c) 23.8 (d) None

HARMONIC MEAN

Q.71. A person travels from city–A to city–B with an average speed of 40 km/h and returns with an average speed of 60 km/h. then what will be his average speed for whole journey.
(a) 48 km/h (b) 45 km/h (c) 50 km/h (d) None

Q.72. Find the HM of 10, 20, 15.
(a) 12.53 (b) 14.9 (c) 13.84 (d) None

Q.73. A person travels from city–A to city–B with an average speed of 60km/hr, and then he travels from city-B to C with an average speed of 80 km/hr, then from C to D with an average speed of 100 km/hr. If the distance between A to B, B to C, C to D are same, then find his average speed for the whole journey.
(a) 76.53 km/hr (b) 75 km/hr (c) 85.68 km/hr (d) None

Q.74. Find the HM of the given observation-

X	20	50	30	40
F	160	200	90	240

(a) 35.34 (b) 32.85 (c) 30.67 (d) None

Q.75. A person travels first 300 km of his journey with an average speed of 60 km/h and then he travels next 500 km with an average speed of 80 km/hr. Find his average speed for the whole journey.
(a) 71. 11 km/hr (b) 81.12 km/hr (c) 61.45 km/hr (d) None

Q.76. A person travels first 300 km with an average speed of 50 km/h. and then he travels next 300 km with an average speed of 80 km/hr. Find his average speed for the whole journey.
(a) 56.6 km/hr (b) 62.5 km/hr (c) 61.5 km/hr (d) None

MISCELLANEOUS QUESTION:-

Q.77. Find the weighted AM of the given table

X	46	67	72	58	53
W	2	2	1	1	1

(a) 58.42 (b) 68.53 (c) 45.79 (d) None

Q.78. There are 50 students in a class of which 40 are boys. The average Wight of the girls is 40, then find average Wight of the boys, if the average Wight of all the students of the class is 44.
(a) 47 (b) 45 (c) 46 (d) None

Q.79. The Average salary of all the employees in a company is `25000. The average salary of male and female Employees is `27000 and `17000 respectively. Then find the no. of male and female.
(a) 60,30 (b) 20,80 (c) 80,20 (d) None

Q.80. The average score of the girls in CA–foundation is 67 and that of the boys is 63. The average score of the whole class is 64.5, then find the percentage of girls and boys.
(a) 63%, 37% (b) 37.5%, 62.5% (c) 62.5%, 37.5% (d) None

Q.81. If the relation between the two variable x any y is $x - y - 10 = 0$. If the mode of x is 23, then find the mode of y.
(a) 13 (b) 14 (c) 10 (d) None

Q.82. If the relation between x and y is 3x–4y = 20 and Q.D. of x is 12. Then find QD_y
(a) 8 (b) 11 (c) 9 (d) None

Q.83. If $Q_1 = 15$, $Q_3 = 40$, find M.D.
(a) 15 (b) 12.5 (c) 27.5 (d) None

Q.84. The Mean and SD of a given set of observation is 1500 and 400 respectively, if there is an increment of `100 in the first year and each observation is hiked by 20% in the second year, then find the new Mean and new Standard Deviation after two years.
(a) 1840, 400 (b) 1500, 480 (c) 1920, 480 (d) None

Q.85. If 5 is subtracted from each observation then its C.O.V. will be 10%. If 5 is added to each observation then C.O.V. is 6% . Then find the original C.O.V.
(a) 6% (b) 7.5% (c) 8% (d) None

Q.86. For a set of 100 observation, taking Assumed mean as 4, the sum of the deviations is –11, and the sum of the square of the deviation is 257. Then find the C.O.V.
(a) 41.21% (b) 16.03% (c) 25% (d) None

Q.87. The mean of 15 numbers is 15, if the two numbers 18 and 12 are excluded, then the mean of the remaining numbers is
(a) 15 (b) 10 (c) 25 (d) None

Q.88. If the value of mode is 25.5 and median is 16.5, the value of arithmetic mean will be
(a) 13.5 (b) 18 (c) 12.5 (d) 12

Q.89. Reciprocal of 0.062 is
(a) .1613 (b) 1.613 (c) 16.13 (d) 161.3

Q.90. A man travels from Kanpur to Lucknow at an average speed of 30 km. per hour and returns at an average speed of 60 km. per hour. What is his average speed?
(a) 40 (b) 45 (c) 50 (d) None

Q.91. In a class, average height of 10 boys is 5ft. 6 inches, whereas average height of 5 girls is 5ft. Average height of all students of the class shall be
(a) 5 ft. 2 inches (b) 5 ft. 3 inches (c) 5 ft. 4 inches (d) 5 ft. 5 inches

Q.92. The A.M. of the two numbers is 6.5 and their G.M. is 6, the two numbers are
(a) 8, 6 (b) 8, 5 (c) 7, 16 (d) 4, 9

Q.93. If the geometric mean of x, 16, 50 be 20, then the value of x is equal to
(a) 4 (b) 10 (c) 30 (d) 80

Q.94. If arithmetic average is 5 and geometric mean is 4, find the Harmonic mean.
(a) 2.5 (b) 3.0 (c) 3.2 (d) 3.8

Q.95. If n = 10, $\sum y$ = 250 and $\sum y^2$ = 6840. Then SD = ?
(a) 7.68 (b) 7.2 (c) 6.6 (d) 6.2

Q.96. Most useful among all measures of dispersion is
(a) Standard deviation (b) Quartile deviation
(c) Mean deviation (d) None of the above.

Q.97. If σ = 16 and $\sum(X - \overline{X})^2$ = 4096, Then the value of n is
(a) 16 (b) 256 (c) 4 (d) none of these.

Q.98. Standard deviation (SD) of a series is 22.5, estimated mean deviation (MD) of the series shall be
(a) 25 (b) 18 (c) 27 (d) None of the above

Q.99. The SD of x is 3, what is the variance of (5-2x)
(a) 1 (b) 6 (c) 36 (d) 20

Q.100. If Q_3 is 14 and Q_1 is 6, coefficient of quartile deviation (Coeff. of Q.D.) shall be
(a) 8 (b) 4 (c) 0.4 (d) 400

Q.101. When QD is 60, then variance (fopj.k) shall be
(a) 90 (b) 81 (c) 8100 (d) 9000

Q.102. What is the value of C.V., if S.D. = 10 and mean = 20
(a) 10% (b) 20% (c) 50% (d) 75%

Q.103. Calculative C.V., if N = 20, $\sum FX$ = 170 and σ = 2
(a) 23.53% (b) 8.5% (c) 25.35% (d) 35-8%

Q.104. What is the value of C.V. if, variance and mean are 144 and 50 respectively
(a) 15% (b) 18% (c) 21% (d) 24%



Answers

MEAN

1	a	11	b
2	c	12	d
3	d	13	a
4	b	14	a
5	c		
6	a		
7	b		
8	c		
9	c		
10	c		

MEDIAN and MODE

15	c	25	a
16	d	26	d
17	b	27	d
18	a	28	b
19	b	29	c
20	a	30	d
21	c		
22	d		
23	a		
24	b		

QUARTILE, DECILE, PERCENTILE

31	b
32	d
33	b
34	a
35	b
36	c
37	a

RANGE & COEFFICIENT OF RANGE

38	b
39	c
40	a
41	a

QUARTILE DEVIATION or SEMI–INTER QUARTILE RANGE & COQD

42	c
43	b
44	a

MEAN DEVIATION & COEFFICIENT OF MEAN DEVIATION

45	b
46	c
47	a
48	b
49	a
50	c
51	a

STANDARD DEVIATION (σ), VARIANCE & C.O.S.D.

52	c
53	a
54	c
55	b
56	a
57	b
58	a
59	c

COMBINED STANDARD DEVIATION:-

60	a
61	a
62	b
63	c
64	a

GEOMETRIC MEAN

65	b
66	c
67	b
68	a
69	b
70	a

HARMONIC MEAN

71	a
72	c
73	a
74	b
75	a
76	c

MISCELLANEOUS QUESTION:-

77	a	87	a	97	a
78	b	88	d	98	b
79	c	89	c	99	c
80	b	90	a	100	c
81	a	91	c	101	c
82	c	92	d	102	c
83	a	93	b	103	a
84	c	94	c	104	d
85	b	95	a		
86	a	96	a		



Chapter – 15 PROBABILITY

Q.1. When a coin is tossed what is the probability of getting Head
(a) 1 (b) 0.5 (c) 0 (d) None

Q.2. When the two coins are tossed together then what is the probability of getting-
(i) Getting both Head (ii) Getting same face on both the coin
(iii) At least one Head (iv) At most one Head
(a) 0.25, $\frac{3}{2}$, $\frac{2}{4}$, $\frac{1}{4}$ (b) 0.25, $\frac{1}{3}$, $\frac{2}{4}$, $\frac{1}{4}$ (c) 0.25, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{3}{4}$ (d) None

Q.3. When the three coins are tossed together then what is the probability of getting-
(i) All Tail (ii) Exactly two Head (iii) At least two Head (iv) At most one Head
(a) $\frac{1}{8}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{1}{3}$ (b) $\frac{1}{8}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{1}{2}$ (c) $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{3}$, $\frac{1}{2}$ (d) None

Q.4. When a Dice is thrown then the probability of getting –
(i) Even number on the Dice (ii) A number divisible by 3
(iii) Divisible by 2 and 3 (iv) By 2 or 3
(a) $\frac{1}{2}$, $\frac{1}{3}$, $\frac{4}{6}$, $\frac{1}{3}$ (b) $\frac{1}{2}$, $\frac{2}{3}$, $\frac{5}{6}$, $\frac{2}{3}$ (c) $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{6}$, $\frac{2}{3}$ (d) None

Q.5. When the two Dices are thrown together then what is the probability of getting-
(i) Doublet (ii) Sum greater than 9 (iii) Sum greater than 9 and Even number on the first Dice
(iv) Sum greater than 9 or Even number on the first Dice
(a) $\frac{1}{6}$, $\frac{1}{6}$, $\frac{1}{9}$, $\frac{5}{9}$ (b) $\frac{1}{6}$, $\frac{5}{6}$, $\frac{4}{9}$, $\frac{4}{9}$ (c) $\frac{5}{6}$, $\frac{1}{6}$, $\frac{1}{9}$, $\frac{2}{9}$ (d) None

Q.6. If one card is taken out from a pack of 52 Cards then what is the probability that-
(i) It is a red card (ii) It is a face card (iii) It is a male face card
(iv) It is a red card and a face card (v) It is a red card or a face card
(a) $\frac{1}{2}$, $\frac{2}{13}$, $\frac{8}{52}$, $\frac{6}{52}$, $\frac{12}{52}$ (b) $\frac{1}{2}$, $\frac{3}{13}$, $\frac{8}{52}$, $\frac{6}{52}$, $\frac{32}{52}$
(c) $\frac{1}{2}$, $\frac{3}{13}$, $\frac{6}{52}$, $\frac{1}{52}$, $\frac{32}{52}$ (d) None

Q.7. A box containing 4 Red, 3 Green & 2 Black balls. If one ball is taken at random then what is the probability of Getting (i) A red ball (ii) A Green or a black ball
(a) $\frac{4}{9}$, $\frac{5}{9}$ (b) $\frac{1}{9}$, $\frac{4}{9}$ (c) $\frac{6}{9}$, $\frac{1}{9}$ (d) None

Q.8. There are 3 Men, 2 Women and 4 Children's in a group. If one member is to be selected at random, then what is the probability of selecting - (i) A Man (ii) A Women or a Man (iii) A Children
(a) $\frac{1}{9}$, $\frac{2}{9}$, $\frac{5}{9}$ (b) $\frac{3}{9}$, $\frac{5}{9}$, $\frac{4}{9}$ (c) $\frac{2}{9}$, $\frac{5}{9}$, $\frac{8}{9}$ (d) None

Q.9. A question of Mathematics is given to the student A and B. The probability that they can solve the problem are 1/3 and 3/5 then what is the probability that-
(i) Any one of them can solve the problem (ii) Both can solve the problem
(iii) At least one of them solve the problem.
(a) $\frac{1}{5}$, $\frac{4}{15}$, $\frac{7}{15}$ (b) $\frac{1}{5}$, $\frac{2}{15}$, $\frac{14}{15}$ (c) $\frac{8}{15}$, $\frac{1}{5}$, $\frac{11}{15}$ (d) None

Q.10. A Man and his Wife appear for an interview for two posts. The probability of selection of Husband is 1/7 and the probability of selection of Wife is 1/5 ,what is the probability that :- (i) Both selected (ii) any one of them selected (iii) at least one of them selected (iv) At most one of them selected
(a) $\frac{1}{35}$, $\frac{10}{35}$, $\frac{11}{35}$, $\frac{34}{35}$ (b) $\frac{1}{35}$, $\frac{12}{35}$, $\frac{11}{35}$, $\frac{33}{35}$ (c) $\frac{2}{35}$, $\frac{10}{35}$, $\frac{21}{35}$, $\frac{34}{35}$ (d) None

Q.11. If the probability of the Horse A winning the race is 1/5 and that of Horse B is 1/6 then what is the probability that one of the Horses will win the race.

(a) $\frac{12}{30}$ (b) $\frac{11}{30}$ (c) $\frac{8}{30}$ (d) None

Q.12. A question of Math is given to the three students A, B and C. If the probability that they can solve the problem are 4/5, 2/3 and 3/7 respectively, find the probability that (i) any one of them can solve (ii) problem will be solved.

(a) 6/35, 104/105 (b) 8/35, 102/105 (c) 9/35, 101/105 (d) None

Q.13. A person 'A' speak truth in 75% cases and an another person B speaks truth in 80% of the cases, in what % of cases are they likely to contradict each other narrating the same incident.

(a) 45% (b) 35% (c) 36% (d) None

Q.14. The odd against a certain event are 5 to 3 and the odd in favaur of an another event are 7 to 5, find the chance that at least one of the event will happen.

(a) $\frac{71}{96}$ (b) $\frac{81}{96}$ (c) $\frac{65}{69}$ (d) None

Q.15. The chances of living a person who is now 35 year old till he is 75 are 8 : 6 and of living another person who is now 40 years old till he is 80 are 4:5 , find the probability that at least one of them will alive next 40 years.

(a) $\frac{56}{126}$ (b) $\frac{96}{126}$ (c) $\frac{96}{120}$ (d) None

Q.16. A question of Maths is given to the two students A and B and the odd in favour of student A solving the problem are 6 : 9 and the odd against the student B solving the problem are 12 : 10 , what is the probability that the problem will be solved.

(a) $\frac{37}{55}$ (b) $\frac{39}{51}$ (c) $\frac{48}{51}$ (d) None

Q.17. The add against a student A solving a problem are 8 : 6 and add in favour of students B solving the problem are 14 : 16 , what is the probability that none of them are able to solve the problem.

(a) $\frac{21}{105}$ (b) $\frac{32}{105}$ (c) $\frac{39}{105}$ (d) None

Q.18. What is the probability that a Leap year selected at random will have 53 Sundays

(a) $\frac{2}{7}$ (b) $\frac{5}{7}$ (c) $\frac{1}{6}$ (d) None

Q.19. What is the probability that a Non-Leap year selected at random will have 53 Sundays.

(a) $\frac{3}{7}$ (b) $\frac{4}{7}$ (c) $\frac{1}{7}$ (d) None

Q.20. A box contains 3 Red, 4 Black and 2 Green balls. If the two balls are taken out together, what is the probability of getting: (i) Both Red (ii) one Red and one Green.

(a) $\frac{1}{5}, \frac{1}{6}$ (b) $\frac{1}{12}, \frac{1}{6}$ (c) $\frac{2}{6}, \frac{4}{7}$ (d) None

Q.21. A box contains 3 Red, 2 Green, 4 Black balls. If the two balls are taken out one by one "without replacement", then what is the probability of getting :- (i) Both Red (ii) One Red and one Green

(a) $\frac{1}{12}, \frac{1}{6}$ (b) $\frac{4}{6}, \frac{2}{6}$ (c) $\frac{1}{7}, \frac{4}{6}$ (d) None

Q.22. A box contains 3 Red, 2 Green, 4 Black balls. If the two balls are taken out one by one "with replacement", then what is the probability of getting:– (i) Both Red (ii) One Red and one Green.

(a) $\frac{5}{27}, \frac{1}{9}$ (b) $\frac{3}{27}, \frac{1}{9}$ (c) $\frac{1}{9}, \frac{4}{27}$ (d) None

Q.23. 4 persons are selected at random from a group of 3 Men, 2 Woman and 4 Children, what is the probability of the selection of (i) two Men and two Woman
(ii) One Men, one Woman and two Children (iii) Exactly two Men.
(a) $\frac{2}{42}, \frac{11}{42}, \frac{45}{126}$ (b) $\frac{1}{42}, \frac{12}{42}, \frac{45}{126}$ (c) $\frac{11}{42}, \frac{12}{42}, \frac{15}{126}$ (d) None

Q.24. Four cards are taken out from a pack of cards, find the probability that 2 cards are Spades and 2 cards are Heart.
(a) 0.022 (b) 0.023 (c) 0.45 (d) None

Q.25. One Bag contain 5 White & 4 Black balls and another Bag contain 7 While & 9 Black balls. If one ball is transferred from the first bag to the second and then one ball is taken out from the second bag, find the probability that the ball drawn from the second bag is a white ball.
(a) $\frac{69}{153}$ (b) $\frac{58}{153}$ (c) $\frac{68}{153}$ (d) None

Q.26. One bag contains 4 White and 2 Black and second bag contains 5 White and 4 Black balls. If the two balls are transferred from the first bag to the second bag and then one ball is taken out from the second bag, find the probability that it is a White ball.
(a) 0.57 (b) 0.23 (c) 0.45 (d) None

Q.27. Two Cards are taken out from a pack of Cards and kept out, and then one card is taken out from the remaining cards. What is the probability that it is an Ace.
(a) 0.07692 (b) 0.17292 (c) 0.07683 (d) None

Q.28. One bag contains 6 Red and 5 Black balls, second bag contains 5 Red and 8 Black balls. If one ball is transferred from bag-I to bag-II, then one ball is drawn out form second bag. What is the probability that it is a black ball.
(a) $\frac{53}{34}$ (b) $\frac{93}{154}$ (c) $\frac{91}{54}$ (d) None

Q.29. A bag contains 5 White and 8 Red balls, two drawings of 3 balls are made such that the balls are replaced before the second trial, then find the probability that the first trial will give the three White balls and the second trial will give three Red ball.
(a) $\frac{560}{81796}$ (b) $\frac{660}{81796}$ (c) $\frac{580}{81796}$ (d) None

Q.30. Again solve the previous question in case of "without replacement".
(a) 0.016 (b) 0.172 (c) 0.018 (d) None

MISCELLANEOUS QUESTIONS

Q.31. The probability that A can solve a problem is 2/3 and that B can solve is 3/4. If both of them attempt the problem, what is the probability that the problem get solved?
(a) $\frac{11}{12}$ (b) $\frac{7}{12}$ (c) $\frac{5}{12}$ (d) None

Q.32. If P(A) = 1/4, P(B) = 1/2, P(A ∪ B) = 5/8, then P(A ∩ B) is :-
(a) 3/8 (b) 1/8 (c) 2/8 (d) None

Q.33. A and B are events and P(A) = 0.4, P(A ∪ B) = 0.7. If A and B are independent, then P(B) is :-
(a) 0.2 (b) 0.3 (c) 0.4 (d) 0.5

Q.34. When two dices are thrown, the probability of getting sum 10 or 11 is :-
(a) 7/36 (b) 5/36 (c) 5/18 (d) 7/16

Q.35. If A and B are two events such that P(A ∪ B) = 5/6, P(A ∩ B) = 1/3, P(A) = 2/3, then A and B are
(a) Dependent events (b) Independent events
(c) Mutually exclusive events (d) None

Q.36. One of the two events A and B must occur. If P(A) = $\frac{2}{3}$ P(B), then the "odds in favour" of B are :-
(a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 3 : 2

Q.37. The probabilities of two events A and B are 0.25 and 0.40 respectively. The probability that both A and B occur is 0.15, the probability that neither A nor B occurs is :-
(a) 0.35 (b) 0.65 (c) 0.5 (d) 0.75

Q.38. One Dice and a Coin are tossed simultaneously. The probability of getting 5 on the top of the dice and Tail on the coin is:-
(a) 1/2 (b) 1/12 (c) 1/6 (d) 1/8

Q.39. If P(A ∪ B) = 0.8, P(A ∩ B) = 0.3, then $P(\overline{A})$ + $P(\overline{B})$ =
(a) 0.3 (b) 0.5 (c) 0.7 (d) 0.9

Q.40. Seven balls are drawn simultaneously from a bag containing 5 White and 6 Green balls. The probability of drawing 3 White and 4 Green balls is:-
(a) $\frac{7}{{}^{11}C_7}$ (b) $\frac{{}^5C_3 + {}^6C_4}{{}^{11}C_7}$ (c) $\frac{{}^5C_2 \cdot {}^6C_2}{{}^{11}C_7}$ (d) $\frac{{}^6C_3 \cdot {}^5C_4}{{}^{11}C_7}$

Q.41. The probability of happening of an event A in one trial is 0.4, the probability that the event A happens at least once in three independent trials is :-
(a) 0.626 (b) 0.784 (c) 0.802 (d) 0.316

Q.42. If 3 Dices are thrown simultaneously, the probability of getting score (sum) 5 is:-
(a) 5/216 (b) 1/6 (c) 1/36 (d) 1/72

Q.43. Two fair coins are tossed at the same time. Find the probability of getting "at least one head" and "at most one head".
(a) 3/4, 3/4 (b) 3/4, 3/2 (c) 5/4, 3/4 (d) 1/2, 3/4

Q.44. From a well shuffled pack of 52 cards, 3 cards are drawn at random. Find the probability that three cards drawn contain "two Kings and one Ace".
(a) 4/5525 (b) 5/5525 (c) 6/5525 (d) 7/5525

Q.45. A Coin is tossed 10 times. The probability of getting exactly six times Heads, is:-
(a) 512/105 (b) 105/512 (c) 100/513 (d) ${}^{10}C_6$

Q.46. If there are 3 Children in a family, then probability that there is One girl in the family, is:-
(a) 2/3 (b) 1/3 (c) 3/28 (d) 3/8

Q.47. The probability of having at least one Tail in 4 throws with a coin is:-
(a) 15/16 (b) 1/16 (c) 1/4 (d) 1.

Q.48. There are four letters and four envelops bearing addresses at random. The probability that the letters are placed in correct envelops is:-
(a) 23/24 (b) 9/24 (c) 1/16 (d) 1/24

Q.49. From a well shuffled pack of playing cards, two cards are drawn one by one "Without Replacement". The probability that both are Aces, is:-
(a) 2/13 (b) 1/51 (c) 1/221 (d) None of these

Q.50. If P(A∩B) = 0.15, P(not B) = 0.10, then P(A/B) is :-
(a) 1/3 (b) 1/4 (c) 1/5 (d) 1/6

Q.51. If A and B are events, such that P(A) = 0.42, P(B) = 0.48 and P(A∩B) = 0.16.
Then P(A∪B) is equal to ?
(a) 0.9 (b) 0.74 (c) 0.2016 (d) None of these.

Q.52. In shuffling a pack of playing cards, four cards are accidently dropped. The probability that the missing cards should be one from each Suit is:-
(a) 1/256 (b) 4/20825 (c) 2197/20825 (d) None of these

Q.53. The problem in mathematics is given to 3 students whose chances of solving individually are 1/2, 1/3, and 1/4. The probability that the problem will be solved at least by one is:-
(a) 1/4 (b) 1/24 (c) 23/24 (d) 3/4

Q.54. If A and B are two events and P(A) = 3/8, P(B) = 1/2, P(A∩B) = 1/4, then the value of P ($A' \cup B'$) is :-
(a) 3/8 (b) 3/4 (c) 1/4 (d) 5/8

Q.55. If A, B and C can hit a target 4 times in 5 shots, 3 times in 4 shots and 2 times in 3 shots respectively, the probability that exactly two of them will hit the target is :-
(a) 13/30 (b) 5/16 (c) 17/30 (d) None of these.

Q.56. A Bag contains 5 Brown and 4 White Socks. A man pulls out 2 socks. The probability that they are of the same colors is:-
(a) 5/108 (b) 1/6 (c) 5/18 (d) 4/9

Q.57. The probability that a man can hit a target is 3/4. He tries 5 times. The probability that he will hit the target at least three times, is:-
(a) 291/264 (b) 371/464 (c) 471/502 (d) 459/512

Answers

1	b	11	b	21	a
2	c	12	c	22	c
3	b	13	b	23	b
4	c	14	a	24	a
5	a	15	b	25	c
6	b	16	a	26	a
7	a	17	b	27	a
8	b	18	a	28	b
9	c	19	c	29	a
10	a	20	b	30	a

MISCELLANEOUS QUESTIONS

31	a	41	b	51	b
32	b	42	c	52	c
33	d	43	a	53	d
34	b	44	c	54	d
35	b	45	b	55	a
36	d	46	d	56	d
37	c	47	a	57	d
38	b	48	d		
39	d	49	c		
40	c	50	d		

CHAPTER – 16 EXPECTED VALUE & THEORETICAL DISTRIBUTION

Probability Distribution & Expected Value

Q.1. A Coin is tossed two times, then what is the Probability Distribution for the number of Heads and hence find the Expected value (Mean) and the Standard Deviation of the number of Heads.
(a) 1, 0.707 (b) 2, 0.709 (c) 1, 1.608 (d) None

Q.2. A Coin is tossed 3 times. Getting Head in any attempt considering success, find the probability distribution for the number of success and hence find the Mean and S.D.
(a) 2, 0.869 (b) 1.6, 0.896 (c) 1.5, 0.866 (d) None

Q.3. A Dice is thrown 2 times. Getting a number greater than four is considering success, find the probability distribution and hence find the Mean & S.D.
(a) 0.67, 0.67 (b) 0.37, 0.67 (c) 0.81, 0.67 (d) None

Q.4. Find E(x) and σ for the given probability distribution table.

X	2	3.5	4.5	5	6
P	0.1	0.2	0.4	0.2	0.1

(a) 4.2, 1.0961 (b) 4.3, 1.0295 (c) 4.5, 1.0145 (d) None

Q.5. Find the Variance of the given probability distribution table.

X	0	1	2	3
P	K	3k	3k	K

(a) 0.75 (b) 0.81 (c) 0.89 (d) None

Q.6. Find the Variance of the given probability distribution table-

X	−3	−2	−1	0	1	2
P	0.1	2k	3k	7k	0.2	0.1

(a) 0.7899 (b) 0.7596 (c) 1.9875 (d) None

Q.7. Find E(x) , V(x) , E(2x −3) , V(2x −3) of the given probability distribution table-

X	−2	−1	0	1	2
P	0.2	0.1	0.3	0.3	0.1

(a) 0, 1.6, −3, 6.4 (b) 0, 1.7, −3, 6.4 (c) 0, 1.6, −2, 5.4 (d) None

Q.8. Find the probability distribution of the number of sixes in the three thrown of a Dice and hence find the E(x) and S.D.
(a) 0.5, 0.645 (b) 0.6, 0.645 (c) 0.7, 0.643 (d) None

Q.9. Find the Mean and the S.D. of a number obtain as a result of throwing a Dice.
(a) 3.6, 0.179 (b) 3.6, 1.754 (c) 3.5, 1.7079 (d) None

Q.10. Expected Value E(x) of the following probability distribution is:-

X	-10	-20	30	75	80
P(x)	1/5	3/20	1/2	1/10	1/20

(a) 31.5 (b) 21.5 (c) 19.5 (d) 17.5

Q.11. If three Coins are tossed. The expectation of the number of Heads is :-
(a) 1/2 (b) 7/2 (c) 5/2 (d) 3/2

Q.12. A Dice is thrown. The expectation of the number on it, is ?
(a) 3/2 (b) 5/2 (c) 7/2 (d) 1/2

Q.13. A player tossed two Coins. If two Heads show he wins `4, if one Head shows he wins `2, but if two Tails show he pays Rs.3 as penalty. The expected value of the gain to him (in rupees) is :-
(a) 1.25 (b) 2.25 (c) 3.25 (d) 1.35

Q.14. A random variable has following probability distribution. Expected value of X is :-

X	4	5	6	8
P	0.1	0.3	0.4	0.2

(a) 4.9 (b) 5.9 (c) 3.9 (d) 6.9

Q.15. A random Variable X has the following probability distribution. Find E(x)

Value of x	-2	-1	0	1	2	3
P (X = x)	0.1	K	0.2	2k	0.3	k

(a) 0.7 (b) 0.6 (c) 0.8 (d) 0.5

Q.16. Three items are drawn at random from a box containing 2 defective and 6 non-defective items. The expected number of non-defective items drawn is :-
(a) 3.25 (b) 2.25 (c) 1.5 (d) 5.25

Q.17. Let X be a random variable assuming values X_1, X_2, and X_3, then the function f defined by $f(X_1) = P(X = X_1)$ is given by the table

X :	–3	6	9
P (X = X_1) :	1/6	1/2	1/3

Find E (2X + 1)
(a) 10 (b) 11 (c) 12 (d) 14

Q.18. Random Variable X has the following distribution, find the value of C

X	1	2	3	4
P (X = x)	C	2C	3C	4C

(a) 0.1 (b) 0.2 (c) 10 (d) 20

Q.19. If X is a Random Variable with given distribution, find the value of K and Variance

X	0	1	2	3
P (X = x)	K	3K	3K	K

(a) 1/8, 22/27 (b) 1/8, 23/27 (c) 1/8, 24/27 (d) 1/8, 3/4.

Q.20. A Random Variable X has the following distribution, find Mean & Variance.

$X (= x_i)$	0	1	2	3
$P (X = x_i)$	2/6	3/6	0/6	1/6

(a) 1, 1 (b) 1, 2 (c) 2, 1 (d) 2, 2

Q.21. The Probability Distribution of Random Variable X is given below. Then, K =

$X = x_i$	1	2	3	4
$P (X = x_i)$	2K	4K	3K	K

(a) 0.1 (b) 0.2 (c) 0.3 (d) 0.4

Q.22. The Mean of the following Probability Distribution is :-

$X = x_i$	-3	-2	-1	0	1	2
$P (X = x_i)$	0.1	2k	3k	7k	0.2	0.1

(a) – 0.25 (b) 0.25 (c) 0.50 (d) none of these.

Q.23. A Random Variable X has the following Distribution. Find Mean & Variance.

$X (= x_i)$	2	5	6	7
$P (X = x_i)$	1/10	K	3/10	4/10

(a) 5.4, 2 (b) 5.8, 2.16 (c) 5.8, 2 (d) None of these

Q.24. A Box Contains 10 tickets, 2 of the tickets carry a price of `8 each, 5 of the tickets carry a price of `4 each and 3 of the tickets carry a price of `2 each. If one tickets is drawn, what is the Mean value of the price.

(a) 3.4 (b) 2.8 (c) 3.1 (d) 4.20

Q.25. The Probability Distribution of a Random Variable X is given below, find Mean.

$X = x_i$	1	2	3
$P (X = x_i)$	1/4	1/8	5/8

(a) 19/8 (b) 5/4 (c) 1 (d) 4/5

Q.26. Find the Standard Deviation for the following probability distribution.

X	1	2	3	-2	-1
P (X)	0.10	0.35	0.20	0.30	0.05

(a) 1.997 (b) 1.897 (c) 1.79 (d) 2.1

BINOMIAL DISTRIBUTION

Q.27. A Coin is tossed 7 times. Find the Probability of Getting Head.(i) Exactly 5 times (ii) At least one time.

(a) 0.163, 0.892 (b) 0.164, 0.992 (c) 0.174, 0.991 (d) None

Q.28. A Dice is thrown 6 times, getting a number divisible by 3 is considering success, what is the probability of getting success. (i) Exactly two times (ii) At least two times (iii) At most 5 times.

(a) 0.4292, 0.64, 0.298 (b) 0.3492, 0.64, 0.798

(c) 0.3292, 0.64, 0.998 (d) None

Q.29. What is the probability of guessing correctly "at least 6" out of the 10 answers in a true / false type objective test.

(a) 0.376 (b) 0.356 (c) 0.981 (d) None

Q.30. The overall percentage of the failures in CA foundation is 30%, what is the probability that out of a group of 6 candidate at least 4 pass the examination.
(a) 0.56812 (b) 0.74431 (c) 0.75612 (d) None

Q.31. In a Binomial Distribution, Mean = 12, SD = 2, Find the value of n and p.
(a) 17, 1/4 (b) 15, 1/2 (c) 18, 2/3 (d) None

Q.32. Find the Binomial Distribution whose Mean is 3 and Variance is 2.
(a) $\left(\frac{1}{2}+\frac{1}{2}\right)^9$ (b) $\left(\frac{2}{3}+\frac{1}{3}\right)^9$ (c) $\left(\frac{1}{3}+\frac{2}{3}\right)^9$ (d) $\left(\frac{1}{2}+\frac{3}{2}\right)^9$

Q.33. Find the Binomial Distribution, if the Mean is 9 and Standard Deviation is 3/2.
(a) $\left(\frac{1}{2}+\frac{1}{2}\right)^{12}$ (b) $\left(\frac{3}{4}+\frac{1}{4}\right)^{12}$ (c) $\left(\frac{1}{4}+\frac{3}{4}\right)^{12}$ (d) None

Q.34. If the sum of the Mean and the Variance of a Binomial Distribution is 4.8 and the number of trial is 5, then find the Binomial Distribution.
(a) $\left(\frac{1}{3}+\frac{2}{3}\right)^5$ (b) $\left(\frac{1}{5}+\frac{4}{5}\right)^5$ (c) $\left(\frac{4}{5}+\frac{1}{5}\right)^5$ (d) None

Q.35. The sum of the Mean and the Variance of a binomial distribution is 24. The product of Mean and Variance is 128. Find the Binomial Distribution.
(a) $\left(\frac{1}{3}+\frac{2}{3}\right)^{20}$ (b) $\left(\frac{1}{2}+\frac{1}{2}\right)^{16}$ (c) $\left(\frac{1}{2}+\frac{1}{2}\right)^{32}$ (d) None

Q.36. An experiment succeed twice as offen as it fails, then what is the probability that in the next five trial there will be exactly 3 success?
(a) 0.329 (b) 0.293 (c) 0.923 (d) None

Q.37. If the 10 coins are tossed 100 times, then how many times would you expect that exactly 7 coins to show Head.
(a) 11 (b) 12 (c) 11.71 (d) None

Q.38. Suppose A and B are the two equally strong tennis players, then which of the two events is more probable.
Event 1:- Player A beats the player B in the three matches out of four.
Event 2:- Player A beats the player B exactly in the five matches out of eight.
(a) First Event (b) Second Event (c) Both Events (d) None

Q.39. In a binomial distribution n = 6 and $4 \times P(x = 4) = P(x = 2)$
then find the probability of success (p).
(a) 1/2 (b) 2/3 (c) 1/3 (d) None

Q.40. In a binomial distribution, n = 6 and $P(x = 2) = 9 \times P(x = 4)$. Then find the value of p and q
(a) 1/4 , 3/4 (b) 2/3, 1/2 (c) 1/3, 2/3 (d) None

MISCELLANEOUS QUESTIONS OF BINOMIAL DISTRIBUTION

Q.41. The Mean of a Binomial Distribution with parameter n and p is :-
(a) n(1– p) (b) np(1– p) (c) np (d) $\sqrt{np(1-p)}$

Q.42. When p = 0.5, the Binomial distribution is :-
(a) Asymmetrical (b) Symmetrical
(c) Both (a) and (b) (d) None of these

Q.43. When p is larger than 0.5, then Binomial distribution is :-
(a) asymmetrical (b) symmetrical
(c) both (d) none of these

Q.44. Mean of Binomial distribution is :-
(a) npq (b) np (c) both (d) none of these

Q.45. Variance of Binomial distribution is :-
(a) npq (b) np (c) both (d) none of these

Q.46. The S.D. of a Binomial distribution with parameters n and p is :-
(a) $\sqrt{\mathrm{np}(1-\mathrm{p})}$ (b) $np^2 (1– p)$ (c) nq (1– q) (d) $n^2 q^2(1– p)$

Q.47. Standard Deviation of Binomial distribution is :-
(a) Square of npq (b) Square root of npq
(c) Square of np (d) Square root of np

Q.48. A Dice is tossed 500 times. The S.D. of the number of 'Sixes' in these 500 tosses is
(a) 50/6 (b) 5/6 (c) 2/3 (d) none of these

Q.49. 6 unbiased Coins are tossed. The Probability of getting At least 4 Heads is :-
(a) 7/32 (b) 11/32 (c) 27/32 (d) 21/32.

Q.50. The Probability of getting exactly 4 Heads in 6 tosses of a Coin is :-
(a) 15/60 (b) 15/64 (c) 13/64 (d) 10/64

Q.51. In a Binomial Distribution, the Mean is 20, Standard Deviation is $\sqrt{15}$, then p =
(a) 3/4 (b) 1/4 (c) 1/2 (d) 1/3

Q.52. If the Mean of a Binomial Distribution with 9 trials is 6, then its Variances is :-
(a) 2 (b) 3 (c) 4 (d) $\sqrt{2}$

Q.53. For a Binomial Distribution n = 10, q = 0.4, then its Mean =
(a) 1 (b) -1 (c) 6 (d) 10

Q.54. For a Binomial Variate X, If n = 5, and P(X = 1) = 8 P(X = 3), then P =
(a) 4/5 (b) 1/5 (c) 1/3 (d) 2/3

Q.55. X denotes the number of sixes in 4 consecutive throws of a Dice, then P(X= 4) is:-
(a) 1/1296 (b) 4/6 (c) 1 (d) 1295/1296.

Q.56. A Coin is tossed 8 times. The Probability of getting a Head three times, is:-
(a) 7/16 (b) 7/64 (c) 7/128 (d) 7/32

Q.57. The Mean number of 'sixes' in two tosses of dice, is:-
(a) 1/2 (b) 1/3 (c) 1/4 (d) 1

Q.58. Out of 128 families with 4 Children each, how many are expected to have at least one boy?
(a) 100 (b) 105 (c) 108 (d) 120

Q.59. What is the probability of making 3 correct guesses in 5 true-false answer type questions?
(a) 0.3125 (b) 0.5676 (c) 0.6875 (d) 0.4325

Note :- Mode of Binomial distribution

(i) When the value of (n + 1)p in a binomial distribution is an integer then the binomial distribution is bimodal and the modal value are –
Mode (Z) = (n+1)p & Mode (Z) = (n+1)p –1

(ii) When (n +1)p is not an integer (fraction) then the binomial distribution is uni-modal and the modal value will be the integral part of the (n +1)p.

POISSON DISTRIBUTION

Q.60. A book of 585 pages contains 43 printing mistake words. What is the probability that 10 pages selected at random will be : $[Given\ e^{-0.735} = 0.4795]$
(i) Find the Mean of the Poisson Distribution (ii) Free from error.
(iii) What is the probability of exactly two printing mistake words (iv) At least one printing mistake
(a) 0.635, 0.4795, 0.1235, 0.7205 (b) 0.735, 0.4795, 0.1295, 0.5205
(c) 0.732, 0.4715, 0.1295, 0.5201 (d) None

Q.61. The mortality rate for a certain disease is 7 in 1000. What is the probability for just 2 death on account of this disease in a group of 400. $[Given\ e^{-2.8} = 0.06]$
(a) 0.2352 (b) 0.2564 (c) 0.2395 (d) None

Q.62. It is given that 3% of the Electric Bulb manufactured by a company are defective then find the probability that a sample of 100 bulb will contain :-
(i) No defective bulb (ii) Exactly 2 defective bulb (iii) At least 2 defective bulb
(a) 0.0408, 0.2286, 0.2968 (b) 0.0518, 0.2286, 0.7368
(c) 0.0508, 0.2286, 0.7968 (d) None

Q.63. A Car agency has two cars which it hires out days by day. The number of car demanded on each day is distributed as Poisson distribution with mean 1.5, then calculate the proportion of days on which 'no car is used' and also calculate number of days on which 'some demand is refused'.
$[Given\ e^{-1.5} = 0.2231]$
(a) 0.2231, 0.1913 (b) 0.1231, 0.1923 (c) 0.2251, 0.1513 (d) None

Q.64. Excel office receives the telephone calls @ of 3 calls per minute on an average, then what is the probability of receiving :- (i) No call in a given minute.
(ii) Exactly two calls in a given minute (iii) At most two calls in a given minute
(a) 0.0507, 0.2286, 0.4318 (b) 0.0508, 0.2286, 0.4318
(c) 0.0108, 0.2286, 0.1318 (d) None

Q.65. In a Poisson distribution, $3 \times P(x = 2) = P(x = 4)$ then, find P(x = 3)
(a) 0.09644 (b) 0.05971 (c) 0.09292 (d) None

Q.66. In a Poisson distribution, $P(x = 2) = 9\ [P(x = 4)] + 90 \times P(x = 6)$, find Mean.
(a) 1 (b) – 1 (c) $\pm$ 1 (d) None

Q.67. The average number of customers who appear at a counter of a bank per minute is 2. Find the probability that during a given minute. (i) No customer appear (ii) At least 3 customers appear.
(a) 0.1271, 0.3141 (b) 0.1371, 0.3143 (c) 0.1341, 0.1143 (d) None

MISCELLANEOUS QUESTIONS OF POISSON DISTRIBUTION

Q.68. If a random variable X follows Poisson distribution, such that P(X = 1) = P(X = 2), then the Mean of the distribution is:-
(a) 1 (b) 2 (c) 3 (d) 4

Q.69. Ten percent of the tools produced in a certain manufacturing process turn out to be defective. The probability that in a sample of 10 tools chosen at random, exactly two will be defective by using the Poisson approximation is:-
(a) 0.184 (b) 0.174 (c) 0.284 (d) 0.384

Q.70. If a random variable X has a Poisson Distribution, such that P(X = 1) = P(X = 2), its Mean and Variance are:-
(a) 1, 1 (b) 2, 2 (c) 2, 3 (d) 2, 4

Q.71. In a Poisson distribution of P(X = 0) = P(X = 1) = K, the value of K is:-
(a) 0 (b) 1/e (c) e (d) None

Q.72. If X is a Poisson variate with P(X = 0) = P(X = 1), then P(x =2)?
(a) e/2 (b) e/6 (c) 1/(6e) (d) 1/(2e)

Q.73. Mean of a Poisson distribution is 1/2, then the ratio of P(X = 3) to P(X = 2) is:-
(a) 1 : 2 (b) 1 : 4 (c) 1 : 6 (d) 1 : 8

Q.74. If the Mean of a Poisson's distribution is λ, then its Standard Deviation is :-
(a) λ (b) $\lambda 2$ (c) $\sqrt{\lambda}$ (d) $1/\lambda$

Q.75. If the Mean of a Poisson variable X is 1, what is P (X = at least one)?
(a) 0.456 (b) 0.821 (c) 0.632 (d) 0.256

Q.76. The standard deviation of x, where x is a Poisson variate satisfying the condition P(X = 2) = P (X = 3) is:-
(a) $\sqrt{3}$ (b) 2 (c) $\sqrt{5}$ (d) 1.5

Note:- Mode of Poisson distribution

(i) When the mean (m) of the Poisson distribution is an integer, then the Poisson distribution is bimodal, it means there are two modes which are -
Mode (Z) = m & Mode (Z) = m–1

(ii) When the mean (λ) of the Poisson distribution is not an integer or it is in fractional form then the Poisson distribution is uni-modal and the modal value a will be integral part of mean. e.g. :- If mean = 2.9 then Z = 2, If mean = 3.1 then Z = 3.

NOTE :–

(01) A Normal distribution is symmetrical distribution.

(02) The shape of normal distribution curve is bell shape.

(03) A normal distribution is a limiting form of Binomial distribution

(04) The conditions for normal distribution are n → ∞ & p and q are nearly equal.

(05) If the mean of normal distribution is μ and S.D. is, then the probability density function is

f(x) = $\frac{1}{\sigma\sqrt{2\pi}} \times e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $(-\infty < x < +\infty)$

If μ is zero, then f(x) = $\frac{1}{\sigma\sqrt{2\pi}} \times e^{\frac{-x^2}{2\sigma^2}}$

(06) Standard N.D. is also known as Unit Normal Distribution or Z-distribution.

(07) The Standard Normal Curve helps us to find the area within the two boundary conditions under the curve.

(08) The top-point of the normal distribution curve (Bell shape curve) represents Mean.

(09) Because of the normal distribution curve is symmetrical the mean, median and mode coincide it means Mean = Median = Mode.

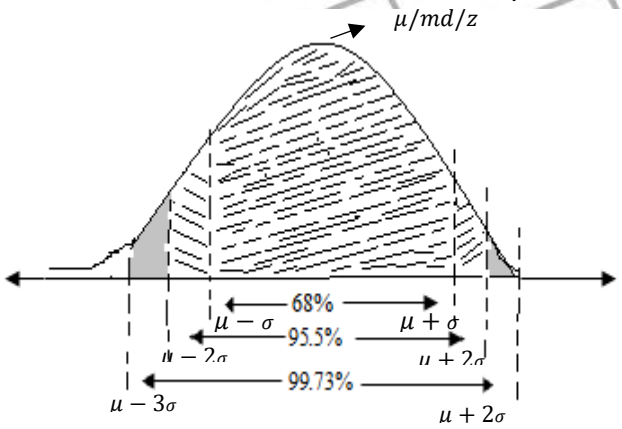
(10) The point of inflexion of the normal curve are x = $\mu + \sigma$ and x = $\mu - \sigma$

(11) About 68% of the area lies between the two point of inflexions that are $\mu + \sigma$ & $\mu - \sigma$

(12) About 95.5% area lies between $\mu - 2\sigma$ and $\mu + 2\sigma$

(13) About 99.73% of the area lies between $\mu - 3\sigma$ and $\mu + 3\sigma$

(14) The total area under the normal curve is equal to unity i.e. 100% area.



(15) For the Poisson Distribution, Variance = μ_2 , Fourth moment (μ_4) = 3. μ_2

(16) For the normal distribution –

(i) QD : MD : SD = 10 : 12 : 15

(ii) $Q_1 = \mu - 0.675\sigma$ & $Q_3 = \mu + 0.675\sigma$

So from (i) + (ii) ⇨ $Q_1 + Q_3 = 2\mu$ ⇨ $\mu = \frac{Q_1 + Q_3}{2}$

(17) A normal distribution curve has a single peak point, so it is a Uni-modal (there is only one mode)

(18) Two tails of the normal curve extend infinitely and never touches the horizontal line i.e. x – axis.

(19) If the mean of x variable is μ_1 and the standard Deviation is σ_1. Similarly the mean of y variable is μ_2 and the S.D. is σ_2 & if Z = x + y follows the normal distribution then the mean of Z will be $\mu_1 + \mu_2$ and SD of Z = $\sqrt{\sigma_1^2 + \sigma_2^2}$

Or

If $x \to N(\mu_1, \sigma_1)$, $y \to N(\mu_2, \sigma_2)$ then Z = (x + y) → $N\left[\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2}\right]$

(20) If the population size is N and sample size is n then the total number of samples that can be taken without replacement is N_{C_n} and with replacement is N^n

Q.1. For the normal distribution with the probability density function $f(x) = ke^{\frac{-(x^2-6x+9)}{8}}$, the mean and S.D. are
(a) (2, 3) (b) (3, 2) (c) (4, 3) (d) (3, 4)

Q.2. The Coefficient of Variation of the probability density function given by $f(x) = \frac{1}{6\sqrt{25}} e^{\frac{-(x-24)^2}{72}}$, $-\infty < x < \infty$ is :-
(a) 50 (b) 25 (c) 24 (d) 76

Q.3. Parameters of the normal distribution $f(x) = \frac{1}{\sqrt{72\pi}} e^{\frac{-(x-10)^2}{72}}$, $-\infty < x < \infty$
(a) (10, 6) (b) (10, 36) (c) (6, 10) (d) (36, 10)

Q.4. The mean deviation of the normal distribution $f(x) = \frac{1}{\sqrt{50\pi}} e^{\frac{-(x^2-30x+225)}{50}}$, is
(a) 4 (b) 5 (c) 10 (d) 8

Q.5. Find Q.D. of the normal distribution $f(x) = \frac{1}{\sqrt{18\pi}} e^{\frac{-(x-10)^2}{18}}$, $-\infty < x < \infty$
(a) 3 (b) $\frac{4}{3}$ (c) 2 (d) $\frac{3}{4}$

Q.6. The mean deviation of the normal distribution is 16. The quartile deviation is
(a) $\frac{40}{3}$ (b) $\frac{20}{3}$ (c) $\frac{100}{3}$ (d) $\frac{50}{3}$

Q.7. If the QD is 18 , then MD is nearly
(a) 10.8 (b) 5.4 (c) 21.6 (d) None

Q.8. If the 1st quartile and mean deviations of a normal distributions are 11.25 and 8 respectively, then the mean of the distribution is
(a) 18 (b) 20 (c) 14 (d) 16

Q.9. If the 1st and 3rd quartiles of a normal distribution are 14.6 and 25.4 respectively, the mean is?
(a) 20 (b) 35 (c) 5.4 (d) 10.8

Q.10. Find the Points of inflexion of the ND $f(x) = \frac{1}{\sqrt{72\pi}} e^{\frac{-(x-18)^2}{72}}$, $-\infty < x < \infty$
(a) (24, 12) (b) (12, 24) (c) (6, 12) (d) (12, 6)

Q.11. The Mean and Variance of a Normal Distribution are 24 and 9 respectively. Then Q_3 is:-
(a) 18 (b) 22 (c) 26 (d) 30

Q.12. If x and y are two independent normal random variables with Mean and SD's (10, 5) and (15, 12), then find the Mean and SD of (x + y)
(a) (27, 15) (b) (10, 27) (c) (25, 13) (d) (12, 25)

Q.13. If x and y are two independent normal random variables with means 18 & 10 and SD's 3 & 4 respectively, then the mean and SD of x – y is
(a) (20, 7) (b) $(8, -1)$ (c) (8, 1) (d) (8, 5)

Q.14. If the points of inflexion of a normal distribution are 18 and 24 then SD of the distribution is
(a) 4 (b) 6 (c) 3 (d) 8

Q.15. In a normal distribution Q_1 is 25 and SD is 6, the median is
(a) 23 (b) 28 (c) 29 (d) 26

Q.16. For the normal distribution $f(x) = \frac{5}{\sqrt{\pi}} e^{-25x^2}, -\infty < x < \infty$. Mean and SD are
(a) $\left(0, \frac{5}{2}\right)$ (b) (1, 10) (c) $\left(0, 5\sqrt{2}\right)$ (d) $\left(0, \frac{1}{5\sqrt{2}}\right)$

Answers

Probability Distribution & Expected Value

1	a
2	c
3	a
4	b
5	a
6	c
7	a
8	a
9	c

MISCELLANEOUS QUESTIONS of Probability Distribution & Expected Value

10	b	21	a
11	d	22	a
12	c	23	b
13	a	24	d
14	b	25	a
15	c	26	a
16	b		
17	c		
18	a		
19	d		
20	a		

BINOMIAL DISTRIBUTION

27	b	37	b
28	c	38	a
29	a	39	c
30	b	40	a
31	c		
32	b		
33	c		
34	b		
35	c		
36	a		

MISCELLANEOUS QUESTIONS OF BINOMIAL DISTRIBUTION

41	c	51	b
42	b	52	a
43	a	53	c
44	b	54	b
45	a	55	a
46	a	56	d
47	b	57	b
48	a	58	d
49	b	59	a
50	b		

POISSON DISTRIBUTION

60	b
61	a
62	c
63	a
64	b
65	c
66	a
67	b

MISCELLANEOUS QUESTIONS OF POISSON DISTRIBUTION

68	b
69	a
70	b
71	b
72	d
73	c
74	c
75	c
76	a

NORMAL DISTRIBUTION

1	b	11	c
2	b	12	c
3	b	13	d
4	a	14	c
5	c	15	c
6	a	16	d
7	c		
8	a		
9	a		
10	b		

CHAPTER – 17 CORRELATION AND REGRESSION ANALYSIS

KARL PEARSON's Coefficient Of Correlation (r)

Q.1. Find the Karl person's coefficient of correlation (r).

X	1	2	3	4	5	6	7	8	9	10
Y	2	4	8	7	10	5	14	16	2	20

(a) 0.58 (b) 0.61 (c) 0.66 (d) None

Q. 2. Find the Karl person's coefficient of correlation (r).

X	65	66	67	68	69	70	71
Y	67	68	66	69	72	72	69

(a) 0.67 (b) 0.57 (c) 0.76 (d) None

Q. 3. Find coefficient of correlation (r).

X	-3	-2	-1	0	1	2	3
Y	8	5	1	0	1	5	8

(a) -1 (b) 1 (c) 0 (d) None

Q.4. Find Co-variance between x and y. { find C.O.V.(x, y) }

X	15	20	25	30	40	50
Y	44	43	45	37	34	37

(a) 40.89 (b) 39.16 (c) -39.16 (d) None

Q. 5. Find Co-variance between x and y. {find C.O.V.(x, y) }

X	1	2	3	4	5	6	7	8	9	10
Y	6	9	6	7	8	5	12	3	17	1

(a) 0.5 (b) 0.4 (c) 0.3 (d) None

Q. 6. The AM of X table is 25 and that of Y table is 18, the sum of the product of the deviation of x and y series from their respective means is equal to 122. The sum of the square of the deviation from their respective means are 136 and 138 respectively for x and y series and the no. of pair of observation is 15, then find r.

(a) 0.89 (b) 0.98 (c) 0.798 (d) None

Q. 7. For the 25 pairs of observations, the following results are obtained:

$\sum x = 125$, $\sum x^2 = 650$, $\sum y = 100$, $\sum y^2 = 460$, $\sum xy = 508$

Later an it was found that the two pairs of observations (6, 14) and (8, 6) were wrong data, while the correct pairs are (6, 8) and (8, 12). Find correct value of r.

(a) 0.76 (b) 0.87 (c) 0.67 (d) None

SPEARMAN'S RANK Co-efficient of Correlation (R)

Q. 8. Find Rank Correlation Co-efficient (R) of the given table-

X	1	2	3	4	5	6	7	8	9	10	11	12
Y	12	9	6	10	3	5	4	7	8	2	11	1

(a) – 0.4545 (b) – 0.5455 (c) 0.4545 (d) None

Q.9. Find Rank Correlation Co-efficient (R) of the given table-

X	78	89	97	69	59	79	68	57
Y	125	137	156	112	107	136	123	108

(a) 0.95 (b) 0.82 (c) 0.61 (d) None

Q.10. Find Rank Correlation Co-efficient of the given table- {# Repeated Rank Case}

X	48	33	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

(a) 0.567 (b) 0.675 (c) 0.733 (d) None

Co-efficient of CONCURRENT DEVIATION (r_c)

Q.11. Find the Co-efficient of Concurrent Deviation (r_c).

X	65	40	35	75	63	80	35	20	80	60	50
Y	60	55	50	56	30	70	40	35	80	75	80

(a) 0.8944 (b) 0.9567 (c) 0.7896 (d) None

Q.12. Find Co-efficient of Concurrent Deviation of given table.

X	150	154	160	172	160	165	180
Y	200	180	170	160	190	180	172

(a) 1 (b) -1 (c) 0 (d) None

REGRESSION ANALYSIS

Q.13. Find the value of r , If $b_{yx} = \frac{3}{2}$ and $b_{xy} = \frac{2}{5}$

(a) 0.467 (b) 0.564 (c) 0.774 (d) None

Q.14. Find the value of r, If $b_{yx} = -0.79$ and $b_{xy} = -1.12$

(a) - 0.94 (b) 0.94 (c) 0.875 (d) None

Q.15. Find b_{xy} and b_{yx}

X	4	2	3	4	2
Y	2	3	2	4	4

(a) 0.50, 0.75 (b) - 0.25, - 0.25 (c) 0.25, 0.25 (d) None

Q.16. If the regression equation of y on x is 64 x – 45 y = 24 and the regression equation of x on y is 5 x - y = 22. Find (i) $\overline{X}$ & $\overline{Y}$ (ii) b_{YX} & b_{XY} (iii) r (iv) If the variance of x is 25, then find the S.D. of y.

(a) 6, 8, 60/43, 1/5, 0.53, 30.25 approx (b) 6, 8, 64/45, 1/3, 0.43, 30.25 approx

(c) 6, 8, 64/45, 1/5, 0.53, 13.4 approx (d) None

Q.17. The two regression equation are 3x + 2y = 26 and 6x + y = 31. Find (i) $\overline{x}$ & $\overline{y}$ (ii) r
(a) 4, 7, – 0.499 (b) 4, 7, 0.499 (c) 7, 4, 0.5 (d) None

Q.18. The two regression lines are 5x + 7y = 22 and 6x + 2y = 20 then find (i) r (ii) If the variance of y is 15, then find standard deviation of x.
(a) 0.487, 2.65 (b) – 0.487, 3.65 (c) – 0.487, 2.65 (d) None

Q.19. For the 100 students of a class the regression equation of the marks in Maths (x) on the marks in Law (y) is 3y – 5x + 180 = 0. The mean marks in Law is 50, and the variance of the marks in Maths is 4/9 times of the variance of marks in Law. Find average score of Maths and also find the co-efficient of correlation between the marks of the two subject.
(a) 66, 0.8 (b) 66, 0.9 (c) 76, 0.9 (d) None

Q.20. $\overline{x}$ = 20, $\overline{y}$ = 15, σx = 4, σy = 3, r = 0.7 Then find the value of y, when x = 24
(a) 19.2 (b) 16.8 (c) 17.1 (d) None

Q. 21. If r = 0.99 and x = 12, Then find the value of y is?

Series	X	Y
Average	7.6	14.8
S.D.	3.6	2.5

(a) 16.783 (b) 17.82 (c) 18.675 (d) None

Q.22. If ∑ x = 55, ∑ y = 74 , ∑ xy = 411 , n = 10 , Find Co-variance (x,y)
(a) 0.4 (b) 0.5 (c) 0.45 (d) None

NOTE :- (i) The slope of the regression Equation of y on x is b_{yx}.
(ii) The slope of the regression Equation of x on y is $1/b_{xy}$.
(iii) When two lines are coinciding i.e. θ = 0° or θ = 180° Then r = + 1 or -1 .
(iv) If θ = 90° i.e. two regression lines are perpendicular, then there is no correlation i.e. r = 0.

MISCELLANEOUS QUESTIONS:-

Q.23. Covariance of the data ∑x = 55, ∑y = 79, ∑ x y = 410, n = 10 is:-
(a) -2.45 (b) 3.45 (c) 2.45 (d) None

Q.24. The Cov (X,Y) of the data: (1, 10), (2, 9) (3, 2), (4, 8), (5, 6), (6, 0) is:-
(a) - 4.42 (b) -3.42 (c) 4.51 (d) None

Q.25. Cov (X,Y) of the data: $\overline{x}$ = 30, $\overline{y}$ = 40, ∑(x-$\overline{x}$)(y – $\overline{y}$) = –235, n = 6 is:-
(a) 39.17 (b) -39.17 (c) 39.71 (d) -39.71

Q.26. The Coefficient of correlation was defined by:-
(a) Pascal (b) Laplace (c) Karl Pearson (d) None of these.

Q.27. For the two variables x and y, the Cov (x, y), Var (x) and Var (y) are 40, 16 and 256 respectively. What is the value of the correlation coefficient?
(a) 0.01 (b) 0.625 (c) 0.4 (d) 0.5

Q.28. If r = 0.6, then the coefficient of Non-Determination is:-
(a) 0.4 (b) - 0.6 (c) 0.36 (d) 0.64

Q.29. If the Rank correlation coefficient between marks is management and mathematics for a group of student is 0.6 and the sum of squares of the differences in ranks is 66, what is the number of students in the group?
(a) 10 (b) 9 (c) 8 (d) 11

Q.30. If the sum of squares of difference of ranks, given by two judges A and B, of 8 students is 21, what is the value of rank correlation coefficient ?
(a) 0.7 (b) 0.65 (c) 0.75 (d) 0.8

Q.31. For 10 pairs of observations, the number of Concurrent deviations was found to be 4. What is the value of the coefficient of concurrent deviation?
(a) $\sqrt{0.2}$ (b) $-\sqrt{0.2}$ (c) $-1/3$ (d) None

Q.32. Eight contestants in a musical contest were Ranked by two judges A and B as:-
Rank by judge A :- 7, 6, 2, 4, 5, 3, 1, 8
Rank by judge B :- 5, 4, 6, 3, 8, 2, 1, 7
The Rank Correlating Coefficient is?
(a) 0.65 (b) 0.57 (c) 0.60 (d) None

Q.33. The Coefficient of Correlation $\rho(x,y)$, when $\sum x = 125$, $\sum y = 100$, $\sum x^2 = 650$, $\sum y^2 = 464$, $\sum xy = 508$, $n = 25$ is:-
(a) 0.3 (b) 0.4 (c) 0.2 (d) None

Q.34. The Coefficient of Correlation $\rho(x,y)$, when
$\sum dx = 0$, $\sum dx^2 = 60$, $\sum dy = 9$,
$\sum dy^2 = 69$, $\sum dxdy = 57$, $n = 9$ is:-
(a) 19/20 (b) 18/20 (c) 17/20 (d) None

Q.35. The Coefficient of Correlation r between x and y, when Cov (x, y) = −16.5, Var (x) = 2.89, Var (y) = 100 is:
(a) 0.97 (b) − 0.97 (c) 0.89 (d) None

Q.36. The Coefficient of Correlation between X and Y is 0.6.
Their covariance = 4.8, Var (X) = 9. Then σ_y is:-
(a) 8/3 (b) 3/8 (c) 8/9 (d) none of these.

Q.37. N = 10, C = 7, then the Coefficient of Concurrent Deviation is:-
(a) 0.745 (b) 0.65 (c) − 0.65 (d) None

Q.38. If r is the Correlation Co-efficient, then:-
(a) $r \geq 1$ (b) $r \leq 1$ (c) $|r| \leq 1$ (d) $|r| \geq 1$

Q.39. The Coefficient of Rank Correlation is calculated by the formula:-

(a) $R = 1 - \frac{2\sum D^2}{n^2-n}$ (b) $R = 1 - \frac{3\sum D^2}{n^2-n}$

(c) $R = 1 - \frac{\sum D^2}{n^3-n}$ (d) $R = 1 - \frac{6\sum D^2}{n^3-n}$

Q.40. Coefficient of Correlation between the two variants X and Y is:-

X	1	2	3	4	5
Y	5	4	3	2	1

(a) 0 (b) -1 (c) 1 (d) None of these

Q.41. For the following data, the Coefficient of Rank Correlation is:-

Rank is Botany	1	2	3	4	5
Rank in Chemistry	2	3	1	5	4

(a) 0.6 (b) 0.93 (c) 0.4 (d) none of these

Q.42. The Karl Person's coefficient is:-

X	4	7	8	3	4
Y	5	8	6	3	5

(a) $\frac{63}{\sqrt{94 \times 66}}$ (b) 63 (c) $\frac{63}{\sqrt{94}}$ (d) None

Q.43. If C = 0, N = 7, then Coefficient of Concurrent Deviation is:-

(a) + 1 (b) – 1 (c) + 0.5 (d) – 0.5

Q.44. With usual notation, if C = 9, r_c = 0.89, then the value of N is:-

(a) 10 (b) 9 (c) 8 (d) 11

Q.45. If b_{xy} = (2/5), b_{yx} = (5/8) and θ be the acute angle between the lines, tan θ is:-

(a) 20/4 (b) 30/41 (c) 6/41 (d) None

Q.46. The Regression Coefficients of y on x is 2/3 and of x on y is 4/3. If the acute angle between the regression lines is θ, then tan θ is :-

(a) 2/9 (b) 1/9 (c) 1/18 (d) None of these.

Q.47. If b_{xy}, b_{yx} and r be Regression Coefficients of x on y and y on x and Correlation Coefficient respectively, then:-

(a) $b_{xy}.b_{yx}$ = r (b) $b_{xy}.b_{yx} = -$ r (c) $b_{xy}.b_{yx} = r^2$ (d) None of these.

Q.48. The intersecting point of the two Regression lines is :-

(a) (0, 0) (b) $(\overline{x}, \overline{y})$ (c) (b_{xy}, b_{yx}) (d) None of these

Q.49. If b_{yx}= 0.8, b_{xy}= 0.46, then r is :-

(a) 0.606 (b) 0.51 (c) 0.69 (d) None of these

Q.50. If the Correlation Coefficient between two variables X and Y is 0.4 and the Regression Coefficient of X on Y is 0.2, then Regression Coefficient of Y on X is:-

(a) 0.4 (b) ± 0.8 (c) 0.8 (d) None of these

Q.51. If b_{yx} = – (4/3), b_{xy} = – (1/3), the value of r is:-
(a) (2/3) (b) – (2/3) (c) (1/9) (d) None of these

Q.52. Regression Equation of Y on X is
8x – 10y + 66 = 0 and if σ_x = 3, then Cov (X, Y) is equal to:-
(a) 11.25 (b) 7.2 (c) 2.4 (d) None of these

Q.53. Coefficient of Correlation is:-
(a) G.M. of the Coefficient of Regression.
(b) A.M. of the Coefficient of Regression.
(c) H.M. of the Coefficient of Regression.
(d) Product of G.M. and A.M. of the Regression Coefficients

Q.54. For a bivariate data, the two lines of Regression are 4x – 5y + 33 = 0 and 2x – 9y + 127 = 0. For this data r =
(a) 2/9 (b) 4/5 (c) 5/4 (d) $\sqrt{(10/36)}$

Q.55. Slope of the Regression Equation of y on x is:-
(a) b_{xy} (b) b_{yx} (c) $1/b_{yx}$ (d) $1/b_{xy}$

Q.56. If $\overline{x}$ = 10, $\overline{y}$ = 50, σ_x = 3, σ_y = 15, r = 0.9 then the estimated value of x corresponding to y = 100 is:-
(a) 19 (b) 20 (c) 18 (d) 21

Q.57. If $\overline{x}$ = 15, $\overline{y}$ = 80, σ_x = 2, σ_y = 16, r = 0.75, then the estimated value of y corresponding to x = 25 is:-
(a) 150 (b) 140 (c) 130 (d) 145

Q.58. Given the following data: n = 7, $\sum dx = -15$, $\sum dy = -0.13$, $\sum dxdy = 18$, $\sum dy^2 = 0.89$. The value of b_{xy} is:-
(a) 18.67 (b) 19.9658 (c) 20.67 (d) 29.6

Q.59. Given data:- $\sum(x-\overline{x})(y-\overline{y}) = 3900$, $\sum(x-\overline{x})^2 = 6360$, $\sum(y-\overline{y})^2 = 2668$, then the Regression coefficient b_{yx} is :-
(a) 0.613 (b) 0.363 (c) 1.363 (d) None

Q.60. Given the following data of a bivariate distribution:-
b_{xy} = 1.36, b_{yx} = 0.613, find Coefficient of Determination
(a) 0.634 (b) 0.834 (c) 0.734 (d) None

Q.61. Given data for a bivariate distribution are: b_{yx} = 0.756, b_{xy} = 0.659. Find Coefficient of Non-Determination
(a) 0.502 (b) 0.402 (c) 0.602 (d) None

Q.62. For the following data σ_x/σ_y = 2/ 3, b_{xy} = 3/5 the Coefficient of Determination is:-
(a) 0.9 (b) 0.79 (c) 0.81 (d) None

Q.63. The two lines of Regression are x + 2y = 7 and 2x + y = 7. The Regression Equation of Y on X is :-
(a) 2x + y = 7 (b) x + 2y = 7 (c) x + 2y = 0 (d) None

Q.64. Likely production corresponding to a rainfall of 40 cms is? Given r = 0.8 and

	Rainfall (in cms)	Qly Output (in quintals)
Average	30	50
S.D.	5	10

(a) 56 (b) 66 (c) 76 (d) 46.6.

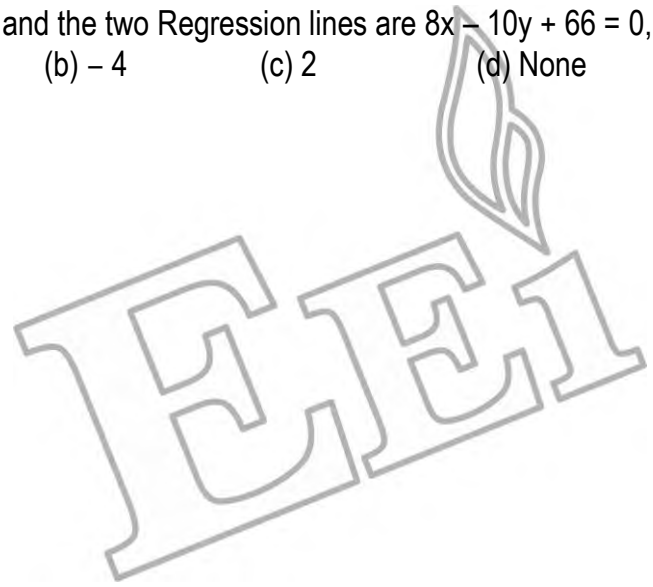
Q.65. The Regression Coefficient b_{xy} for the following data:-
(5, 2), (7, 4), (8, 3), (4, 2), (6, 4)
(a) 0.5 (b) 0.6 (c) 0.7 (d) 1

Q.66. The Coefficient of Correlation between x and y for the following data is:-

X	1	3	4	5	7
Y	3	7	9	11	15

(a) 2 (b) 3 (c) 1 (d) None

Q.67. Given $\sigma_x = 3$, and the two Regression lines are $8x - 10y + 66 = 0$, and $40x - 18y = 214$, then $\sigma_y = ?$
(a) 4 (b) – 4 (c) 2 (d) None



Answers

KARL PEARSON's Coefficient Of Correlation (r)

1	b
2	a
3	c
4	c
5	b
6	a
7	c

SPEARMAN'S RANK Co-efficient of Correlation (R)

8	a
9	a
10	c

Co-efficient of CONCURRENT DEVIATION (r_c)

11	a
12	b

REGRESSION ANALYSIS

13	c
14	a
15	b
16	c
17	a
18	c
19	b
20	c
21	b
22	a

MISCELLANEOUS QUESTIONS:-

23	a	33	c	43	b	53	a	63	b
24	a	34	a	44	d	54	d	64	b
25	b	35	b	45	b	55	b	65	d
26	c	36	a	46	c	56	a	66	c
27	b	37	a	47	c	57	b	67	a
28	d	38	c	48	b	58	b		
29	a	39	d	49	a	59	a		
30	c	40	b	50	c	60	b		
31	c	41	a	51	b	61	a		
32	b	42	a	52	b	62	c		

Price Relative / Index Number = $\frac{P_1}{P_0} \times 100$

Where, P_1 = price of current year
P_0 = price of base year

Q.01. Find the price relative of the given table, taking 1998 as the base year.

Year	1998	1999	2000	2001	2002	2003	2004	2005
Price	6	7	7	8	10	14	12	13

Price Relative $[P_1/P_0 \times 100]$

$\frac{6}{6} \times 100 = 100 \Rightarrow \frac{7}{6} \times 100 = 116.67 \Rightarrow \frac{7}{6} \times 100 = 116.67 \Rightarrow \frac{8}{6} \times 100 = 133.34$

$\frac{10}{6} \times 100 = 166.67 \Rightarrow \frac{14}{6} \times 100 = 233.34 \Rightarrow \frac{12}{6} \times 100 = 200 \Rightarrow \frac{13}{6} \times 100 = 216.67$

Q.02. Find link relative or chain index of the given table-

Year	1998	1999	2000	2001	2002	2003	2004	2005
Price	6	7	7	8	10	14	12	13

[Chain index]

$\frac{6}{6} \times 100 = 100 \Rightarrow \frac{7}{6} \times 100 = 116.67 \Rightarrow \frac{7}{7} \times 100 = 100 \Rightarrow \frac{8}{7} \times 100 = 114.2$

$\frac{10}{8} \times 100 = 125 \Rightarrow \frac{14}{10} \times 100 = 14 \Rightarrow \frac{12}{14} \times 100 = 85.71 \Rightarrow \frac{13}{12} \times 100 = 108.33$

Types of Index Number

(i) Price Index Number
(ii) Value Index Number
(iii) Quantitative Index Number
(iv) Special purpose Index Number

(I) PRICE INDEX

(1) Laspeyre's Index Number – Gave importance to the quantity of base year

$$P_L = \frac{\sum P_1 q_0}{\sum P_0 q_0} \times 100$$

(2) Possache's Price Index – Gave importance to quantity of current year

$$P_P = \frac{\sum P_1 q_1}{\sum P_0 q_1} \times 100$$

(3) Dorbish and Bowley's Price Index –

$$P_{DLB} = \frac{\left[\frac{\sum P_1 q_0}{\sum P_0 q_0} + \frac{\sum P_1 q_1}{\sum P_0 q_1}\right]}{2} \times 100$$

Dorbish and Bowley is the Arithmetic Mean of Laspeyre's and passache's Index.

$$P_{DLB} = \frac{P_L + P_B}{2} \times 100$$

(4) Fisher's Ideal Price Index –

$$P_F = \sqrt{\frac{\sum P_1 q_0}{\sum P_0 q_0} \times \frac{\sum P_1 q_1}{\sum P_0 q_1}} \times 100$$

"Fisher's ideal price index is the GM of Laspeyer's and passache's Index."

$$P_F = \sqrt{P_L \times P_P} \times 100$$

(5) Marshall – Edgewarth Price Index –

$$P_{ME} = \frac{\frac{[\sum p_1q_0 + \sum p_1q_1]}{2}}{\frac{[\sum p_0q_0 + \sum p_0q_1]}{2}} \times 100 = \frac{\sum p_1q_0 + \sum p_1q_1}{\sum p_0q_0 + \sum p_0q_1} \times 100$$

(6) Walsch price Index

$$P_W = \frac{\sqrt{\sum p_1q_0 + \sum p_1q_1}}{\sqrt{p_0q_0 + \sum p_0q_1}} \times 100 = \frac{\sum p_1\sqrt{q_0q_1}}{\sum p_0\sqrt{q_0}q_1} \times 100$$

(7) Kelly's Method

$P_K = \frac{\sum P_1 q}{\sum P_0 q} \times 100$ Where, q = $\frac{q_0 + q_1}{2}$

For quantitative index formulas of price index will be used instead of p, q will be used and instead of q, p will be used, 1 and 0 will be same.

Q.03. Find the price index of laspeyre pass ache dorbish & bowley fisher and kelley's price index.

Commodity	2000		2005	
	Price	Qty.		Price
A	15	15	A	15
B	20	5	B	20
C	4	10	C	4

P_0	q_0	P_1	q_1	p_0q_0	p_0q_1	p_1q_0	p_1q_1
15	15	22	12	225	180	330	264
20	5	27	4	100	80	135	108
4	10	7	5	40	20	70	35
				365	280	535	407

Laspeyre's Index = $\frac{\sum p_1q_0}{\sum p_0q_0} \times 100 = \frac{535}{365} \times 100$ = 146.57

Passache's Index = $\frac{\sum p_1q_1}{\sum p_1q_0} \times 100 = \frac{407}{280} \times 100$ = 145.35

Dorbish Bowley = $\frac{P_L + P_P}{2} = \frac{146.96+145.35}{2}$ = 145.96

Fisher's Index = $\sqrt{P_L \times P_P} = \sqrt{146.57 \times 145.35}$ = 145.95

Marshall – Edgeworth = $\frac{\sum p_1q_1 + \sum p_1q_1}{\sum p_0q_0 + \sum p_1q_1} \times 100$ = 535 = $\frac{535 + 407}{365 + 280} \times 100$

= $\frac{942}{645} \times 100$ = 146.04

Walsch price Index = $P_w = \frac{\sqrt{\sum p_1q_0 \times \sum p_1q_1}}{\sum p_0q_0 \times \sum p_1q_0} \times 100 = \frac{\sqrt{535 \times 407}}{\sqrt{365 \times 280}} \times 100$

= $\frac{466.63}{319.68} \times 100$ = 145.96

Kelley's Index,

q	p_1q	p_0q
13.5	297	202.5
4.5	121.5	90
7.5	52.5	30
	471	322.5

$P_k = \frac{\sum p_1q}{\sum p_0q} \times 100 = \frac{4 + 1}{322.5} \times 100$ = 146.04

Q.04. Find the fisher's ideal price index of given table.

Commodity	p_1	Total value	p_0	Total value	q_0	q_1	p_0q_0	p_0q_1	p_1q_0	p_1q_1
A	10	560	6	300	50	56	300	336	500	560
B	2	240	2	200	100	120	200	240	200	240
C	6	360	4	240	60	60	240	240	360	360
D	12	288	10	300	30	24	300	240	360	288
E	12	432	8	320	40	36	320	288	480	432
							1360	1344	1900	1880

Fisher's Index = $\sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100 = \sqrt{\frac{1880}{1360} \times \frac{1900}{1344}} \times 100$ =139.79

Quantity Index – $Q_F = \sqrt{\frac{\sum q_1p_0}{\sum q_0p_0} \times \frac{\sum q_1p_1}{\sum q_0 \times p_1}} \times 100$

$= \sqrt{\frac{1344}{1360} \times \frac{1880}{1900}} \times 100 = \sqrt{0.977} \times 100 = 98.88 \times 100 = 98.88$

Test of Adequacy

(i) Unit Test

This test is satisfied by all the Index number formula except "Simple (un weighted) Aggregative Index formula."

Index = $\frac{\sum p_1}{\sum p_0} \times 100$

(ii) Time Reversal Test

The condition of the time reversal test is:

$p_{01} \times p_{10} = 1$

Where, p_{01} = 1 on 0 [Current year on Base year]

p_{10} = 0 on 1 [Base year on Current year]

Note:- The time reversal test is satisfied by –

(i) Marshall – Edgeworth formula
(ii) Fishers Ideal price Index formula
(iii) Walsch formula
(iv) Kelley's formula
(v) Simple Aggregative Index formula
(vi) (without frequency) Simple geometric mean of price relative

Index = $\frac{\text{Gm of } p_1 \text{ table}}{\text{Gm of } p_0 \text{ table}} \times 100$

(vii) (with frequency) Weighted GM of price relative.

(iii) Factor Reversal Test –

The condition of the factor reversal test is

$P_{01} \times Q_{01}$ = Value Index

$P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_o}$

Note:- The factor reversal testis satisfied by only **"Fisher's Ideal price Index formula."**

(iv) Circular Test

The condition of the circular Test is –

$P_{01} \times P_{12} = P_{20} = 1$

"The circular test is an extension form of time reversal test."

The circular test is not satisfied by laspeyre, passache, and Fisher.

C.P.I [Consumer price Index] Cost of Living Index.

CPI = $\frac{\sum p_1 q_0}{\sum p_0 q_o} \times 100$

Industrial production Index Number –

IPI = $\frac{\sum IW}{\sum W}$ Where, I = $\frac{q_1}{q_0}$

W = Relative importance of different output

Income Index Number –

II = $\frac{\text{Income oc Corrent year}}{\text{Income of base year}} \times 100$

Q.05. The monthly income of an employee was ₹ 8000 in 2014. The consumer price index in 2014 160. If the CPI in the year 2017 is ₹ 200 if he has to be rightly compensate then the additional dearness alliance to be paid to him in 2017.

Sol:- Salary

Index 160 → = ₹ 8000

Index 200 → = $\frac{8000}{160} \times 200$ ₹ 10,000

Additional dearness alliance = 10,000 – 8000 = ₹ 2000

Q.06. $\sum W \log P = 199.49$

Food	Rent	Cloth	Fuel	Other
30%	15%	20%	10%	25%

$\sum PW = 10101.5$

Sol:- Index = $\frac{\sum PW}{\sum W} = \frac{10101.5}{100} = 101.015$

Q.07. What are the merits and demerits of graphic /free hand curve method?

2004		2005
(i) Wheat ₹16 per 20 kg	→	0 per 20 kg
⇨ $\frac{16}{20}$ per kg.	→	⇨ $\frac{20}{20}$ per kg.
(ii) Cloth ₹ 2 m	→	₹ x per meter
(iii) House Rent = ₹ 30	→	Increase by15⇨₹45
(iv) Other ₹ 10	→	doubled ⇨ ₹ 20

$\left[\frac{\frac{20}{20}}{\frac{16}{20}} + \frac{x}{2} + \frac{45}{30} \times \frac{20}{10}\right] \times 100 = 640$

$\frac{5}{4} + \frac{x}{2} + \frac{3}{2} + \frac{2}{7} = 6.4$

5 + 2x + 6 + 8 = 25.6

2x = 25.6 – 19 ⇨ x = 6.6 ⇨ x = 3.3

Hence the increase in price of cloth from 2004 to 2005 is ₹ 3.3 ₹ 2 = 1.3